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CENSUS OF INDIA 1961

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ASSAM

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E. H. PAKYNTEIN

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Superintendents, that the Census provided a unique opportunity for conducting and documenting a survey of this kind. As such a survey was quite outside the usual terms of reference of Census work it was thought prudent cautiously to feel one's way with the thin end of the wedge of what would, it was hoped, prove to be an exciting pursuit. It was therefore considered the wiser course to wait until the State Census Offices felt so interested that they would no longer take the inquiry as an imposition but rather want to do it on their own and ask for the necessary staff and equipment. This office too, in its turn, could make use of the interval to organise and elaborate the design of inquiry in order to feed the appetite that work in progress would serve to whet. Because it was a labour of love, sought to be unobtrusively thrust on one's colleagues and because the inquiry itself was so vast that normally it would demand in any country as big a set-up, if separately established, as the Census organisation itself and that over a much longer period, and because it was almost a pioneer venture, nothing like it having been undertaken since the 1880's, it was decided to move towards a build-up by stages, to let the inquiry unfold itself only as fast as my colleagues chose to ask for more.

Thus, in the first circular of 18 February 1960, it was suggested that the inquiry might be conducted through the agency of the Development Department, the State Director of Industries, the Director of Tribal Welfare, the Registrar of Co-operative Societies, and other organisations concerned with the promotion of household industry. A draft questionnaire containing 30 questions in three parts was recommended for canvassing. It was suggested that information on this questionnaire, village by village and area by area, might either be obtained through the regular departmental channels of the State Government, or through the newly set up Census organisation, or through the hierarchy of the newly-created Panchayets. Stress was laid on the need of photographic documentation and illustration of designs, shapes and forms not only by photographs but with the help of line drawings or sketches together with a full description of the materials used.

Almost the whole of 1960 and the first half of 1961 were spent in organising and taking the census count, although several States even during this period had not allowed the grass to grow under their feet but made exploratory studies and decided in their minds how the inquiry should be organised. A series of regional conferences held in Trivandrum, Darjeeling and Srinagar in May and June 1961 revealed much enthusiasm among State Superintendents to proceed with the survey, but the need of separate staff and equipment was felt at the same time as the realization dawned that this was much too serious an inquiry to be treated casually and left to be achieved through the usual administrative channels and State Census Superintendents proceeded to augment their staff with qualified research and investigating officers, technical persons, photographers, artists, draughtsmen and other trained personnel.

This was followed by rapid progress in coordination between the Central and State Census offices in the matter of exchange and processing of information, documentation and investigation, of assisting each other with trained investigators and in editing and finalizing drafts, layouts, presentations.

Mention has been made of a questionnaire in three parts and thirty questions. The idea was to make a beginning with empirical, analytical studies based on a structured questionnaire which would replace general descriptive accounts that had obtained so far. The primary aim was to obtain a picture as much of the artisan himself as of his craft, to obtain a perspective of the artisan and his craft in his social and economic setting, the extent to which tradition bound him and the winds of change ruffled him, the extent of his mobility and immobility, the conditions of market, credit, new contacts and designs in which he operated, the frame of new as well as traditional producer-customer relationships in which he still worked, and how far he was ready to pierce his own caste-tribe socio-economic cocoon and make a break through to new opportunities promised by the Five Year Plans. The aim was to hold up the mirror to hereditary skills struggling with the dialectics of tradition and change.

Thus the first part of the questionnaire, purporting to be a village schedule, sought to take account of the size and population of the village, its remoteness from or proximity to centres of trade and commerce, in short, the degree of isolation in which the artisan worked, and the relative strengths of various communities in the village which would afford clues to social interdependence and the prevalence of the *jajmani* system. The second part was devoted to artisan communities in the village: the several castes of artisans, the number of families in each, the total number of workers, males and females, the extent of cooperative activity among them, the extent of dependence upon employers and of wage or contract labour. There were questions on the raw materials used, the means of their procurement, the possible extent of dependence on others for raw materials, the extent of the material that artisans can handle within the limits of their skill. There were other questions on the exchange and flow of designs, the use of colours, the ancientness of the craft and legends associated, the colonization of the craftsman, on patrons and customers and on social and economic contact with the world inside and outside the village. There were specific questions on the workshop itself and particularly the tools and the source of supply of these tools, because it was felt that tools decide everything and are the surest index of inertness or flexibility. Separate blocks of questions were designed to bring out the ramifications of artisan castes throughout the country and the ways they sustained themselves, the type of clientele they catered for, the extent to which they operated on money or barter or service, how specialized their craft was, how wide the market, how dependent they were on their socially preordained clientele and how restricted the latter was by the seemingly unalterable laws of social custom; the extent to which they could operate in the open market, the range of their wares and the sizes to which these were ordinarily restricted either by the limits of their own skill or the length of their customers' pursestrings. Inquiries were to be made about the operation of middlemen and of co-operative societies, the people who gave new designs and demanded new products. Finally the several stages of production of the articles themselves were to be fully described including the final and finishing stage and a list of very skilled craftsmen of each community was to be furnished. The third part was devoted specially to tribal communities and designed

to find out how self-sufficient or dependent they were on the production and supply of manufactured goods, the extent to which they produced themselves or depended on others, their contacts with other communities and the specific forms of production and commerce through which these contacts were maintained.

Particular emphasis was laid on the need of obtaining as full an account as possible of unique regional design differentiations as they reflect not only the very culture patterns of the country but the persistent inventive faculties of the craftsmen. The importance was emphasised of giving full attention to articles of domestic use as it is in their shapes, designs and forms that the culture patterns and traditional skills persist most tenaciously.

Simultaneously with the investigation of specific crafts, State Superintendents proceeded to compile a comprehensive list of all types of handicrafts obtaining in their State. As for the specific crafts to be investigated several tables were devised from the structured questionnaire in order to guide investigators toward pointed observation and analysis, to enable them to write, not just general descriptions, but with their eye on the object and on facts.

Investigations conducted between September 1961 and May 1962, including a study group of all States and the Social Studies Division in December 1961 at Delhi, stimulated many of the States into going in for a much enlarged schedule. The revised village schedule itself, the counterpart of the first part of the February 1960 schedule, contained 19 large sections containing elaborate and probing questions. The Family Schedule for practising artisan families similarly contained 19 main questions each subdivided into many questions. The Family Schedule for non-practising artisan families contained 21 questions. There were schedules for the study of co-operative societies, of production-cum-training centres, and of consumer's preference. This enlarged schedule of investigation, in the formulation of which the states themselves actively assisted, was greatly welcomed. The surveys that will appear in this series will therefore consist of two main types: (a) those based on the original short schedule and (b) those based on the much enlarged schedule. In some cases Census Superintendents felt enthused enough to scrap the work based on the original short schedule and do it over again on the enlarged schedule. In the meantime much experience was gained on the analysis of facts and figures to clothe each observation with plenty of authentic information so that the reader could make his own judgement instead of being expected to see all the time through another pair of eyes.

This programme of survey of handicrafts and household industries has been fortified by several ancillary surveys, each one of which would deserve major attention. Along with the survey a compilation has been made of all handicraft centres in each state and an inventory prepared of skilled craftsmen. Photographic and other documentation has been built up to constitute what may now be regarded as the most considerable repository in the country. Elaborate and accurate maps of craft centres in taluks, tehsils and districts are either ready or under preparation. A full census of all fairs and festivals, weekly hats and markets, throughout India, has been taken and is being published for the first time.

Andhra Pradesh has embarked upon a project of chronicling the social and religious antiquity and uniqueness of every fair and festival. A separate volume will be devoted to each district which promises to be of the utmost value to sociologists and orientalists. A full and complete inventory, replete with sketches and measurements of every object, has been prepared of exhibits in museums of tribal crafts in India. There has been a fairly satisfactory survey of houses and buildings, indigenous architectural designs and use of local building materials of the whole country. All this has been entirely a labour of love, patiently organised and executed under great strain and in disregard of health and comfort, for which I take this opportunity of expressing my appreciation and grateful thanks to my colleagues.

New Delhi
July 30, 1964

Asok Mitra,
Registrar General, India

PREFACE

With a view to promote the growth of small industries and handicrafts, the national planners established the following boards during the First Plan :

1. The Khadi & Village Industries Board,
2. The All-India Handicrafts Board,
3. The All-India Handloom Board,
4. The Central Silk Board,
5. The Coir Board and
6. The Small Industries Board.

The 1961 Census introduced a special series of questions on household industry, input of family and hired labour and the periods over which household industry is conducted. It was also necessary to give a proper description of what was produced and where, and to make assessment of the limits of rigidity within which traditional skill operates. This was sought to be obtained by studying the caste, occupational, social and economic stratifications, the limitations of credit and marketing facilities, the dominance of custom over contract and such other pertinent factors.

In February 1960, the Registrar General suggested to the State Census Superintendents that the Census had a unique opportunity for conducting and documenting the handicrafts survey. In the first circular of 18 February 1960, it was suggested that the enquiry might be conducted through the agency of the Development Department, the State Director of Industries, the Director of Tribal Welfare, the Registrar of Co-operative Societies and other organisations concerned with the promotion of household industry. A draft questionnaire containing 30 questions in three parts was recommended for canvassing. It was suggested that information on this questionnaire, village by village and area by area, might either be obtained through the regular departmental channels of the State Government or through the newly set-up Census organisation or through the hierarchy of the newly created Panchayets. Stress was laid on the need of photographic documentation and illustration of designs, shapes and forms not only by photographs but with the help of line drawings or sketches together with a full description of the materials used. During 1960 and the first half of 1961, the census count did not allow much scope for the Census Superintendents to do much for the handicrafts survey, but after a series of Regional Conferences held in Trivandrum, Darjeeling and Srinagar in May and June 1961, it was found that separate staff and equipment was necessary and that the State Census Superintendents should proceed to augment their staff with qualified research and investigating officers, technical persons, photographers, artists, draughtsmen and other trained personnel.

In Assam, I contacted the Registrar of Co-operative Societies, the Director of Panchayets, the Director of Industries and the Development Commissioner to find out whether they could help me in making the survey according to the suggestions of the Registrar General. At a meeting held in the office room of the Deputy Development Commissioner to which the Director of Industries also came, the whole matter of survey of Rural Handicrafts was discussed threadbare and it was decided that the survey according to the questionnaire could be done only in development blocks where there were Industries Extension Officers who could utilise the Gram Sevaks for making the survey and fill-

ing in the questionnaires. The questionnaire was slightly modified by me and approved by the Registrar General. Thereafter, I had about 10,000 copies of these Schedules printed and they were given to the Director of Industries for circulating them to the Extension Officers of the various development blocks which were then functioning. After more than a year and a half, the filled-in schedules were received only from 24 blocks.

In June 1962, I appointed two Industries Investigators who were sent to me on deputation by the Director of Industries. The first thing that I did was to ask these two Industries Investigators to examine all the schedules that we had received. It was found after scrutiny that the entries in the schedules were incomplete, insufficient and inaccurate and that it was next to impossible to get any clear-cut idea about the existing crafts of Assam from such wrongly filled-in schedules. It was also found that the tables could not be prepared from these schedules. The next alternative was to extract some information from the Houselists and this had already been done when Table E-III was prepared. Table E-I requires the number of persons employed and not sex-wise. The sex-wise figures can be collected from the Household Schedules, but even then the sex-wise figures could be obtained only in respect of the members of the family working and not of the hired workers. Moreover, it was calculated that if we wanted to extract only these figures from the Household Schedules, a sizable number of sorting staff would be required and my Tabulation Officers estimated that a sum of about Rs.50,000/- would have to be expended for this. Moreover, 20% of the Household Schedules were then in the Central Tabulation Office. I therefore thought that the idea of collecting the sex-wise figures from the Household Schedules should be dropped. It was also not advisable to refer back to the Director of Industries or Block Development Officers because it might take another two or three years to get the informations and even then, the informations might still be incorrect. In view of these facts, I had no other choice except that of compiling the Table-I as decided in the Study Camp held in New Delhi in December 1961, that is, that 'industries having less than 5 workers may be considered on an ad hoc basis as household industries, and as it is not possible to obtain figures of males and females separately from the houselists, only a number of total persons engaged in the industries should be furnished'.

The preparation of monographs did not however present a problem for us in Assam. According to the Registrar General's advice, monographs should relate to selected industries which are important for their artistic or special appeal to the people. Nevertheless, in view of the circumstances stated above, I thought that by trying to attempt too many, we might not able to do well anywhere in respect of the household industries of Assam. I therefore decided that we should start with six industries and see how a good write-up could be written about each of them after making thorough studies of these crafts and industries. Accordingly, I selected the following industries in Assam for specialised study and monograph writing :

1. Handloom Weaving.
2. Sericultural Weaving (Endi & Muga).
3. Bamboo and Cane works including all subsidiaries under these broad heads.

4. Pottery :
   (a) Earthen Vessels.
   (b) Toys and Images.
   (c) Modern Decorative Articles.

5. Blacksmithy.

6. Wood Works :
   (a) Carpentry.
   (b) Wooden toys and utensils
   (c) Huganal and Kharam.

7. Later on, I also decided to have a special survey report of the Silk Industry of Sualkuchi in the Kamrup district.

It may also be noted that although there has been a lot of fanfare about cottage industries in Assam, the fact is that even those industries which used to flourish in the past are now slowly decaying away. It was reported by Dr. B. K. Roy Burman that in Sarthebari, the machine imported for helping

bell metal works could not be used for lack of operators. Similarly, in the umbrella handle-making institute in Dudhnai, no trainee was seen by an officer of the Registrar General's office (Dr. B. K. Roy Burman, Officer on Special Duty) who visited it, although the building is there. It is also reported that bell metals have no market because they are too costly and cannot compete with cheap manufactured articles of better artistic designs. The finish of the bell metals and ivory works in the Barpeta subdivision is also reported to be very poor compared with similar products in other parts of India. We will therefore have to make an evaluation of all the factors about each category of household industry with suggestions as to how the quality and marketing facility can be improved. In other words, while making a case study of each item of cottage industry, we should also discuss it in the light of present day economics apart from the historical aspects which may be of sentimental value only.

My Industries Investigators who had already made a specialised study of these Handicrafts also visited villages where they are mostly manufactured for case studies. After the field studies had been completed, draft monographs were prepared in coordination with the Registrar General's office (Dr. B. K. Roy Burman, Officer on Special Duty, Handicrafts and Social Studies Section) in the matter of exchange and processing of information, documentation and investigation in editing and finalizing drafts layouts and presentations. Table-1 is added to the report and I have used the two reports published by the Director of Statistics in 1957-58 of the field studies he made of the rural crafts in Assam through some selected Industries Officers in the year 1953-54 and I have acknowledged that fact in the Bibliography. My thanks are also due to the Directors of Co-operative Societies, the Block Development Officers and a host of other officers as well as the artisans who have rendered valuable assistance in connection with the survey.

Field Investigators :

1. Shri A. Bhattacharjee (28.5.62—12.7.62)
2. Shri A.M. Saikia (11.6.62-4.2.63)
3. Shri S.C. Nath (13.8.62-28.2.65)
4. Shri A.K. Endow (14.3.63-31.12.65)

Sarvashri S.C. Nath and A.K. Endow conducted case studies in the field and prepared the preliminary drafts while Shri A.M. Saikia conducted field studies and did some spade-work in drafting. Shri A. Bhattacharjee also made some contribution. Shri K.S. Dey, Tabulation Officer scanined the draft and made some corrections. Shri F.B. Lyngdoh, ACS, supervised the work throughout and gave guidance to the Investigators and he also made some corrections in the draft. The final editing and drafting was done by me.

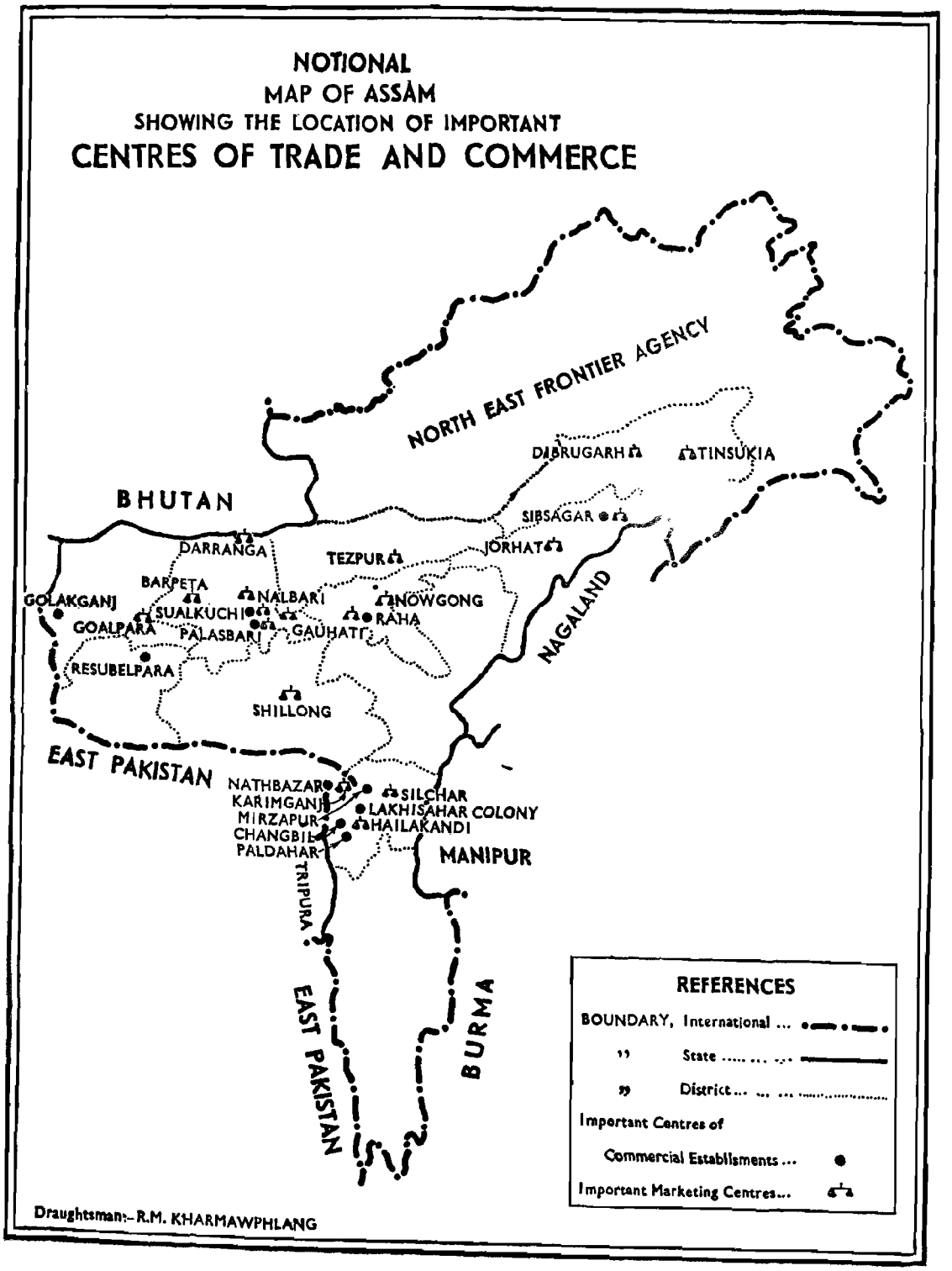
Draughtsmen :

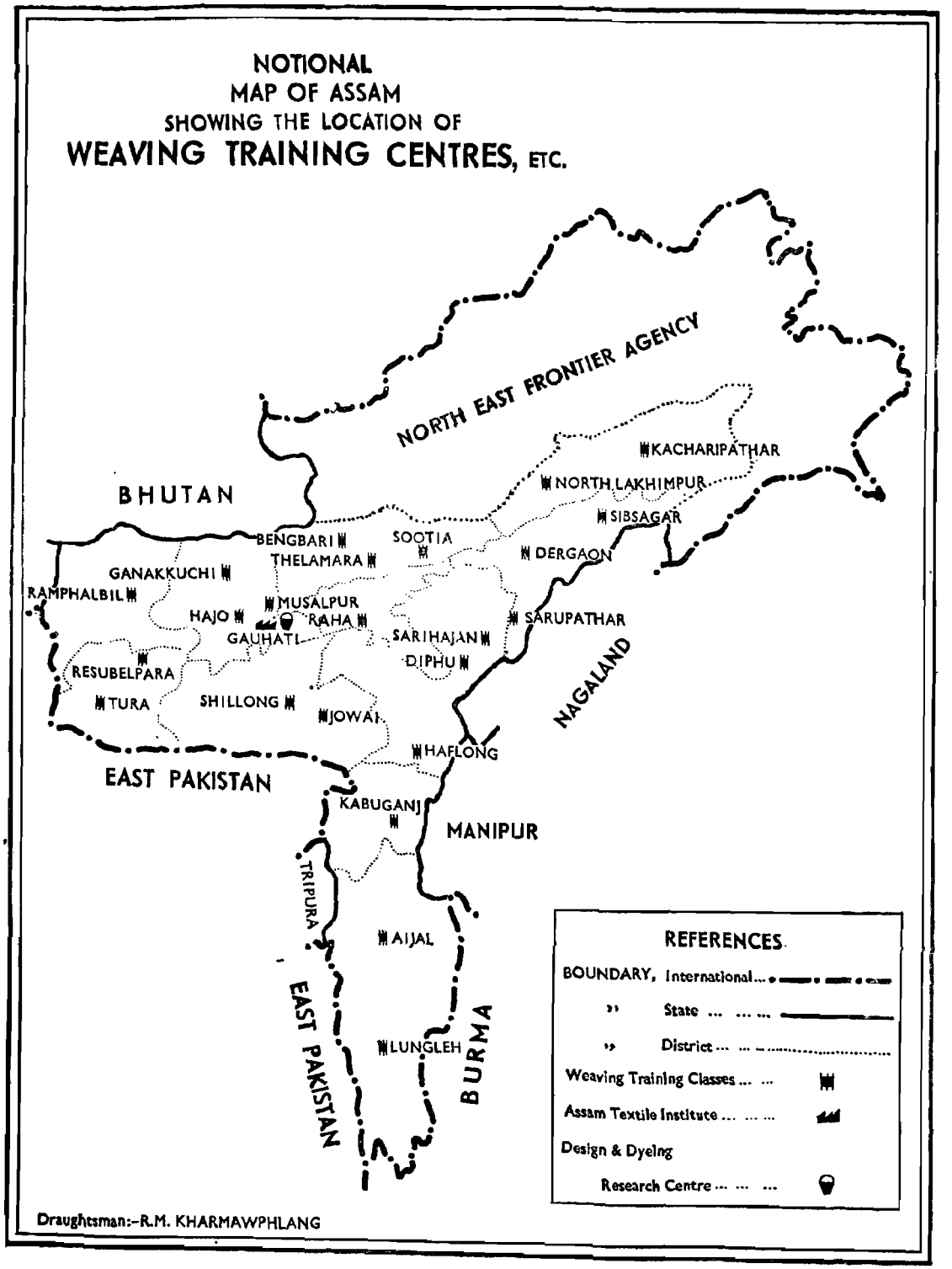
1. Shri Thantlinga.
2. D. Fraser (16.10.63—22.11.64)

Photographer :

S. K. Dutta (9.6.62—15.12.65).

E.H. PAKYNTEIN


NOTIONAL
MAP OF ASSAM
SHOWING THE LOCATION OF IMPORTANT
CENTRES OF TRADE AND COMMERCE
NORTH EAST FRONTIER AGENCY
DIBRUGARH
TINSUKIA
BHUTAN
SIBSAGAR
DARRANGA
TEZPUR
JORHAT
BARPETA
NALBARI
NOWGONG
GOLAKGANJ
SUALKUCHI
RAHA
GOALPARA
PALASBARI
GAUHATI
NAGALAND
RESUBELPARA
SHILLONG
EAST PAKISTAN
NATHBAZAR
SILCHAR
KARIMGANJ
LAKHISAHAR COLONY
MIRZAPUR
HAILAKANDI
CHANGBIL
PALDAHAR
MANIPUR
TRIPURA
EAST PAKISTAN
BURMA
REFERENCES
BOUNDARY, International ...
" State ...
" District ...
Important Centres of
Commercial Establisments ...
Important Marketing Centres...
Draughtsman:- R.M. KHARMAWPHLANG


NOTIONAL
MAP OF ASSAM
SHOWING THE LOCATION OF
WEAVING TRAINING CENTRES, ETC.
NORTH EAST FRONTIER AGENCY
BHUTAN
KACHARIPATHAR
NORTH LAKHIMPUR
SIBSAGAR
SOOTIA
BENGBARI
THELAMARA
DERGAON
GANAKKUCHI
RAMPHALBIL
MUSALPUR
HAJO
RAHA
GAUHATI
SARUPATHAR
SARIHAJAN
DIPHU
NAGALAND
RESUBELPARA
TURA
SHILLONG
JOWAI
HAFLONG
EAST PAKISTAN
KABUGANJ
MANIPUR
TRIPURA
AIJAL
EAST PAKISTAN
LUNGLEH
BURMA
REFERENCES
BOUNDARY, International
" State
" District
Weaving Training Classes
Assam Textile Institute
Design & Dyeing
Research Centre
Draughtsman:-R.M. KHARMAWPHLANG

CHAPTER I

Handloom Weaving

Introduction : Handloom Weaving of Assam is characterised by its distinctiveness, and although most of the products are of purely utility purposes, some of them which are used for certain occasions are of exquisite beauty, durable quality, delicate weave, dainty designs and delightful colours. The beautiful handloom fabrics show the creative genius of individual weavers whom tradition has made brilliant artisans. The muga saris, decorated blouse piece, Sualkuchi silk, *mekhela*, *chaddar*, *riha* and *laisamphi* (*khes*) are some of these beautiful weaves. Handloom is infact the symbol of the largest and oldest cottage industry of Assam and there is hardly any village with no loom. It is normal in any village of Assam to see the handloom weaver preparing the warp of his fabric under the cool shade of a spreading tree. He has no school ing, yet tradition has taught him to produce fabrics of delicate textures and designs.

It has been the age-old tradition in Assamese society that a grown-up girl should make a present of self-woven '*bihuan*' (*gamosa*) to her beloved as a token of love and also to her elders as a symbol of respect. Under ordinary circumstances, a girl will not be chosen by any bridegroom unless she knows the art of weaving . An Assamese folk song in this respect depicts such a picture —

"Aru beli kaisila hasati dim buli
Aibeli katisa paji,
Tomare hasati Amake nalage
Tirota ani lom kaji"

which literally means—"previously you promised me a *gamosa* (towel); but now you are preparing the slivers only; so I do not want your *gamosa*, rather I shall marry an expert girl (weaver)".

History and origin : "The handloom industry has always occupied an important place in every Assamese household, which probably contained as new a handloom, besides other articles for weaving and spinning. Handloom of a crude type are to be found even today among certain tribes. Cotton shrubs were grown for the manufacture of cotton cloths. The use of 'karpasa' (cotton) garments is also shown by the 'Kalika Purana' of the 10th century A.D. and the 'Harsa Charita'. During the Ahom period the weavers had a good reputation. They were even employed for the supply of royal robes to the Ahom kings. The usual process of manufacture involved treating the threads with some gummy substance, prepared generally from pounded rice to make them hard before their use in the loom. There was an extensive supply of cotton clothes, and the art reached a stage of perfection. The '*Kalika Purana*' proves also the use and manufacture of woolen garments (*kambala*), bark cloths (*valka*), silk (*kosaja*) and hemp cloths (*sanavastram*). Bark cloths were made of fibres of trees and plants. The process is best known to most tribes of Assam, who manufacture excellent fabrics of various designs even to day.

It is likely that the art of sericulture, weaving etc., was introduced into Assam at an

early period by the Bodos and the allied tribes. The place names like *Jonga*, *Donga*, etc. occurring in the *Arthasashtra* associated with the industrial products of Kamrupa, which have a Bodo origin, only support our contention. It is rightly pointed out that coloured cloths are more extensively used and manufactured by the tribes than by the people of the plains. Some Nagas have been expert dyers and produce extremely brilliant colours. The Manipuris have long been known as skilful and artistic dyers, and they may have been better in thisre spect than any people of Eastern India".

Formerly, it was rare to have an Assamese family without a loom and the latter was as precious to the Assamese maiden as life itself. There is a folk song that the loom also wept along with the members of the family on the departure of the bride after marriage :—

"Bargharat kandile make bapeke
Moralat kandile bhani
Barir pisefale kande tatar shale
Aideuk bia dibar shoni".

which literally means—on hearing the finalisation of the marriage of the damsel of the house, parents started weeping in the dwelling house and the younger sister wept in the marriage pandal while the loom also takes part in weeping behind the dwelling house.

This industry lost much of its eminence with the advent of the British when the Indian market was flooded with mill-made cheap foreign textiles. Some of the traditional weavers began to leave their trade. But though bereft of much of its past glories, the industry has been continuing to occupy a prominent place in the day to day life of the people of Assam. The people belonging to all castes and communities are found engaged in this trade in almost all the districts although it is not so widely practised in the United Khasi and Jaintia Hills. The majority of the weavers take this trade as a subsidiary means of occupation in the midst of their domestic works and agricultural off-time. Now in Assam, this industry may be regarded as a woman's craft (except in Cachar where it is generally carried on by the males) because 82 per cent of the weavers are found to be females.

It is very interesting to note that during the Ahom rule, the kings ordained that before sunset every woman must spin a certain number of hanks of yarn and the defaulters had to face corporal punishment. In his book "An Account of Assam" Dr. J.P. Wade observed: "Warlike cloth is made in the following manner. At midnight the cotton is ginned, pressed into rollers, spun into thread, manufactured into cloth and worn by the warriors in the morning".

Occupational trends . It is seen from our early references that handloom weaving is extensively practised in this State. The industry has no functional caste and people belonging to all castes, communities and tribes are found engaged in it. Handloom weaving is universally practised by the hill tribes in all the districts except the district of United Khasi and Jaintia Hills where it is gradually dying out mainly because of the fact that the working women prefer to engage themselves in out door activities. In the other three hill districts, Garo Hills, United Mikir & North Cachar Hills and Mizo Hills, it was found that every tribal woman, not to speak of every family, has her own loom. In the plain districts also the vast majority of the families, irrespective of castes or communities, are engaged in this industry. It must, however, be made clear that by far the large majority of the families, both in the plain and in the hill districts, carry

* The History of Civilization of the People of Assam by P. C. Choudhury, 1959, pp. 363-364, 369-370.

on weaving exclusively for home consumption. Others carry on the industry mainly for providing the requirements of the family and sell only the surplus, if any. Thus a comparatively small number of families are found engaged in professional or commercial lines. It is found that a majority of the commercial establishments are owned by professional weavers who migrated mostly from East Pakistan while a few establishments are owned by indigenous weavers.

Nature of the establishments : In the the sphere of weaving in Assam, generally three types of establishments can be seen :--

(a) The first type of establishment is composed of amateur weavers who produce cloths to meet the requirements of members of their family only.

(b) The second type of establishment is carried on more or less with a commercial bias. Besides fulfilling the family requirements the weavers sell their surplus products in the local market, if any. The weavers constituting this group are part-time workers.

(c) The last type of establishment is a purely commercial one. Weavers composing this group are all full-time workers and either work individually or collectively under co-operative societies and factories.

Of late a good many of the weavers have come to realise the importance of the commercial type of establishment and they have come forward to change the type of their establishments from the domestic to the commercial one. In many places, weavers have organised co-operative societies. Both financial and technical assistance are being rendered by the State Government and the All India Handloom Board to deserving societies. Training is also being imparted by Government sponsored institutes to the artisans as well as to the managerial staff of the co-operative societies. Both loan for block and share capital and grant-in-aid for working capital and subsidy on sales are advanced to the needy and deserving societies. In order to facilitate supply of tools, accessories and raw materials as well as disposal of the produce at a better price, 32 emporia have also been set up by the State Government. Commercial establishments of the following places have earned a good reputation for their quality products : (1) Sualkuchi, (2) Raha, (3) Palasbari, (4) Nishangram, (5) Resubelpara, (6) Sibsagar, (7) Lakhisahar Colony, (8) Nath Bazar, (9) Changbil, (10) Mirzapur, (11) Paldahar, and (12) Golokganj. Almost all the commercial establishments and co-operative societies have taken up improved tools and accessories such as fly shuttle loom, warping drum, dobby, jacquard machine etc. for economic and other reasons. The introduction of Chittaranjan semi-automatic looms in some of the establishments has brought great advantage and considerable improvements in the commercial groups. Nevertheless it remains a fact that the majority of the weavers of the first two groups above are still using primitive and uneconomic looms till today and there can be no doubt that weaving as an industry with outdated looms is commercially a failure, the price obtained for finished products being out of all proportion to the time spent on production.

Workshop : In the types other than commercial, looms are generally installed within the four walls of the dwelling huts or in the verandah of the artisans' house. Weavers engaged in the loin looms sometimes work in open places also, whereas in the case of commercial establishments or co-operative societies and factories, looms are generally installed in a common workshed for the purpose of efficient and co-ordinated work and necessary supervision.

Technique of Production

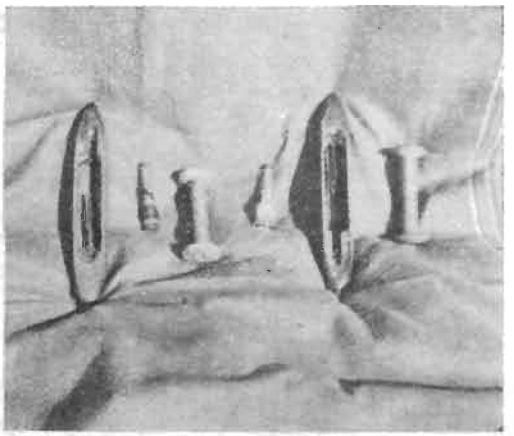
Raw materials : Cotton, muga, pat, eri and silk yarn are the basic raw materials used in weaving. Various counts of yarn ranging from 10s to 80s are generally used by the weavers who obtain these mostly from the merchants while only a very few do the spinning themselves. Almost the entire bulk of cotton yarn as well as a substantial part of the silk yarn consumed by the weavers are obtained from outside the State. Generally, all these yarn are imported by the merchants from whom the weavers obtain their requirements. The merchants do not generally supply yarn on credit. Some of the Government emporia are reported to be supplying yarn to weavers for manufacturing certain specified cloths and for such works the weavers get only weaving charges. A few co-operative societies supply yarn to the weavers, but such supplies are not sufficient to meet their entire requirements. At present, spun yarn is made available for khadi weaving societies by a few spinning societies run by the Assam Khadi and Village Industries Board. Though cotton is grown in the hills of Assam and different materials for dyes are also found growing in its forests, imported yarn in all the requisite shades available with the village shopkeepers is generally used. It is also gathered that some of the tribal weavers are still practising cotton ginning and spinning with traditional implements.

Cotton : The area and production under cotton has been estimated at 41,571 acres and 6,827 bales during 1964-65. The variety of cotton grown in this State is of short staple. The main cotton growing areas are Garo Hills and United Mikir and North Cachar Hills. These two districts cover about 90 per cent of the cotton-growing area of the State.

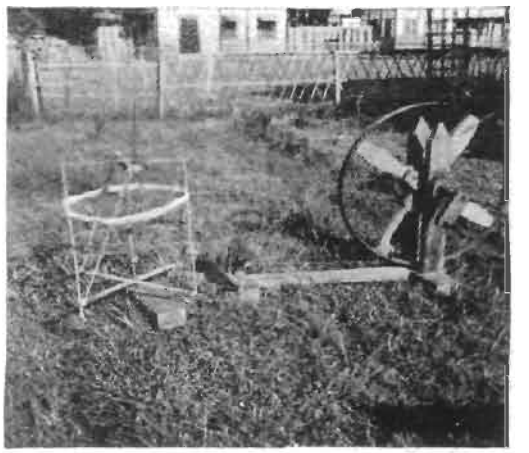
Dyeing or colouring : Dyeing is closely connected with handloom weaving and as such it can be regarded as an important ancillary industry to weaving. The weavers of the State generally do not prefer to purchase imported dyed yarn from the merchants but locally dye their yarn. They think that the colour of imported dyed yarn is not fast and it is not possible to get yarn of the required count with desired shades. Some weaving establishments have their own dyeing arrangements. But the majority of the unorganised weavers have to depend for their requirements on commercial dyeing establishments which are found mostly in urban and some advanced rural areas. As regards the availability of the materials, no difficulty was reported in respect of either yarn or dye-stuff.

Though commercial dyeing establishments in the plains are sufficient to meet the local demand, it is very rare to find such establishments in the hill districts. But it is wrong to assume that the hill tribes do not use dyed yarn. In fact, they are found to weave more colourful fabrics than their counterparts in the plains. The hill tribes of the State have their own indigenous processes of dyeing which they inherit from their forefathers and as a matter of fact they are also self-sufficient in this branch. As such, there is a very limited scope of commercial dyeing establishments in the hills.

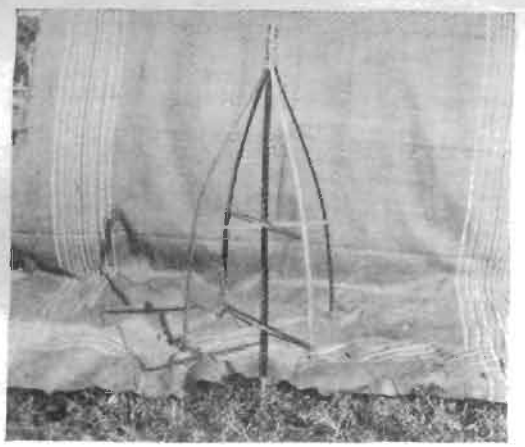
The hill tribes have their own indigenous methods of dyeing. Their dye-stuffs mainly consist of barks, seeds, flowers and leaves of trees. Details of such methods or dye-stuffs are however jealously guarded by the tribes as secrets and are not given out to outsiders. Even then it is gathered that they prepare the colour out of the indigenous dye-stuff by getting the latter boiled into a vessel of water. When the colour is obtained from the barks etc., the yarn required to be dyed is then put into the vessel and boiled for at least 20 minu-



Shuttle, warp and weft bobbins



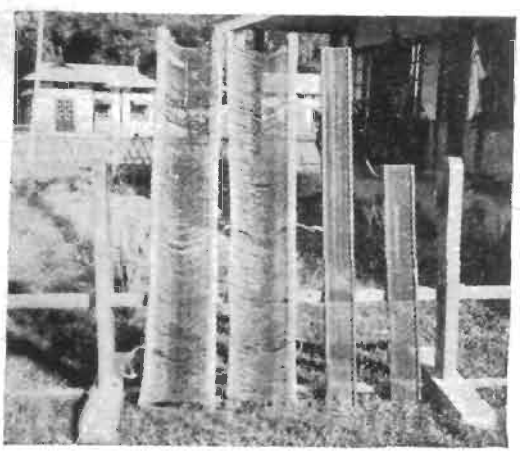
'Sereki' and 'Charkha'



An 'ugha' or 'Paghe'



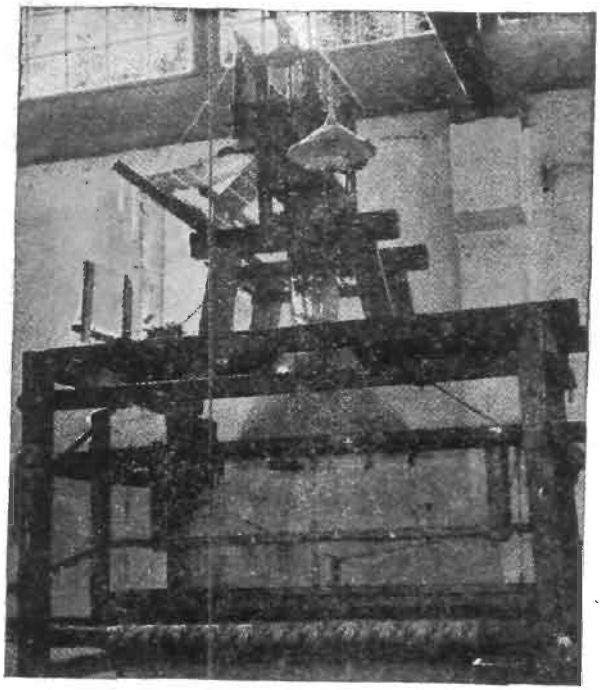
A hand-creel



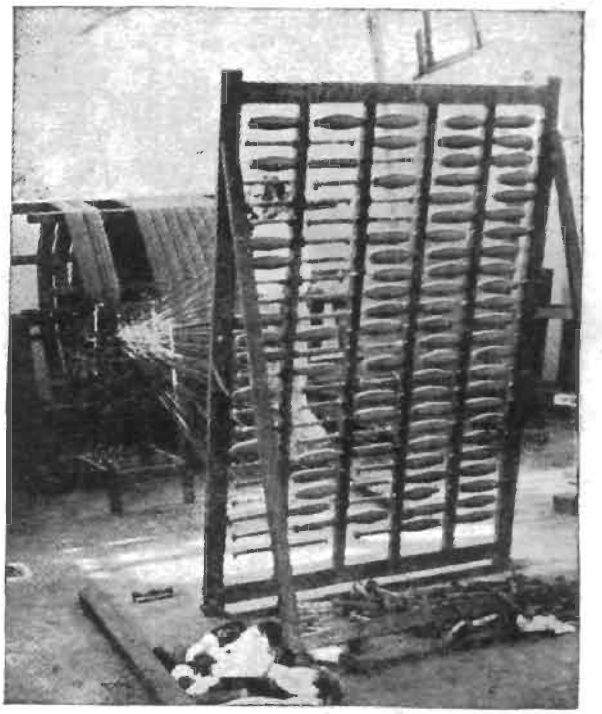
Healds and reeds



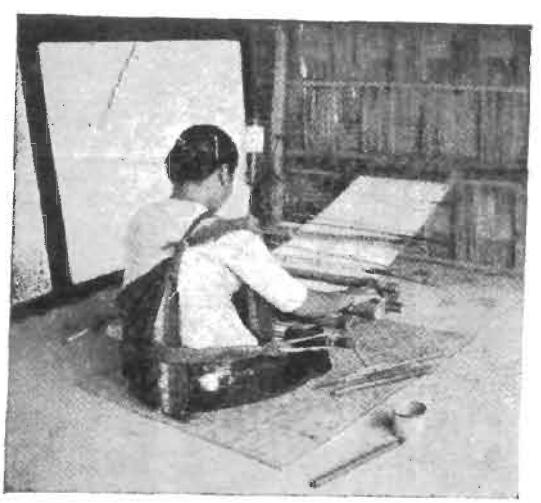
An Assamese weaver on throw-shuttle loom



Part view of a Jackard Machine



Drum warping in progress



A Mikir weaver on her loin loom



A Bengali weaver on his pit-shuttle loom

tes. While this boiling is carried on, a little quantity of salt is also mixed into the boiled dye and yarn to make the colour fast. Then the yarn is taken out and dried in the sun. Some common dyeing materials used by the tribal dyers are reported to be babla (*acacia arabica*), khayer (*acacia catechu*), supari (*areca catechu*), lac (*coccus lacca*), kusum (*carthamus tincterius*).

Different types of looms and components : The principal components of a loom are locally termed as (1) *ugha*, (2) *sereki*, (3) *rash*, (4) *makoo*, (5) *beam*, (6) *jator*, (7) *karoni* etc. But there is no common nomenclature for all the parts and accessories and the names widely vary from district to district. While the mechanism of weaving has been standardised, each region of handloom weavers has its own system, adapted to the local circumstances, and also uses such parts and accessories as have been enjoined by local traditions. The following are the different types of looms commonly used by the weavers in Assam.

Throw shuttle loom : This type of loom is found in almost every Assamese household. The loom is fitted to four bamboo or wooden posts fixed in the ground. Of the primary motions, shedding (opening of the warp threads) is effected by a set of healds operated by the foot, while the shuttle is thrown across the shed by one hand from one side and caught by the other at the opposite side of the cloth. The beating up of the weft is done by a reed commonly made of bamboo fitted on to a sley which is given a to-and-fro motion also by the hand.

Loin loom : This type of loom is in common use amongst the hill people. The warp of manageable length and breadth is prepared and fastened at one end generally to a wall of a house while the other is tied to the waist of the weaver with a cotton or leather belt. No reed is used, and the shed is effected by half-heald process operated by the hand. The weft yarn is inserted in the shed by means of one bamboo tube and a wooden string is used for beating up of the weft.

Fly-shuttle loom : This is an improved type of frame loom which considerably increases the output of the weavers. The loom is made of wood in such a manner that it can be fitted up and dismantled easily. The shedding is done in the same way as in throw shuttle looms, but the shuttle is driven across the shed by a string pulled by the hand. The beating up of the weft is done by a reed made of steel fitted on to a sley which is given an oscillating motion also by the hand.

Pit loom : This type of loom is found to be in use mostly amongst the Bengalee weavers who migrated from East Pakistan. The general features of such looms are almost similar to that of fly-shuttle looms, the only difference being that a square pit is dug in the ground where weavers put their legs whereas the beam and warp is set to the frame just on the ground level. Treadles are operated by the weavers with the foot which are placed a few inches above the bottom of the pit.

The following are the components of a fly shuttle loom :

(i) frame, (ii) sley, (iii) shuttle box, (iv) picker, (v) sley cap (vi) side bar, (vii) top bar, (viii) shuttles, (ix) handles, (x) treadle, (xi) beams, (xii) heald, (xiii) reed, (xiv) beam-winding rod etc.

Of the minor parts and accessories of the loom, mention may be made of levers, needles, shafts, lease rods, swifts, reeds, winding groves, spindles, spinning wheels, bobbins, pier, bobbin-frame, lease-taker, brush etc.

Loom preference-its reasons : The use of primitive and antiquated equipments by a

majority of the weavers is commonly attributed to their lack of knowledge as well as of finance. In some places where the weavers know about improved equipments and techniques they expressed their inability to afford the same due to lack of finance. Fly shuttle looms are definitely more efficient than loin looms while semi-automatic looms are still more efficient. But the ordinary fly shuttle loom is found to be unsuitable for use in the hill villages for it cannot be properly accommodated in the rural huts, its portability is also not so easy and it is not possible to preserve the texture or more intricate designs of the loin-loom cloth on it.

The loin loom has not lost its popularity because of its advantages which can hardly be beaten by any of the improved looms. These are (1) the compact texture of the cloths woven in a loin loom, (2) the ease with which this loom can be erected in any convenient place or rolled up with the warp on when not in use and (3) the simplicity of the loom which enables its parts to be replaced easily and practically without any cost. But weaving on loin loom has some other disadvantages, elimination of which by slight improvements on the existing methods will do much good. Weaving on loin loom cause much strain at the back of the weaver. Due to this reason, almost all the loin loom weavers gradually suffer from acute back pain along with other complications as they grow older and older. Thus, ultimately they are compelled to give up weaving. The weaker section also cannot weave in the loin loom for a long period because they cannot stand the strain of loosening and tightening of the warp during weaving. Moreover, a pregnant woman is usually not allowed to weave in the loin loom which is considered injurious to her health as well as to that of the baby in the womb.

Improved looms and accessories : Ever since the beginning of this century, numerous improvements in looms and accessories have been effected by incorporating the principles of the working of foreign looms. There are several types of improved looms in current use in Assam. The Chittaranjan semi-automatic loom, now very popular, is capable of weaving 30 to 40 yards in eight hours with 32s to 40s yarn. The improved fly-shuttle frame looms have a capacity of 10 to 15 yards in eight hours while a pit loom with throw-shuttle producing 5 yards in 8 hours. The prices of the different looms are Rs.60.00, Rs.150.00 and Rs.300.00 for a pit loom with throw-shuttle, frame loom and semi-automatic loom respectively.

Considerable improvements have also been effected in the attachments. Multiple shuttle-boxes containing shuttles with yarn dyed in different shades are convenient for use when yarn of more than one colour is necessary for weaving. Dobby attachments help in weaving designs on the borders of dhotis and sarees. For ornamental designs on the body, the borders and on cross borders, the jacquard machine is of great advantage.

Manufacturing processes : The art of weaving commences with the preparation of cotton. This work is generally done by the womenfolk of the weaver's family. In order to give an impression on the different processes of actual weaving in fly-shuttle loom, a very brief account is narrated below.

Wetting : First of all, the hanks of yarn are to be unfastened and properly examined. The loose hanks are then placed in a reservoir of water for one to three days, according to necessity. The water of the reservoir has to be changed every day.

Winding : Wetting finished, the yarn is to be wound on the reels. It is also done



Winding on a sereki



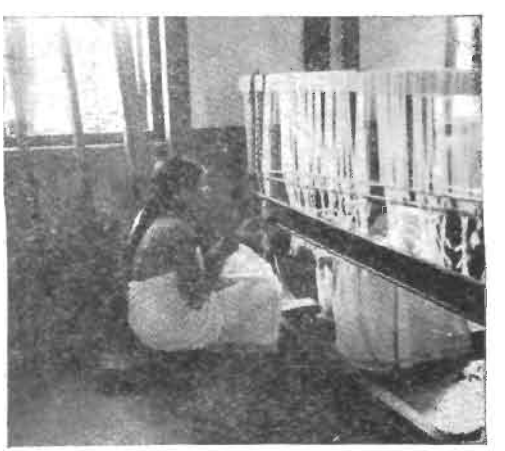
Bobbin winding done by an old type charkha



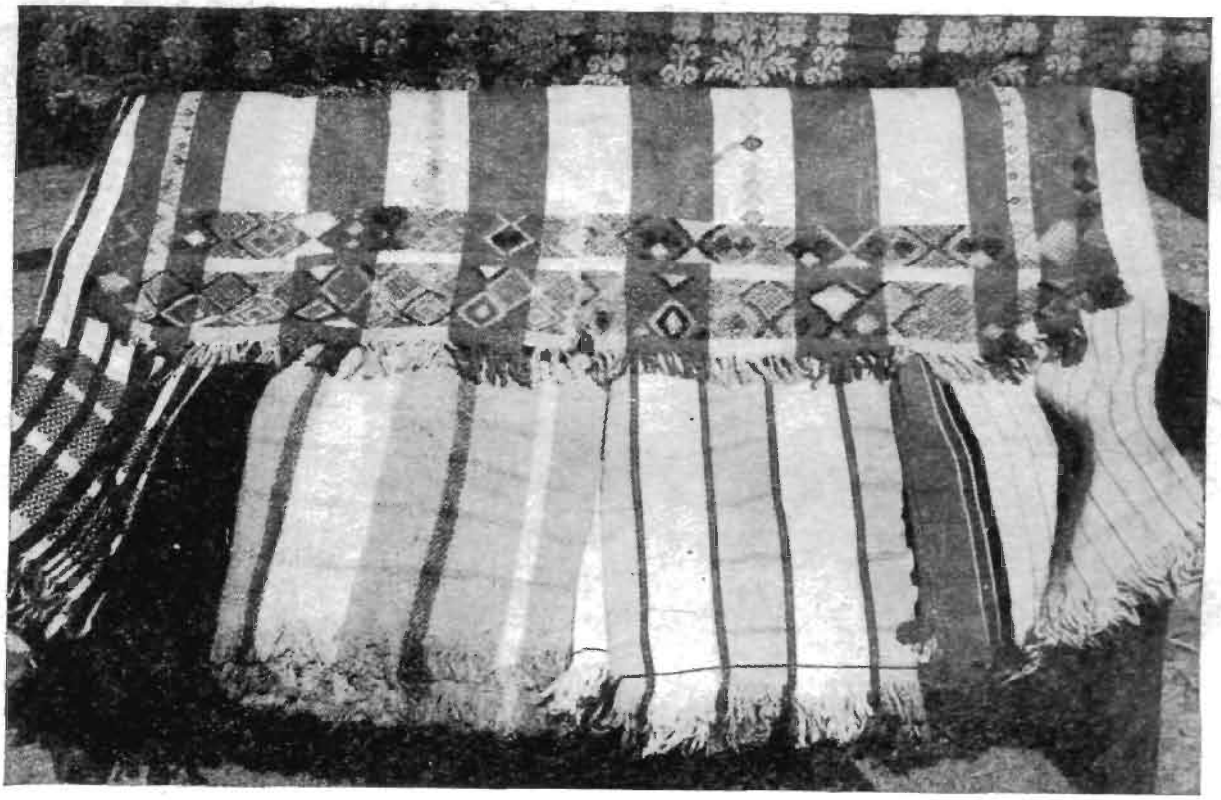
Bobbin winding with modern accessories performed by a trajnee



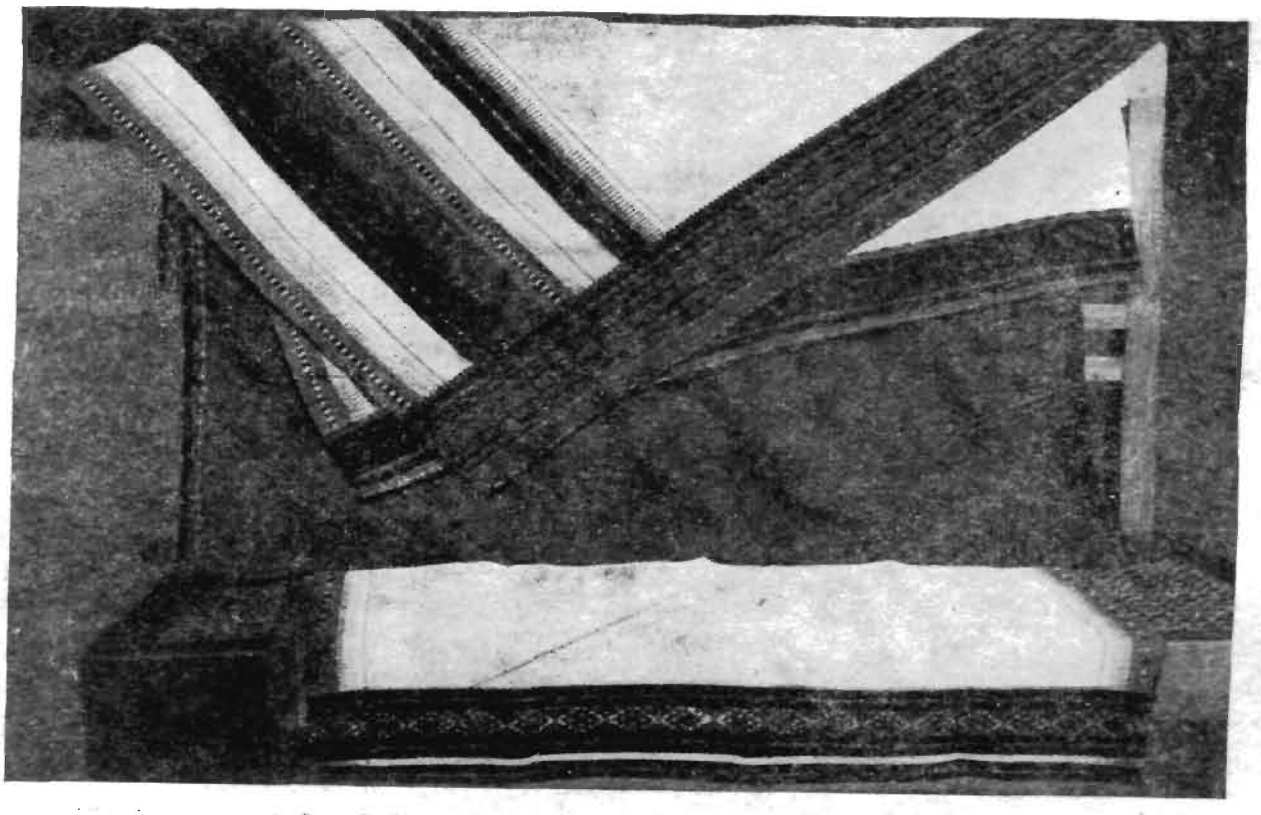
Warp is finally checked by a Miri-weaver



Denting is performed by a trainee



Some colourful Mizo fabrics



A few Gard fabrics of unmatched beauty and elegance

prior to sizing and during winding operation special care being taken so that individual thread does not get mixed up and sometimes piecing is also done to join breakages, if any.

Then follows the preparation of the warper's bobbin.

Warping : It is performed in different systems at different places and in various circumstances. The following are the main different systems found in the State :1. Stick warping, 2. Zig-zag warping, 3. Peg warping and 4. Drum warping.

Stick warping : It is generally done at an open space. The weaver is to move from one stick to another and back repeatedly with filled-up bobbins arranged in a cage-creel.

Zig-zag warping : When there is no space sufficient enough for working, warping is done in a limited space, even inside a house, by putting the sticks in a zig-zag way.

Peg warping : For reasons stated above, warping is sometimes done by putting pegs on the wall of a house.

Drum warping : It is perhaps better in any circumstance and is a time-saving device too. It can be done indoor and in a small space. A book creel is used for placing a good number of bobbins, say 80 to 120 and warping is carried on with many threads at a time. The drum is a big wooden reel of 1½ to 5 yards circumference which is made to revolve vertically or horizontally, fixed in a suitable frame-work. The bobbins are placed in a book-creel by the side of the drum. The ends are taken from the bobbins and drawn separately through eyes of a row of needles, arranged in a "heck". Such drawing helps in forcing the lease. The warp is to be wound on the vertical drum spirally and the heck slides up and down. When the warp is cut off and unwound, it is brought to the loom in the form of a ball for the next operation.

Denting : The next process of drawing the warp ends through the reeds is known as denting.

Beaming : Beaming is the next process. The warp is now ready be stretched to the full length. The ends of the threads nearer the reed are fixed to a rod which is placed in the groove with the beam. The beam is then wound.

Heald knitting : The beam so prepared is then put on a frame and a yard of it is stretched and held tight by fixing it to two temporary posts. It is essential that the threads should be in uniform tension. Lease rods are placed in the warp causing a slight opening in it. One end of the thread with which the healds are to be knitted on the warp is then passed through this opening from the right to the left. The actual knitting is done with a tapering piece of wood.

Looming the warp : The next operation is looming the warp. The warper's beam being ready, it is placed in the appropriate position in the loom.

Shedding : Looming of the warp being accomplished, preliminaries to actual weaving should be taken in hand. As one of the treadles is depressed by the foot, the healds connected with it go down while those connected with other treadles go up forming an opening in the warp known as 'shed'. The shuttle moves through this shed and weaves.

Now having discussed the different processes of weaving, let us see how the loom is actually operated by a weaver. There are

three principal motions which govern the art of weaving, namely, the shedding motion, the picking motion and the beating motion. The shedding motion consists of depressing the treadle by one foot which results in making an opening in the warp thread. The picking motion consists in propelling the shuttle by giving an oblong pull to the handle. The beating motion means drawing the sley forward with the object of beating the weft yarn to the fell of the cloth. Dexterity in weaving depends on how quickly these three motions are affected in a concerted manner.

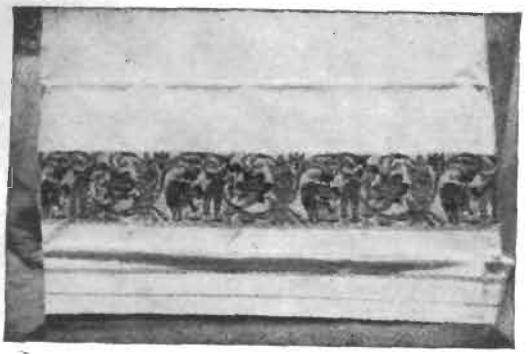
Designs : We can get a fairly good idea of Assam handloom from the following remark once made by Mahatma Gandhi: "Assamese women are born weavers, they can weave fairy tales in their cloths". It is worthwhile mentioning in this paragraph that the handloom fabrics of Assam are noted for their various artful designs and brilliant colour combinations. The artisans of the plains and hills generally attempt to bring out such types of design in their cloths which reflect their respective religions, social and cultural behaviour. Their respective motifs of design have bearings on their customs, traditions, religions as well as aesthetic senses. The designs used in Assam are mostly symbolical and consist of abstract figures of animals, human beings, creepers, flowers, birds, diamonds, cross border and heavenly phenomena. The Assamese weavers produce beautiful designs of different colours and shades in the border of their *mekhela*, *chaddar*, *riha* and *gamosa*, and sometimes use golden and silver threads in the border to produce greater effect. The Assamese girls weave a special type of *gamosa* known as '*bihuan*' with a unique type of design representing heavenly phenomena, which is offered as a gift to the elders as well as to the lovers just on the eve of the Bohag Bihu (celebration of New Year's Day) as a token of love and respect. The motif of weaving a '*bihuan*' has been very beautifully expressed in the following Assamese couplet :

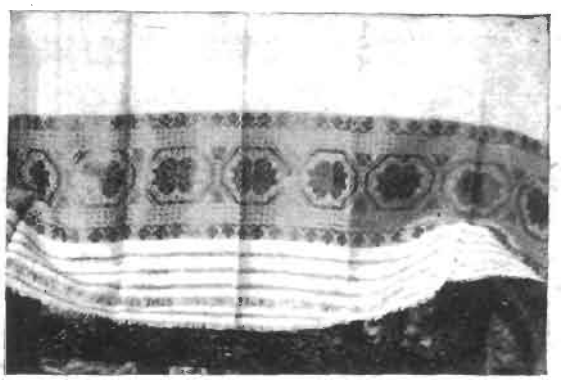
"Moromar digh di senehar bani lai
Hepahar asure boa
Soponor fulere fulam more bihuan
Mon di abeli loa".—

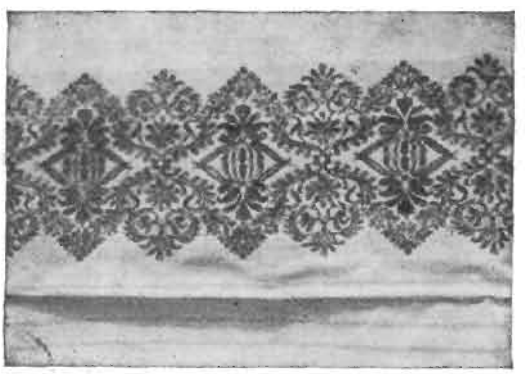
which literally means--"Oh my darling, accept the '*bihuan*' (Assamese towel) now with all your heart, because I have made it with the warp of love and the weft of affection, which is being bordered with my eagerness and embroidered with dreamy flowers".

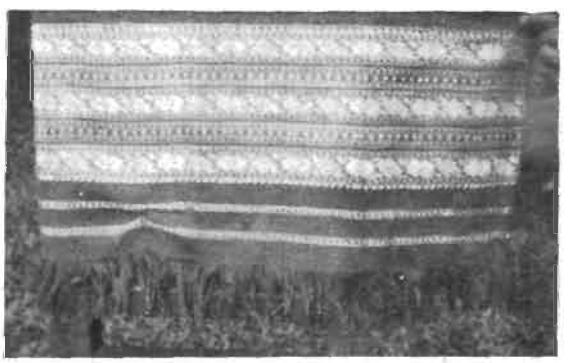
The tribal people of Assam have their own distinctive designs of unique artistic expression which are strictly local in their character. Even now, amongst the different tribes of Assam, a few primitive but attractive types of designs are practised, which at once become the envy and wonder in the world of arts and crafts. The Assam tribal handloom fabrics present a wide range of brilliant colours and superb designs.

Sources of designs : The task of making design in weaving requires a high degree of skill on the part of the weavers. Such skill is found to be hereditary or traditional in character. The designs used in weaving in the State of Assam by different tribes and communities are primitive and local, carrying some peculiar significance and meaning on the social, cultural and religious life of the people. To find out the source of various tribal and local designs, it can be stated that due to constant working on a particular craft, the expert weavers sometimes invent new techniques and designs by dint of the energy of their brain and the urge for artistic innovations. The traditional artisans generally try to produce such types of design, strictly local and typical in character, by symbolical

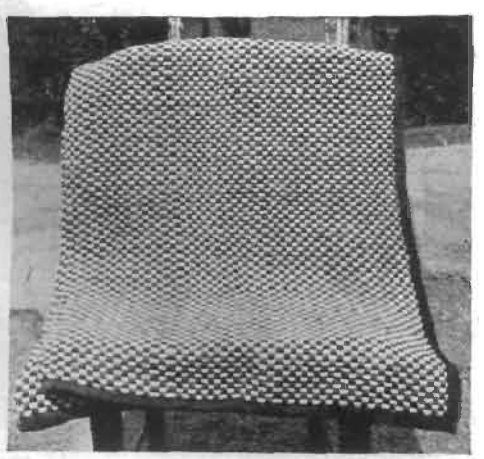








Several popular & attractive designs of Assamese handloom fabric

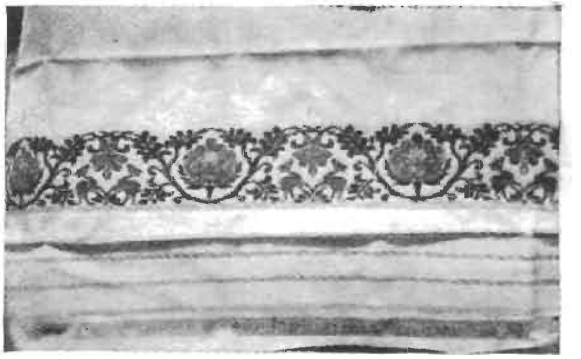


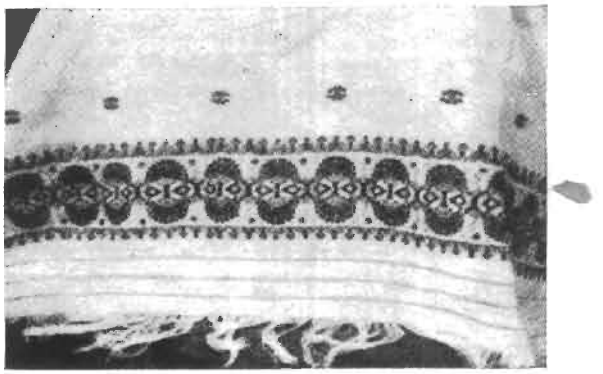
A 'Laisengphee' or wrapper

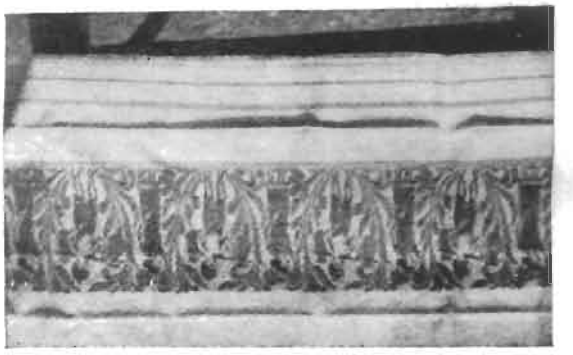


Typical design used in Mekhela & Chaddar













Several popular & attractive designs of Assamese handloom fabric

expression of various ideas and imaginations that they could conceive in their mind.

Finished products : The handloom fabrics produced in Assam have practically nothing in common with those manufactured in other parts of Indla. So, a brief account of them are given below.

Mekhela : *Mekhela* is a traditional garment made of either cotton or muga or pat and used by the Assamese womenfolk in general. This type of handloom cloth can nowhere be seen except in Assam. A piece of cloth of 3 yards by 1 yard is sufficient to make a *mekhela*. There are different varieties of *mekhela*, such as *uka mekhela*, *pator mekhela*, *mugar mekhela*, *mugbania mekhela*, *phulbahar mekhela*.

Chaddar : This garment is used by the Assamese women along with the mekhela o the body. This piece of cloth is used to cover the upper portion of the body. The usual size is 3 yards long by 1 to 1½ yard wide.

Riha : It is the name of a particular garment used by the Assamese women along with the *mekhela* and the *chaddar*. *Uka riha* is a plain one. *Paridia riha* has borders on both sides of a coloured silk or cotton cloth, whereas *deel bahar riha* is richly decorated with appropriate designs of flowers and creepers and sometimes with the figures of birds and animals.

Pashra : *Pashra* is woven either with cotton or eri and sometimes with both. It is used as a wrap during the cold weather.

Barkapar : -(literally means-large piece of cloth). It is also made of either cotton or eri yarn and sometimes with both. It is quite similar to *pashra* but double in size. The usual size is 21 feet long and 5 feet wide.

Churia : *Churia* or dhoti is woven out of cotton yarn and used by the males. Two varieties of *churias* are used, namely, *barchuria* of the size of 15 feet by 4 feet and *soruchuria* of 3 feet by 4 feet.

Cheleng : *Cheleng* is a type of cloth used by men as a holiday dress. It is used to cover the upper portion of the body. *Uka Cheleng* has muga lining on cotton fabric. *Gunakata pator cheleng* is made of pat thread with borders of gold or silver threads. *Mugabania cheleng* is made of both muga and cotton yarn and sometimes with pat and muga yarn.

Gamosa : It is generally woven by the Assamese weavers for use as a towel. The body is plain, having borders of coloured threads specially red or green on all sides. There are special types of *gamosa* known as '*bihuan*' and '*moksa*'. *Moksa* is prepared in the Kamrup district for keeping betel-nuts. The *moksa* is much smaller in size than the general *gamosa*.

Tribal products and others : Some special handloom cloths woven and used by the Garos :

Dakmonda : It is a beautifully decorated piece of cloth used by the Garo women as a substitute for a sari. The Garo weavers generally use coloured cotton yarn for this cloth. Two or three decorated borders are used on one side of the cloth.

Guna reking : It is a small piece of cloth used by the non-Christian Garo women of the interior places instead of the *dakmonda* : size 1½ yard by ½ yard.

Nengeta : Used by the Garo males of interior places : size - 3 feet by 3 inches.

Some of the special handloom cloths woven and used by the Mikirs and Bodos-

Pini : (*mekhela*) : all the *pinis* are black coloured.

Pachclin : (*chaddar*) : very beautifully decorated.

Wankok : (cotton belt) : used by woman on the loin, size-2½ yards × 2 inches.

Khonjari : (wrap) : used by old women as wrapper.

Rikong (nengota) : used by the male Mikirs.

Dakhona : garment used by Bodo women.

Some of the special handloom cloths woven and used by the Kacharis :

Righu : used by the Kachari women as a *mekhela* (4½ feet by 2 feet).

Rijimphai : (5½ × 2′) - used by women just on the breasts.

Haraimin : (3′ × 8′) - it is like a *chaddar* used by the women as *orna* (scarf).

Rikuksha : (8 × 2′)- used by the women for covering their heads.

Rithap : Fine quality eri *chaddar*.

Rimchow : Plain cotton *chaddar* used by the males.

Some of the special handloom cloths manufactured and used by the Mizos or Lushais.

Puanchu : This piece of cloth is used by the Mizo women. Superbly decorated and very durable, the border is designed with figures of diamonds, birds, flowers, creepers etc. The highly decorated *puan* is very costly and the price may go upto Rs.200/- per piece.

Kawrehui : beautifully and highly decorated blouse piece woven by the Mizos on the loin loom. The best quality blouse piece costs Rs.25/- each.

Puanlukhum : used as a scarf (*orna*) by the Mizo bride during marriage.

Puandum : 'Puan' means cloth, 'dum' means 'black'. This cloth is used to cover the dead or for covering the coffin for the dead.

Hand-bag : The Mizos manufacture artistically decorated hand-bags which are used by students for carrying books as well as by the general people for carrying various articles.

Some of the special handloom cloths manufactured and used by the *Khasis* :

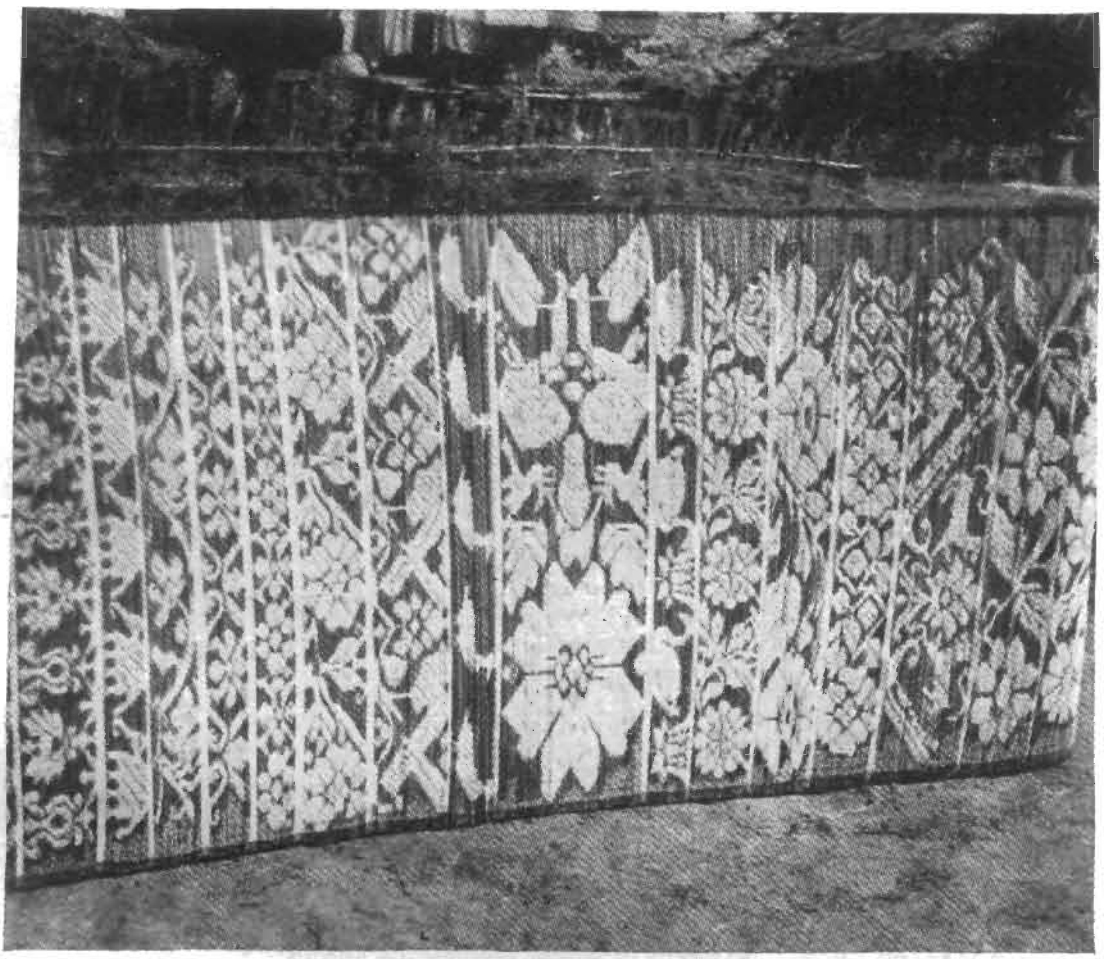
Ryndia stem : It is generally made of red and yellow coloured yarn and used by the Khasi women instead of the *mekhela* or the sari.

Thohsaru : It is woven with cotton or silk yarn of black and white colour. It is used like the '*ryndia stem*'.

Ryndia tlem : This cloth is a gent's wear for the Khasis and is used mostly in the rural areas.

The *Manipuri* weavers of the Cachar District weave the following cloths :

Laisingphi or Khes : (Wrapper) - this is a Manipuri term used for a quilt-like winter cloth of plain weaving; it can be used as a wrapper and is very durable. A special feature of this fabric is that the cloth is wadded with cotton during the process of weaving.



A specimen of designed catalogue maintained by tribal weavers of Assam

Fanek : -used by the Manipuri females instead of sari, a multi-coloured piece of cloth.

Pelei : dancing apparel for Manipuri girls Embroidered with artistic designs of high skill, its cost varies from Rs.100/- to Rs.200/- according to the quality and design.

Other products of the Manipuri weavers are bed sheets, bed covers, shirtings, saris, dhutis, mosquito nets and cheap quality *gamosas*.

The Bengalee weavers of Assam, mostly those who have migrated from East Pakistan, are found manufacturing cloths of various designs, using dobby and jacquard machines for commercial purposes. Their products are saris, dhutis, scarfs, towels, bed covers, bed sheets, lungis etc.

Economics of the Industry

Cost of production : It is very difficult to find out the cost of production of and the profit on the sale of handloom cloth because of the fact that handloom weaving in Assam is mainly a subsidiary and off-time occupation. The majority of the weavers in Assam take to weaving after all the domestic works are over and there is no fixed time for this purpose. The cost of a cloth depends on the quality of yarn and design used. The cost of handloom cloth is made up of the cost of yarn, dyestuff, sizing materials etc., plus the labour charges (see Appendix 'E').

Earnings of commercial weavers : The average earnings per month of the skilled and semi-skilled artisans are reported to be as follows :

	Full time worker	Part time worker
Skilled artisans	Rs.90/-	Rs.40/-
Semi-skilled artisans	Rs.55/-toRs.60/-	Rs.25/-

Employment, production and sales : It is gathered that the handloom industry (including sericulture) in Assam provides total and partial employment to nearly 12 lakhs of people (1957-58)*. The production of handloom cloths (other than Khadi) has gone up and reached the second plan target of 54 million yards.

**It is also reported that the total sales of Khadi fabrics in Assam amounted to Rs.450,546/- in 1961-62. Out of this, khadi worth Rs. 362,022 was sold by the Assam State Khadi and Village Industries Board, Gauhati, in the year 1961-62. The Assam Pradesh Congress Committee Khadi Bhandar, Gauhati, effected a sale of Rs.13,205/-. The Assam Samagra Seva Sangha near Gauhati sold khadi worth Rs.1,530/- in the same period The sales of Rampur Endi Production Centre amounted to Rs.38,397/-. The Government Khadi and Endi Production Centre, Raha (Nowgong) effected sales worth Rs.35,390/- during the year 1961-62.

Finance : Weavers of the unorganised sector are reported to be facing difficulty regarding proper finance and have to depend on their own limited resources because credit facilities are very unsatisfactory. Sometimes traders and middlemen take advantage of this situation and exploit the weavers by advancing money against their finished products. The prices offered as such are naturally very low. The financial condition of the few co-operative societies is not very encouraging. The State assistances or aids from the Panchayats

*"Assam Gazette", June 12, 1962.

**"Jagriti" Seventh Anniversary Number—1962, by Khadi and Village Industries Commission, Bombay.

or District Councils are also generally confined to co-operative societies and training-*cum*-production organisations. One of the common complaints of the weavers is about the inadequacy of finance. They do not have sufficient fund to invest in better and improved equipments. In the circumstances, they can hardly be expected to provide the entire capital necessary to run such an industry on commercial lines. This insufficiency of funds might have resulted in low production and lack of enterprising commercial outlook with the majority of the weavers in the State. During the Second Plan period, the State Government spent nearly Rupees 8.87 lakhs for the expansion and development of handloom weaving and introduced some 150 power looms in different parts of the State.

Marketing and its problems : In respect of marketing of handloom fabrics of Assam, it was found that the role of the producers is mainly confined to the local markets due to the fact that the majority of the products are suitable only for local uses. Even today most of them have not adopted the change in tastes, habits and fashions of the consumers which is essentially required for capturing a greater market outside.

Apart from the sales to consumers, a substantial part of the products sold in local markets are purchased by traders and middlemen who resell the products in different markets. In some cases weavers were found in the role of traders. The weavers have no marketing organisation of their own, except a few co-operative societies, and as a result they are often forced to sell their products in buyers' markets whereas they have to obtain their raw materials from the sellers' markets. As such, it can be safely assumed that the weavers are doubly exploited by the traders and merchants. For better marketing of handloom and other products of cottage industries, Government emporia are set up at important places of the State and outside, but these have not been of much help because it was reported that officially the emporia connot purchase any and every products offered for sale by the weavers. They only make purchases of certain products at specified rates and even for these purchases they do not make payments to the weavers in cash. As a consequence the weavers do not like the idea of marketing their products through the Government emporia. It is heartening to note that in order to encourage and popularise locally manufactured handloom cloths a rebate system on the purchase of handloom cloths manufactured by co-operative societies has been introduced by Government. Some well-organised co-operative societies and factories have come forward to solve the marketing problems by opening up their own selling centres at Shillong, Gauhati, Nalbari, Barpeta, Silchar, Karimganj, Hailakandi, Sibsagar, Sualkuchi and other prominent places of the State. It is also reported that a large quantity of endi cloth is sold to Bhutias at the Daranga Mela, near the Bhutan border which is annually held during the winter. The table below prepared by the Department of Economics and Statistics, Assam, on the basis of data obtained during a sample survey conducted in the year 1954-55, shows the average value of products manufactured and marketed per establishment :

TABLE NO.1.

Average value of products manufactured and marketed per establishment. (Figures in brackets indicate percentages to average value of products manufactured per establishment).

Industry	Average value of products manufactured per establishment (in rupees)	Average value of products retained for home consumption per establishment (in rupees)	Average value of products remaining unsold per establishment (in rupees)	Average value of products sold per establishment (in rupees)				
				In local markets	In markets within the districts	In markets within the State	In markets within Indian Territory	Total sold
1	2	3	4	5	6	7	8	9
Hand loom Weaving	2,525 (100.0)	41 (1.6)	83 (3.3)	1,982 (78.5)	241 (9.5)	178 (7.1)	*	2,401 (95.1)

*Negligible

Government assistance : At one period it seemed as though this craft which had enjoyed a place of fame and glory in the cultural life of Assam might become completely extinct. The industry experienced a severe competition from the cheap mill-made textiles which challenged the very existence of the handloom weavers. But thanks are due to Government which has shown a very keen interest in trying to restore this decadent industry to its pristine glory through the introduction of various developmental schemes. Considerable emphasis has been laid by the State Government and other agencies for the expansion and development of handloom industry during the last two Plan periods.

Training facilities and financial help : The training programme taken up by the State Government has been intensified during the plan period. The number of peripatetic weaving parties and the staff under them have been increased to enable them to exercise close and effective superivision of the weaving activities in general and that of the weavers' co-operative societies in particular. Increased grants-in-aid, subsidy and industrial loan had been sanctioned to enable the artisans to take up the industry on commercial lines for their economic upliftment. During the second Plan period, an amount of * Rs.153,000/- was sanctioned as industrial loan. Over and above, a scheme for the supply of improved looms and accessories at subsidised rates was also introduced.

A few schemes being implemented under the auspices of the All India Handloom Board with the active participation of the State Government are listed below :

(1) mobile van, (2) establishment of emporia, (3) extension of dye-house and (4) research, designing and dyeing.

Besides these programmes, the Assam State Khadi and Village Industries Board has also come forward with many schemes for the development and expansion of spinning and weaving among the unemployed and under-employed people of the rural and urban areas. The schemes of this Board also provide facilities (such as financial, technical, managerial, supply of implements and marketing,etc.) to the members of khadi co-operative societies.

* 'Assam 1961', by the Publicity Department, Shillong

Schemes under the Third Five Year Plan : Altogether seven schemes involving a total outlay of *Rs.29.50 lakhs have been taken up for implementation during the Thrid Five Year Plan. The schemes have been worked out with particular emphasis on thc training programme for providing technical know-how to the industry, establishment of weavers' extension service units for standardisation and quality control, conversion of throw-shuttle looms to fly-shuttle looms for augmenting production and lowering the cost of production and on research for evolving improved designs and techniques of production.

Conclusion : Handloom weaving has been practised in this State since the misty past. For various reasons, economic and otherwise, the industry declined during the British rule. Until lately the fabrics produced by the handloom weavers of Assam reached an excellence which had never been equalled, far less excelled. By now, this noted craft of Assam has definitely been revitalised through the efforts and economic safeguards provided by Government.

The handloom weaving industry has made a remarkable stride during first three Five Year Plan periods as a result of implementation of various developmental schemes taken up with the funds obtained from the Government of India as well as the State Government. The industry has thus been able to play a vital role in uplifting the socio-economic condition of a sizable number of artisans in the State. With the advancement of research works undertaken, many more developments of this craft are still to be seen. On the basis of past experience, however, we can look to the future with confidence and hope.

*'Assam 1961', by the Publicity Department, Shillong

APPENDIX A

A brief report on the handloom weaving industry of Mynso village in the United Khasi-Jaintia Hills District :-

About twenty miles away, north-east of Jowai, lies a village named 'Mynso' which is the subject of our study for its singular attachment to the handloom industry in the United Khasi-Jaintia Hills. Jowai, a small town and subdivisional headquarters, is only 40 miles east of Shillong, the district headquarters of the United Khasi-Jaintia Hills. The village Mynso is named after a small river draining outside the village and the word 'Mynso' literally means 'danger'.

The village is about 3 square miles with a total population of about 1,642 and the approximate number of family is 372. All the villagers belong to the Pnar community.

Handloom weaving is the chief craft of the village. Out of the total population, only 95 women are actually engaged in this craft. Compared to this trade, others employed in the manufacture of wooden, iron and cane-bamboo products are insignificant. It is not possible to say exactly as to when did the community come to this village. 'It is gathered that the village Mynso had been in existence before the great earthquake occurred in the year 1897. Jaintia had been an independent kingdom (hima) till 1835 under a king (syiem). The people here belong to the Indo-Mongoloid stock with an Austro-Asiatic dialect.

Weaving is an occupation of women in this region and thus it happens that only women are creative in the handloom industry. In the past, every house had one loom and all women knew the art of making cloths. The situation is quite different now. All traits of human life are always in a constant state of flux and changes are bound to occur. The trade of men and its course of development is determined by the general cultural history of the people. The looms used by the weavers here are almost like the loin looms, except for the only variation that the weavers do not tie this type of loom unto their loin but fix it with ropes and bamboo sticks on the ground. The loom is tied to the bamboo posts with ropes and placed almost horizontal to the ground.

The weavers do not purchase any yarn. Cotton is available in sufficient quantity in the neighbouring areas of United Mikir and North Cachar Hills District. The villagers purchase the cotton and spin the required quantity of yarn themselves.

The tools required for spinning are given below. The price is also mentioned against each item.

1.	Tylliat (ginning tool) of wood ...	Rs. 8.00
2.	Ryntieh (bow for carding) of bamboo	Re. 1.00
3.	Shyrkha (charka for spinning) of wood with iron spindle-18 inches	Rs.10.00
4.	Kla (winding tool) one set of 4 pieces	Re. 1.00
		Rs.20.00

Warping is done in an open space where the local weavers use one bamboo splint with four small eyes through which four threads are passed. The cross-wise arrangement of every alternative thread is made by this bamboo split. The length of the cloth depends on the distance they make in between two bamboo sticks according to their need. The whole cross-wise arrangement of thread is sized with rice solution. When the warp gets dry after sizing, the whole arrangement is fixed in the loom.

The following are the parts of a complete loom. The local price of each part is also mentioned.

1. Tainti (cloth beam) ... Re.1.00
2. Tadonghushor (shuttle with bobbin) Re.0.50
3. Shun (reed) ... ... Re.1.00 to Rs.2.00
4. Wait (splint of bamboo-length 24" and breadth 3½") Re.0.25
5. Lawi (heald) Re.1.00
6. Bhin (used to keep warp yarn in uniform tension) ... Rc.0.25
7. Myllon (two bamboo rods put at the end of the yarn) Re.0.06
8. Khaliabthad (bamboo rod) Re.0.12
9. Khangdung (to fix the loom so that it can be tightened up with a rope made of wood, with a slight inward bend like a bow) ... Re.0.50
10. Rope (to tie with item No.9 length 15 yards) Rs.3.00

All these parts are made in the village.

Weaving is a part-time trade of the village womenfolk and not run on a commercial basis. There are of course some weavers who weave to earn cash. The process of weaving is very slow and laborious for which they cannot produce cloths in large quantity. The maximum yardage of cloth one can produce in a day is roughly three. The capacity of the reed is also very limited. It is only 24 inches maximum. The weavers of this village can weave only plain cloths without any design. It is reported that only two old women of the village can make some kind of design on cloths which is used for some religious purpose in the Khasi Hills. But during the period of investigation they could not show any sample of such design.

The method of weaving is still indigenous and the cloths produced are not of good quality although they are very durable. The handloom weaving in the village is carried on without any sizable capital and division of labour. All the processes from the spinning of yarn upto the producing of cloths are generally done by a single weaver. For the purpose of dyeing the weavers sometimes use an indigenous colour prepared by themselves.

Though there are ninety five weavers altogether, only a few of them work in the loom regularly. The number of regular weavers is really very negligible, due to the fact that a weaver cannot make any handsome profit and is unable to compete with mill-made cloths. The Khasi people became accustomed to mill-made cloths since the days of British rule. Though it has not died out completely, the handloom weaving is really in a decaying stage in Mynso village.

APPENDIX B

STATEMENT SHOWING THE NUMBER OF REGISTERED CO-OPERATIVE SOCIETIES AND NUMBER OF ARTISANS WORKING UNDER THEM

Name of the district	Number of registered weaving co-operative societies up to 1st July, 1961	Number of artisans	Males	Females
1. Goalpara	114	1,897	1,115	782
2. Kamrup	823	6,823	2,903	3,920
3. Darrang	146	2,899	1,237	1,662
4. Nowgong	161	2,848	945	1,903
5. Sibsagar	238	4,808	1,306	3,502
6. Lakhimpur	188	4,427	295	4,132
7. Cachar	149	4,397	3,474	923
8. United Mikir and North Cachar Hills	33	731	248	483
9. Mizo Hills	5	284	60	224
10. United Khasi-Jaintia Hills	9	111	77	34
11. Garo Hills	34	782	344	438
Total	1,900	30,007	12,004	18,003

APPENDIX C

STATEMENT SHOWING THE NUMBER OF HANDLOOM WEAVERS IN DIFFERENT DISTRICTS (EXCLUDING WEAVERS UNDER THE CO-OPERATIVE FOLD)

Name of the district	No. of weavers outside the co-operative fold	No. of full-time weavers			No. of part-time weavers		
			Males	Females		Males	Females
1	2	3	4	5	6	7	8
1. Goalpara	3,550	766	104	662	2,784	141	2,643
2. Kamrup	47,555	1,381	800	581	46,174	68	46,106
3. Darrang	2,429	600	150	450	1,829	79	1,750
4. Nowgong	4,253	1,160	590	570	3,093	289	2,804
5. Cachar	7,639	3,474	2,922	552	4,165	2,282	1,883
6. Sibsagar	50,425	106	26	80	50,319	29	50,290
7. Lakhimpur	36,936	..	..	..	36,936	..	36,936
8. Garo Hills	2,141	197	..	197	1,944	..	1,944
9. United Khasi-Jaintia Hills	1,259	25	.	25	1,234	..	1,234
10. United Mikir & N. Cachar Hills	1,568	30	..	30	1,538	20	1,518
11. Mizo Hills							
Grand Total :—	157,755	7,739	4,592	3,147	150,016	2,908	147,108

* The figures given as such are compiled for the bracketed districts.

APPENDIX D

STATEMENT SHOWING THE NUMBER OF KHADI SPINNERS AND WEAVERS IN THE STATE

Year	No. of Khadi spinning and weaving institutes functioning	No. of Khadi spinners	No. of Khadi weavers	Yardage of cloth produced
1	2	3	4	5
1957-58	15	5,300	1,005	32,342½ sq. yds.
1958-59	15	20,645	6,049	50,838¼ sq. yds.
1959-60	15	13,357	4,574	56,156¼ sq. yds.
1960-61	15	13,357	4,017	55,423 sq. yds.
1961-62	12	10,644	3,689	58,390 sq. yds.
Total :				253,150 sq.yds.

APPENDIX E

STATEMENT SHOWING THE COST OF HANDLOOM FABRICS IN DIFFERENT DISTRICTS OF ASSAM- 1961 CENSUS

Name of the district	Name of the fabric	Size	Cost of thread, dye, etc.	Time taken to make by Full time artisan	Time taken to make by Part time artisan	Sale price	Remarks
1	2	3	4	5	6	7	8
1. Lakhimpur	(1) Muga riha	12′ × 2′0″	Rs.11.00	3 days in T S. loom 1 day in F.S. loom	6 days in T.S. loom 3 days in F.S. loom	Rs.16.00 Rs.20.00	Generally coloured riha is not used except that of natural colour.
	(2) Pat (silk) riha	-do-	-do-	-do-	-do-	Rs.16.00 to Rs.20.00	-do-
2. Sibsagar	( ) Cotton chaddar 40s yarn (with simple cross border)	6 2½ cubits	Re. 1.50	2 days in T.S. loom 3 hours in F.S. loom	4 days in T.S. loom 6 hours in F.S. loom	Rs.3.00	Generally coloured chaddars are not used.
	(2) -ditto-with 80s yarn	-do-	Rs. 3.00	2½ days in T.S. loom 3½ hours in F.S. loom	5 days in T.S. loom 7 hours in F.S. loom	Rs. 5.00	-do-
3. Cachar	(1) Sari 40s yarn (with simple border)	10 × 2⅓ cubits	Rs. 3.50	4 hours in F.S. loom	8 hours in F.S. loom	Rs. 6.00	Only yarn dyed to fast colour is used for the manufacture of sarees.
	(2) Sari 80s yarn (with simple borders)	12 × 2⅓ "	Rs. 5.00	6 hours in F.S. loom	16 hours in F.S. loom	Rs. 8.50	Most of the weavers of this district are professional and are whole time workers.
4. Goalpara	(1) Bed cover (cotton)	90″ 45″	Rs. 6.75	4 hours in F.S. loom	2 days in F S. loom	Rs. 8.00	Only for plain border and cross border, 1½ lbs.2/20s dyed yarn is required.
5. Kamrup	(1) Muga mekhela	100″ 35 ′	Rs.16.00	1 day in F.S. loom	3 days in F S. loom	Rs.20.00	12 tolas of muga yarn is required for ½″ floral design on the borders.
6. Darrang	( ) Endi chaddar	108″ × 54″	Rs.16.00	1 day in F.S. loom	4 days in F.S. loom	Rs.25.00	1½ lbs. Eri yarn required for plain and ordinary eri chaddar.
7. North Cachar	(1) Cheleng (cotton)	108″ 46″	Rs. 2.25	4 hours in F.S. loom	2 days in F.S. loom	Rs. 3.50	With plain border and cross, ½ lb. 40s yarn required.
8. Garo Hills	(1) Garo Dakmonda (designed)	6′ 3′	Rs. 6.00	24 hours in T.S. loom	50 hours in T.S. loom	Rs.15.00	Floral spot with cross borders and diamond border.
	(2) Garo Dakmonda (Plain)	-do-	Rs. 5.00	10 hours in T.S. loom	18 hours in T.S. loom	Rs.12.00	Plain sheet with border of diamond design.
9. Mizo Hills	(1) Puanchu (floral designed)	7½ × 4′	Rs.50.00	Not known	Not known	Rs.100.00	Rs.20/- approximately required to purchase woolen yarn for floral designs.
10. United Mikir & North Cachar Hills	(1) Rijumphai (designed)	2yds × 1yd	Rs.11.00	10 days in T.S. loom	30 days in T.S. loom	Rs.30/- to Rs.50/-	..
11. United Khasi & Jaintia Hills	(1) Ryndia stem (endi)	10½′ × 3½′	Rs.23.00	20 days in loin loom	3 months in loin loom	Rs.35.00	Manufactured from endi yarn. Purchased from the plains people and woven by Mikir ladies on loin looms in some places at Bhoi area and Jowai subdivision.

APPENDIX F

TREND OF PRICES OF HANDLOOM PRODUCTS

Name of the District	Products	Unit	Price during (in rupees)							
			1939		1953		1954		1961	
			Urban	Rural	Urban	Rural	Urban	Rural	Urban	Rural
1	2	3	4	5	6	7	8	9	10	11
1. Goalpara	(i) Sari	Pair	..	5.00	..	8.75	..	8.50	12.00	N.A.
	(ii) Gamosa	No.	0.37	0.37	1.00	0.81	1.00	0.75	1.12	0.87
2. Kamrup	(i) Chaddar (cotton)	No.	2.12	2.00	6.00	6.00	6.00	6.00	6.50	6.50
	(ii) Mekhela (cotton)	No.	1.50	1.50	3.00	3.00	3.00	3.00	3.00	3.00
3. Darrang	(i) Chaddar (cotton)	No.	1.25	1.25	3.50	3.50	3.50	3.50	2.50	2.62
	(ii) Mekhela (cotton)	No.	1.00	1.00	3.00	2.75	3.00	2.75	3.19	3.25
4. Nowgong	(i) Chaddar (cotton)	No.	1.69	..	4.62	..	4.50	..	4.00	3.75
	(ii) Mekhela (cotton)	No.	1.12	..	2.75	..	2.75	..	3.12	3.00
5. Cachar	(i) Sari (cotton)	Pair	5.00	5.00	8.00	8.00	8.00	8.00	14.50	14.50
	(ii) Dhuti	Pair	8.00	8.00	16.00	16.00	16.00	16.00	19.00	19.00
6. Sibsagar	(i) Chaddar (cotton)	No.	1.75	1.75	6.00	6.00	6.00	6.00	..	..
	(ii) Mekhela (cotton)	No.	1.50	1.25	3.50	3.00	3.50	3.00	..	..
7. Lakhimpur	(i) Chaddar (cotton)	No.	1.75	1.50	4.12	3.00	4.12	3.00	4.75	3.50
	(ii) Bed sheet	No.	2.00	2.00	5.00	4.50	5.00	4.50	5.50	5.50
8. Garo Hills	(i) Dakmonda	No.	7.00	4.00	12.00	8.00	12.00	8.00	16.00	15.00
	(ii) Bed cover	No.	4.00	..	10.00	..	10.00	..	12.00	10.00
9. United Khasi & Jaintia Hills	(i) Bed cover	No.	..	..	7.00	7.00	7.00	7.00	10.00	11.00
	(ii) Towel	No.	..	..	..	2.00	..	2.00	3.00	3.50
10. Mizo Hills	(i) 'Puan'	No.	5.00	..	20.00	..	20.00	..	35.00	..
	(ii) Eptachei (Lushai chaddar)	No.	2.50	..	8.00	..	8.00	..	10.00	..
11. United Mikir & N.C. Hills	(i) Kachari chaddar	No.	..	2.00	..	6.00	..	6.00	..	..
	(ii) Kachari mekhela	No.	..	1.75	..	5.75	..	5.75	..	..

Sources : The Department of Economics and Statistics and the Department of Industries, Assam, Shillong.

APPENDIX G

TREND OF PRICES IN RAW MATERIALS IN ASSAM

Name of the district	Description of raw materials	Unit	Price during			
			1939	1953	1954	1961
1	2	3	4	5	6	7
1. Goalpara	20s yarn	Bundle	Rs. 9.00	Rs. 22.50	Rs. 22.50	Rs. 29.00
2. Kamrup	20s yarn	Bundle	Rs. 5.00	Rs. 22.50	Rs. 22.50	Rs. 22.00
3. Darrang	20s yarn	Bundle	Rs. 7.00	Rs. 21.00	Rs. 20.75	Rs. 26.50
4. Nowgong	40s yarn	Bundle	Rs.10.00	Rs. 26.00	Rs. 26.00	Rs. 35.00
5. Cachar	20s yarn	Bundle	Rs. 10.50	Rs. 22.00	Rs. 20.00	Rs. 25.00
6. Sibsagar	30s yarn	Bundle	Rs. 8.75	Rs. 25.00	Rs. 25.00	..
7. Lakhimpur	20s yarn	Bundle	Rs. 5.00	Rs. 21.25	Rs. 20.00	Rs. 27.00
8. Garo Hills	30s yarn	Bundle	Rs. 8.75	Rs. 30.00	Rs. 30.00	Rs. 50.00
9. United K-J. Hills	20s yarn	Bundle	..	..	Rs. 30.00	Rs. 35.00
10. United Mikir & N. Cachar Hills	20s yarn	Bundle	Rs. 5.00	Rs. 20.00	Rs. 24.00	..
11. Mizo Hills	(i) 20s yarn	Bundle	..	..	..	Rs. 30.00
	(ii) Cotton	Maund	Rs. 4.00	Rs. 13.00	Rs. 14.00	Rs. 40.00

Sources : The Department of Economics and Statistics and the Department of Industries, Assam, Shillong.

APPENDIX H

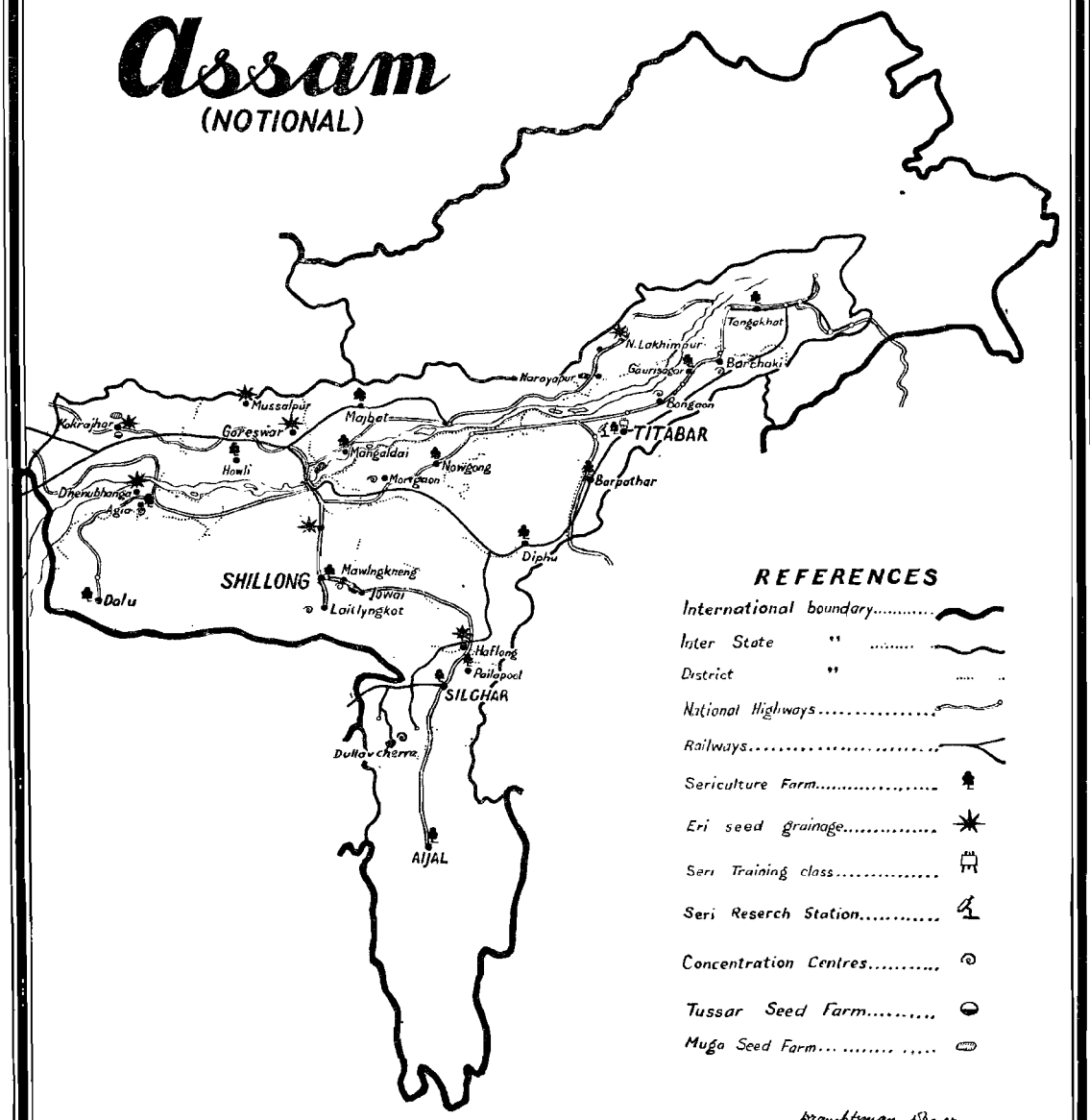
LIST OF GOVERNMENT SPONSORED HANDLOOM WEAVING TRAINING CENTRES

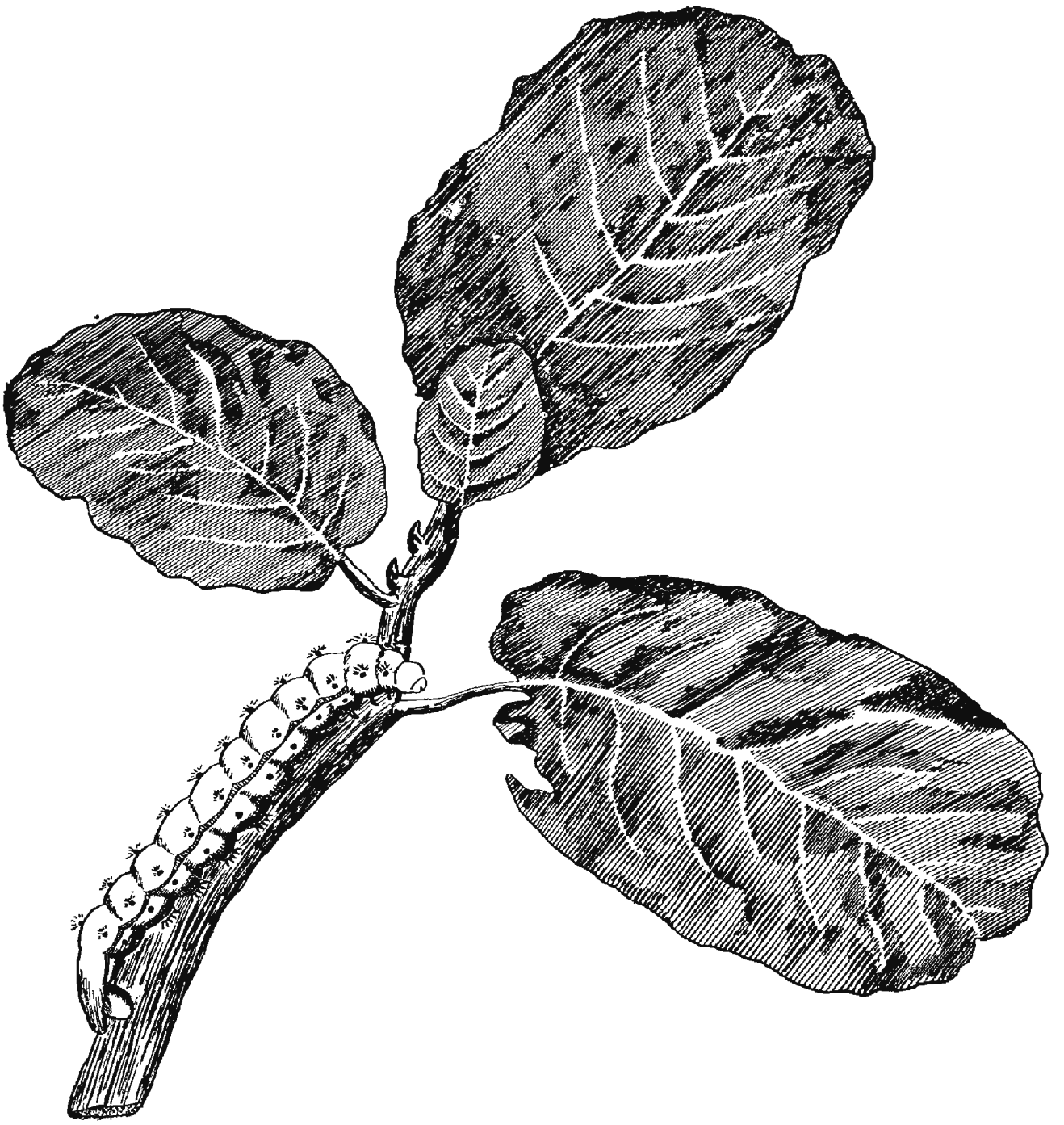
1.	Weaving Training Class	North Lakhimpur, Dist. Lakhimpur
2.	,, ,, ,,	Kacharipathar, Dist. Lakhimpur
3.	,, ,, ,,	Sarupathar, Dist. Sibsagar
4.	,, ,, ,,	Sibsagar, Dist. Sibsagar
5.	,, ,, ,,	Dergaon, Dist. Sibsagar
6.	,, ,, ,,	Sootea, Dist. Darrang
7.	,, ,, ,,	Thelamara, Dist. Darrang
8.	,, ,, ,,	Bongabari, Dist. Darrang
9.	,, ,, ,,	Raha, Dist. Nowgong
10.	,, ,, ,,	Mussalpur, Dist. Kamrup
11.	,, ,, ,,	Hajo, Dist. Kamrup
12.	,, ,, ,,	Ganakkuchi, Dist. Kamrup
13.	,, ,, ,,	Ramphalbil, Dist. Goalpara
14.	,, ,, ,,	Resubelpara, Dist. Garo Hills
15.	,, ,, ,,	Tura, Dist. Garo Hills
16.	,, ,, ,,	Shillong, Dist. U.K. & J. Hills
17.	,, ,, ,,	Jowai, Dist. U.K. & J. Hills
18.	,, ,, ,,	Sarihajan, Dist. U.M. & N.C. Hills
19.	,, ,, ,,	Diphu, Dist. U.M. & N.C. Hills
20.	,, ,, ,,	Haflong, Dist. U.M. & N. C. Hills
21.	,, ,, ,,	Aijal, Dist. Mizo Hills
22.	,, ,, ,,	Lungleh, Dist. Mizo Hills
23.	,, ,, ,,	Kabuganj, Dist. Cacnar
24.	Assam Textile Insutute	Gauhau, Dist. Kamrup

BIBLIOGRAPHY

1. The History of Civilization of the People of Assam by Dr. P.C. Choudhury, Department of Historical and Antiquarian Studies in Assam, Year 1959.
2. Census of India, 1951, by A.K. Mitra, Vol. VI.
3. Handloom Weaving in Assam by Mr. Medhi, Retd. Deputy Director of Sericulture and Weaving, Assam.
4. A Monograph on the cotton fabrics by H.F. Samman, 1888.
5. 'Assam'—1960 & 1961 by the Publicity Department, Government of Assam.
6. Hunter's Statistical Account of Assam by Trubner & Co., London, 1879.
7. A Review of Industrial and Economic Situation of Assam, July 1958, by the Director of Industries, Government of Assam.
8. A Survey of Cottage Industries in Assam, Vol.I and II, by the Department of Economics and Statistics, Assam, 1958.

Sericulture Map of Assam (NOTIONAL)





Muga Worm

CHAPTER II

Sericulture

Introduction: Assam is the leading State in India in so far as the production of non-mulberry silk is concerned, producingab out two-third of India's total output. Assam's association with the traditional muga dates back to time immemorial and she is still enjoying the monopoly in the production of muga silk. Till recently, eri rearing was practised in Assam only, although it is now being carried on in some other parts of the country also.

Sericulture is a traditional and age-old industry extensively practised in Assam. It is, next to weaving, practised during agricultural off-seasons, thus providing some employment to about 8 lakhs of people and thereby influencing the socio-economic set up of rural Assam. In the recent past, pat rearing was exclusively undertaken by a particular caste known as 'jugi' or 'katoni' as primary occupation. The members of this caste were also cautious enough not to divulge their indigeneous rearing techniques to other communities of the villages. Thus a wrong prejudice grew in the minds of caste Hindus that the rearing of pat was a traditional craft of low caste people. With the passing of time, however, this idea has gradually been disappearing and the culture of eri and muga has been taken up and practised by other sections of the people also in their off-time. Now the expansion of the craft can be visualised as the rearing of silk is popularised and intensified in the rural areas of Assam both in the plains and hills districts.

The silk fabrics of Assam produced out of eri, muga and pat yarn are durable and attractive and have always been acclaimed as forming the national dress of the Assamese people. Every Assamese woman irrespective of caste and creed is found to wear pat or muga attire in marriage and other festivals except those who are poor and cannot afford to have it. But it is said that during the reign of Ahom kings the pat and muga cloths were associated with special glamour and significance, and wearing of the same stuff was strictly confined to the members of the royal and noble families only.

History and origin : *"The art of sericulture and the rearing of cocoons for the manufacture of various silk cloths were known to the Assamese as early as the '*Ramayana*' and the '*Arthasastra*'. The former mentions *Magadha*, *Anga*, *Pundra* and the country of the 'cocoon rearers'. (*Kosa Karanam Bhumih*), which was no other than Kamrupa, lying to the east of Pundra. The classical writers, beginning at least with the first century A.D., made important mention about the production of silk and the silk trade in and through Assam. Schoff, on the basis the *Periplus*, contends that the silk industry originated in China and travelled from there to Assam and other parts of India. In the opinion of Watt, it originated in Manipur in Assam. He also adds that this place was the home of the silkworm—that the real mulberry silk

*Abstract from : The History of Civilization of the People of Assam..pp. 364-68 by P. C. Choudhury

insect originated in Manipur and went from there into China. The evidence from the 'Arthasastra', the 'Harsacharita' and the classical writers among others prove that in the art of the rearing of silk cocoons and the weaving of the finest silk textiles, the weavers of Kamrupa had a reputation equal to those of China. The tradition has been continued to the present times. This is confirmed by observations of some later writers. Qazim, for instance, writes that the silk of Assam was very excellent, resembling that of China. Taverniere remarks that the silk of Assam was produced on trees and the stuffs made of them were very brilliant. The industry, therefore, was continued throughout the centuries and Assam's silk fabrics are much esteemed in India."

It is reported that there is a legend as to the origin of sericulture. It is said that a Brahmin girl gave birth to three children without having a husband. So the society outcast her for the loss of her chastity. She had to live in the jungles for a long time in distress with her three children and went to a holy hermit who being sympathetic to her, applied some spiritual power and transformed her three children into three beautiful worms namely eri worm, pat worm and muga worm.

It is recorded that long long ago, a Chinese lady was married to some Tibetan prince who brought three species of silk worms of different varieties amidst her other dowris from her parents' house and used to rear them in Tibet. From Tibet, it probably came to Assam. *The Chinese record states that the ancient route for trade in silk and other goods with Bhutan and Tibet was through Udalguri in Darrang District of Assam. So from Tibet it easily came to Assam. The cultivation of different varieties of silk worms was first taken up by the Bodo community and then it gradually spread amongst other communities.

This trade appears to have been flourished and developed under the patronage of the Ahom kings. The kings took care and evinced keen interest for the development of this traditional craft. Due to the adequate measures taken by the Ahom kings, the rearers became skilful. The Sibsagar district which was the capital of the Ahom kings became the centre of all varieties of silk.

**During the British rule, attempts to develop this industry on commercial lines were made by several tea planters with the help of the Government upto 1885, but all of them were a total failure. This failure was commonly attributed to the lack of knowledge of technique in rearing, spinning and reeling.

This industry had to face much bottleneck during the British rule when the Indian market was flooded by foreign mill-made cheap artificial silk cloths; yet it had not lost its own glories and unique position it was holding in the past. Even now Assam is occupying a unique position in the production of two indigenous silk fibres, eri and muga, in the world. Many foreigners are surprised when they chance to have a look at the muga and pat fabrics of Assam because here is something new and distinctive about the materials and designs.

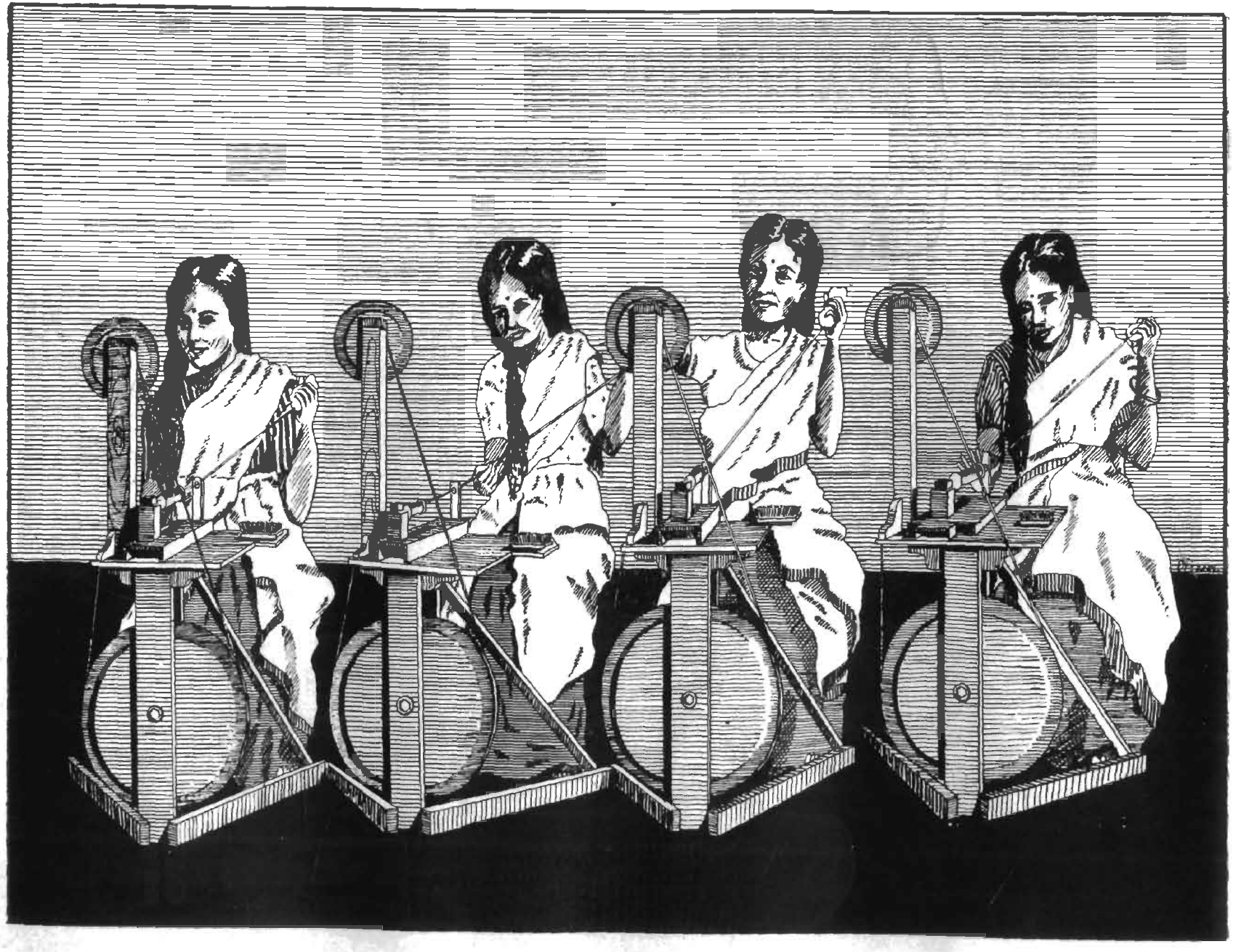
Nature of establishments : Sericulture is mainly a household industry. The main rearer is generally a female who takes this as her off-time occupation, though she gets

*Abstract from – 'Facts about Assam Silk' by the Sericulture & Weaving Deptt., Assam, Shillong.

**Abstract from a monograph on the Silk Cloth of Assam by B. C. Allen, I.C.S.



11. Children spinning cocoons on a takli as a pastime



Endi spinning with 'Das' type 'Charkha'

help from other members of the family also. Most of the rearers practise this trade to fulfil the requirements of the members of the family. But there is another type of workers who take this trade more or less on commercial lines though the practice is confined to only a few. Besides fulfilling the family requirements, such people sell the surplus, if there be any.

Though the majority of the rearers falls under the above two groups, there is another group carrying on the industry on purely commercial lines. At present, cocoon growers' co-operative societies have been organised in different parts of Assam with a view to get some common facilities and organising the trade on a strong footing to stabilise the industry. Both financial and techincal help are being extended by the Government to the deserving societies.

It is also observed that the rearing is conducted in most of the cases in the dwelling houses of the rearers, and sometimes in the open compounds of their hutments during fair weather. But nowadays, some rearers are constructing net fly-proof model rearing houses for the purpose of scientific rearing, which has proved to be of great success in yielding higher output in so far as pat silk is concerned.

Tools and implements : The tools and implements required for this industry are very few and simple and can easily be obtained locally at a moderate price. Most of the appliances are prepared by the artisan himself or by the family members. A list of the tools and implements used are given below :—

Tools & implements :	**Approximate cost :**
1. Bamboo trays (dala)	Rs. 2.00 to Rs.3.00 each.
2. Bamboo chandrakiful	Rs. 6.00 to Rs.8.00 per 100 yds.
3. Thread net	Re. 0.75 each.
4. Charkha (nidhiram)	Rs. 48.00 each.
5. Takli	Re. 0.50 each.
6. A boiling pan	Rs. 4.00
7. A complete loom	Rs. 50.00 to Rs. 75.00

Bhir Charkha : (Muga reeling charkha) It consists of a short wooden stick fixed with a fly-wheel on a stand. It is made to rotate with the palm of the hand but some improvement has been made to revolve it by means of a cycle free-wheel fixed on the wooden stick with a chain. One end of the stick outside the stand is fitted with a bamboo bobbin which serves as a reel. The reeler, by pedaling with his left leg, revolves the fly-wheel with the bobbin fitted on and the palm of his right hand holding the thread rolls it on the fore-arm of the right hand for twisting while the thread is kept going to the bobbin from the reeling basin. The twisting is also done by means of a spindle in improved reel.

Das Charkha : (Eri spinning charkha) It consists of wooden frame with three wooden legs as stand. The frame is operated by foot. The motion is transmitted to a bobbin loosely fitted in spindle from the main wooden wheel through two ancillary wooden wheels of different sizes by passing cotton cords through the groves of those wheels and the bobbin. This is fitted with automatic take-up and traverse which is operated by means of a can fitted on axle of the main wheel, a lever and a cast ratchet wheel. From the hands of the spinner the thread passes over a brass-cap

and is then wound on to the bobbin. The cap serves both as flyer and as a traverse guide.

Dala : (Bamboo tray) :– It is a bamboo tray, round or rectangular in shape, used to keep the worms for feeding purpose.

Improvements on spinning devices : Till today, takli locally known as takura, is used by most of the rearers for spinning the yarn. This process appears to be a laborious one and takes much time. The main argument put forward for use of the takli is that an appreciable uniformity of yarn can be maintained through this process and the spinning on takli can be done along with other domestic works just like knitting of woollen garments. However, it is encouraging to note that one Shri Nidhiram Das of Assam has invented an eri spinning machine which is named Nidhiram Charkha after his name. The rate of production on this charkha is much higher than that of the takli. The charkha costs about Rs.48.00. Besides Nidhiram Charkha, another improved type of charkha named Jaipur Charkha is also gaining popularity amongst the rearers for the purpose of eri spinning. Both these charkhas have a good deal of advantages over the takli.

Kinds of silk worms : *"The varieties of silk from Assam are *edi or erandi* (attacus ricini), made from the silk of the worm of the same name; *muga* (antheroea Assamoea) from a cocoon of the same name and *pat* (patta).

The rearing of edi cocoons takes a long time before they provide silk fit for spinning, the cocoons are fed as the name indicates, on castor plants. The endi cloths are usually white with a yellowish tinge, smooth as well as rough and very warm, used during winter. As the Latin name indicates, the muga silk is chiefly associated with Assam. The muga has varieties like the *campa*, the cocoons of which are fed on the *campa* tree (michelia champaka) and the *mejankari* or *adakari*, feeding on the plants of the same name. The muga cloths are usually yellowish with a tinge of gold and are often dyed red with lac. The pat silk is the product of *bombyx textor* and *bombyx croesi*; the cocoons are fed on the mulberry trees. Of all the silk cloths, the pat fibres are the smoothest and the finest, with a mixture of white yellowish tinge".

Rearing of silk worms according to areas : The concentration of rearers of a particular silk worm in different areas of the State depends on rainfall and climatic conditions of different parts of the State. The eri culture is done in areas where rainfall is fairly heavy and the atmosphere is humid, whereas mulberry silk worm thrives only under certain temperature and humidity. The composition of sericulturist groups in a few districts of the State as revealed through our field investigation is presented below.

Darrang : Sericulture is widely practised in the Darrang district by the Kachari (Bodo) and Assamese women who regard it as their traditional industry. This industry is a source of subsidiary means of occupation in this part of the State. Out of the three varieties of silk worms only eri and pat worms are widely reared in this district. Muga worms, it is learnt, are reared by a few members of the Nath or Yogi community.

Goalpara : The Rabha women are chiefly found to be engaged in rearing silk worms. Next come the Assamese womenfolk. Rearing of eri worms is extensively practised by the latter. But in very recent times, a few Assamese women have also taken to rearing pat and muga worms.

*Abstract from : The History of Civilisation of the People of Assam by P.C.Choudhury P. 365--366



Endi worm

Garo Hills : This district is inhabited mainly by the Garo and Rabha communities. Sericulture is widely practised here also. It is gathered that eri rearing is mainly done by the Rabha women as a subsidiary means of occupation. It is also reported that with the advent of the community projects, sericulture has been popularised amongst the Garo community through demonstrations and various forms of assistance rendered. Now-a-days, some of the Garo women have also taken to rearing pat and eri worms.

Nowgong : In the Nowgong district, sericulture is widely practised in the Raha area by the Nath community. This particular community is known here as 'Yogi' or 'Katoni'. They have taken up rearing of all the three varieties of silk worms, eri, pat and muga. A few of the Assamese and Kachari people are also found to be carrying on the rearing of silk worms.

Sibsagar and Lakhimpur : During the reign of the Ahom kings, both these districts were centres of all varieties of silk. The name of sericulture is traditionally associated with these two districts. The Assamese, Miri and some other tribal communities are also found to be engaged in this age-old craft. But amongst the silk rearing communities, the Nath community of these districts also deserves special mention, and its members are found to be very skilful in the art of rearing all the varieties of silk worms. It may be mentioned here that the bulk of the muga silk produced in Assam is being contributed by the rearers of the Sibsagar district.

Kamrup : Kamrup district is also a major silk-producing area of Assam where all the three varieties of silk worms are reared. Sualkuchi, a silk-fabric producing town of this district, has been taken up for detailed study of the industry where the subject will be dealt with more elaborately.

Cachar : Until 1953, the practice of this trade was rather scattered and unorganised in this district. But thanks to the co-ordinated effort of the Development Blocks and the Sericulture and Weaving Department of the State, this industry has grown to an appreciable extent in this part of the State. The main centres of silk worm rearing in this district are Pailapool, Dargarbond and Darmikhal villages. The Manipuri and Bengali communities mainly rear pat cocoons and the Kachari women are mostly engaged in rearing eri cocoons. During the field study, two cocoon marketing co-operative societies, one at Darmikhal and the other at Dargarbond village, have been found in this district. It is an interesting feature to note that most of the rearers have got model rearing houses constructed with some financial assistance obtained from the Development Blocks. Mulberry cuttings for cultivation and seeds for rearing are being supplied by the Sericulture and Weaving Department of the Government of Assam.

United Mikir & North Cachar Hills : In this district eri is extensively produced by the Mikir and Kachari people. With the encouragement of the Government Sericulture Farm, a small quantity of mulberry silk is also being produced.

United Khasi & Jaintia Hills & Mizo Hills: A large quantity of eri is produced in the United Khasi & Jaintia Hills and the product of mulberry silk is also being increased with the advent of several sericulture farms. The Mizo Hills district is still in its infancy in so far as the development of sericulture is concerned. The cultivation of mulberry is however being slowly popularised there by different Government agencies.

It is also observed that the rearers of all the districts of Assam maintain small gardens of mulberry, castor and champa (Adakari & Mejankari) for feeding the worms of pat, eri and muga respectively. Some of the rearers are reported to bring the leaves of castor and champa from the nearby jungles where they are found in abundance.

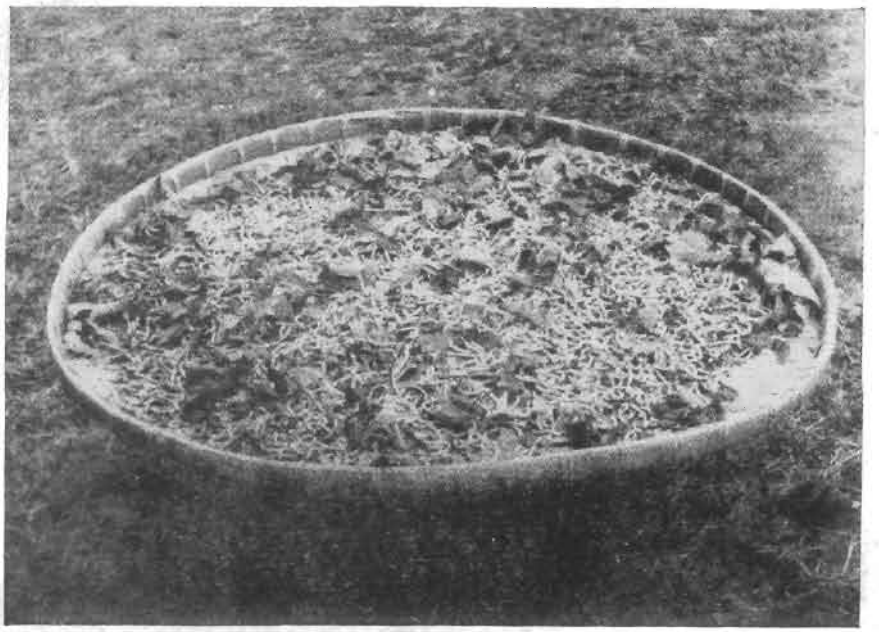
Technique of Production

Rearing of endi worms : Main food plants (1) Castor (Ricinis communis)
(2) Keseru (Heterepanax fragons)
(3) Tapioca

The endi worm derives its name from its main food, viz. the leaves of castor plant known as 'endi pat' in Assam. The leaves of Keseru and some other trees also serve as its food. The castor plant grows abundantly in the gardens of the rearers and also in the waste land near their houses. In the earlier stages, the worms are supplied with tender leaves and when they grow up, old leaves are supplied to them. Tapioca leaves are also occasionally used for feeding endi worms. It is a kind of shrub with decorative leaves and stout tubers underneath the soil.

Generally four to five broods can be reared in a year. The cocoons which are preserved for breeding purposes are kept in a bamboo basket having a narrow mouth locally known as 'khaloi' or 'kok'. The mouth is sealed with a piece of cloth to protect the cocoons from damage likely to be caused by rats and other insects. After 12 to 15 days in the hot season and 20 to 25 days during the winter moths emerge. After a day the female moths which can easily be distinguished by their larger bodies are tied to some bamboo sticks in such a way that they can easily move for mating and laying eggs. The males are left united and yet they do not attempt to fly away leaving the females. After mating, each female lays about 200 eggs. The life of the moth also ends here. The eggs are collected by the rearer and kept inside a folded piece of cloth with a loose knot at the top which is hung on the wall of the house for the purpose of hatching. The time for hatching varies according to the season and the temperature. Generally in the hot season the hatching takes not more than a week while in winter it takes about 15 days. In temperate weather about 10 days are required for this purpose. The knots in which the eggs are kept are opened just when the tender and the young worms begin to come out from the cell of the eggs. These worms are then transferred to a bamboo tray on which the young worms will be supplied with tender leaves of castor plants. These tender leaves are chipped into small fragments to make them soft. As the worms grow day by day, older leaves of castor plant or keseru are given to them. In some places it is seen that a piece of bamboo of nearly 5 to 6 feet in length is suspended horizontally with the help of ropes tied on both ends to some parts of the roof or ceiling and then older leaves are made into bundles and placed on the bamboo piece. The older worms are left on these leaves to allow them to take their feed freely.

The life of the endi worm undergoes four moultings. It is about a quarter of an inch in length at the first stage with light blackish colour. As it grows, its original colour is changed to orange colour with 12 rings on the body. In the third stage it is changed to white colour, and in the last stage the body becomes dark green. When it attains the last stage the worm is about 3 inches in length. As soon as worms attain maturity they cease to take their food and climb up to the top of the bundles of the leaves. These matured worms are then placed in a big bamboo basket filled with dried plantain or other leaves where



Pat or mulberry worms being fed on a 'dala'



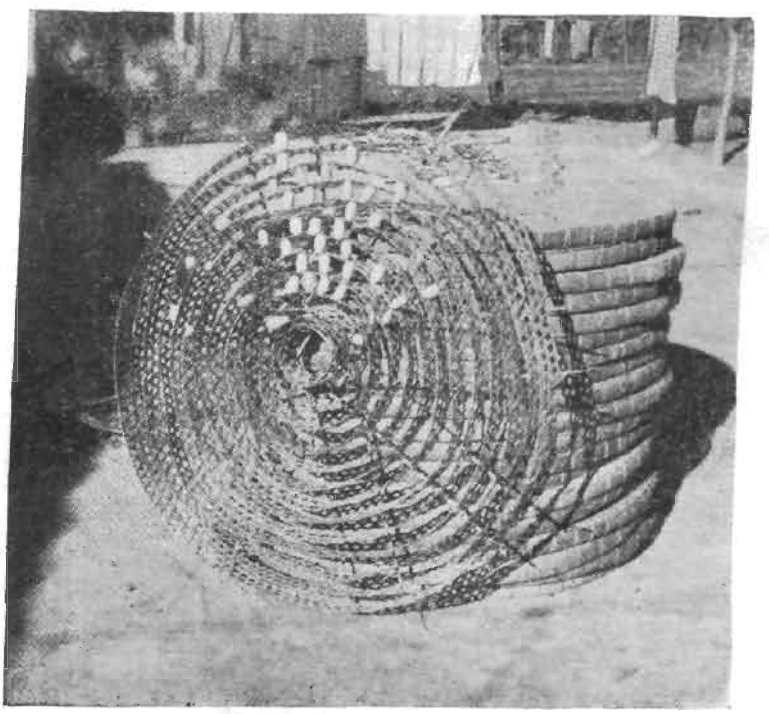
Rearing of pat worms conducted in a model rearing house (K. & J. Hills)



'Polu' (Endi) feeding of Castor leaves



Stage set for Cocoon (Endi) spinning



Mulberry cocoons seen in the Chandrakiful

they begin to spin on the same day. It takes them four to five days to complete the cocoons by spinning.

It is reported that a large number of worms die of peculiar diseases unknown to the rearers. Apart from these diseases, flies, rats and a few other insects are great enemies of the worms. Darrang, Goalpara, Kamrup, Garo Hills and North Cachar and Mikir Hills are the main endi producing areas.

Rearing of muga worm : Main food plants (1) Som (machilus odoratissima)
(2) Sualu (tethramthera monopetala)

During the field investigation, it was gathered that five broods of muga worms are generally reared every year. The different broods have different names in relation to the corresponding months in which they are reared up. They are Katia—reared in the month of Kartika, Jethua--reared in the month of Jaistha, Ahorua—reared in the month of Asahara, Bhadia—reared in the month of Bhadra and Jorua—reared in the coldest month. But the same set of rearers do not rear all the five broods throughout the year.

The cocoons which are preserved for the purpose of breeding are placed in a bamboo tray and hung up in the house. Proper care is taken to prevent them from damage likely to be caused by insects or other enemies. After about 15 days in the hot season, and 20 to 25 days in the cold season, the moths emerge. The female moths, recognizable by their larger bodies, are tied to a bamboo stick with a thread in such a way that they can move easily. The male moths are left free to facilitate them to move at liberty for mating. Each female lays about 250 eggs in three days and these are collected for rearing purposes. The eggs laid after these three days are rejected as they are regarded to be invalid. The eggs so collected are then transferred to a bamboo basket filled with pieces of cloths for hatching. The room in which the eggs are kept for hatching should be kept warm during winter. The hatching takes about 7 to 10 days according to the variations of seasonal temperatures. When the young worms begin to come out from the cells of the eggs they are generally placed on the tender leaves of a champa or sualu tree, as the case may be, in the open air. The silk worms which feed on the leaves of the champaka tree are found to yield finer varieties of silk. The muga worm is reared outdoors upon the trees in the open air but hatched and spinned indoors.

During the life time of the worms, they are to pass through four moultings which have got equivalent local names such as chaiura, duikata, tinichal kata and maika. In the first stage, the worm is found to be a quarter of an inch in length. As the worm increases in size, day by day, the colour also changes and the body becomes light. In the final stage, it becomes as thick as the forefinger with light green colour. Its body is composed of ten rings.

The muga worms take their food from early morning till evening with an interval at midday. At night they come down and take shelter as we do. They are also found to descend the tree as and when the leaves of that tree is exhausted. In such a case, the worms are transferred to some other tree.

The worms require about 25 to 40 days to attain maturity. It is observed that the worms are regarded to have attained maturity as soon as they cease to take their food and begin to descend the tree. The mature body of the worm gives a peculiar sound if slightly tapped. At this stage the worms are transferred to a basket made of bamboo filled with dried plantain or other leaves where they

begin to spin. Two or more worms generally do not choose the same leaf for spinning. From the date of their transfer to the basket it takes four to six days for complete formation of the cocoons.

As the worms are reared in the leaves of the trees in the open air, they are exposed to the attacks of crows, kites, bats, owls and flies. These enemies sometimes cause great damage and the rearers have to take much precautions and care to prevent them from doing so. Besides these factors, some diseases are also prevalent among the worms of which no remedial measure is known to the rearers. A good number of the worms die of 'phula rog'. When the worms suffer from this type of disease, the body of the worm is swollen and the colour is also changed and becomes yellowish. Besides, the worms cease to take their food as soon as they are attacked by this disease. To get rid of this disease, the rearers sprinkle some water with 'tulasi pat' (leaves used for Ayurvedic medicines—regarded sacred by the Hindus) on the diseased worms. Muga is extensively grown in the districts of Lakhimpur, Sibsagar, Kamrup, Goalpara and Garo Hills. The reeling of Muga is mostly done on 'bhir' as well as with improved reeling machine.

Rearing of pat worms :

Main food plant : Mulberry.

The varieties of pat worms found to be reared in Assam are locally known as Barpalu and Sarupalu. The worms mainly sustain upon the leaves of mulberry trees and are reared in the dwelling house or model rearing house of the rearer. The mulberry trees are planted in some selected plot of land near the house of the rearer. The rearer has to take much care for his mulberry garden and has to fence it also to protect it from cows and goats.

Healthy cocoons are generally selected for breeding. The cocoons are normally kept in a bamboo tray and placed in a rack made of bamboo specially meant for the purpose. This is done to avoid damage of the cocoons from rats or other insects. Generally the moths emerge within a period of 10 to 20 days according to the variation of seasons. Then both the male and female moths are kept in a bamboo tray for about four days for the purpose of mating after which the female moths are separated and transferred to a piece of cloth where they use to lay eggs. After about two days the moths die. The eggs are put again into another piece of cloth for hatching which is then kept in a bamboo basket. After a few days the young worms come out from the cell of the eggs.

The young worms are then transferred from the piece of cloth to a bamboo tray. They are supplied with tender leaves of mulberry trees. The leaves of mulberry trees are chipped into very small pieces and supplied to the worms for consumption as food. As they grow up, older leaves are given to them.

It is observed that the pat worms, like endi and muga, also have to pass through four moultings. During these processes, they change their skin as well as colour. The worms generally take 30 to 40 days to attain maturity. After that, they are transferred to a bamboo tray locally known as 'dala'. This dala is divided into several compartments with bamboo mats.

Chandrakiful : Normally each compartment is alloted to two or three worms where they begin to spin. About five days are required to complete the spinning of cocoons.

These worms have also numerous enemies such as rats, birds and insects. To avoid from all these, the State Government is

rendering all sorts of technical and other guidance.

Method of spinning : Endi spinning : To make the cocoons ready for spinning, the chrysalis still alive inside the cocoons should be destroyed by exposing them to the sun or by boiling the cocoons in water. Then the cocoons are cut open and the chrysalis extracted. In some places, the chrysalis are extracted only by cutting the cocoons open without boiling or exposing them to the sun. The next stage is to boil the cocoons to make them softer for easy spinning. For this purpose the cocoons are placed in a cauldren or 'karrai' filled with water and boiled for about two to three hours by adding some soda to it. Proper care is also taken to guard against over boiling. After this process is over, the cocoons are kept on a sieve to allow them to be cooled down and then washed well in clean water to remove other particles.

Two different methods of spinning are prevalent in different parts of the State. In some places, 'takli' locally known as 'takura' is used, while in other places, the spinning charkha (an improved labour-saving device) is employed. For spinning on the takli, the cocoons are placed on a vessel filled with water and each cocoon is taken up separately. Thus the open mouths of the cocoons broadened with the hand are placed one by one on one end of a bamboo stick of about 8 inches in length. Some fibres of the cocoons so arranged are then connected with the top of the takli, which will in turn, be caused to move with one hand for twisting the fibres and finishing the same into thread. The uniformity of the thread is preserved by rubbing it between two fingers while twisting is in progress. The joining of thread ends with fibres of the cocoons is also done in the same way (i.e. by rubbing). The twisted thread is wound on the lower part of the handle of the takli and made like a bobbin. Wet cocoons are generally used for spinning on the takli. The process of spinning on the charkha is quite different and only dried cocoons are used for the method of spinning. In the beginning, the spinner takes some cocoons in his hand and some fibres of a cocoon are attached on the spindle head of the charkha, while it is operated by paddling. The spindle of the charkha with a rope-wheel attachment, when in operation, rotates at high speed and as a result the thread is twisted and wound on the spindle itself in the shape of a bobbin. During this process, rubbing of the twisted thread with the fingers of the other hand is required to preserve uniformity of the thread. The fibres of the cocoons in hand will be connected one after another with the end of the thread already twisted and wound on the spindle and the production of thread will continue as before.

Muga spinning : The muga thread is generally not spun/it is rather reeled. For this purpose, the cocoons are boiled in a vessel for about an hour in an alkaline solution and then they are allowed to cool down. The process of reeling muga thread is much easier than that of spinning endi thread. Generally, two persons are required for reeling. One collects the ends of the fibres out of eight to ten cocoons and passes them over to the other person over a bamboo bar already placed for that purpose. The other person rubs the ends of the fibres to make a single thread and reels the same on an instrument just like charkha (made out of wood) known as 'bhir' in the local language. While reeling, the fibres come out from the cocoons automatically and form into thread. The reeled thread is wound on a wooden or bamboo instrument locally called 'chereki'. Thus it is seen that the muga thread is not spun; but occasionally it is spun with 'takli' or 'Das type' charkha to make it stronger and finer.

Pat spinning : The pat thread is sometimes spun, but generally reeled in the same procedure as muga reeling. To make the cocoons ready for reeling, the chrysalises alive inside the cocoons are to be destroyed either by exposing them to the sun or by boiling. As the cocoons have a yellowish tinge, the cocoons are placed in a vessel with water and boiled by adding some amount of potash to it to make the thread white and soft. During the period of boiling, the cocoons are stirred with a bamboo stick and as a result the filaments of the cocoons are attached to the stick. The ends of the filaments thus attached are taken to the reeling machine where the thread is reeled. Spinning of pat thread can also be done with any one of the following devices, Takli, Das type charkha, Joypur charkha.

Weaving of silk cloths : The *modus operandi* of silk weaving is almost the same as that of cotton weaving. Three different methods of warping can be seen in different parts of the State. They are (1) ground warping, (2) peg warping and (3) drum warping. All the different stages of manufacture of handloom cloths have been discussed in the chapter on handloom weaving. The art of silk weaving requires a high degree of skill and dexterity on the part of the weavers. In most cases, this is found to have been traditional and hereditary.

Finished products : The following list will give us an impression about the different types of silk fabrics being produced by the traditional artisans throughout the State.

Eri products :

1.	Eri Borkapor	used by males as dhoti and wrapper respectively.
2.	Eri Pashra	
3.	Eri Chaddar	

Pat and Muga products :

1.	Mekhela	Description of these fabrics have already been given in the chapter on Handloom Weaving.
2.	Chaddar	
3.	Riha	
4.	Cheleng	
5.	Churia	
6.	Blouse piece etc.	
7.	Sari	
8.	Dhara & Jainsem	Silk fabrics produced at Sualkuchi and used by the Khasi and Jaintia ladies.

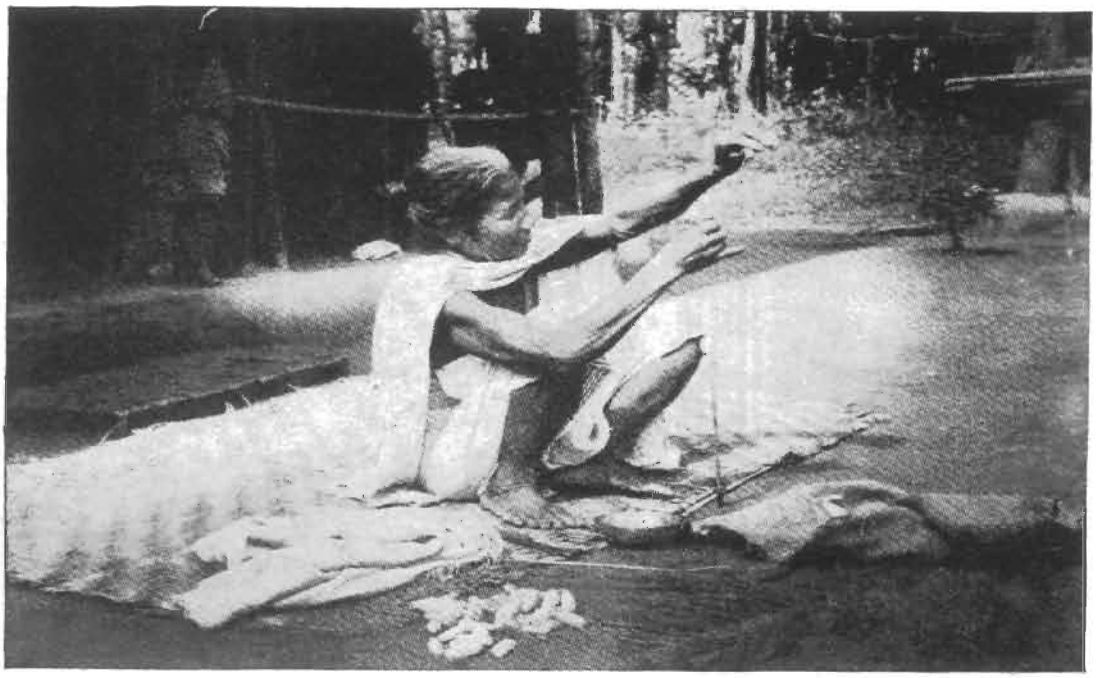
Let us now discuss some salient features of different varieties of silk cloths produced in Assam.

The endi cloth produced in Assam is warm, durable and easily washable. This cloth is generally used as a wrapper during winter. An old Assamese proverb amply justifies the utility of endi clothes—

'dair pani erir kani', which means 'endi cloth warms a person as the water of curd cools him. Palasbari, Kokrajhar, North Lakhimpur and Haflong are a few important endi-producing centres.

Muga fibres appear to be yellowish with a tinge of gold. Cloth prepared out of muga thread is much lighter than eri cloth and gives a pretty gloss after wash. The muga fabric is attractive, durable and fine and hence the price of this fabric is higher than that of endi fabric. Sualkuchi, Sibsagar and Jorhat are the main producing centres of muga cloths.

The pat cloth is an article of luxury. It is finer and whiter than both endi and muga fabrics. It is also very soft, durable and surely the costliest amongst the three varieties of cloths produced in Assam. A local saying stated below will clearly show the supremacy of pat silk over others—



An old Kachari woman spinning Endi silk on a 'takli'



General view of a Govt. mulberry farm



A tribal woman with her Endi yield bound for market place

'kona houk khora houk bhuyar puali fota houk singa houk patar tongali'
—(Blind or lame he may be he is still a Bhuyan's son; it is still a scart of pat silk; though tattered and torn).

Economics of the industry : In this section we shall deal with other aspects of the industry, such as cost of production, source of marketing, financial and technical assistance on the basis of information and data available from different sources.

Cost of production : Cultivation of silk worms and silk weaving is not a principal or main occupation of the people; it is rather a subsidiary occupation and carried on during off-time of the artisans. The hours of work devoted to silk rearing or silk weaving is not uniform in cases of all rearers or weavers. Also the hours of working in the particular line differ from day to day for the same person. So it is very difficult to find out the cost of production. Besides there is another difficulty in that regard. The products, whether of yarn or silk fabrics, are not of uniform quality, however, an attempt is being made to give an idea about the earnings of rearers per establishment. For this purpose, two tables extracted from the Report on a survey of Cottage Industries in Assam (conducted in 1954-55) published by the Department of Economics and Statistics, Assam, are furnished below :—

Table I

Average cost and receipt per establishment (silk rearing)

(Figures in brackets indicate percentage of total costs and receipts)

(In Rs.)

Industry	Average cost per establishment								Receipt		
	Depreciation of fixed capital	Value of short live equipments	Wages paid to hired-workers	Value of raw materials consumed	Value of fuel and other materials consumed	Other costs	Total cost (excluding family labour	Value of products manufactured	from servicing and work done for others	Total receipt	Average cost as percentage to average receipt
1	2	3	4	5	6	7	8	9	10	11	12
Silk Rearing	8/-	14/-	..	30/-	3/-	..	55/-	182/-	..	182/-	30.2
	(14.5)	(25.5)		(54.5)	(5.5)		(100.0)	(100.0)		(100.0)	

Table II

(Figures in brackets indicate percentage to average value of products manufactured per establishment)

(In Rs.)

Industry	Average value of products manufactured per establishment	Average value of products remaining for home consumption per establishment	Average value of products remaining unsold per establishment	Average value of products sold per establishment				
				In local markets	In markets within the district	In markets within the State	In markets within Indian territory	Total sold
1	2	3	4	5	6	7	8	9
Silk Rearing	182/-	27/-	29/-	120/-	5/-	1/-		126/-
	(100.0)	(14.9)	(15.9)	(65.9)	(2.8)	(0.5)		(69.2)

Marketing : Those rearers who do not spin themselves generally sell their raw cocoons to the spinners who in their turn sell the yarn to the weavers. During the field study, it was revealed that although Palasbari of Kamrup District is renowned for its production of endi fabrics, cocoons are not produced in this town. The spinners and weavers of this place purchase their requirements from other villages, Dhenubhanga and Dudhnai. A few years ago, it was reported, there was no organised markets for cocoons in the State. The marketing of cocoons was then a monopoly business of middlemen and traders. They used to advance some money to the rearers during lean seasons against a certain amount of their production of cocoons. The producers had no other way out but to sell their cocoons at a price dictated by the middlemen, which was invariably a low price. The middlemen in their turn fetched a higher price by selling off the cocoons to the spinners and weavers. Thus, we see that in the past, middlemen or traders enjoyed the lion's share of the profit obtained from marketing of cocoons, depriving the producers, that is, the rearers of their due. But now, in the sphere of marketing, the picture has completely changed. It is heartening to note that with a view to safeguard the interest of the cocoon producers, spinners and weavers from the grip of the middlemen, the Government has started three cocoon marketing centres in the important cocoon growing areas of the State. These centres purchase raw cocoons at a fixed price from the rearers and sell them in time to the reelers and spinners at a reasonable price.

As regards marketing of silk cloth, the problem is no longer as great as it was a few years back. The various silk fabrics of Assam have attained much reputation and popularity in the markets within and outside the State. The bulk of the silk fabrics produced in Assam are sold in local markets, fairs and hats. Special mention may be made about the Daranga Mela, which is held in the Kamrup District near the Bhutan border during the winter season every year. Large quantities of endi cloths are sold in this mela to Bhutias who come from across the border. Marketing of silk products is also done through the State Government Emporia, Khadi Bhandars and co-operative sale centres within and outside the State. Besides these, some selling agents are also found in the prominent towns of Assam dealing in silk fabrics produced in Assam.

Assistance : After independence, the national Government has realised the importance of this age-old craft and is trying to render all sorts of facilities and assistance to the rearers. Along with the State Government, the Central Silk Board is also playing a vital role towards the proper implementation of schemes relating to the development of sericulture. Besides financial assistance, supply of mulberry cuttings and distribution of implements at subsidised rates are made available to the rearers through development blocks. The following schemes deserve special attention.

Sericulture Farms : With a view to supplying disease-free seeds and mulberry grafts to the rearers, eleven such farms have been started in different parts of the State (See Appendix C).

Basic Muga Farms : To meet the growing demand for muga seeds, three such farms have also been started in prominent muga growing centres of the State (See Appendix D)

Eri Seed Grainage : Eight such grainages have also started functioning in suitable places in order to increase the supply of

disease-free eri seeds to the rearers. (See Appendix E).

Marketing, Training and Research : For the purpose of facilitating marketing of cocoons at reasonable prices, ten cocoon growers, co-operatives have so far been formed in the whole of Assam. (See Appendix F). To train up personnel for different development programmes, a certificate course of one year's training is provided in the Sericulture Training Institute, Titabar. The Sericulture Research Station, Titabar, has been addressing itself to the problems of eri and muga with limited investigations in mulberry silk.

Chowki Rearing Centres : Another scheme known as chowki rearing is being implemented by the Government. In these centres, the silk worms are reared by trained departmental personnel to a certain stage and then distributed to the rearers of the neighbouring areas.

Production : The amount of raw silk produced in the State during 1956-57 and 1957-58 is stated below :—

	1956-57	1957-58
1. Eri cut cocoons ...	482,000 lbs.	500,000 lbs.
2. Muga raw silk ...	184,600 lbs.	190,000 lbs.
3. Mulberry raw silk ...	27,600 lbs.	28,000 lbs.

Prospects of the industry : The silk fabrics of Assam have got both aesthetic and utilitarian values and there is immense possibility of expansion of the demand for fine varieties of these fabrics, both inside and outside India. This industry is mainly a subsidiary occupation providing part-time employment in the off-season to the cultivators and the members of their families.

The climatic conditions all over the State are favourable and permit the rearing of all the three varieties of silk worms. This trade does not require full time attention of the rearers. The idle manpower may be mobilized and employed in the pursuit of silk worm rearing which requires a very small amount of investment. The leisure time of the cultivators (both male and female) can be fully and profitably utilised in rearing silk worms. The children members of the cultivators' families may also be employed in practising this trade as it does not require physical strength on the part of the rearer.

Conclusion : Silk is Assam's traditional fabric and clothing. Silk rearing and weaving is an age-old occupation of the people. The special variety of silk (called muga) which has natural golden yellow or light brown colour is a durable and fascinating stuff. This variety of silk is not found in any other part of the country and is less known outside Assam. This should find great favour outside the State and abroad.

Endowed with all the advantages like temperate climate, density of population, suitable soil, traditional skill, agrarian economy etc., the State of Assam is pre-eminently suited for the growth of sericulture. The various developmental schemes taken up and implemented during the First and Second Five Year Plan periods are having a definite impact on the production of raw silk in the State. The improved methods (agronomical) have been advocated in the matter of cultivation of food plants, selection of suitable strains and their propagation have been effectively done. With regard to the silk worm rearer as well, improved varieties have been evolved and popularised amongst the sericulturists, thus reaping the benefit of very low incidence of diseases and high output of cocoons and raw silk in the State.

On the close study of the craft from various angles, we are definite about the fact that it has a very wide scope of expansion and development in the State of Assam in relation to other parts of the country. If the developmental schemes chalked up by the Government are availed of and utilised by the people to the best possible extent, it will no doubt prove to be of immense value in raising the economic condition and per capita income of the vast multitude of semi-employed rural population. The anticipated increase in output at the end of the Third Plan period will place the industry on a strong footing—augmenting a still stronger and brighter future.

We may now close this chapter with a felicitious comment of the Planning Commission:-

*"The problems of development of an underdeveloped economy is one of utilizing more effectively the potential resources available to the community, and this involves economic planning. But the economic condition of a country at any given time is a product of the broader social environment, and economic planning has to be viewed as an integral part of a wider process aiming not merely at development of resources in a narrow technical sense, but at the development of human faculties and building up of an institutional frame-work adequate to the needs and aspirations of the people".

APPENDIX A

FACTS ABOUT ASSAM SILK

1. Production of silk (in pounds)

Year	Eri cut cocoons	Muga raw silk	Mulberry raw silk
1953	383,000	100,500	22,944
1954	395,000	121,000	23,620
1955	477,000	157,000	26,120
1956	482,000	184,600	27,600
1957	500,000	190,000	28,000
1958	537,820	207,000	31,457
1959	500,000	186,000	30,800
1960	450,000	120,000	25,000
1961	513,000	125,000	29,000

2. There are about 32,000 handlooms engaged in silk weaving in the State.

3. Acreage under mulberry cultivation has increased from 1,600 acres in 1953 to 2,300 acres by the end of 1961.

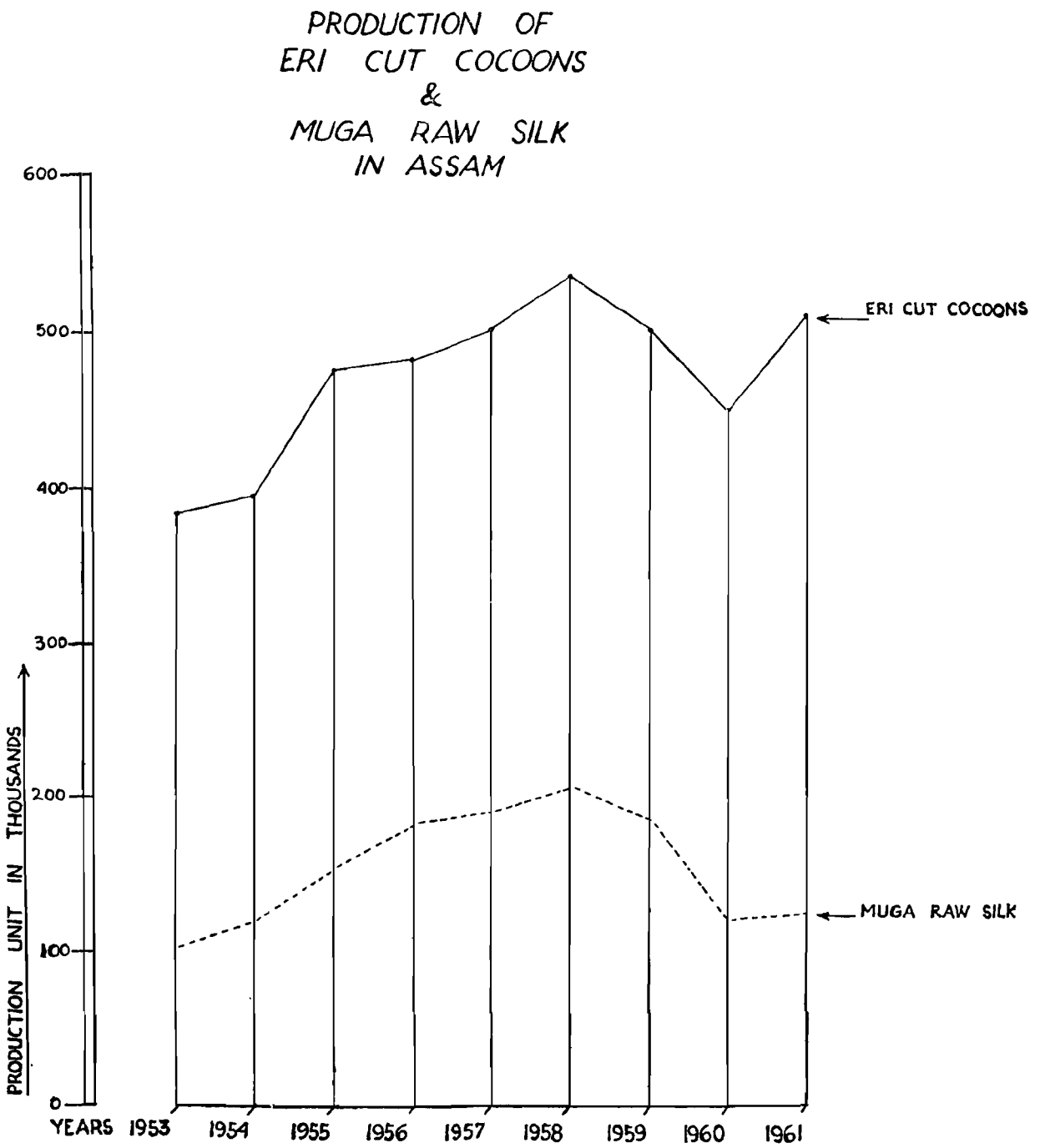
4. Assam is exporting a large quantity of eri cocoons outside the country and considerable quantity of eri fabrics outside the State through different trade channels to Sikkim and Tibet via Kalimpong.

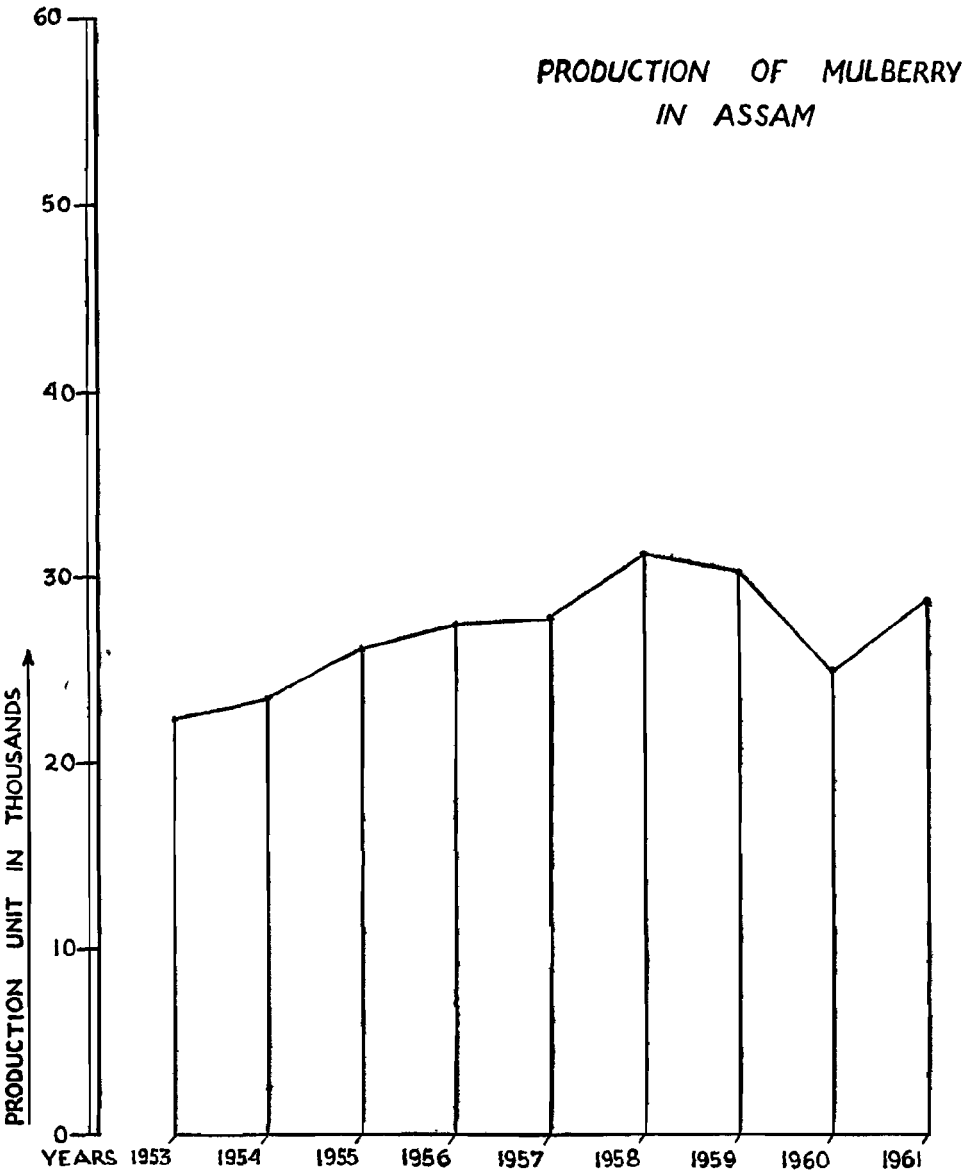
5. A good quantity of muga raw silk is sent outside Assam for use in embroidery, ornamental work and fishing line.

6. The Sericulture Research Station has been established during 1953-54 at Titabar at a total cost of Rs. 146,000/- on a 50: 50 basis between the Central Silk Board and the State Government.

7. Rearing of Tussar silk worm has been introduced since 1957 amongst the tribal

*First Five Year Plan, p.l.


PRODUCTION OF
ERI CUT COCOONS
&
MUGA RAW SILK
IN ASSAM
600
500
400
300
200
100
0
PRODUCTION UNIT IN THOUSANDS
YEARS
1953
1954
1955
1956
1957
1958
1959
1960
1961
ERI CUT COCOONS
MUGA RAW SILK


PRODUCTION OF MULBERRY
IN ASSAM
60
50
40
30
20
10
0
PRODUCTION UNIT IN THOUSANDS
YEARS 1953 1954 1955 1956 1957 1958 1959 1960 1961

people living near forest reserves where 'asan' (tarminalia tomentosa) food plants of Tussar silk worm are available.

A Tussar Seed Cocoon Rearing Farm at Kokrajhar has been started (1959) by the Government with a view to producing seed cocoons for tussar producers. 'Arjun' and 'Asan' seedlings have been transplanted in the said farm from the nurseries. A few broods of tussar was also raised successfully on existing 'asan' plantations with seed cocoons obtained from Bihar.

8. The following table gives the price trend in the Assam silk market for March, 1961 with comparative rates for the previous month of the same year and for the corresponding month in the previous year.

Rs. Per Kg.

	March 1960	February 1961	March 1961
Eri silk			
i. Eri cut cocoons	7.70 to 13.30	11.00 to 15.40	11.00 to 15.40
ii. Eri yarn	55.00 to 77.00	55.00 to 66.00	55.00 to 66.00
Muga silk			
i. Reeling cocoons (per 1000)	12.00 to 15.00	15.00 to 20.00	15.00 to 20.00
ii. Raw silk (a) warp	77.00 to 99.00	77.00 to 99.00	77.00 to 99.00
(b) weft	66.00 to 77.00	72.60 to 77.00	72.60 to 77.00
iii. Silk waste	4.40 to 6.60	4.40 to 8.80	4.40 to 8.80
Mulberry silk			
i. Reeling cocoons	4.40 to 5.50	3.85 to 5.50	3.85 to 5.50
ii. Charkha raw silk (a) warp	77.00 to 88.00	72.60 to 77.00	72.60 to 77.00
(b) weft	66.00 to 77.00	63.80 to 72.60	63.80 to 72.60
iii. Silk waste	3.30 to 5.50	3.85 to 5.50	3.85 to 5.50

(Source : "Silk News Letter" of the Central Silk Board)

9. Different months and seasons of breeding of eri, muga and pat.

(a) **Eri** :—

(i) March ... April Brood.
(ii) April ... May ,,
(iii) May ... June ,,
(iv) July ... August ,,
(v) September ... October ,,

(b) **Muga** :—

(i) May ... June Brood.
(ii) June ... July ,,
(iii) August ... September ,,
(iv) October ... November Brood
(v) December ... January ,,

(c) **Pat** :—

(i) Spring ... 2 crops (early and late spring)
(ii) Summer ... 1 crop.
(iii) Autumn ... 2 crops (early and late autumn)

10. The rearers get the saplings of mulberry plant seeds, cocoons, etc. from the Government Farms and Grainages. The prices of the same are listed below.

Variety		Price	
(i)	Mulberry cutting	Free supply	
(ii)	Pat Seed Cocoons	Multivoltine,	Re.0.25 per 100
(iii)	Layings	Univoltine	Re.0.75 ,, 100
		Multivoltine,	Re.0.75 ,, 100
		Univoltine,	Rs.1.50 ,, 100
(iv)	Eri seed layings		Re.0.03 ,, gram

11. Followings are the improved varieties of silk worms evolved and popularised through government agencies.

(a) **Univoltine** :
- (i) Chinese white
- (ii) A.L. Bivoltine
- (iii) Chinese Golden
- (iv) Assamese Borpat white
- (v) Assamese Borpat yellow

(b) **Multivoltine** :
- (i) Assamese sorupat white
- (ii) Assamese sorupat moria
- (iii) Mysore local

(c) **Hybrid** :
- (i) Assamese sorupat white A.L. Bicoltine
- (ii) Assamese sorupat white Chinese white
- (iii) Mysore local A.L. Bivoltine
- (iv) Assamese sorupat moria A.L. Bivoltine
- (v) Assamese sorupat moria Chinese white

APPENDIX B

Trend of prices of raw materials in Assam

Name of the District	Description of raw material	Unit	Prices during			
			1939	1953	1954	1961
1. Goalpara	(i) Pat cocoon	Sr.	..	..	..	3.50
	(ii) Eri cocoon	,,	..	..	..	6.00
2. Kamrup	(i) Eri cocoon	Sr.	5-8-0	5-8-0	6-0-0	9.00
	(ii) Muga cocoon	1000 nos.	6-0-0	15-0-0	15-0-0	10.00
3. Darrang	(i) Eri yarn	Sr.	5-0-0	15-0-0	15-0-0	15.00
	(ii) Pat yarn (raw)	,,	20-0-0	60-0-0	60-0-0	60.00
4. Nowgong	(i) Eri cocoon	,,	2-8-0	6-0-0	6-0-0	7.00
	(ii) Pat yarn (raw)	,,	20-0-0	60-0-0	60-0-0	60.00
5. Cachar	(i) Pat cocoon	,,	..	..	..	3.50
6. Sibsagar	(i) Muga cocoon	1000 nos.	6-0-0	10-0-0	12-0-0	15.00
	(ii) Eri cocoon	Sr.	..	..	..	6.00
7. Lakhimpur	(i) Eri yarn	,,	3-0-0	12-0-0	12-0-0	14.00
	(ii) Muga cocoon	,,	6-0-0	20-0-0	20-0-0	20.00
8. U.K. & J Hills	(i) Pat cocoon	..	..	..	..	4.00
9. Garo Hills	(i) Muga cocoon	1000 nos.	..	..	5-0-0	15.00
	(ii) Eri cocoon	Sr.	..	..	4-0-0	6.00
10. United Mikir & N.C. Hills	(i) Eri cocoon	Sr.	2-0-0	3-8-0	3-8-0	6.00
	(ii) Pat yarn (raw)	,,	..	6-0-0	60-0-0	60.00
11. Mizo Hills	(i) Pat cocoon	..	..	..	..	4.00

APPENDIX C

List of Government Sericultural Farms in Assam

Source :- Department of Sericulture and Weaving, Assam, Shillong.

Name of the Farms with location	Date of inception
1	2
1. Government Sericultural Farm Titabar.	1919
2. Government Sericultural Farm, Shillong.	1925
3. Government Sericultural Farm, Senchoa.	1942
4. Government Sericultural Farm, Gourisagar.	1945
5. Government Sericultural Farm, Mangaldai	1948
6. Government Sericultural Farm, Jowai.	1949
7. Government Sericultural Farm, Hawli.	1952
8. Government Sericultural Farm, Diphu.	1952
9. Government Sericultural Farm, Aijal.	1952
10. Government Sericultural Farm, Pailapool.	1956
11. Government Sericultural Farm, Agia.	1956

APPENDIX D

List of Basic Muga Seed Farms in Assam

Source :— Department of Sericulture and Weaving, Assam, Shillong.

Name of the Farms with location	Date of inception
1	2
1. Basic Muga Seed Farm, Khanapara	1954
2. Basic Muga Seed Farm, Kokrajhar	1957
3. Basic Muga Seed Farm, Narayanpur	1957

APPENDIX E

List of Government Eri Seed Grainages in Assam

Source :- Department of Sericulture and Weaving, Assam, Shillong.

Name of the Seed Grainages with location	Date of inception
1	2
1. Government Eri Seed Grainage, Haflong	1952
2. Government Eri Seed Grainage, Mussalpur	1953
3. Government Eri Seed Grainage, Dhenubhanga	1953
4. Government Eri Seed Grainage, Kokrajhar	1954
5. Government Eri Seed Grainage, Ouguri	1955
6. Government Eri Seed Grainage, N. Lakhimpur	1956
7. Government Eri Seed Grainage, Nongpoh	1957
9. Government Eri Seed Grainage, Goreswar	1957

APPENDIX F

List of Cocoon Marketing Societies in Assam

Source :— Department of Sericulture and Weaving , Assam, Shillong.

Name of the Societies	Location	Date of inception
1	2	3
1. Dengaon Cocoon Marketing & Silk Growers' Society	Dengaon (Diphu)	3-1-1962
2. Dhakuakhana Cocoon Marketing and Silk Growers' Society	Dhakuakhana (N. Lakhimpur)	28-3-1962
3. Nitaipanidihing Silk Growers' Co-operative Ltd.	Nitaipanidihing (Sibsagar)	22-1-1962
4. Mohondijoa Cocoon Marketing & Silk Growers' Co-operative Ltd.	Mohondijoa (United Mikir & N.C. Hills)	26-2-1962
5. Nongpoh Silk Growers' Co-operative Society Ltd.	Nongpoh (Shillong)	12-3-1962
6. Behali Reshom Utpadam Samabay Samity	Behali (Tezpur)	31-3-1962
7. Baithalangso Cocoon Marketing & Silk Growers' Co-operative Society Ltd.	Baithalangso (United Mikir & N.C. Hills)	20-3-1962
8. Boko Reshom Producers' Samabay	Boko (Kamrup)	16-3-1962
9. Joypur Cocoon Marketing & Silk Growers' Co-operative Society.	Joypur (Cachar)	24-7-1962
10. Mukdangra Cocoon Growers' Co-operative Ltd.	Mukdangra (Tura)	28-12-1962

N.B.—The societies mentioned above were organised some 6/7 months earlier to the date of registration as noted against each.

BIBLIOGRAPHY

1. The History of Civilization of the People of Assam by P. C. Choudhury, 1959.
2. Hunter's Statistical Account of Assam.
3. 'Silk Cloth of Assam' by B.C. Allen, 1911.
4. 'Glimpses of Silk Industry in Assam' by Sericulture and Weaving Department, Government of Assam, 1955.
5. 'Facts about Assam Silk' by Sericulture and Weaving Department, Government of Assam.
6. 'Silk News Letter' by the Central Silk Board.
7. 'Assam', 1960 and 1961, by the Publicity Department, Government of Assam.

CHAPTER III

SILK WEAVING AT SUALKUCHI

Introduction : Sualkuchi village together with Sualkuchi Parbat and Bamun Sualkuchi have recently been grouped to form the new township of Sualkuchi on the north bank of the river Brahmaputra. The town lies about 29 miles from Gauhati in the north-west direction, and is connected with the latter by a good motorable road through Hajo, and by a motor launch service during the fair season. It is not served by any railway.

A traveller who approaches this small town, finds himself entering into a series of factories scattered throughout the town. As he moves on, the click-clack sound of the shuttle also moves along with him from house to house. Hardly can a single house be found without a loom. This town is undoubtedly one of the most famous silk fabric producing centres of Assam. Silk weaving is not uncommon in other parts of Assam, but Sualkuchi claims a technique, quality and reputation of its own which are unique in so far as muga and pat silk fabrics are concerned.

History & Origin : The History of the Sualkuchi silk industry has been traced back to about 340 years ago. It is believed that by about 1620 A.D. when western Kamrup was taken over by the Mohammedan rulers of Dacca, they brought a Hindu zamindar named Satrajit of Bhusana from East Bengal and gave him an estate at Sualkuchi. Probably that zamindar brought some weavers from other parts of the State and gave them necessary patronage to carry on their trade. The present weavers who are the direct descendants of the founders of the industry state that their ancestors used to tour other parts of the State and it was quite likely that they had picked up the techniques and the designs from those areas, and with their own dexterity developed them here.

Sualkuchi silk is as old as its people. Although the origin of silk weaving at Sualkuchi is still obscure, there is no doubt about its antiquity. During the olden days, the industry appears to have been patronised mainly by the nobilities and the ruling kings of Assam, and the type of fabrics produced were beautifully decorated with designs and were very costly being more intended to serve the individual needs and tastes of the well-to-do. However, the products of this industry enjoy a great reputation outside Assam also for their delicate designs, quality and durability. A huge quantity of muga saris is being regularly exported to West Bengal from this silk cloth-producing centre. The industry seemed to have faced a period of adversity perhaps due to the disturbed fortunes of the ruling kings and nobilities in olden times. It could not thrive well during the British regime as the latter did not take any interest for the progress of this industry in the State.

After Independence, the efforts and interests taken by the State Department of Sericulture and Weaving, the All India Handloom Board and the All India Silk Board, both sericulture and weaving industries have received a new lease of life and have begun to make good progress. Concentrated and vigo-

rous activities have been carried out for the development of sericulture and weaving industries in the State since the inception of these agencies.

Area, Households, Population and their Economic Activity: Sualkuchi town covers an area of 0.52 sq. miles and has 2,061 households with a population of 12,087 of whom 6,445 are males and 5,642 females according to the 1961 Census.

The population of the town is composed of (i) Brahmins, (ii) Baishyas, (iii) Kalitas, (iv) Koches, (v) Kumars, (vi) Keots, (vii) Kaibartas and (viii) Katonis but the majority of the households belong to the Baishya and Kalita Castes.

The following table gives the distribution of population according to the nine industrial categories of workers and non-workers.

Table No. I

Categories	Persons	Males	Females
Total population	12,087	6,445	5,642
Total workers (I—IX)	5,706	3,177	2,529
I. As Cultivator	228	213	15
II. As Agricultural Labourer	6	6	—
III. In Mining etc.	64	64	—
IV. At Household Ind.	3,943	1,530	2,413
V. In Manufacturing	238	214	24
VI. In Construction	3	3	—
VII. In Trade & Commerce	642	616	26
VIII. In Transport etc.	91	91	—
IX. In Other Services	491	440	51
X. Non-workers	6,381	3,268	3,113

There are eight registered co-operative societies organised in this town with the object of silk spinning and weaving. Six of them were found actually functioning and the remaining two were not functioning at the time of investigation. The total number of members enrolled in all these societies is estimated to be 630 at that time. The names of all the registered societies are given below:—

(1) Assam Co-operative Silk House Ltd. (functioning).
(2) Assam Co-operative Resham Samabay Pratisthan (functioning).
(3) Madhya Sualkuchi Resham Samabay Samity Ltd. (functioning).
(4) Sualkuchi-Bhatipara Muga Spinners' and Weavers' Co-operative Society (functioning).
(5) Sualkuchi Muga Spinners' and Weavers' Co-operative Society (functioning).
(6) Bamun Sualkuchi Silk Samity Ltd. (functioning).
(7) Purba Sualkuchi Resham Samity Ltd. (not functioning).
(8) Sualkuchi Pat-Muga Palupoha Samabay Samity Ltd. (not functioning).

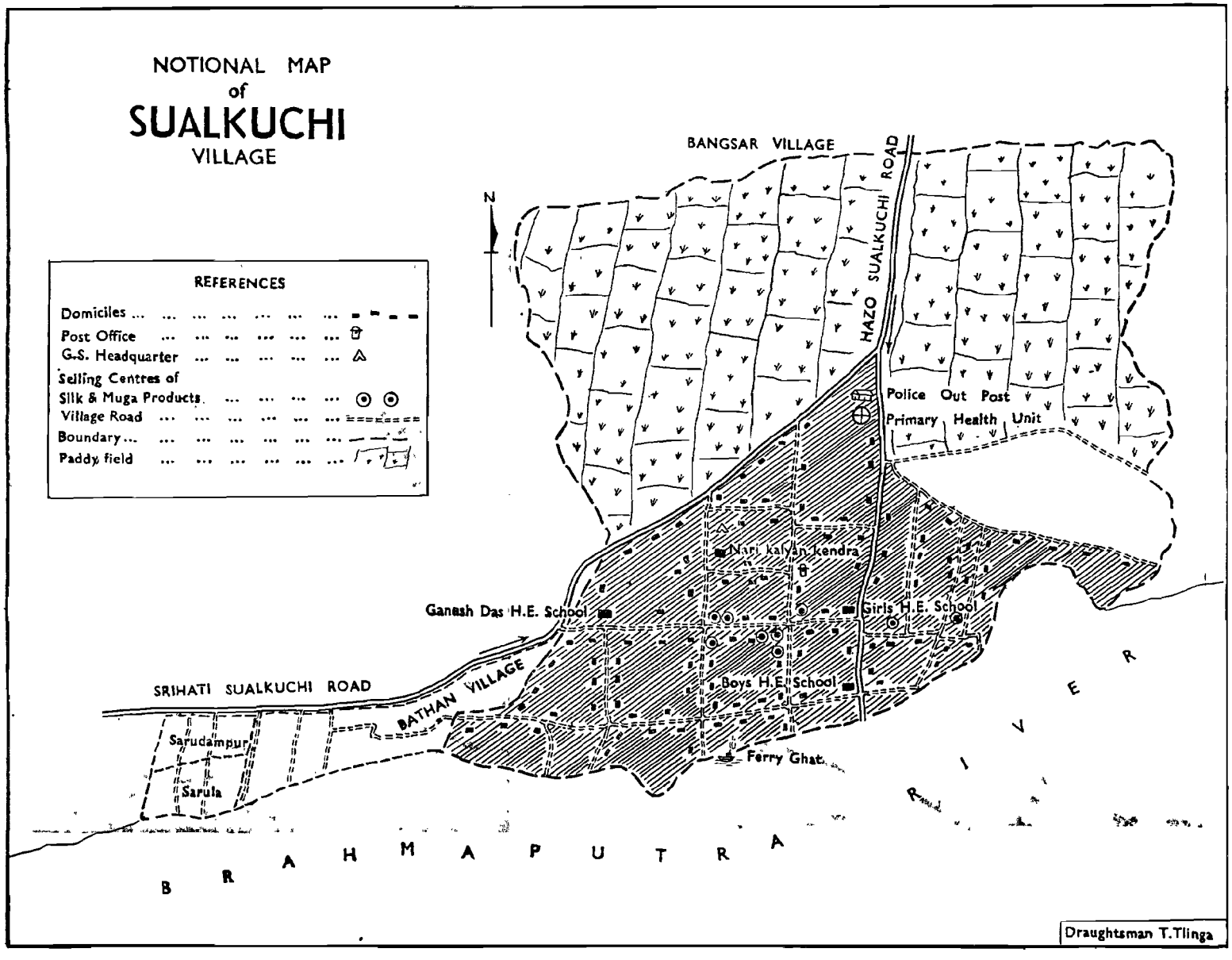
Though a good number of co-operative societies are functioning at Sualkuchi, the vast majority of the weavers numbering about 1,000 are still working outside the co-operative fold. Some of them are working on wage basis in local weaving factories while the others work individually.

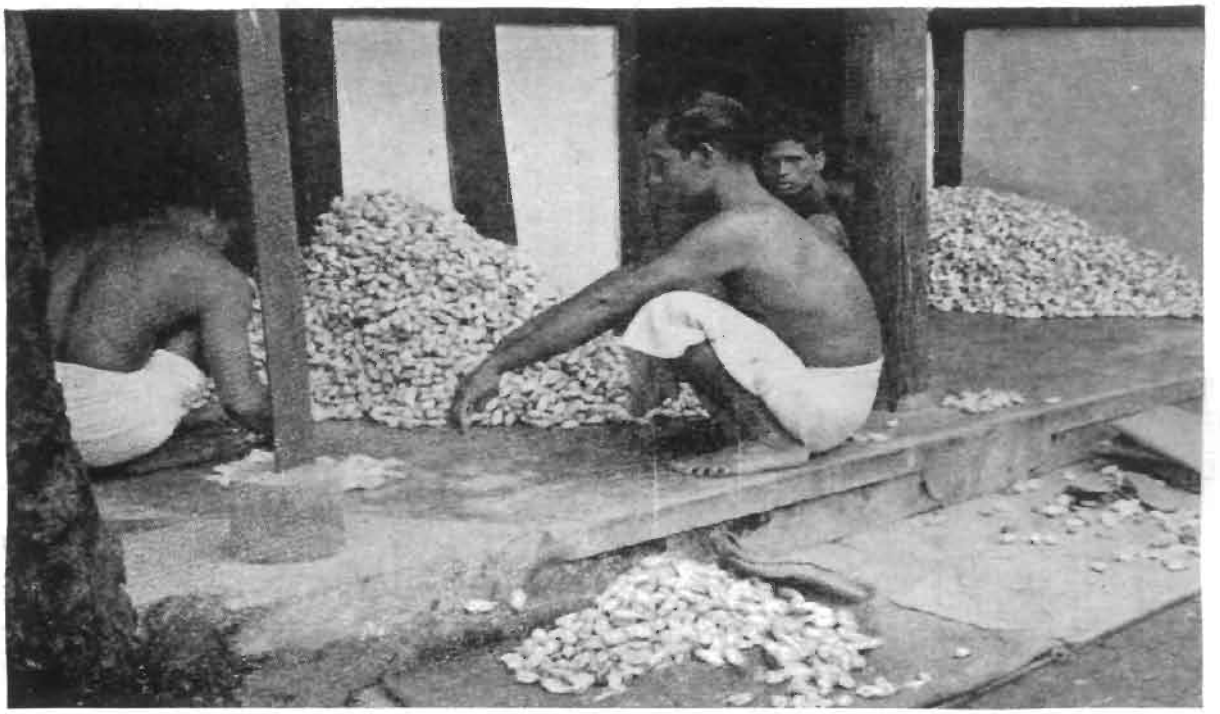
Nature of establishment: The weavers who work under the co-operative fold collect muga cocoons or pat yarn from their respective societies, and in the case of muga cocoons, they themselves spin the yarn and weave the particular cloth as ordered by the societies. On delivery of the finished products to the societies, they are paid specified rates on different cloths. Most of the weavers of this group work in their own houses.

A second group of weavers work in some factories within the town itself. They are provided with looms in the factory and the raw material is also supplied by the proprietor. They are paid on piece basis.

A third group of weavers work neither under the co-operative fold nor in factories,

NOTIONAL MAP of SUALKUCHI VILLAGE





Silk cocoons are usually supplied to the weavers from Co-operative Stores



Yarn is washed before finally used for weaving

but in their individual houses. They procure the raw materials from the local merchants either on cash payment or on credit. Their products are also sold to the local merchants with a very low margin of profit.

TECHNIQUE OF PRODUCTION

Raw materials : At Sualkuchi, muga and pat cloths are generally woven. Weaving of endi cloth is almost absent there. The chief raw materials required for muga and pat cloths are yarns of the respective varieties. The weavers used to procure muga cocoons from different districts of the State such as Garo Hills, Lakhimpur and Sibsagar and spin the yarn themselves. But as regards pat, they import yarn of this variety exclusively from Mysore State which is very famous for its pat silk production. It is also gathered that for the purpose of embroidery and designs, they require a small quantity of dyed yarn of different shades, which they generally procure from the local market.

The present costs of yarn of both these varieties are stated below :—

1. (a) Muga cocoons
   —Rs. 20.00 per 1000 cocoons.
   (b) Muga yarn
   —Rs. 20.00 to Rs. 40.00 per lb.
2. Pat yarn
   —Rs. 49.50 to Rs. 56.58 per lb.

The process of muga spinning as practised in this town is almost similar to that of other parts of the State, and has been elaborately described in the monograph at chapter II.

Preparatory processes : The art of silk weaving commences with the preparation of the warp. Spinning of muga yarn is generally done by the womenfolk and children of the weavers' families. The operations involved in the preparation of warps for most of the silk fabrics comprise of not less than four chief divisions, namely, 1. Bobbin winding, 2. Warping, 3. Beaming and 4. Looming. The first process commences with the preparation of warper's bobbin. It is generally done on traditional charkhas.

For silk weaving at Sualkuchi, it is seen that drum warping is very popular. It is perhaps better in any circumstances and, moreover, a time-saving device too. It can be done indoors and in a small space. A book creel is used for placing a good number of bobbins, say 80 to 120, and warping is carried on with many threads at a time. The drum is a big reel of $1\frac{1}{2}$ to 5 yards in circumference which is made to revolve vertically or horizontally, fixed in a suitable framework. The bobbins are placed in a book-creel by the side of the drum. The ends are taken from the bobbins and drawn separately through eyes of a row of needles arranged in a "heck". This arrangement helps in forming lease. The warp is to wound on the vertical drum spirally and the heck slides up and down. When the warp is cut off and unwound, it is brought to the form of a ball for the next operation.

Then follows the operation of beaming. The act of spreading the warp yarns and winding evenly and tightly upon a weaver's beam is called beaming. The warp ends are passed through a reed and the sheet of yarns is wound in tension upon the beam.

Drafting, denting, tying up of healds, etc., come under looming or getting.

Drafting: Drawing the warp ends through the eyes of the cotton healds in a pre-destined plan according to the nature of the weave is called drafting. There are various kinds of drafting: straight, pointed and mixed. Four

healds in a set are numbered 1, 2, 3, 4 from the weaver's side. For plain weaving the warp threads are to be drawn in the order 1, 3, 2, 4. That is the first thread through an eye of the first heald, the second thread through that of the third heald, the third thread through that of the second heald and the fourth thread through an eye of the fourth heald.

Denting: The process of passing the warp threads through the opening of the reed (dent) by two in general and more or less, according to the character of a pattern is known as denting.

Tie up and shedding: The healds are generally tied up round a top pulley or roller, so that when a treadle is pressed, the connected shaft is pulled down, thereby the tied up shaft is pulled up.

Looming of the warp being accomplished, preliminaries to actual weaving should be taken in hand. As one of the treadles is depressed by foot, the healds connected with it go down, while those connected with other treadles go up, forming an opening in the warp known as shed. The shuttle moves through this shed and weaves.

The weaving of silk at Sualkuchi is done on fly-shuttle looms. Prior to the introduction of fly-shuttle looms, the local weavers used throw-shuttle looms and the designing was also done by them with the hands. They now use dobby machines, attached with the fly-shuttle looms, for the purpose of making designs. It is rather encouraging to notice that a complete switch over to fly-shuttle looms has taken place replacing the old throw-shuttle looms, which is not economic for commercial production. A detailed description on the actual weaving process has been given in the chapter on Handloom weaving.

Designs: Though silk weaving in Assam is confined mainly to the production of some local cloths, *mekhela, chaddar* etc., the silk weavers of Sualkuchi produce the most beautiful and some of the best fabrics. These are liked all over the State for their delicate weave, dainty designs and delightful natural colours. The designs used in silk weaving are mostly symbolical and consists of abstract figures of animals, human beings, flowers, creepers, birds, diamonds, cross borders and sometimes heavenly phenomena. The local weavers create beautiful designs of different colours and shades in the border of their *mekhelas, chaddars* and *rihas* and sometimes use golden and silver threads in the border to produce greater effect. The local weavers are quite good in bringing out a great variety of colourful patterns and designs, mostly based on their local traditions. It is understood that the All India Handloom Board has undertaken a research for improving and elaborating the existing patterns and designs by retaining their symbolic meaning nd traditional motif.

Finished products and their uses: The silk abrics produced at Sualkuchi are not common vith those of the fabrics manufactured in other parts of India. So a brief account of these inished products are given below:—

Mekhela: Assamese women's wear with exquiitely beautiful designs, made of either pat or nuga. The size of the cloth is 3 yards by 1 ard.

Chaddar: Women's wear with various designs, made of both pat and muga yarn. The size of the cloth is $2\frac{1}{2}$ yards by $1\frac{1}{2}$ yards.

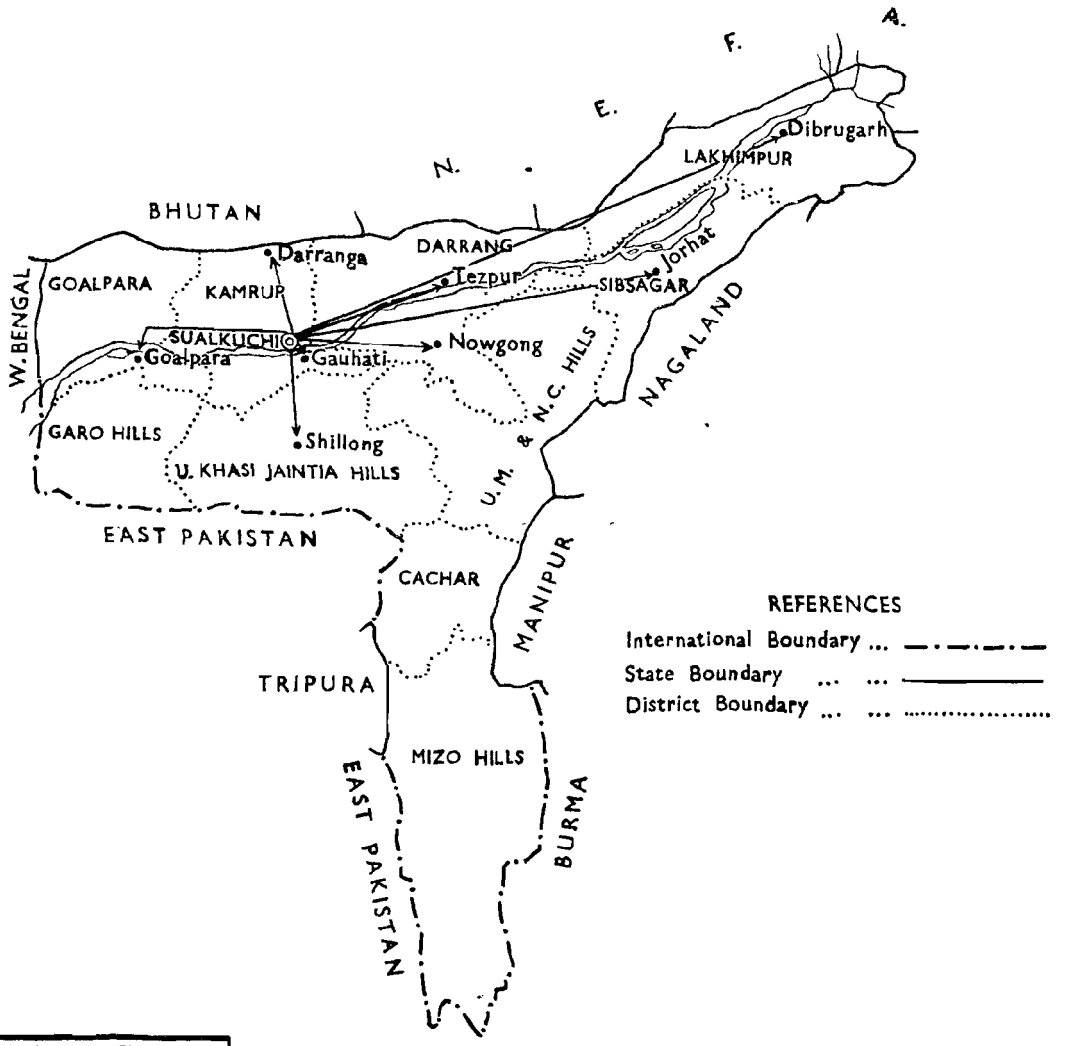
Riha: This is also a decorated piece of cloth used by Assamese womenfolk along with the *mekhela* and the *chaddar*. It is made of both pat and muga silk. The size of the cloth is $108'' \times 25''$.

Dhara: Used by Khasi ladies of Assam, made only of muga. The size of the cloth is 3 yards by 27 inches. It is beautifully decorated in the border.

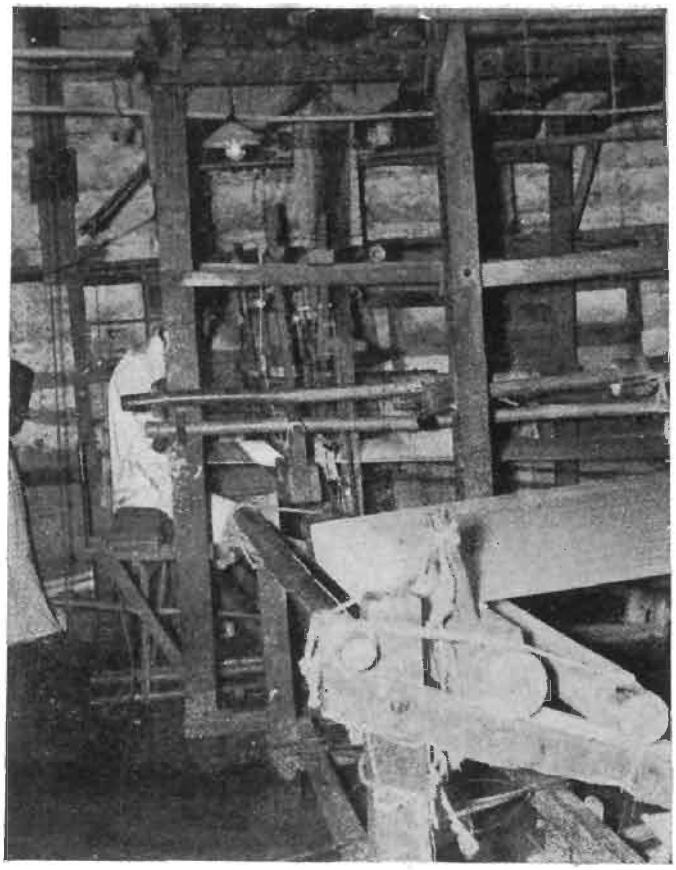
Jainsem: This garment is also used by Khasi

ASSAM

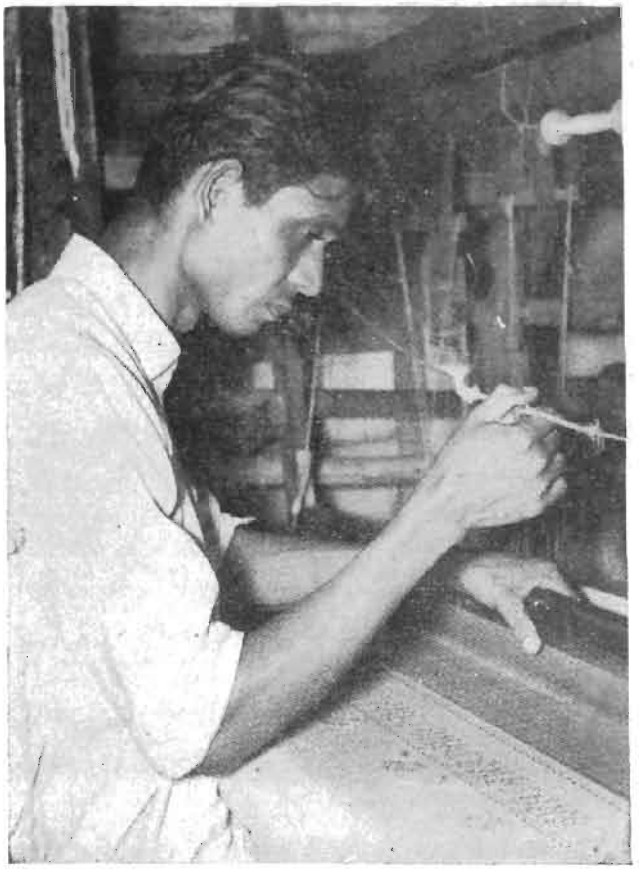
SHOWING THE DISTRIBUTION OF SUALKUCHI SILK PRODUCTS AT IMPORTANT MARKETING CENTRES (NOTIONAL)



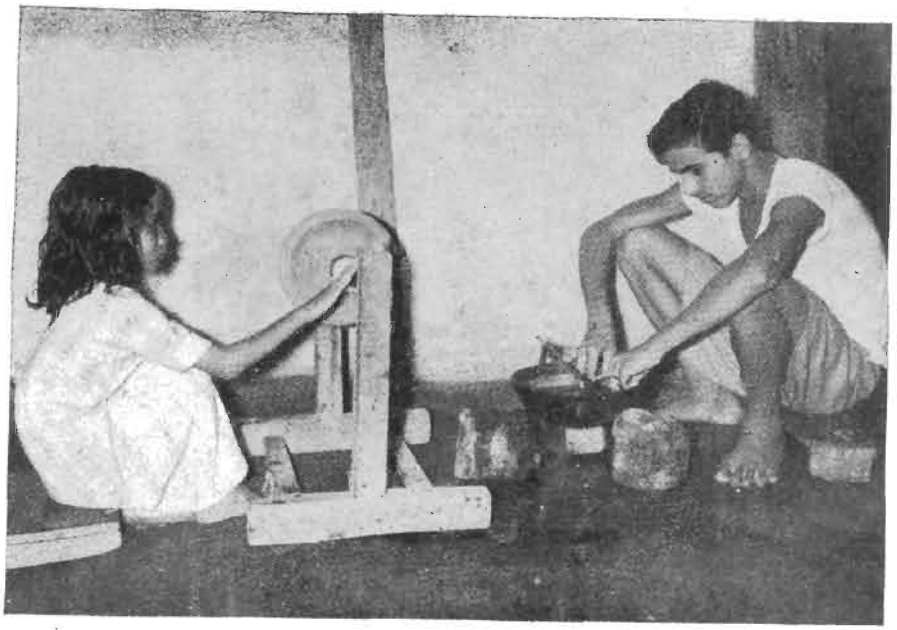
Draughtsman T.Tlinga



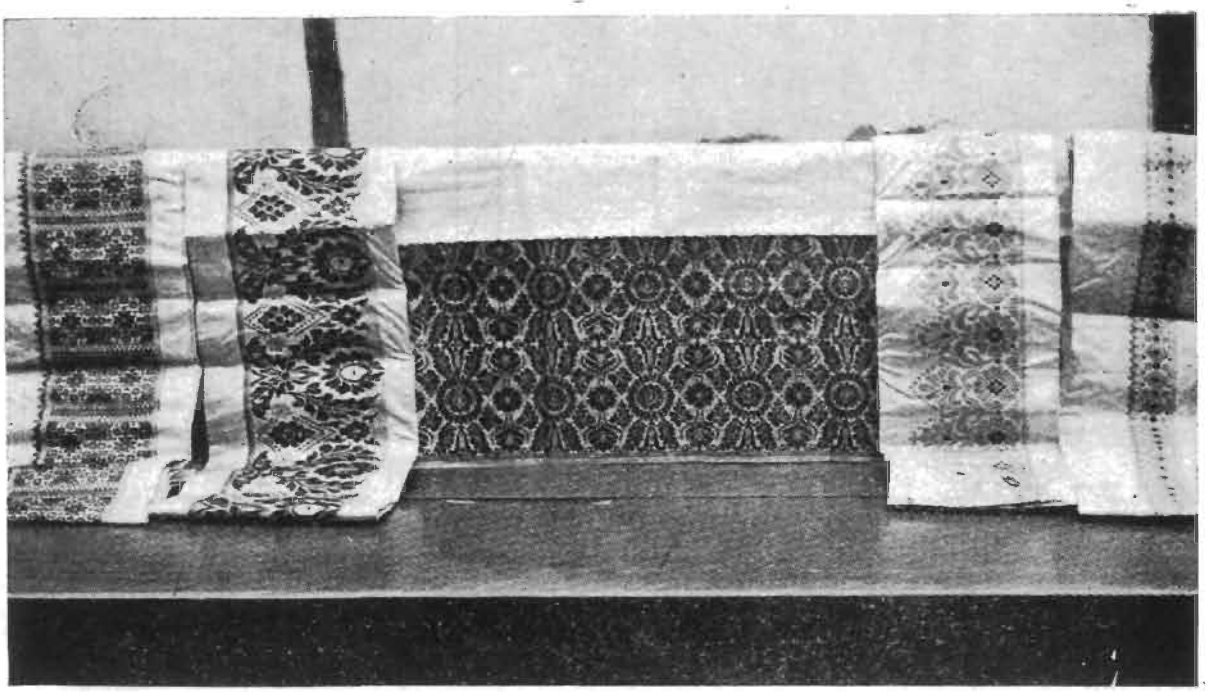
General inside view of a weaving factory at Sualkuchi



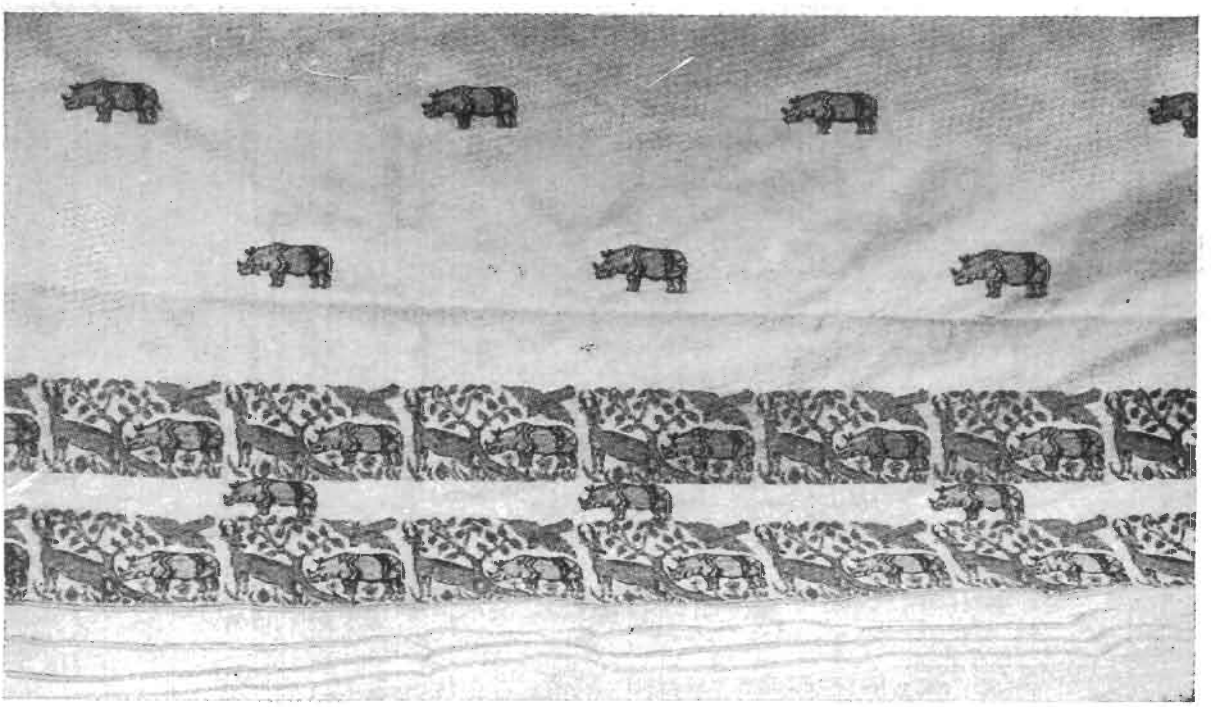
A Sualkuchi silk weaver at his fly-shuttle loom



Reeling of 'muga' on the 'bhir'



A selection of some colourful designs from Sualkuchi



Reproduction of typical Assamese design on Sualkuchi Silk



Some special Sualkuchi designs on silk fabrics

ladies. It is made of only muga threads with excellent designs in the border. The size is 3 yards and $2\frac{1}{2}$ yards.

Shirting and blouse pieces: Made of both pat and muga silk. For the purpose of shirting, generally plain cloths are woven. But elegant and decorative blouse pieces are also manufactured now-a-days.

Sari: There is a particular demand for red-bordered muga saris among the Bengalee womenfolk of Assam and West Bengal. This sari is used by Bengalee ladies during some religious ceremonies. Now-a-days, Sualkuchi weavers are also manufacturing this particular variety of muga sari. The size of this cloth is $5\frac{1}{2}$ yards by $1\frac{1}{2}$ yards and approximate cost varies from Rs. 45.00 to Rs. 60.00.

Workshop and tools implements: The workshop facilities and tools and implements required for this industry are very simple. Practically there is no common workshop for the weavers of Sualkuchi. The majority of the weavers carry on the industry in their own houses. A very negligible number of artisans who are working in some small factories carry on their work in the factory house where about eight persons can be accommodated at best. The houses are mostly made of corrugated iron sheet roofs with mud-plastered bamboo walls. A few spinning charkhas for muga yarn and a complete set of fly-shuttle loom with dobby machine attachment are adequate enough to carry on this industry.

About 39% of the total weavers are working under the co-operative groups, and the rest are working independently. Unlike other large scale factories, the environment is healthy and homely.

ECONOMIES OF THE INDUSTRY

Social & economic aspects of the industry: Sericulture in Assam is a traditional cottage industry which influences the socio-economic set-up of the rural life and culture of the Assamese people. The rearing of silk worms, culture of their food plants, reeling, spinning and weaving of silk has made sericulture a semi-agricultural occupation providing a subsidiary source of income to the agriculturists. The people of Sualkuchi by pursuing the silk weaving industry not only produce clothings to meet their own requirements but also earn a subsidiary income.

Again, due to the regular floods, agricultural production from the low-lying land is very uncertain. Consequently, many people have chosen to engage themselves in weaving as their primary occupation, while some others are engaged in the production of the raw materials and of finished products.

The majority of the artisans at Sualkuchi are followers of Sankardev while the rest are followers of Dev Damodar. For the purpose of religious congregation the people generally assemble in different *namghars* located within the town itself. Besides the Bihu festivals, the national festivals of Assam, the major festivals observed in this locality are the "Sankardev Tithi", "Asokastomi" and the "Tithi of Dev Damodar". During these festivals, big fairs are also held at Sualkuchi when a great quantity of the local silk fabrics are sold.

The use of silk cloths has a very important place in the social and religious life of the people of Assam. It is always with a sense of pride that the mother gives a three-piece silk garment to her daughter at the time of the latter's marriage and a complete silk dress to her son-in-law while welcoming him to the marriage 'vedi' or platform.

Cost of production: The cost of a particular silk cloth is almost entirely made up of the cost of muga or pat thread used, the cost of dyeing and the cost of the golden thread (jari),

plus the labour charges. A detailed table on the cost of production has been incorporated in the chapter on Sericulture in Assam.

Production & employment: There are about 1,200 handlooms engaged in silk weaving at Sualkuchi. The total value of silk cloths produced here annually is estimated to be worth rupees twenty lakhs. The total number of skilled and semi-skilled artisans engaged in the industry is said to be nearly 1,600. Generally, the weavers have peak seasons during May to June and September to November when the demand for silk fabrics is high.

Earnings of a worker: The average earnings per month of the skilled and semi-skilled weavers are reported to be as follows:

	Full time worker	Part-time worker
1. Skilled	Rs. 90.00 to Rs. 120.00	Rs. 40.00 to Rs. 60.00
2. Semi-skilled	Rs. 50.00	Rs. 25.00

Marketing: More than fifty per cent of the total annual products are sold through the sales depots run by different cooperative societies and Khadi Bhandars located in different parts of the State, viz., Gauhati, Shillong, Tezpur, Jorhat and Sualkuchi. Besides these agencies, marketing of local silk products are also done by the merchant community in different markets and fairs of the State.

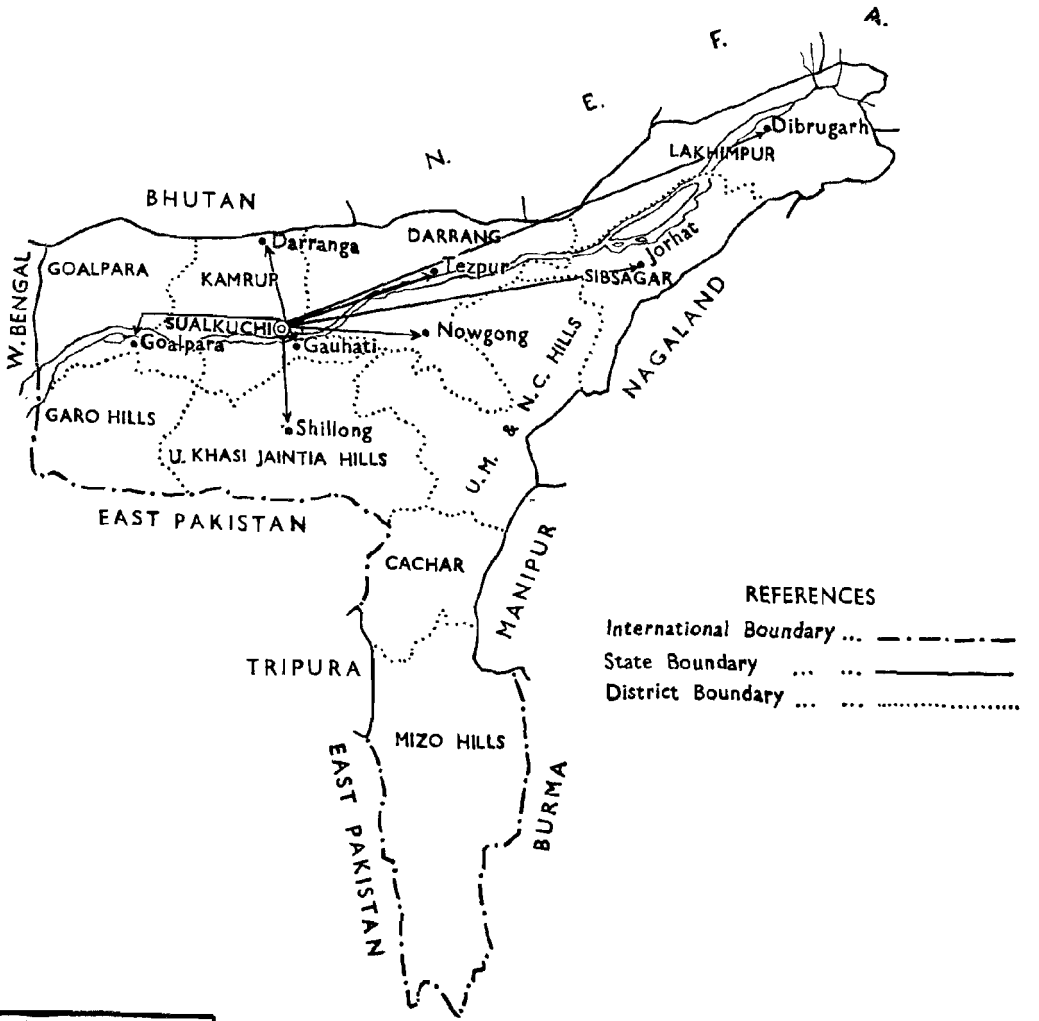
Silk products are also marketted throughout the State and in other States by the sales centres sponsored by the All India Khadi and Village Industries Commission, the Government Emporia and the State Marketing Corporation. The Sualkuchi silk products like dhara, jainsem, etc. are generally sold in the Khasi & Jaintia Hills district. It is also reported that a good quantity of silk products are being exported to foreign countries in different manners.

Government assistance for the industry: At one time the industry faced a severe competition from mill-made silk cloths. The popular Government appreciated the needs and difficulties of this reputed but declining industry and provided adequate and immediate protective measures. The different types of assistance rendered by the Sericulture and Weaving Department and other agencies, such as the Co-operative Department and the All India Handloom Board, by advancing grants and loans for working capital of the co-operative societies and other organisations have given a new lease of life to this decaying industry. Improved tools and accessories are also provided by the Sericulture and Weaving Department at a subsidised rate, i.e. only 20 per cent of the actual cost are to be paid by the weavers. In the sphere of technical assistance, the All India Handloom Board is taking a keen interest to find out ways and means for improvement of the existing designs and products to suit the varied tastes of the different sections of consumers. The facilities rendered as such by different agencies will go a long way in revitalising this age-old traditional industry of Sualkuchi.

Present condition of the co-operative societies dealing with silk: Earlier in this report, a discussion has been made about the condition of the co-operative societies already in existence at Sualkuchi. Out of the eight registered societies, two are found to be non-functioning. Though a good number of co-operative societies are functioning, more than a half of the weavers' community are still left outside the co-operative fold. Attempts are being made to bring all artisan members within the co-operative fold so that they can also avail the facilities through co-operative societies, but this depends upon the measure of success by which co-operative societies can inspire the villagers.

ASSAM

SHOWING THE DISTRIBUTION OF SUALKUCHI SILK PRODUCTS AT IMPORTANT MARKETING CENTRES (NOTIONAL)



Draughtsman T.Tlinga



A prominent sales centre at Sualkuchi



Male and female weavers of Sualkuchi

The statement in Appendix A gives the reader an idea about the functioning and present position of different co-operative societies of Sualkuchi.

Difficulties in procuring raw materials : It is rather astonishing that while the silk weaving industry of Sualkuchi is consuming a great quantity of muga cocoons and pat silk yarn, there is no cocoon rearing industry within its boundary or the neighbouring villages. It has to procure raw materials from other parts of the State and sometimes from outside the State, resulting in high cost of raw materials. It is now high time for the local producers and the Government alike to give a serious thought to the matter and study the feasibility of encouraging the local people to take up rearing of different varieties of silk cocoons. This will also open another field of employment for the local people.

It is also gathered that there are no organised cocoon markets in Assam at present. The price of cocoon rises very high during off-season and the spinners and weavers have no alternative but to make their purchases paying unduly high prices to the middlemen who collect the cocoons during seasons and hoard them.

The Government will do well also to try to find out ways and means to solve this problem and to arrange supply of cocoons to spinners and weavers at reasonable prices.

Conclusion : Besides providing employment to the local people, this industry earns a good fame for the State for its craftsmanship. It is in a position to satisfy people of different tastes in and outside the country. If proper publicity is made, it may find some foreign markets from where a good amount of foreign exchange can also be earned. It should be admitted that in spite of their artistic merits, the silk products have a limited market because of their high cost of production. The prices of better quality products are beyond the reach of the middle class. Unless the cost of production is lowered, this industry will continue to satisfy only the well-to-do people with a very limited market. In the context of the above observations, the question arises whether it will survive and flourish as craft in 'stomach' conscious society. The Government is, of course, aware of the position and is taking steps to meet this challenge. Immediate action on the following directions will help in this regard.

1. *Introduction of modern method and technique*:—Handloom should be replaced by powerlooms. Artisans should be well trained to handle powerlooms. Financial help for incurring initial expenditure at the time of switch over to powerlooms should be given.
2. *Procuring of raw materials*:—Artisan families should be encouraged to rear eri, muga and pat cocoons which will give part-time employment and at the same time eliminate the function of the middlemen resulting in the reduction of the cost of raw materials. The Government should also make arrangement for supply of the raw materials at a reasonable cost.

3. *Introduction of new designs—items*:—Various markets of the country should be studied and new items of produce and designs should be introduced to suit the tastes of the consumers from different parts of the country.
4. *Marketing of finished products*:—Government should collect and supply the market intelligence from other States so as to enable the artisans to meet the changing taste and growing demands of the consumers. Government should also take necessary steps through various agencies for wide publicity to popularise the silk products of Assam.

APPENDIX A

Sl. No.	Name of the Co-operative Society	Year of Registration	No. of shareholders	No. of artisan members	Total amount of paid up capital (share value Rs. 10/- each)	Amount of loan received upto 1963	Source of loan received	Grants	
								Amount of grant received upto 1963	Source of grant received
1	2	3	4	5	6	7	8	9	10
1.	Sualkuchi Muga Spinners and weavers Co-operative Society Ltd.	1954	81	81	Rs.2058.00 Govt. loan Rs.2550.00 by members	1. 1962-Working Capital Loan-Rs.20,600/- 2. 1963-Share Capital Loan-Rs.3,525/- Working Capital Loan Rs.8,400/-	All India Handloom Board ,, ,,	Rs.2,400/-	All India Handloom Board
2.	Sualkuchi Bhatipara Muga Spinners Co-operative Society Ltd.	1954	52	52	Rs.298.00 by members	Nil	..	..	
3.	Assam Samabay Resham Pratisthan Ltd. Head Office-Gauhati; Branches 1. Jorhat 2. Nowgong 3. Tezpur 4. Sualkuchi.	1949	152	100	Rs.18,675/-	Working Capital Loan-Rs.85,000/- Share Capital Loan-Rs.21,600	All India Handloom Board.	Rs.4,000/-	,
4.	Bamum Sualkuchi Silk Samabay Samiti Ltd.	1957	59	59	Rs.650.00	1959-Working Capital Loan-Rs.9,000/-	,,	..	
5.	Madhya Sualkuchi R.S.S.		127	127	Rs.29,062/-	..	..		
6.	Assam Silk House	1941	66	50	Rs.9,950.00	Rs.40,000/- Rs.78,000/-	Sericulture & Weaving Deptt. Govt. of Assam All India Handloom Board.		
7.	Purba-Sualkuchi Samabay Pratisthan.	..	..	..	..	..	..	.	..
8.	Sualkuchi Patmuga Palupoha Samabay Samity Ltd.	..	..	..	..	..	.	..	..

APPENDIX A—(*Concld.*)

Purpose for receiving the grants	Amount of rebate on muga-silk cloth sale (year-wise)	Amount of loan refunded	Extent of business done so far	Profit ( )	Other funds of the Society	Remarks
11	12	13	14	15	16	17
1. To provide godown facilities for storing muga cocoons.	N.A.	N.A.	Rs. 115,697.52	1961-62 (+)Rs. 1,286.57	..	..
2. ..	,,	,,	..	..	..	N.F.
3. To improve the housing condition of the Jorhat Branch of the Society.	,,	,,	Rs. 651,180.31	(+)Rs.23,972.55	Reserve Fund Rs.17,319.82 Charity- Rs. 4,301.68 Bad Debt- Rs. 5,572.76	..
4. ..	,,	,,	Rs. 26,913.44	(+)Rs. 757.45		
5. ..	,,	,,	Rs. 25,503.97	..	..	..
..	,,	,,	Rs.2,185,771.31	N.A.	..	Branches :— 1. Fancy Bazar, Gauhati. 2. Shillong. 3. Tezpur. 4. Sualkuchi. Head office :- Panbazar, Gauhati
7. ..	,,	,,	..	..	..	N.F.
8 ..	,,	,,	..	..	..	N.F.

Note —N.A.—Not Available
N.F.—Not Functioning

CHAPTER IV

POTTERY

Introduction : Pottery is one of the age-old crafts of Assam carried on by the most antiquated method of production. In the plains districts, pottery is a hereditary occupation of the people belonging to the communities like Kumar, Hira, Paul, Rudrapaul, etc.

In the 1891 Census Report by Sir E. A. Gait, it is mentioned that the real Kumar or Kulala is a caste of the Novasakha group, but the author was inclined to suspect that many of the persons returned as Kumars in the upper districts of the Brahmaputra Valley were really potters of the Kalita and Keot castes. The author added that the latter was generally the accepted view, although some of the native officers who reported on the subject denied the claim of the Kumar Kalita to rank as a Kalita. What, however, they really meant was, the author fancied, that a Kumar Kalita does not rank on the same level as an ordinary Kalita.

The same author, in an article in the Journal of Indian Art and Industry published in 1898, observes that there are two distinct classes of persons engaged in the manufacture of pottery in the Brahmaputra Valley, the Hiras and the Kumars. (The word "Kumar" is derived from "Kumbhakar" and means maker of pots). The word "Kumar" on the other hand, so far as the Brahmaputra Valley proper is concerned, is not used to denote persons of any particular caste or sub-caste but is applied to persons of several different castes (chiefly the Kalita, Keot and Koch) who make, or whose ancestors are remembered to have made, articles of earthenware. Thus, there are Kumar Kalitas, Kumar Keots, and Kumar Koches, and the people so designated continue to retain their old caste status.

In few rare instances, persons other than Hiras and Kumars are found manufacturing earthen vessels in the Brahmaputra Valley. In the Tezpur subdivision, a few Domes make large earthen pots which are used for boiling gur; they rank equally with other Domes, and only make these utensils as a secondary occupation, their main means of livelihood being the catching and selling of fish. Near Dibrugarh, a few families of up-country men from Arrah are potters by profession. They are known locally as Hira Kumars. Such up-country potters have settled recently at Golaghat*

The present position of the Kumars is not uniform throughout the Brahmaputra Valley. They are treated more or less as members of a scheduled caste in the Kamrup district while they rank almost equally with ordinary Kalitas and others in the upper districts, although matrimonial relation with Kumars is not favoured.

* Reference:—Manufacture of Pottery in Assam by E. A. Gait.

In the Cachar district, the word 'Kumar' denotes what it does in Bengal, the separate potter caste i.e. one of the nine professional castes (Navashakha) which are said to be descendants as the offspring from Viswakarma, the divine artisan. According to the Brahmabaibarta Purana, the union of Viswakarma and a Sudra woman resulted in the birth of nine castes. Malakar, Karmakar (blacksmiths), Sankhakar and Kumbhakar (potters) are considered to be the more important ones. In the Cachar district, some of the craftsmen are engaged in making only images of gods and goddesses. At present, the image-making is their subsidiary occupation, their main occupation being 'Grahapuja' along with the studies of astronomy. This class is known as Grahabipra or Ganak (astrologers) and the members of their families prepare images of Hindu gods and goddesses during different religious occasions. According to the estimates of the numerical strength of some backward classes prepared by the Census Organisation in 1954, Grahabipra or Ganak was shown as one of the backward classes in Cachar. According to Bhabishya Purana, the union of Surja-Deva (sun) and a Brahmin woman resulted in the birth of this class. It is written in the 139th chapter of Bhabishya Purana that they were brought from Shaka-dwipa by Lord Krishna's son, Shamba.

In Garo Hills, pottery is practised by the non-Garos but they are persons of the same caste as the Kumars of Cachar or Sylhet. In the Khasi & Jaintia Hills, the caste system is unknown and there the pottery craft is carried on by any one choosing to learn the work.

Before the advent of scientific development, these village potters were economically self-sufficient, but due to the introduction of ceramic and alluminium utensils, the condition of the trade has deteriorated to a great extent

History and Origin : @ From time immemorial the pottery industry came into being by a natural evolution. The story of the first pot begins with a 'keeper' and 'things to keep'. According to Hindu mythology, the need for such a thing first became necessary to contain the 'nectar to make the gods immortal for all times' and the container was known as Amrita Kumbha. It was then that Viswakarma, the celestial artificer, took from each of the assembled gods a bit of "kala" (spirit and moulded it into a pot.

As a single pot could not possibly serve the needs of the gods, Lord Siva created an organised caste of potters upon whom he bestowed the exclusive privilege of making pots. As the first potters were created by Siva, the potters considered themselves to be under his guidance. Apart from this mythological background, antiquated practice of the pottery industry in India could be amply proved by the excavation of pottery articles from the ruins of Mohenjodaro and Harappa which flourished from 2850 B.C. to 2300 B.C.

"Earthen vessels made on the wheel have been used in Hindusthan without doubt ever since the Aryan invasion. The first known mention of them is in the Rig Veda, and by the tradition of these provinces, their invention is attributed directly to the gods. One of these legends tells how Kopa Bhagat Rishi, the fabled progenitor of the potter caste, was ordered by Parameswar to make earthen vessels. The Rishi's primitive method involved the baking of the pots for a full year, but on one memorable

@ :—Shrihatter Itibritta by Achhyut Charan Choudhury, *Tatwanidhi.*

day, the Sukat Chauth or birthday of Gonesha, after he had filled his kiln, a cat gave birth to some kittens in one of the pots. Hereupon the saint prayed earnestly for the safety of the kittens, who were preserved by Parameswar's causing the vessels to be baked in the miraculously short time of one day. The residence of Kopa Bhagat is said to have been at Jagannath, which is accordingly regarded by potters as their headquarters. Another account is that in old days the people used to work up the clay for pots with their spittle, and Parameswar deeming this unclean revealed the secret of the wheel".**

"A study of pottery and potters' life in India by a two-man Anthropological Survey of India team brings to light the variations in the technique of pottery manufacture prevalent in different parts of the country and the social laws associated with the life of potters. The team has collected data from 140 out of the 322 districts in this country during the past 20 months on the manufacturing processes of pottery and on the potters' castes.

Pottery manufacture as studied by the team shows mainly two distinct processes prevalent in the country. Techniques which developed in South India have spread through the ages obliquely to Eastern India i.e. Orissa, West Bengal and Assam. The form of the potter's wheel, the method of firing and other techniques followed in this region resemble those of South India.

Potters of the South-Eastern area work with the pivoted spoke wheel, while in North-west India the socketed block wheel is used. The stone socket on which the wheel revolves is buried underground in the South-Eastern area and in the North-west the stone socket is attached underneath the clay wheel and the wooden pivot is buried in the ground. In the central area where techniques of South-east and North-west zones have blended, the stone sockets system (north-west) and the wooden spoke wheel (south-east) are found".*

As regards the history and origin of the pottery industry in Assam, it has been possible to gather that the use of pottery wares was prevalent as far back as the 5th & 6th Century A.D.

"The extensive remains of temples and buildings give ample evidence of working on stone and bricks. This is proved also by a number of epigraphs. The art of brick making is mentioned in the Sualkuchi grant of Ratnapala. It was highly developed at a subsequent time, particularly during the Ahom period.

The art of making pottery was known from very early times. The Nidhanpur grant mentions *kumbhakaragarta* (potter's pit), and the Kamauli grant refers to the *Kumbhakaras,* who were professional pottery makers. Some of the best specimens of pottery, with artistic and decorative designs, belonging to the 5th-6th century A.D., have been discovered from Dah Parvatia; some specimens have also been found in Tezpur and near Sadiya. The clay seals of Bhaskaravarman also point to the fact that the art of clay modelling was developed. Moreover, Bana mentions among the presents of Bhaskara 'drinking vessels embossed by skilful artist, molasses in earthen pots and 'cups of ullaka diffusing a fragrance of sweet wine'.***

It is very difficult to give a definite note on the existence of pottery during pre-historic

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* Statesman, Calcutta 24th August, 1963.

** Extract from the 'Monograph on the Pottery of North Western Provinces and Oudh' by H. R. C. Dobbs, C.S.

*** 'The History of Civilisation of the People of Assam' by Dr. P. C. Choudhury, pp. 377-378.

epoch, as no systematic study has been made, but random finding of small-sized potteries of the pre-historic times, collected during various excavations and diggings leave no doubt that the art of pottery also penetrated in Assam long ago. An account of a recent findings of potteries near Sarania Hills, Gauhati, is as follows: —

"The pottery finds at the foot-hill region of Sarania near Gauhati, Assam, consisted of 14 whole pots and a few pieces of potsherds. These were found inside an old and abandoned well, eqposed while digging a fresh one in the month of May, 1960, at a depth of 12 to 15 feet from the surface.

All the pieces of pottery are wheel made, materials used for the pottery being coarse clay. Almost all the pots are slipped and varnished. Three big-sized pots are of lota type; and the same are carinated. One of these is of ash-grey colour and the rest are of pale cream and orange colour. There are two small pots of lota type, one being carinated. There are two spouted jars of miniature size. There is one long-necked beaker of light-grey colour. The collection includes three miniature pots, two with narrow neck and one with broad neck. There is one lid of light grey colour. An earthen amulet-like object is a peculiarity of this collection. This is a hand-made pottery of greyish black colour. A perforation throughout the length indicates it probably to be a net-weight. A wide-mouthed bowl with corded mark decoration on the outer surface is obviously a cooking vessel. It is made of fine levigated clay. Colour is black. The find includes five potsherds. One potsherd of red-polished ware shows incised decorations of circles and dots arranged in bands. The peculiarity of this potsherd is a mica wash on the outer surface. Another potsherd is of a big storage jar. It is very thick and made of coarse clay".*

Nature of Establishment: It has been observed that most of the potters used to live in the village in a group. The concentration of the potters in a particular place depends mainly on the availability of a good type of clay and other facilities which are favourable for running the trade. The potters also reside in towns or in suburban areas. As the town potters do not have other sources of income, they solely depend upon the sale of their products. The town potters of this State fall into two different groups, the first group manufactures the pottery utensils while the second group the images of different deities and dolls and toys. During Durga Puja, and Saraswati Puja, the image makers (Rudrapaul and Acharjee) earn a lot, almost sufficient to meet the family expenditure throughout the year. They also make dolls and toys which they sell in meals and fairs held on different occasions.

The village pottery, in most cases, is run on the household basis. In such households, an extra thatched-house is constructed for carrying on their business. A kiln for baking the pottery articles is also prepared jointly by two or three families. The craftsman of the family is engaged in fashioning the vessels and other male members are engaged in preparing the clay while the women and children do the beating and polishing etc. Most of the village potters are cultivators and they engage themselves in agricultural operations according to calendar. It is a good step that at the instances of the State Industries Department and the State Khadi and Village Industries Board, the pottery industry is being organised and brought under the co-operative fold in order to render the potters financial and technical aids.

** Note: —By Prof. M. C. Goswami, M.A.,B.T.(Cal.), M.Litt(Cantab), G.U.

Technique of Production :

Raw materials : (i) *Clay* : –"The raw materials required for this industry is mainly clay which the potter fashions into beautiful and useful pots. But all types of clay are not suitable for the preparation of pottery utensils. There are various types of clay required for various purposes. Clay is the basic raw material from which our houses are built up in the villages. In the towns, houses are built of bricks and roofed with tiles, both of which are again made from clay. Pure white clays are used in making the porcelain cups and saucers. Some special kinds of clay are used for whitening and giving weight to finished cotton and other textile goods, and for making good papers. It is also used as a filler in the manufacture of linoleum, oil cloth, rubber, imitation leathers, insecticides etc. Smaller quantities are used in cosmetics and medicines. Clays are also used in the clarifications of some food products. There are some types of clays known as 'fire clays' which are used in the manufacture of refractory bricks. These are used to line furnaces.

Fire Clay: -This type of clay commonly occurs in association with the coalseams of Upper Assam but has nowhere been worked so far. Fire clay from near Jowai in the Jaintia Hills has been reported to be of good quality.

Kaolin or China Clay:—Kaolin is recorded from various parts of the Garo, Khasi & Jaintia Hills, below the Cherra sandstone deposits. Kaolin is also reported from Dora river in Lakhimpur and also from the Mikir Hills. A fine white clay has been reported from the Nambor river and from Barpather in Sibsagar district. These localities are yet to be examined.

Lithomarge: -Lithomarge occurs abundantly in association with the coal beds of the Garo Hills. Good exposures occur near Tura, especially north-west of the Rongrengiri coalfield and many places along the southern slopes of the Garo and Khasi Hills ranges below the Nummulitic limestone horizon. The lithomarge with low iron and alkali content appears to be suitable for use as fire clay but so far nothing has been done to utilise it as such. Some of the clays or shales of Upper Assam have been conveniently used for the preparation of oil-well drilling mud.

Near Sohrarim, and in a few other places in Khasi Hills, the sandstone contains a fair amount of white and light coloured clay as *matrix*. This can be recovered by washing and used in the ceramic industry".*

Potter's Clay: –The claycy loam is best suited for pottery. It requires the experienced eye of an artisan to locate the variety of clay. The clay which is suitable for the pottery is found anywhere in Assam, especially in the alluvial tracts. The quality of clay is not uniform in all the districts. In the district of Sibsagar, the clay is exceptionally good, while the clays of Barpeta in Kamrup district are highly sandy. The clays of Cachar and Goalpara districts are also good for the purpose of the pottery industry. The clay used by the potters of the Brahmaputra valley is of a blackish colour and is locally known as 'kumar mati' or 'atta-letia mati'. The word 'atta-letia mati' derives its significance from the fact that it is pasty like gum which is known as 'atta' in the Assamese language. This type of clay is generally found at the bottom of rivers or ponds. A peculiar type of clay used by the Hiras, known as 'hira matti', is available only in Kamrup district. This type of clay is very stiff and grey in colour and due to its stiffness makes better earthenware than kumar-mati. In Cachar and Garo Hills, the clay used is known as 'athali mati', that is stiff clay.

* Reference: Mineral Resouces of Assam', published by the Directorate of Information & Public Relations, Assam.

The clay used by the potters of Khasi & Jaintia Hills are of two types—one is of dark blue (khyndew iong) and the other brownish (khyndbthuh).

(ii) *Gerimati*:—A kind of red-earth or red ochre known as Gerimati or Gerumati available everywhere in Assam, is used by the potters for preparing colours of their earthenwares in the indigenous method.

(iii) *Khoir or Kath (acacia catechu)*: –This is used along with Gerumati for preparing the colour. It is generally imported from outside the State.

Apart from these materials, the bark of the mango tree (*mangifina indica*), basaka leaves, washing soda, etc. are used for preparing the colours. Other materials such as cloth, bamboo, straw, jute, dhara (bamboo mat) and various colours are required for the preparation of images. These materials are available locally.

Preparation of clay : The potters usually collect the clay by digging the surface of the ground nearly two to three feet deep where they generally find the required quality of soil. The lump clay is then loosened with the help of a spade in order to separate any foreign particles or rubbishes such as stones, kankar, etc. Water is then added to make the clay muddy and the same is kept under a cover for a period of 2 to 3 days. Then the fine sands are mixed with the clay at the rate of two seers of sand with one maund of clay. But this proportion is variable according to the stiffness or elasticity of the soil. With a certain amount of testing and experience, the quantity of water and sand required for developing correct plasticity can be very easily estimated. After mixing the requisite sand and water with the clay, the same is kneaded with the feet twice and left to remain as it is for the day. On the following day the lump of clay is cut into small pieces by a sharp bamboo split to mix and knead again. This process continues till it gets the required plasticity.

The potters residing in low lying areas generally collect their soil during the winter and store it, to be used in the rainy season when the land is submerged. The dried up clay obtained as such is soaked under water in a pit before it is being processed as explained earlier. Now, the pasty clay is shaped into balls of different sizes to be used for the manufacture of different types of earthenwares, big or small. These balls are then transferred to the wheel of the potters and it is ready for preparing the earthenwares. Paddy husk and bits of jute fibres are mixed with porous clay for making images.

Preparation of colour : The colour used by the potters for the manufacture of all the household articles including tiles and bricks is prepared in a very indigenous method. Gerumati (red ochre) the main ingredient of this colour added with other materials in the following proportion is boiled in water for about three hours. Now the undissolved and rough ingredients are separated from the liquid colour by the process of filtration with the help of a piece of cloth. This liquid is used as colour for their pottery articles and is applied before baking.

1.	Gerumati (*red ochre*)	1 seer
2.	Koir of Kath (*acacia catechu*)	2 chataks
3.	Bark of mango trees	1 chataks
4.	Basak leaves (a kind of shrub	2 chataks
5.	Washing soda (calcium carbonate)	½ chatak

Method of production of earthenwares : All the earthenwares, except very large vessels, toys, a few ornamental wares and some special articles such as pipe-bowls and water goblets; are fashioned on the wheel. In Assam the wheel is termed 'chak'. The common chak is made of either clay, stone or wood but most commonly of clay.

Flat articles made by Kumars, such as



Carrying clay from the field for pot making



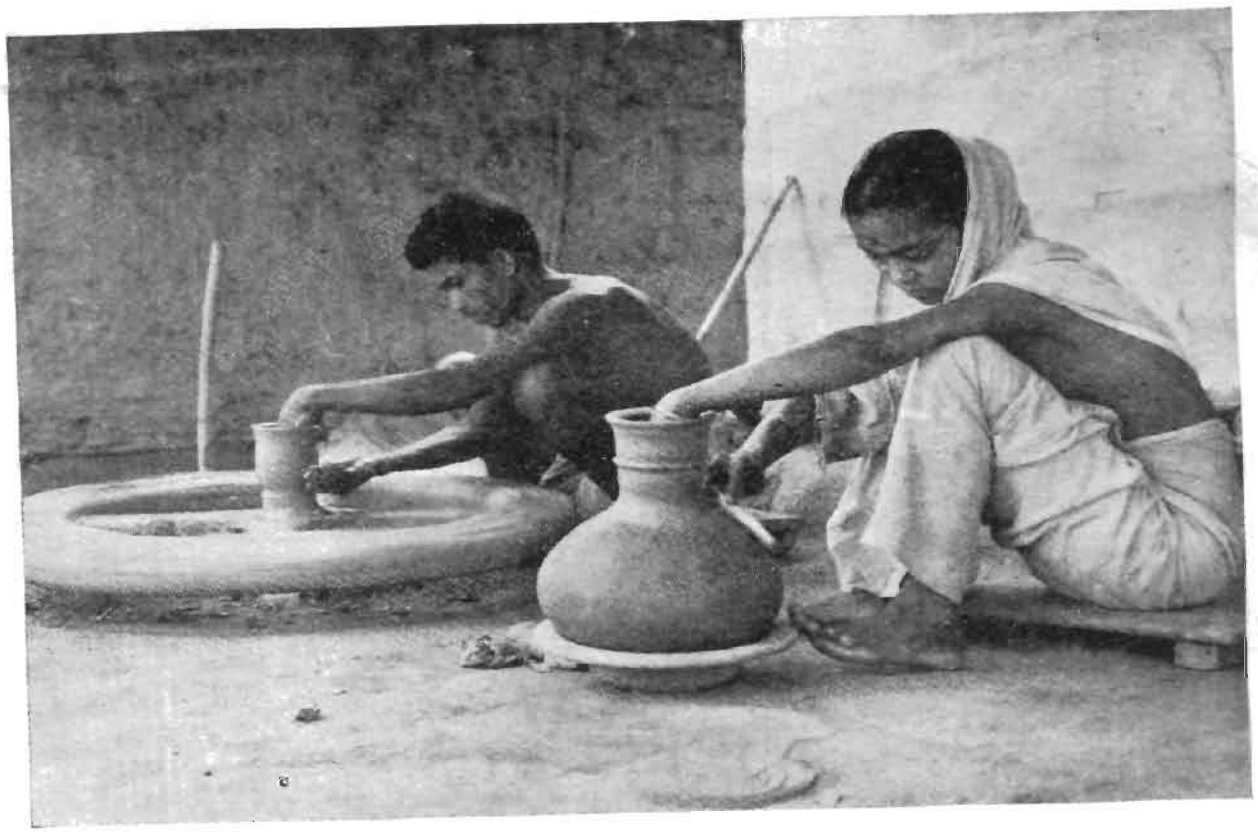
Kneading of potter's clay



Preparation of clay for tile making



Spreading the clay in the sun



Male and female potter work shoulder to shoulder



A chain wheel being
worked by the village potters

plates etc. and all utensils manufactured by the Hiras of the Brahmaputra Valley and the potters of Khasi and Jaintia Hills are roughly moulded on a flat disc or wood and not on the wheel However, it is noticed in some places that the Hiras use the wheel even for the manufacture of such earthenwares for higher turn-over.

As soon as the clay balls are ready, the potter starts the wheel with his hands, takes up the turning stick and deftly fits it into one of the holes on the circumference of the wheel as they pass round. Grasping the upper end of the stick with his right hand and the lower with his left, he gives it a few vigorous turns, and the wheel is set spinning so evenly and steadily for at least five minutes that it shows no sign of falling. The rough shape of the vessel is now revolving on the top of the ball of clay. It may be a *gharra* (water jug), a *handi* (vessel for storing milk or water) or any other vessel. The emerged shape of the vessel is then separated with the help of a cutting string. It is then kept for drying for a period of 24 hours after which the final shaping and the remaining work is done by hand, with the help of a pestle locally known as 'boila' and a wooden mallet known as 'matha'. This work is generally done by women of the potter's household. After the final shaping, the vessel is set aside to dry for another period of 4 days. When the vessel is completely dried up, it is coloured with an indigenous reddish or yellowish solution.

All the vessels manufactured as above are piled around a 'bhati' or kiln systematically for baking. The unbaked vessels are always placed with their mouths downwards, slightly inclining towards the centre. A few small pieces of wood are set among the different piles of vessels. The whole cone-shaped pile is then covered with straw and plastered with clay (generally mixed with cowdung). Then the 'bhati' is set on fire. After 8 hours of burning at low temperature, the volume of burning, for another four hours, is increased to raise the temperature. When the fire has burnt itself out, the potter waits until it is quite cool, which is usually after twelve hours, and then opens the kiln and takes out the baked earthenwares. These are now ready for sale.

The 'bhatis' are generally fired at night. The quantity of fuel required is eight to ten maunds approximately per 'bhati'. In a single firing, articles of different varieties worth Rs. 45 to Rs. 50 - can be baked in a 'bhati'. Percentage of loss by cracking varies from 5% to 25% in normal cases.

Bricks, tiles & glazed pottery wares : It is heartening to note that the State Khadi and Village Industries Board, Assam, has taken up a few bold schemes to uplift the economic condition of the rural potters. It is reported that as a result of a rapidly decreasing demand for ordinary pottery wares due to the availability of cheap metal wares in the market, most of the potters under the co-operative fold are changing over to the production of glazed pottery wares, bricks and tiles which have better prospects for sale. Technical guidance including training and financial assistance on these schemes are generously coming from the Assam Khadi and Village Industries Board.

The process of brick and tile-making is very simple and does not involve the use of any complicated machinery or skilled labour. Plastic and vitrifiable clays—the raw materials required for brick and tile making—are available in abundance all over the State, particularly in the plains districts. Brick and tile manufacturing can therefore be taken up as an important village industry which can provide seasonal employment to potters as well as to farmers, when they are not engaged in their fields, and enable them to increase their per capita income. The process involved in brick

or tile manufacturing is almost similar to pot-making except in the method of moulding and the shape of the kiln. In both the cases, all operations are generally done manually.

First of all, the soil for bricks or tiles should be dug and brought to the site where these are to be moulded and burnt. Some water is sprinkled on the heap of soil so collected and left over for some days. After this the necessary quantity of sand, if required, and water should be mixed with the soil very well. The mixing of the soil is done by kneading and with the help of spades or with the help of animals. In case of large scale manufacture, pugging of soil is usually done with machines. The object of kneading or pugging is that the soil may become homogenous and develop the required plasticity. It is necessary that the soil must be mixed very thoroughly.

The wooden mould is then dipped in water for sometime and some quantity of sand is sprinkled inside the mould so that the brick should not stick to the mould. A lump of clay is then taken and rolled in ash or sand, and with a jerk the lump is thrown into the mould in such a manner that it is completely filled with the soil. After that the lump is pressed properly, with oil or some lubricant being used on the metal before moulding, and the pressing is done by hand operation or pedalling. After that the surplus soil is removed by scraping and the top surface levelled. For scraping, metal plate or wire is used and thereafter the brick or tile is detached from the mould and kept at a place for drying. For drying, the objects are arranged in a row which are kept at equal distances so that they receive air and sun for proper drying. The bricks or tiles are dried in the sun till they become sufficiently hard and strong and then are ready for burning in the kilns. For complete drying in the sun it takes about 6 to 8 days.

After drying the objects are taken to the kiln and arranged in it for burning. After construction of the kiln, the space which is left in the middle of the kiln should be filled with dry wooden logs so that it is half-full with fuel. After this, fire is to to be ignited. The fire should be slow and dull as long as the dampness of the objects is not driven out. After covering the kiln with a layer of 25 mm of soil, more fuel is added gradually and the fire is burnt very briskly. The fire is kept burning in this manner for abou 10 days. If the objects are not properly burnt, baking is continued for one or two days more After proper burning of the objects, the kiln is allowed to cool. Generally, it is found that 15- 20 per cent of the bricks or tiles are either broken or rendered unfit for use during burning operation. In an ordinary kiln, 1,800 to 6,000 bricks can be burnt, whereas 8,000 to 10,000 tiles can be burnt in special kilns.

Panibhora potters in Cachar and Gauripur potters of Goalpara are producing tiles in large quantities. It is gathered that the tiles produced in Gauripur are exported up to Tinsukia and other towns of Assam by rail and river routes. At present, they are facing difficulties in exporting their tiles due to the non-availability of railway wagons regularly.

During the field survey, it was gathered that the potters of Panibhora village have organised one Co-operative Society which was registered in 1959 with a paid-up share capital of Rs. 225/- only. The society received a loan of Rs. 12,686.00 and a grant of Rs. 6,390.00 from the State Khadi and Village Industries Board, Assam. Out of this amount, the Society purchased one acre of land at a cost of Rs. 2,417.58 and a tile making machine (Raniganj type) at Rs. 2,061.00 only. A workshop-cum-office building was constructed at a cost of Rs. 7,322.23 and a bhati-shed at Rs. 1,205.00 only. Tiles manufactured during 1962-63 (from September to April) were 40,000 only against the production target of 100,000 during the same



Drying of roofing tiles in the sun



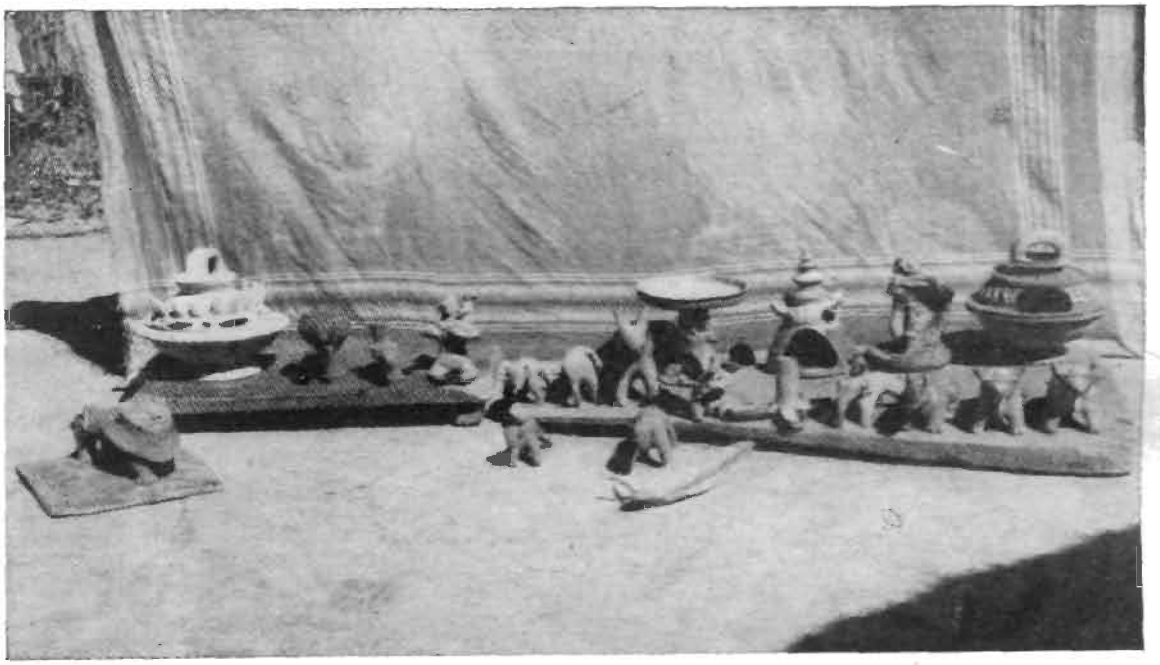
Drying of ridging tiles in the sun



Hand-press used for tile making



Preparation of slabs on the wooden moulds



A few typical earthen dolls and toys of Assam



Assamese 'Bihu' dance—an exquisite piece of clay modelling

year. The selling price of the tiles was as follows:

(a *Roofing tiles*—
1st class—Rs. 225.00 per thousand
2nd class—Rs. 180.00 to Rs. 200.00 per thousand
(b) *Ridging tiles*—
Rs. 50.00 per thousand

It was reported that one person of this society was trained in glazing technique of pottery articles under the auspices of the All India Khadi and Village Industries Commission.

Glazed Pottery: To attract the bulk of customers in the market, glazing of a few handy and decorative pottery wares is being initiated by the Assam Khadi and Village Industries Board in collaboration with the Khadi Commission. Materials for pottery glazing can be easily available from the market. A few of the raw materials required for glazing are listed below:—

1. Glass Powder	2. Borax frit
3. China clay	4. Felspar
5. Flint	6. Ball clay
7. Zinc oxide	8. Tirconium oxide
9. Tin oxide	10. White lead

White lead is generally used to obtain extra-whiteness in the glazed wares. Leadless glazing is used on red-clay-wares. The temperature required for melting the glazing is 1,050 C. Besides glazing, some potters have taken up to lacquering which is also found to be very successful and profitable. Lacquering is a simple method of rendering pottery articles impermeable and attractive. Manufacture of glazed pottery wares does not involve any complicated procedure except soaking of the objects in glazing liquid or lacquer prepared beforehand. Such soaking of the objects is done prior to burning. Such glazed or lacquered pottery wares include cups, plates, water jugs, tea pots, flower vases, ash-trays, agarbatti stand, etc.

Dolls, toys and image making: Whenever we come across any mela or fair, either in urban or rural areas, it is a common experience to witness a few stalls selling earthen images of deities and varieties of tiny dolls and toys. The producers of these articles are no other than of the 'acharjee' and 'potter' communities of the locality. Earthen dolls, toys etc., are in great demand on occasions like 'rathayatra', 'jhulan', 'durga puja' and 'kali puja'.

The different types of dolls, toys and images sold in the local markets are models of different deities and human beings, different types of animals and fishes, birds, vegetables, fruits etc. The method of production is very easy and simple. The makers generally use plastic clay for the manufacture of these articles. The prepared clay is taken by the 'potter' or 'acharjee', as the case may be, and cast into moulds prepared by hand (made of earth or timber by themselves) for a few minutes and then taken out for drying in the open. After a satisfactory drying they usually colour the articles with different shades of colour paint purchased from the market or sometimes they themselves prepare the colour out of some indigenous raw materials.

Deities of Durga, Kali, Ganesha, Lakshmi, Saraswati and Manasha etc. are prepared by the Rudrapaul (potter) and Acharjee communities on different occasions. The clay required for this purpose is prepared in a different manner. They mix paddy husk, bits of jute fibres etc. with the clay to make it more plastic. A skeleton structure of the image is made by light wooden splints along with bamboo splits. This skeleton is then stuffed with straw to make a rough pattern. The clay is laid on the straw with hand which is covered with fine cloth very thinly plastered with clay to avoid cracks. The model is then kept for a few days for drying and when it is completely dried, the colours of different shades and varieties are applied as required for a parti-

cular image or images. The hair of the image is made out of jute fibres and is dyed in black colour. The face of the image is made from a cast which is generally made of clay. The colour used by the potters is generally prepared by themselves, but the basic colour is procured from the local market.

Tools and Equipments : As stated earlier, there is an extra thatched hut in every potter's house for carrying out the wheel work in the production of pottery goods. Two or three potters possess jointly one bhati for baking their earthen materials. The following are the main tools and implements required for the preparation of pottery articles.

1. *Potter's wheel*—The wheel of the potter resembles the bullock-cart wheel, while the wheel used by potters from Bihar (residing in Assam) is like a flat disc. The potter's wheel is used for preparation of earthenwares except big vessels, dolls and toys. This wheel is known as 'chak' both in Bengali and Assamese and can be made either of clay, stone or wood. The clay wheel is generally prepared by the potters themselves and it takes them about three or four days to do so. This is the simplest type of wheel used in Assam and lasts for about 4 to 5 years. The diameter of the wheel is usually more than 3 feet. In the centre there is a solid disc of tamarind or some other hard wood, some 3 inches in diameter, to which the outer rim is joined by four wooden spokes each of which is about 6 inches in length. The outer rim which is about 6 inches broad is made of split bamboo bound with cane and covered with a thick plaster of clay mixed with fibres of sage palm (chaur). The object of this rim is to increase the weight of the wheel and thereby add to its momentum. The wheel is operated by hand with the help of a stick and supported on a wooden axis turning on a pivot fixed in the ground.

2. *Kiln or bhati*—A kiln for baking common earthenwares is locally known as 'bhati' or 'point' in Bengali and 'paghali' in Assamese. A thatched structure measuring 20′ × 12′ is sufficient for such a bhati. It consists of a circular structure, usually lined with bricks, about 3′ deep and about 1′ in diameter. A hole is made at a distance of about 3′ from the pit and is connected with the bottom of the pit for administering fuel. Just from the upper edge of the pit, four pillars made of straw plastered with cowdung and clay are raised about 3′ and joined together at a common point forming an arch. The unbaked vessels are systematically placed in the space enclosed by these pillars with their mouths downwards, slightly inclining towards the centre. When all these unbaked vessels are covered by a layer of straw and clay, a bottomless 'kalasi' is placed above the 'bhati' to serve as a chimney.

3. *Athail (Bengali), Athali (Assamese)*- An earthen plate with a hollow surface, having a diameter of 6″ to 8″ and 3½″ inches in depth in which the final shape is given to all articles, whether originally shaped on the wheel or on the rolling board. The *athail* is generally placed on a wooden or earthen stand.

4. *Rolling Board*—This is locally known as 'para' -an earthenware tray on which utensils are roughly shaped when the wheel is not used. The earthenwares like 'sara', 'basan', 'sanak', 'patil' etc., are prepared on this *para*.

5. *The mallet*- This is known as 'boila' in Bengali and 'boilapeteni' in Assamese, and is used for hammering the pottery utensils for giving the proper shape. This is made of clay.

6. *Firani*- It is nothing but a wooden plank measuring one square foot in size. The rotating of the *para* is done by placing it over the *firani*.

7. *The Polisher*—In both Assamese and Bengali it is known as 'chaki' and it is used for polishing the earthenwares. It is made of fine clay.

8. *Matha*—It is a piece of plank measuring 8″ × 5″ with a handle. Whenever the beating is done with the *boila* from inside the utensil, this is used for giving suppor from the outside. In Assamese it is known as *bat*.

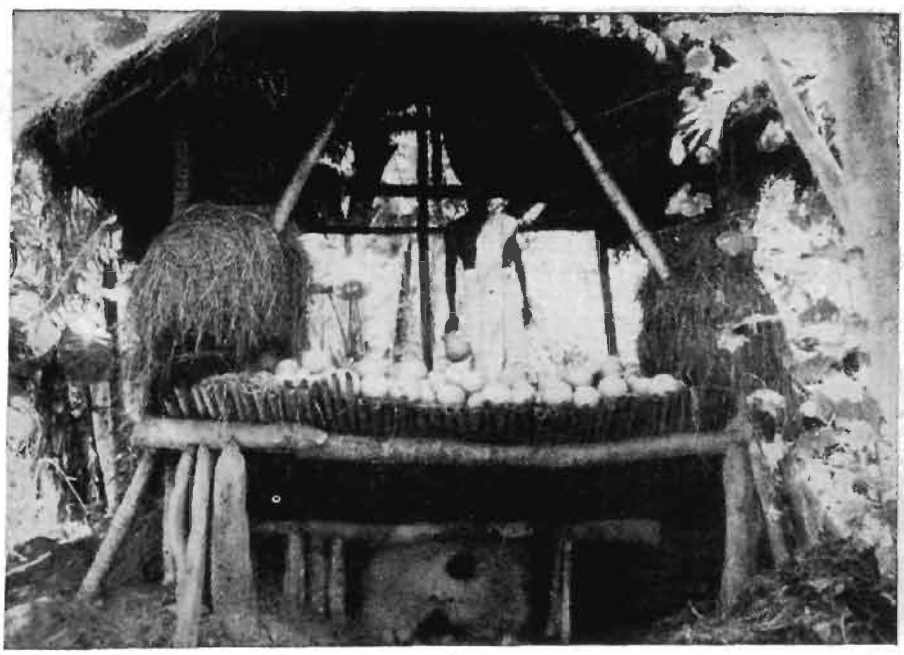
9. *Uchha*—It is a wooden pole with a blade at





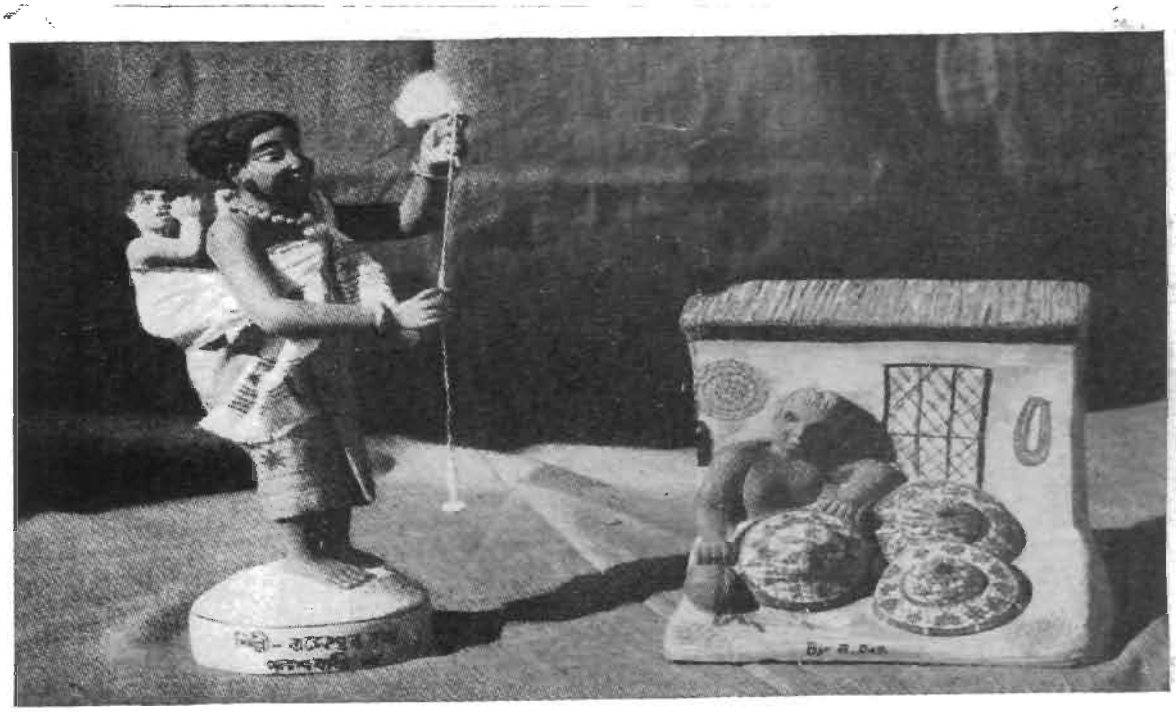
'Bhati' used for making earthen rings

Above & Below : Two typical country kilns used by potters in Assam

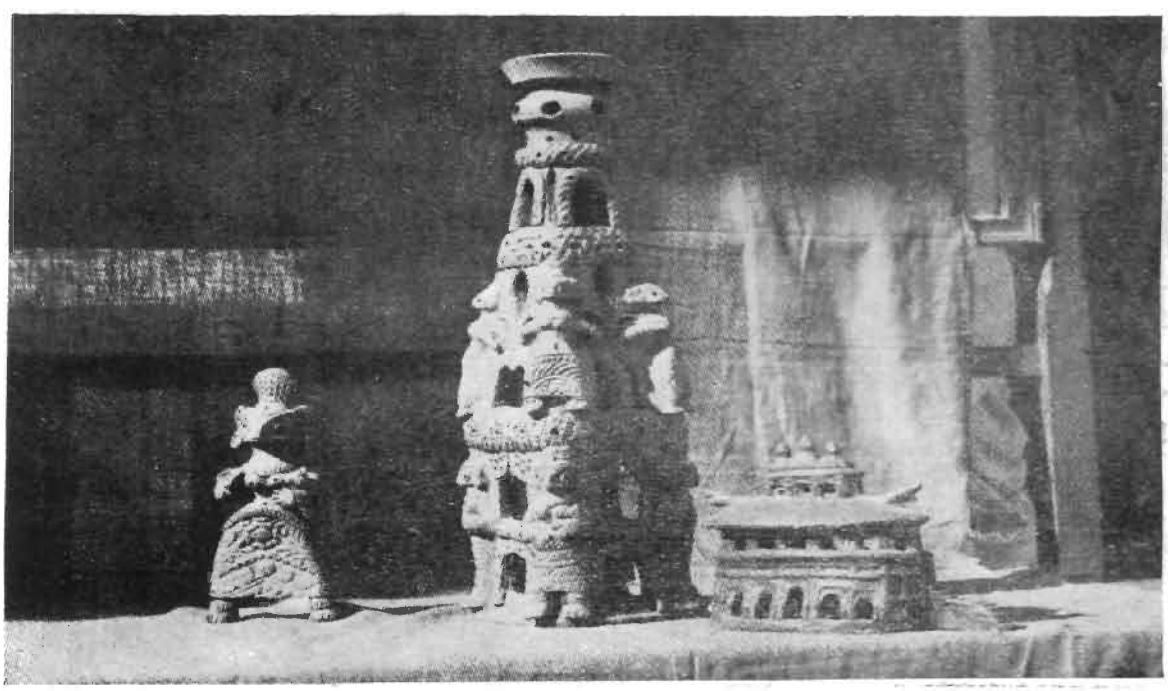




'Bhati' shed used for tile making



An Assamese spinner and a 'japi' maker—two clay models



Some beautiful terracotta models of the past

one end and used for collection of clay from the bottom of the pit. The length of such *uchha* is about $4\frac{1}{2}$ feet.

Finished products and their uses : Pottery goods are generally used as domestic utensils and also for performing various ceremonial functions and worships. Even in the early part of this century, earthen plates and glasses were used by the Hindus for taking meals at the time of big festivals instead of banana leaf which is in use now-a-days. These earthenwares are destroyed at once after use. In many railway stations, earthen cups are used instead of glass or ceramic cups. It is also good from the hygienic point of view The earthen vessels may broadly be divided into two groups, namely—(a) earthenwares used for domestic purposes and (b) earthenwares used for ceremonial purposes. Attempt is being made to give a clear idea about the different vessels in so far as their uses are concerned. The following are the most common products of the pottery industry.

Earthenwares used for domestic purposes :
1. *Kalasi*—It is a vessel used by the people for storing drinking water and for other domestic purposes. The size and shape of *kalasis* differ from district to district. Liquid molasses are also preserved in *kalasis* by the peasants.

2. *Hari or Matka*– This is a big type of vessel used mostly for storing purposes. Fishermen use this type of vessel for preparing a particular type of dry fish known as *'sidal'*. The dry fish stored in such *matka* can be preserved for a pretty long time because it is possible to seal the mouth of the *matka* with some amount of pasty clay. The rural people use it instead of trunk or suitcase for keeping their cloths and also for storing rice. In Garo Hills, 'handi' or 'hari' (dikha in Garo) of different shapes is used for preparing the local liquor (arak). Another 'hari' known as Rang-dikha is used by the Garos for storing rice, paddy, wine, etc.

3. *Basan or Kahi or Hanak*—It serves the purpose of a plate or dish. Muslim peasants generally take their meals in such utensils.

4. *Lota*—It is a miniature type of *kalasi* used for taking water out of a larger vessel for drinking, washing, etc.

5. *Bati*— Bati is something like a cup. It is generally used for keeping curries, dal etc., by the Muslim peasants.

6. *Saru*—This is used by Mohammedans for cooking rice. There are two other types of *sarus* known as (a) 'Patsaru' and (b) 'punasaru'.

(a) *Patsaru* is used for frying rice for preparing 'handoh' (a kind of Assamese tiffin). (b) *Punasaru* is used for cooking a special type of rice locally known as 'bora' (Assamese), or 'birein chawl' (Bengali), without adding water. This *punasaru* is perforated at the bottom and is placed on another vessel which is filled with water. Rice is kept in the *punasaru* and is covered by a 'dhakni' or cover in such a way that no vapour can go out. The joining surfaces of both the vessels is also to be sealed by a paste-like clay. Now it is to be kept on a *chulla* (oven) for applying heat. The rice is thus prepared with the help of vapour.

7. *Dhakni or Sara* (Cover)- This 'dhakni' is used for covering all kinds of foodstuffs at the time of cooking mostly by the Muslims.

8. *Gamla*—This is a big type of vessel used by washermen for washing cloths and also by the cultivators as a feeding trough for cattle.

9. *Dema*—This is a big vessel which resembles the 'saru'. The mouth of this vessel is a bit bigger and is used by fishermen for keeping fishes with water in living condition. The fingerlings are also carried in these vessels for sale. These are also used for many other domestic purposes.

10. *Badna*—This is a jar with a spout, generally used by the Mohammedans for carrying water. It serves the purpose of a *lota*.

11. *Dhupdani*—This is used by the Hindus for burning incense (dhunas) at the time of any puja.

12. *Tekeli*- The 'tekeli' is used for preparing 'tekeli pitha' (a kind of cake prepared out of rice in the Assam Valley). It is also used for preserving 'handoh', rice powder, etc.

13. *Kota*- This kind of vessel is used for preparing curds. The name 'kota' is known in the Lower Brahmaputra Valley but the curd prepared in the Cachar district is kept in an earthen *basan* called 'patil' (a pot).

14. *Chilim*—'Chilim' or tobacco bowl is used as the topmost part of a 'hukha' (country pipe for tobacco smoking).

15. *Earthen lamps (chaki or pradip)*—These are abundantly used in rural areas by the peasants instead of brass and tin lamps.

16. *Fuldani*- –This is nothing but a flower vase used for decoration purposes.

17. *Tob*—This type of earthen pot is used for planting small flowers and coloured leafy trees. Generally these *tobs* are placed on the verandah of houses.

18. *Dhol*- It is an earthen drum fitted with leather and used as musical instrument. This type of instrument is used during pujas, Bihu festivals etc.

19. *Nagra*- –This is another type of musical instrument used at the time of *Namkirtans*, pujas etc

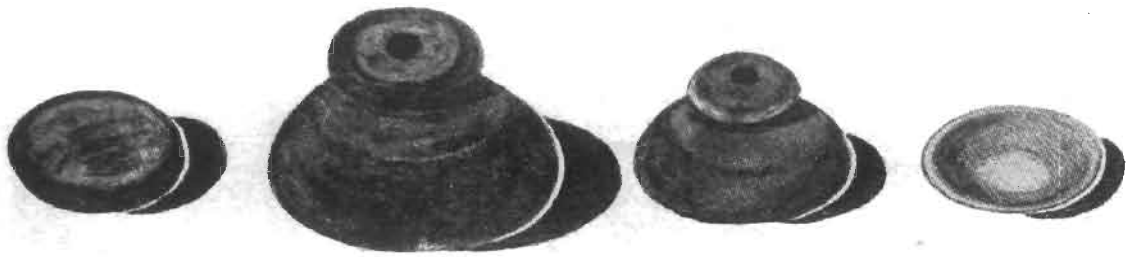
20. *Baiya* (Tabla)—This is a musical instrument which looks like a small 'nagra' and is used along with a 'daina'. The body of the *baiya* is made of clay while that of the *dama* is of wood.

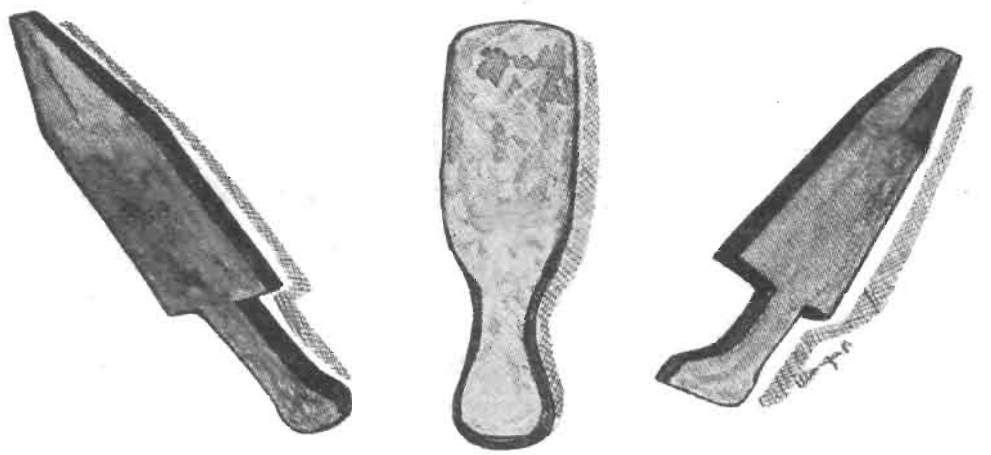
21. *Earthen Rings*- In some districts of Assam, the use of earthen rings instead of R.C.C. rings for the construction of ring wells is in vogue. The cost of these rings is very low in comparison with that of R.C.C. rings. The cost per ring (36" to 40" diameter and 6" height) is Re. 1/- only.

Earthenware used for ceremonial purposes : Earthen vessel plays a vital role in the performance of ceremonial functions all over India, especially amongst the Hindus. In connection with earthenware articles, various superstitions are observed which are very interesting to note. In so far as Assam is concerned, earthenware is used by the different sections of the people in various functions and ceremonies right from birth till the death of a person.

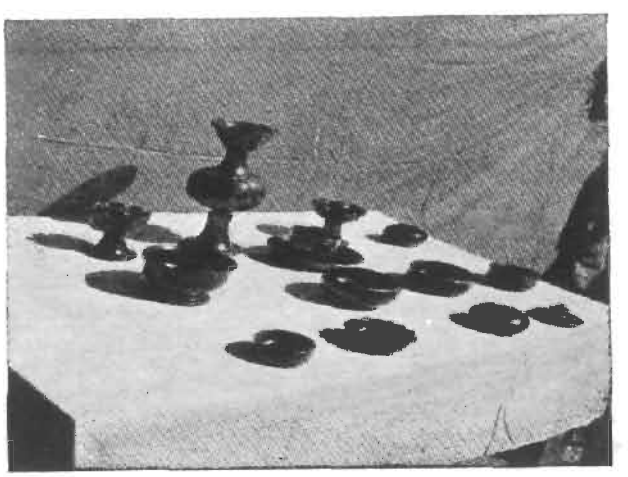
Seeing an earthen 'kalasi' or 'ghara' filled with water is treated as a symbol of good luck but to see an empty 'ghara' or 'kalasi' while going on business is a sign of misfortune. In the absence of an image of particular deity, the Hindus worship that particular god or goddess by placing an earthen 'ghara' or 'ghata' full of water over the uppermost portion of a plain tain leaf with a small heap of paddy over it. A bunch of five mango-leaves is then placed on the mouth of the 'ghata'. This is locally known as 'ghat puja'. In the same way, when *ghats* are used in any ceremony or function they are known as 'mangal ghat'. On such occasions these 'ghats' are generally placed just at the entrance of the house where the function is held.

A particular type of earthen vessel (saru is used by a certain section of people for throwing all dirts and wastes at the time of childbirth. As it is a custom to use an earthen pot, no other vessel of this type is used. The pot is kept by the side of the *poati* (the woman who





SOME HAND TOOLS LIKE MOULDS, MALLETS AND POLISHER USED IN ASSAM



Some glazed pottery works of Assam

SOME EARTHENWARES OF ASSAM



has just delivered a baby) and thrown away after five days from the date of delivery. This 'saru' is locally known as 'kumaru' in Kamrup district only when it is used for that particular occasion.

Earthen vessels decorated with different colours are also used during the course of marriage by the Hindus. The following are the articles required for this purpose—(1) One 'pradip' placed on a stand is used as a lamp and is burnt on mustard oil; (2) One 'basan' or 'patil' (saru) full of paddy known as 'purna-patra'; (3) One conical pot used at the time of playing 'kari' or 'kara' by the bridegroom and the bride; (4) One 'kalasi' and four 'gharas' used for carrying water from tanks for the purpose of ceremonial baths of the bride and the bridegroom.

The next important use of the earthen vessels comes after death. A 'kalasi' is taken to the burning ghat along with the dead body and broken after being filled with water as soon as the funeral activities are over. For the Hindus, those who bury the dead body, an earthen 'kalasi' is used at the time of the burial. In their case, the 'kalasi' is kept upside down by the side of the grave.

In almost all the rural areas of Assam Valley there is a superstition that no lamp with kerosene oil can be put by the side of a person suffering from pox but only an earthen lamp (pradip) with mustard oil.

ECONOMIES OF THE INDUSTRY

Cost of production : The cost of production of pottery articles includes the costs of clay, fuel, glazing, lacquering, materials used (if any) and labour charges. Generally, the ordinary earthenwares are sold at a very cheap price. Only a little bit of lacquering or glazing can fetch a better price. It may be mentioned here that the more weighty a pot is the more is the quantity of clay utilized and firewood consumed.

Earnings of a worker : The average monthly income of a potter is reported to be as follows:

	Full time	Part-time
1. Highly skilled worker	Rs. 60/- to Rs. 75 p.m.	Rs. 30 - p.m.
2. Semi-skilled worker	Rs. 40/- p.m.	Rs. 15/- p.m.

The rate of wages for the workers engaged in image and doll making are given below:—

Process	Rate of daily wage
1. Making of bamboo and straw frames	Rs. 4.00
2. Modelling of clay figures	Rs. 4.00
3. Painting	Rs. 2.00 to Rs. 3.00
4. Dressing and ornamenting the images or dolls	Rs. 1.50

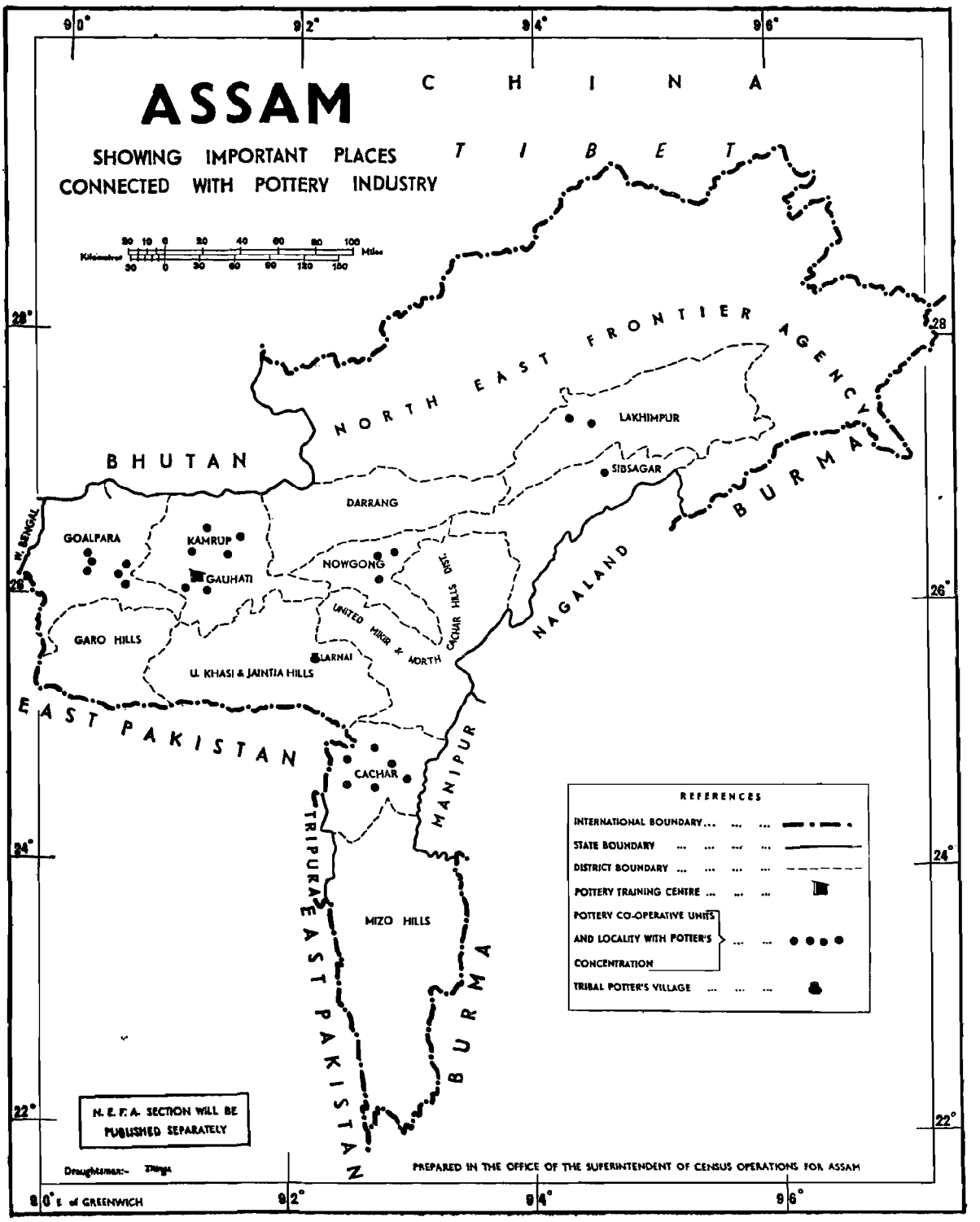
Marketing facilities : It is observed that the barter system still prevails in the rural areas particularly in case of buying and selling of pottery articles. The pottery goods are sold to the peasants by traveling peddlars in exchange of paddy. In the urban areas, however, the potters generally dispose of their produce to the merchants at wholesale rates who in turn market the products. But in some cases, the potters themselves sell their articles in the nearby local markets. Fine dolls and toys and other decorative potterywares are sometimes sold through the different Government emporia of the State.

Consequently, the cost of production becomes higher. Again, buyer's preference is always for light things as they are better looked and better baked. The production of light or heavy articles depends on the skill of the craftsman.

The following table gives the cost of production per establishment on the average for all the plains districts separately for pottery, brick making and image and doll making.

Selected Centres in districts	Depreciation of fixed capital	Value of shortlived equipment	Total value of materials consumed	Wages paid to hired workers	Other costs	Total cost	Cost per establishment (average)	Total value of products	Total receipts	Receipt per establishment (average)	Excess (+) of receipt over cost per establishment (average)
1	2	3	4	5	6	7	8	9	10	11	12
					Table 1 (Pottery)						
Goalpara	203	920	5,455	767	..	7,345	199	22,671	22,671	613	+ 414
Kamrup	106	90	2,755	188	..	3,139	131	13,854	13,854	577	+ 446
Darrang	101	25	1,852	..	..	1,978	198	8,204	8,204	820	+ 622
Nowgong	63	14	1,094	..	354	1,525	305	5,058	5,058	1,012	+ 707
Cachar	269	3	1,387	..	445	2,104	111	7,469	7,469	393	+ 282
Sibsagar	306	..	725	..	333	1,364	52	7,936	7,936	305	+ 253
Lakhimpur	13	132	370	..	188	703	234	2,697	2,697	899	+ 665
Total	1,061	1,184	13,638	955	1,320	18,158	146	67,889	67,889	547	+ 401
					Table No. 2 (Brick making)						
Goalpara	15	69	118	1,130	66	1,398	1,398	1,800	1,800	1,800	+ 402
Kamrup	..	..	..	..	..	..	..	..	..	..	..
Darrang	62	946	6,912	12,468	658	21,046	10,523	26,120	26,120	13,060	+2,537
Nowgong	..	..	..	..	..	..	..	..	..	..	..
Cachar	9	57	970	2,380	1,768	5,184	1,728	6,735	6,735	2,245	+ 517
Sibsagar	..	..	..	..	..	..	..	..	..	..	..
Lakhimpur	..	..	..	..	..	..	..	..	..	..	..
Total	86	1,072	8,000	15,978	2,492	27,628	4,605	34,655	34,655	5,776	+1,171
					Table No. 3 (Image and doll making)						
Goalpara	66	122	1,250	430	380	2,248	562	6,804	6,804	1,701	+1,139
Kamrup	192	90	2,292	864	350	3,788	1,894	7,246	7,246	3,623	+1,729
Darrang	131	201	1,672	..	830	2,834	1,417	3,676	3,676	1,838	+ 424
Nowgong	20	8	106	..	..	134	134	360	360	360	+ 221
Cachar	74	186	2,269	392	602	3,523	881	6,182	6,182	1,546	+ 665
Sibsagar	11	12	816	980	548	2,367	1,184	4,636	4,636	2,318	+1,136
Lakhimpur	146	10	928	..	155	1,239	620	5,090	5,090	2,545	+1,925
Total	640	629	9,333	2,666	2,865	16,133	949	33,994	33,994	2,000	+1,051

Source :—Report on 'A Survey of Cottage Industries in Assam' 1958.


ASSAM
SHOWING IMPORTANT PLACES
CONNECTED WITH POTTERY INDUSTRY
CHINA
TIBET
NORTH EAST FRONTIER AGENCY
BHUTAN
BURMA
LAKHIMPUR
SIBSAGAR
DARRANG
W. BENGAL
GOALPARA
KAMRUP
GAUHATI
NOWGONG
NAGALAND
UNITED MIKIR & NORTH CACHAR HILLS DIST.
GARO HILLS
LARNAI
U. KHASI & JAINTIA HILLS
EAST PAKISTAN
CACHAR
MANIPUR
TRIPURA
MIZO HILLS
EAST PAKISTAN
BURMA
REFERENCES
INTERNATIONAL BOUNDARY
STATE BOUNDARY
DISTRICT BOUNDARY
POTTERY TRAINING CENTRE
POTTERY CO-OPERATIVE UNITS AND LOCALITY WITH POTTER'S CONCENTRATION
TRIBAL POTTER'S VILLAGE
N. E. F. A. SECTION WILL BE PUBLISHED SEPARATELY
Draughtsman:- Zinga
PREPARED IN THE OFFICE OF THE SUPERINTENDENT OF CENSUS OPERATIONS FOR ASSAM
90° E of GREENWICH
90°
92°
94°
96°
28°
26°
24°
22°

Production and sale under organised sector : It is noticed from a report of the Assam State Khadi and Village Industries Board for the year 1960-61 that unglazed pottery worth Rs. 11,885/- and building materials worth Rs. 33,760/- were produced during the period under review. Sales worth Rs. 10,667/- of unglazed pottery and Rs. 11,715/- worth of building materials were effected. Full time employment was provided to 232 persons while 165 persons were engaged as part time workers. Wages paid amounted to the tune of Rs. 22,791/-.

Assistance available under different schemes: In order to stabilise the economic condition of the potters of this State, various schemes have been chalked out by the Assam State Khadi and Village Industries Board in collaboration with the Cottage Industries Department, Assam. The State Industries Department is assisting the potters financially by way of sanctioning grants-in-aid and by advancing working capital loans to individual potters as well as Potters' Co-operative Societies. Activities of the Assam State Khadi and Village Industries Board have been slowly expanding in the field of pottery industry. The State Board is helping the potters in respect of financial, technical and managerial facilities through registered co-operative societies. Various improved pottery tools and implements are made available at 50 per cent subsidised rates. Loan for share capital (in the case of a registered co-operative society) and loans and grants for working capitals and grants for the staff, etc. are provided by the State Board. Every year a few pottery artisans are sent outside the State with stipends for acquiring knowledge in the improved methods and techniques on commercial pottery wares. Apart from this, the Board has organised a demonstration squad on pottery with technical personnel who are going round the interior places of the State to demonstrate on the up-to-date technique and improved methods of production. Potters are also reported to be taking very keen interest in such demonstrations. Further it is reported that out of the 51 existing Potter's Co-operative Societies, 41 units have already received financial assistance from the commission through the Khadi Board for implementation of different schemes in Community Projects and non-Community Project areas of the State.

Conclusion : It is the comparative cheapness of the earthenwares that has enabled the pot-making industry to survive through the ages. Those types of earthen vessels used only for serving were almost replaced by modern plates long ago. Likewise, the earthen pots are also being replaced by copper pots. It is only the cooking vessels that could not be easily replaced as the cost of bronze vessels was prohibitive. But the influx of cheap aluminium vessels has been rapidly replacing the earthen cooking vessels also thereby making the survival of the pot-making industry more precarious in that it has to depend solely on the incapacity of the poor to go in for any other type of vessel. The uses of the pots and vessels mentioned above are now confined to the poor Mohammedans of the rural areas only. This calls for a re-orientation of the industry to divert the skill of the potter to more attractive products which can stand competition in an open market. It is encouraging to note that the State Government through the Industries Department and the Khadi and Village Industries Board started this work in the right direction. Technical and financial help are being extended to the Potters' Co-operative Societies and individual potters for construction of improved kilns, supply of plaster moulds, replacement of potters' wheel by the mechanised wheel etc. for improving the pot-making industry. Pot makers are also being encouraged to take up doll, brick and tile manufacturing. But many things have yet to be done and it seems essential that the future of the pot-making industry be ensured by organising new co-operative societies and revitalising the existing ones on a sound basis and business lines.

THE POTTERY INDUSTRY OF LARNAI VILLAGE IN THE UNITED KHASI AND JAINTIA HILLS

INTRODUCTION

The village Larnai is situated about 11 miles away north-west of the Jowai town with a mythological background after its name. The word 'khiaw larnai' connotes 'earthen pot'. The myth says that long ago, God appeared before a potter and asked him how he gathered the knowledge of making earthen articles. In reply, the potter said that he acquired it from the Almighty God. God was pleased with the potter as the latter was found faithful and blessed him that whatever article made by him out of clay would turn into an article of brass and be unbreakable. After a few days, God appeared again before the same potter and asked how he got the knowledge of making pottery. This time the potter boasted of his art and replied that it was his own brain which invented the art of making pottery and showed his pride of it. God became dissatisfied with his answer and cursed that the article made by the potter out of clay would be of mere clay and breakable and not of brass as blessed before. Since then the pottery is there and the village is also named after '(khiaw) larnai'.

Lying over an area of $1\frac{1}{2}$ sq. miles and having 53 households, the village Larnai has a population of 221 (106 males, 115 females). The inhabitants of the village belong to the Pnar community, 48 or 21.72 per cent of whom are engaged in pottery. Pottery here is the occupation of women only and they constitute 41.74 per cent of the total female population of the village.

The potters of Larnai make their pottery without the aid of a potter's wheel and the clay used by them is of two types. The first and perhaps the most important one is a black clay locally known as 'dawiong' and the other is greyish-green known as 'dawkhluit'. 'Dawiong' or the black sticky clay is found only in the bed of a very small stream flowing through the Sung valley, and it is reported that this type of clay is found only in this place. Since it is the most important raw material for pottery, it provides an explanation for the absence of the industry in the other parts of the district and hence for the monopoly of the industry for many generations by the people of Larnai which is only 6 miles from the location of the clay. 'Dawkhluit' is obtained from the surrounding slopes of the Sung valley. The potential reserve of this type of clay is, however, much more great than that of the former. This raw material looks like sandstone with greyish-green colour. The important characteristics of this type of clay is that it is very heavy and sticky.

PROCESS OF PRODUCTION

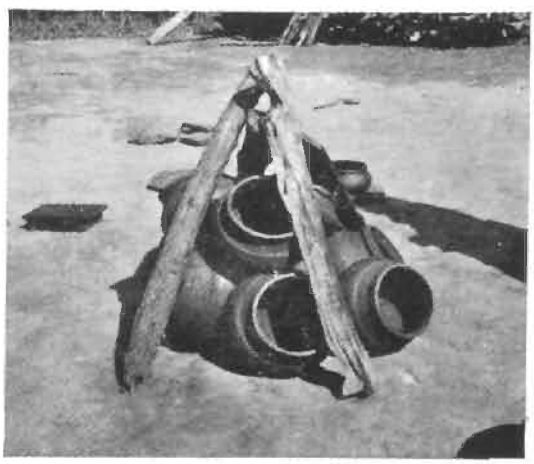
Preparation of clay: The proportion of mixing different types of clay is 3 parts of black sticky clay (dawiong) and 1 part of grevish-green sandstone clay (dawkhluit). To this mixture a little ash is added. The usual amount of ash is about 1/10th of the total mixture. Here it may be pointed out that if a quantity of black soil is brought to the village at a time, the owner has to keep it in a wet dug-hole so as to maintain its sticky condition. In case it gets dried up, the whole quantity of clay will be of no use for the purpose.

When the clay and ashes are taken in right proportions the mixture is placed in a





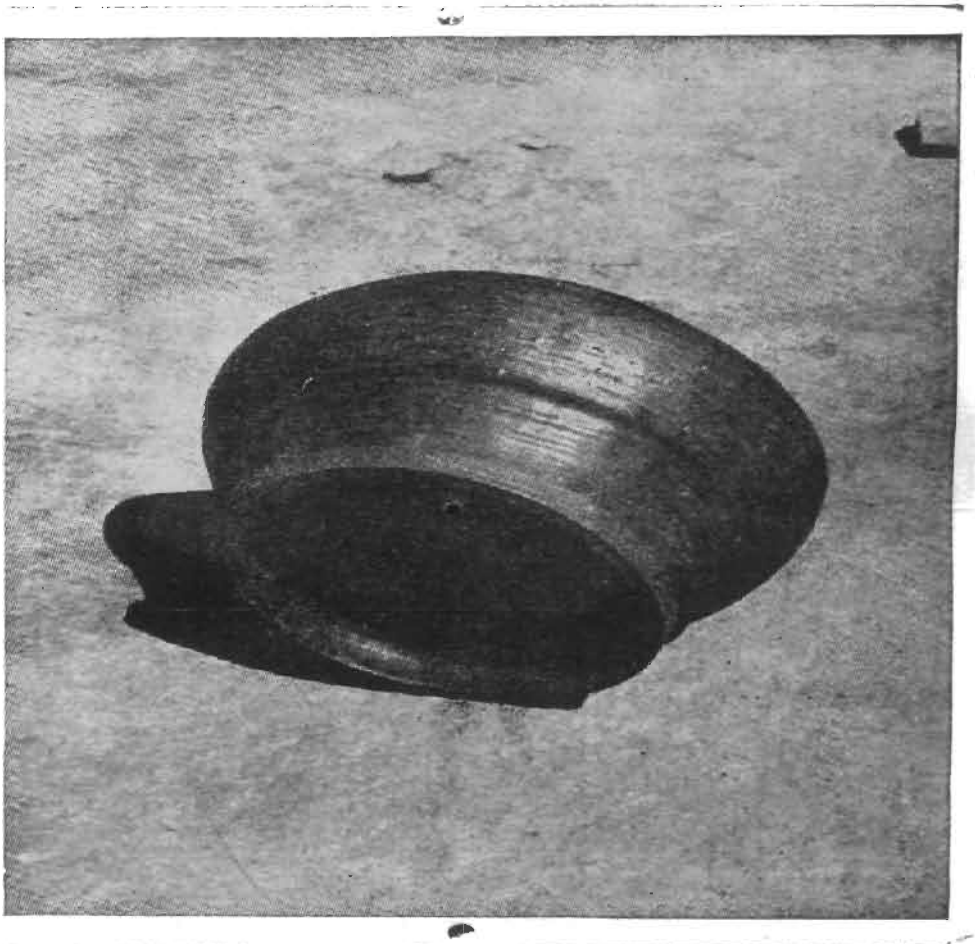




Larnai potters at work—different stages upto baking of pots



'Kum-Kyiad' from Larnai



'Ukhra'—also from Larnai

wooden mortar locally known as 'u-thlong' and some water is added. Then it is crushed with the help of a wooden pestle locally known as 'u-synrai' till all the ingredients are completely mixed up. From time to time, little quantity of water is added so as to make it sticky. Now the clay is taken out from the mortar and placed on a piece of a big raw hide where it is again crushed by the pestle and made into a thick sheet by rolling with the pestle.

Preparation of the earthenwares : *1st stage*—The clay sheet is now cut into long pieces locally known as 'pyrthap'. The 'pyrthap' is then placed on a wooden plank, the size of which is about 2′ long and 1′ wide. This wooden plank is known as 'thap lynter' where it is again spread by beating it with a wooden stick locally known as 'shoh pyrthap'. On beating, the clay sheet takes a rectangular shape.

2nd stage—Now this rectangular-shaped clay sheet is bent to make a ring locally known as 'khiaw' but in the 'khiaw' stage, both the ends of the clay sheet are not sealed. The diameter of the ring and the thickness of the sheet of clay varies according to the size of the pots. When both the ends of the 'khiaw' are sealed by pressing with the fingers along with water, it is known as 'wiang'.

3rd stage—(*shaping*)—A round stone known as 'mo-shoh-at' is held by the left hand to support the 'wiang' from inside while the beating is done with a flat piece of wood locally known as 'siang' by the right hand. This is done to give a final shape to the eathenware. To seal the bottom of the earthenware, a flat stone is used in place of a round one. This flat stone is known as 'mo-shoh-sdiap'.

4th stage—(*burning and colouring*)—All the earthenwares so made are dried in the sunshine for one or two days. When these are dried, they are kept in a heap in the open and covered with dry firewood. The wood is now set on fire. When the earthenwares are fully burnt, they are taken out of the fire from time to time with the help of a stick and a liquid colour already prepared is applied with the help of a brush locally known as 'nob'. The products are now ready for sale.

TOOLS AND EQUIPMENTS

The following are the tools and equipments required for this craft.

1. *Hoe*—This is locally known as 'wakhu' and is used for digging the clay from the clay hole. The hoe used in this district is a bit different from that of other districts. It consists of three parts namely (i) a blade made of iron (wakhu), (ii) an angle (sop) made of iron and (iii) fitted with a wooden handle (sping).

2. *Shovel*—This is locally known as 'lyngka' and also used for digging clay from the clay-hole.

3. *Carrying basket*—This is known as 'khoh', fitted with a brace-piece (star) and used for carrying the clay. These carrying baskets are made of cane and bamboo.

4. *Mortar*—This is locally known as 'thlong', made from the trunk of a tree.

5. *Pestle*—This is made from a single piece of solid wood which is about 5 feet long with a diameter of about 3 to 4 inches. This is known as 'u-synrai'. Mortar and pestle are used for crushing the clay collected from the field and also for crushing the bark of certain tree (soh khmi) from which the red colour juice is obtained for colouring the finished articles.

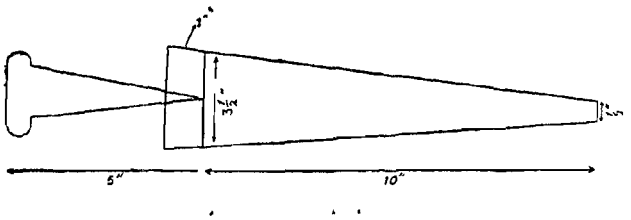
6. *Buffalo skin*—This is locally known as 'snieh' and is used for further crushing and rolling the clay which is ready for preparing the articles.

7. *A piece of rectangular wood*—Measuring 10″ long, 2″ broad and 1″ thick which is filled with a handle of about 1″ in diameter.

This is locally known as 'shoh pyrthap' and is used as beater.

8. *A piece of plank*—Measuring 24" long, 12" broad and 1½" thick locally known as 'thap lynter'—used for making the clay sheets.

9. *Wooden hand-beater*—Locally known as 'siang-shoh-at', the common type of which measure as follows:



Each potter generally keeps more than one of such beaters of different sizes and weights. It is used for shaping the pottery articles.

10. *Round flat stone*– This is locally known as 'mo-shoh-at' or 'mo-shoh-sdiap' of different sizes and shapes, used for supporting the sheet of clay from inside by the left hand while beating is done by the right hand with 'siang-shoh-at' from outside.

11. *Brush*—This is made of the bark of a tree (nob) and is used for applying the liquid colour on the pots after burning.

FINISHED PRODUCTS AND THEIR MARKETING

The most common earthen articles produced by the potters of Larnai are described below:

1. *Khiaw ranai*—This is a cooking pot used in Khasi and Jaintia Hills. It is a round vessel with a short neck, of a height ranging from 5" to 8" and a diameter ranging from 8" to 20".

Cost	Big size	...	Rs. 5.00 each
	Medium size	...	Rs. 2.00 each
	Small size	...	Rs. 1.25 each

2. *Kum*- It is a big round vessel used for brewing of local liquor, storing water or other kinds of liquids. It is also used in the process of storing dry fish. It is a long-necked vessel with a usual height of 2½' and diameter of 20". The cost of such vessel varies from Rs. 12/ to Rs. 15/ each. Another type of 'kum' used in the process of distillation of local liquor is much smaller than the above type. The vessel has round, broad base sloping gradually to the narrow and round neck. The average size of this vessel is about 6" high and 5" diameter at the base with a mouth of about 2½" in diameter.

The cost of such kum is Rs. 3.00 each.

3. *Bhot*—This pot is used for distillation of local liquor. The shape and size of this type of vessel is just like that of a small 'kum' but shorter in height. On the broad base of the vessel two spouts are fixed. During the process of distillation, the boiling vessel is always covered with the 'bhot' turning upside down. The condensed vapour of liquor comes out through these spouts, at the end of which two receivers are placed so as to collect the distilled spirit.

The cost of each *bhot* varies from Rs. 4/- to Rs. 5/-.

1. Some other earthenwares are also prepared by them but these pots have got resemblance with the potteries in the plains and need not be described here again.

Marketing: Marketing of the pottery wares produced at Larnai village is known to

be confined within the Khasi and Jaintia Hills. There is good demand for these cheap pottery wares throughout the rural areas of the district. All these products are sold in village markets by the artisans themselves or sometimes through middle-men.

CONCLUSION

It is very significant to note that Khasi and Jaintia Hills is the only exception amongst tribal areas of Assam where the indigenous method of earthenware manufacture is still practised by the local artisans. It is not uncommon to find a few potters in the Garo Hills and United Mikir and North Cachar Hills Districts, but these are reported to have migrated from other districts of the State. The existence of the pottery industry in the Khasi and Jaintia Hills District which is being carried on by the Pnar potters for generations clearly shows the antiquity of the craft in this self-developed tribe. This age-old craft can still survive with slight modifications and improvements in the method of production and if liberal patronage is available from the local people and the Government.

APPENDIX A

POTTERS POPULATION OF ASSAM AS PER 1961 CENSUS IS INDICATED IN THE APPENDIX BELOW, DISTRICT-WISE.

State/Division/District	Males	Females	Total
Assam	6,420	7,353	13,773
Assam Plains Division	3,121	3,555	6,676
1. Goalpara	664	653	1,317
2. Kamrup	771	1,462	2,233
3. Darrang	270	96	366
4. Lakhimpur	165	12	177
5. Nowgong	288	151	439
6. Sibsagar	696	750	1,446
7. Cachar	267	431	698
Assam Hills Division	88	117	205
8. Garo Hills	76	109	185
9. United Khasi & Jaintia Hills	8	16	24
10. United Mikir & North Cachar Hills	6	1	7
11. Mizo Hills	Nil	Nil	Nil

APPENDIX B

STATEMENT OF POTTERY CO-OPERATIVES OF ASSAM

Name of district	Number of Co-operative Societies	Number of paid-up share capital	Amount of paid-up share capital	Total number of workers	Extent of business done during 1961-62	Amount of loans and grants received from various sources
1	2	3	4	5	6	7
1. Goalpara	6	48	645.00	48	7,332.00	4,900.00
2. Kamrup	5	267	1,386.00	210	104,478.68	31,085.00
3. Darrang	Nil	..	.	..	..	..
4. Sibsagar	1	38	435.00	30	393.00	15,150.00
5. Nowgong	2	81	605.00	81	*2,855.00	7,500.00
6. Lakhimpur	2	36	360.00	25	N.A.	N.A.
7. Cachar	6	121	1,370.00	105	N.A.	8,200.00
8. United Khasi & Jaintia Hills	Nil	..	..	..	..	..
9. United Mikir & North Cachar Hills	Nil	..	..	..	..	..
10. Garo Hills	Nil	..	.	..	.	..
11. Mizo Hills	Nil	.	..	..	.	..
Total of all districts	22	591	4,801.00	499	115,058.68	66,835.00

*Extent of business done by one Society only. Figures of all columns exclude three societies.

N.A. — Stands for 'Not Available'.

APPENDIX C

PRODUCTS AND MARKETING

Selected Centres (in Districts)	Total value (in Rs.) of products	Value (in Rs.) of products retained for home consumption	Value (in Rs.) of products remaining unsold	Value (in Rs.) of product sold				
				In local market	Within the district	Within the State	In Indian territories	Total
1	2	3	4	5	6	7	8	9
Goalpara	22,671	..	1,111	18,245	3,315	..	..	21,560
Kamrup	13,854	..	1,805	10,365	1,684	..	..	12,049
Darrang	8,204	.	555	5,542	2,107	..	..	7,649
Nowgong	5,058	.	737	4,058	263	..	..	4,321
Cachar	7,469	..	382	5,549	1,538	..	..	7,087
Sibsagar	7,936	..	333	2,000	4,336	1,267	..	7,603
Lakhimpur	2,697	..	61	2,636	..	..	..	2,636
Total of Plains districts	67,889	..	4,984	48,395	13,243	1,267	..	62,905
Garo Hills	3,788	..	314	2,804	670	..	..	3,474
United Khasi & Jaintia Hills	1,900	..	..	950	950	..	..	1,900
Mizo Hills	..	..	..	..	..	..	..	..
United Mikir & North Cachar Hills	984	..	68	916	..	..	..	916
Total of Hills districts	6,672	..	382	4,670	1,620	..	..	6,290
Total of all districts	74,561	..	5,366	53,065	14,863	1,267	..	69,195

Source : A Survey of Cottage Industries in Assam by the Department of Economics and Statistics, Assam, Shillong.

APPENDIX D

TREND OF PRICES OF POTTERY PRODUCTS

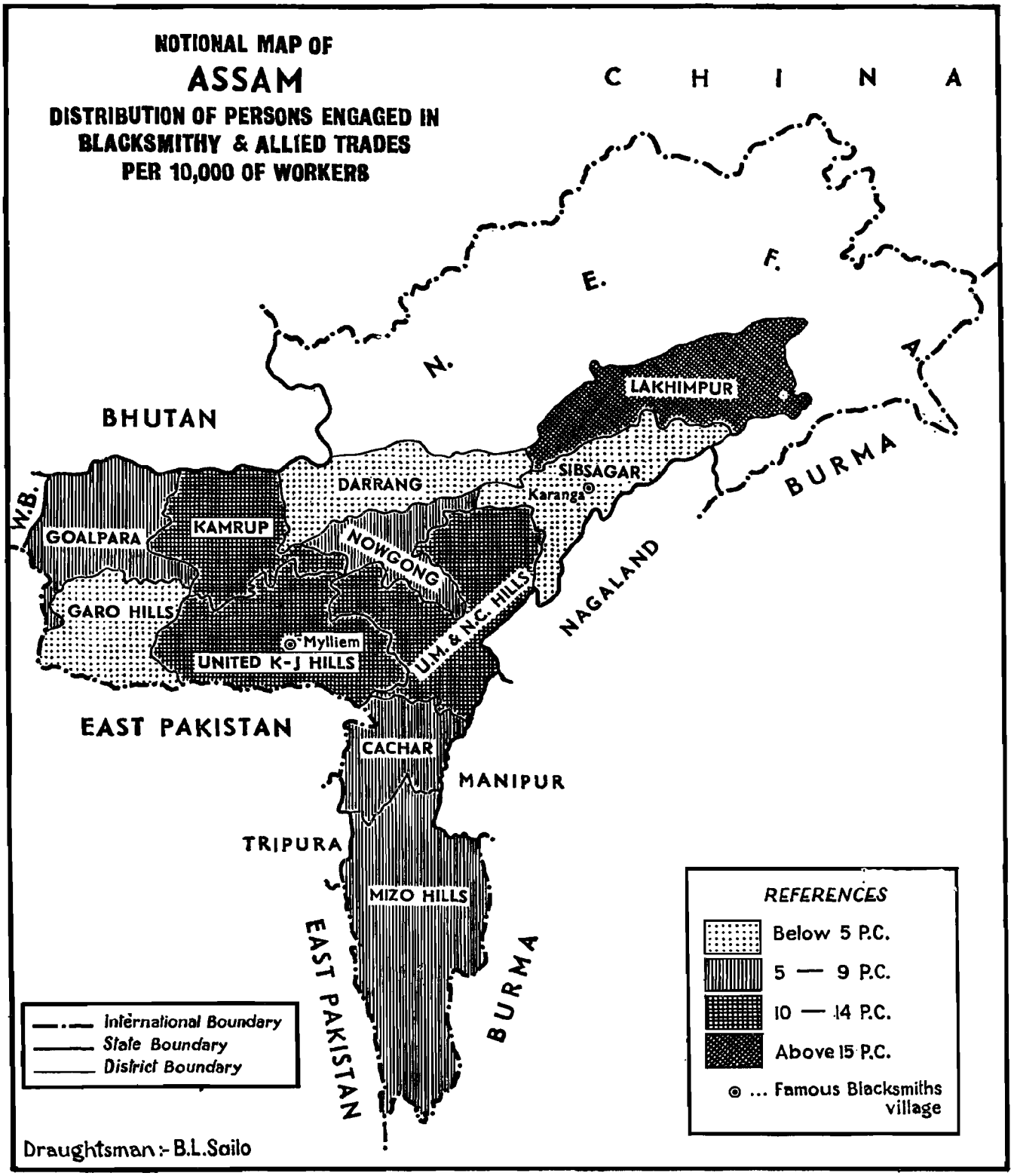
Name of the District	Products	Unit	Price During (In Rs.)							
			1939		1953		1954		1961	
			Urban	Rural	Urban	Rural	Urban	Rural	Urban	Rural
1. Goalpara	Pitcher	No	..	0-4-0	..	0-4-6	..	0-4-6	..	..
2. Kamrup	Pitcher	No	0-1-6	0-1-6	0-4-0	0-5-0	0-4-0	0-5-0	..	..
3. Darrang	Pitcher	No	0-1-0	0-1-0	0-3-0	0-3-0	0-3-0	0-3-0	..	..
4. Nowgong	Pitcher	No	..	0-0-6	..	0-2-0	..	0-2-0	..	..
5. Cachar	Pitcher	No	..	0-2-0	..	0-4-0	..	0-4-0	..	..
6. Sibsagar	Pitcher	No	0-1-6	0-1-0	0-5-0	0-4-0	0-5-0	0-4-0	..	..
7. Lakhimpur	Pitcher	No	0-1-6	0-1-0	0-4-0	0-4-0	0-4-0	0-4-0	..	..
8. Garo Hills	Pitcher	100	...	12-0-0	..	20-0-0	..	20-0-0	..	..
9. United Khasi & Jaintia Hills	Earthen pot	No	...	0-2-0	..	0-12-0	..	0-12-0	..	..
10. Mizo Hills	Nil		Nil	Nil	Nil	Nil	Nil	Nil	..	..
11. United Mikir & North Cachar Hills	Pitcher	No	..	..	..	2-8-0	..	2-8-0	..	..

TERMINOLOGY

English	Pnar
Potter's clay	
(a) Black clay (b) Greyish-green clay	Dawiong Dawkhluit
Wooden mortar	U thlong
Wooden pestle	U synrai
Clay sheet	Pyrthap
Clay ring (before sealing)	Khiaw
Clay ring (after sealing)	Wiang
Hoe	Wakhu
Shovel	Lyngka
Carrying basket	Khoh
Braw piece	Star
Buffalo skin	Snieh
A piece of rectangular plank (10'' x 2'' x 1'')	Shoh pyrthap
A piece of plank (24'' x 12'' x 1½'')	Thab lynter
Wooden hand beater	Siang-shoh-at
Round flat stone	Mo-shoh-at or Mo-shoh-sdiap
Brush	Nob
Cooking pot	Khiaw Ranai
Big round vessel	Kum
Liquor distillation pot	Bhot
A flat piece of wood	Siang

BIBLIOGRAPHY

1. Manufacture of Pottery in Assam by E.A. Gait.
2. A Monograph on the Pottery of North-Western Provinces and Oudh by H.R.C. Dobbs, C.S.
3. The Statesman, Calcutta, 24th August, 1963
4. Srihatter Itibritta by Achhyut Charan Chowdhury, Tatwanidhi.
5. Extract from the Report supplied by Shri M.C. Goswami M.A.,B.T. (Cal) M. Litt. (Cantab), Head of the Department of Anthropology, Gauhati University.
6. History of Civilization of Assam by Dr. P.C. Chowdhury M.A. (B.H.U.)., Ph.D. (London).
7. Mineral Resources of Assam, Published by the Directorate of Information and Public Relation, Assam.


NOTIONAL MAP OF
ASSAM
DISTRIBUTION OF PERSONS ENGAGED IN BLACKSMITHY & ALLIED TRADES PER 10,000 OF WORKERS
C H I N A
N. E. F. A.
BHUTAN
LAKHIMPUR
SIBSAGAR
Karanga
DARRANG
BURMA
W.B.
GOALPARA
KAMRUP
NOWGONG
U.M. & N.C. HILLS
NAGALAND
GARO HILLS
Mylliem
UNITED K-J HILLS
EAST PAKISTAN
CACHAR
MANIPUR
TRIPURA
MIZO HILLS
EAST PAKISTAN
BURMA
International Boundary
State Boundary
District Boundary
REFERENCES
Below 5 P.C.
5 — 9 P.C.
10 — 14 P.C.
Above 15 P.C.
... Famous Blacksmiths village
Draughtsman :- B.L.Sailo

CHAPTER V

Blacksmithy

Introduction : Blacksmithy is an important and commonly met household industry in Assam. It is invariably found as the hereditary occupation of the households concerned. The son works as an apprentice and receives training under his father and the skill is thus carried down from generation to generation.

In the plains districts, the ironsmith is known by the term 'Kamar'—one of the nine professional caste-groups (Naba-Saka). The Kamars do not like to be designated as such but prefer to continue to retain their old caste status which they claim to be descended from Viswakarma. A growing tendency is noticed among the Kamars to change their original caste names for better ones. This is because they do not rank equally with people of higher castes and suffer from some sort of inferiority complex. They form a class of their own for matrimonial relations and other social purposes. A Kalita does not generally favour to enter into a matrimonial alliance with a Kamar.

In the hill districts, however, the caste system is unknown, and there the industry is carried on by any one choosing to learn the work.

In the olden days the ironsmith used to make his own anvil, bellows, hammers, chisels, etc. that is, all the tools of his craft. He used to make all the agricultural and domestic tools required by the society. Besides, he manufactured countless articles of daily use.

Time has brought about many far-reaching changes in our social and economic life but some of the tools and implements have come down through the ages to us exactly in their original shapes. The present usual products of blacksmithy all over the State are simple agricultural implements like ploughshare, sickle, hoe and various implements of daily use such as 'dao', knife, axe, etc. Apart from the manufacturing of these products, the blacksmiths also undertake repair and other servicing works.

Nature of Establishment : Blacksmithy establishments are found in both rural and urban areas but generally the industry is more concentrated in urban areas. Usually the rural establishments are scattered and one establishment often serves the need of several villages. But occasionally, concentrations are also found in rural areas such as village Karanga in Sibsagar district and Mylliem in United Khasi and Jaintia Hills district. Such areas are found to be inhabited by a large number of hereditary blacksmiths for generations. For example, the blacksmiths of Karanga have been working in the village from several generations and are only branches of the same family. It is said that in the days of Ahom rule, the blacksmiths of Karanga were specially appointed for manufacturing artilleries. The blacksmiths of

Karanga take pride in their ancestors whom they claim to be the manufacturers of the biggest cannon still found displayed in the compound of the subdivisional court building at Sibsagar.

It is found that most of the artisans are working in an unorganised way. According to a survey of cottage industries in Assam conducted by the State Statistical Bureau in 1954-55 "out of 669 establishments listed in the selected centres, 97.9 per cent were family establishments and only 1.8 per cent were co-operative organisations. Only two establishments were found to be partnership organisations. The position has not been improved much since then. Only recently a few cooperative societies have been organised and the impact of their activities on the artisans are yet to be seen".

The urban units generally provide whole-time employment and principal occupation while their rural counterparts provide only part-time and subsidiary occupation to the workers. According to the 1961 Census, there are 4,813 persons engaged in blacksmithy as full-time workers in the State of Assam. Out of these workers only 21 are found to be females. In fact it is a man's job and women are generally not engaged in blacksmithy.

The blacksmith's workshop or workshed is locally known as 'Kamarshala' or 'Kamarshal' (Kamar means blacksmith and shal or shala means workshed). It is a very simple type of construction with thatched roof raised on a bamboo structure and is generally located within the compound of the artisan's house. A 'kamarshala' requires a space of only about 10 sq.ft. and no wall. At least two persons are generally required to do a job in the 'Kamarshala', one doing the job of a pivot while the other assists him. Some minor operations like polishing and sharpening of small tools or articles can, however, be done individually.

TECHNIQUE OF PRODUCTION

Raw materials : The chief raw materials for the industry are steel and soft iron. These are imported from outside the State by the merchants and the blacksmiths obtain their requirements from the local merchants. A substantial part of the raw materials is, however, supplied by the scraps of motor vehicles obtained from war dumps. A considerable part of the requirement of steel in the industry is met by old and worn out springs of motor vehicles which also come from the dumps of scrap iron. The springs are usually collected out of the dumps and sold in the markets at much higher prices than the scraps. It is found that the blacksmiths prefer to purchase the scraps because these are much cheaper than new iron. Sometimes the blacksmiths purchase big dumps of scraps either alone or jointly with others at cheap prices and the supply from such purchases last for years. These scraps are available in local markets, railway and tea garden workshops. The artisans under cooperative societies get the supply at a cheaper price than the unorganised ones. The cooperative societies purchase the dumps of scraps in auction from railway or tea garden workshops at cheap price and supply to the artisans.

The chief fuel is coal but some of the blacksmiths use charcoal as well which is available from local merchants all over the State. Rural blacksmiths sometimes burn logs of wild timbers and prepare charcoal for their own use. The average price per quintol of charcoal varies from Rs. 35.00 to Rs. 40.00 in different districts of the State.

Tools and implements : The tools and implements used by the blacksmiths are very

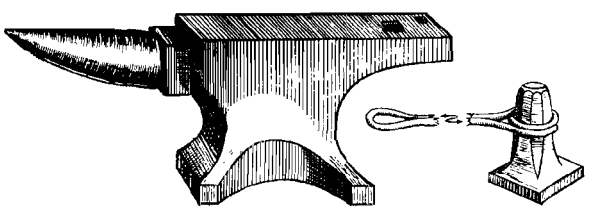


Fig. Anvil.



Fig. Flatter.

Fig. Blacksmith's Hammer.

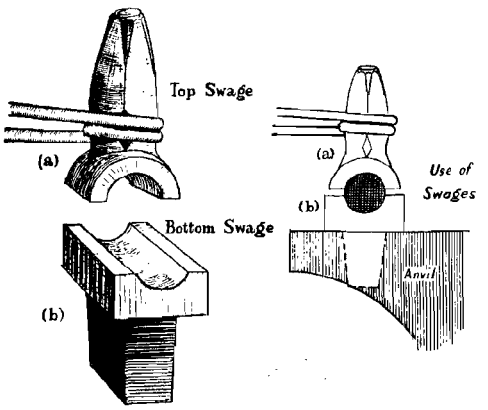


Fig. Blacksmith's Swages.

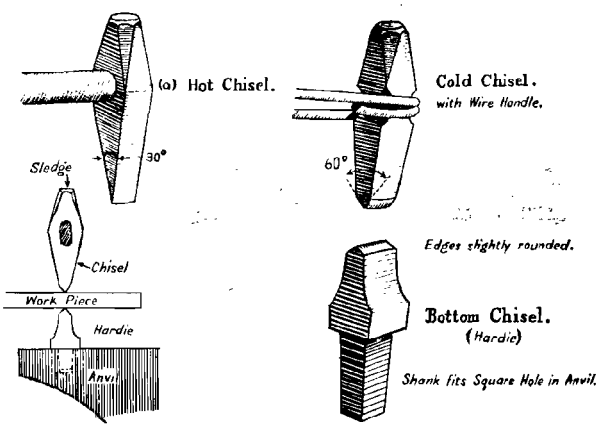


Fig. Blacksmith's Chisel.

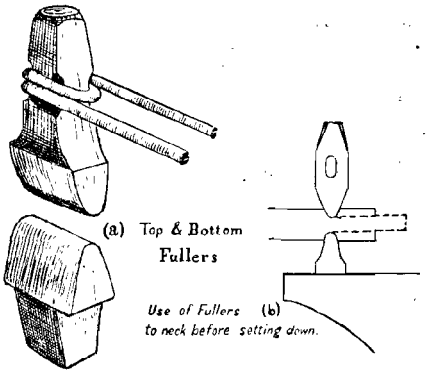
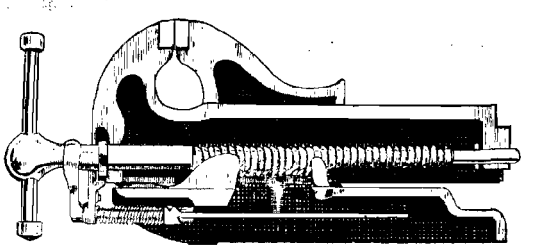


Fig. Blacksmith's Fullers



Section of quick operating type.

Fig. Fitter's Vise.

Improved handtools now being introduced

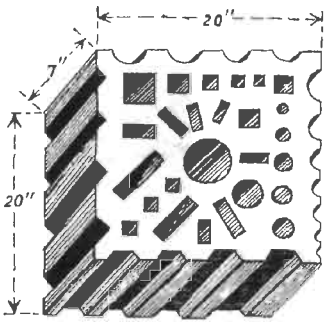


Fig. Swage Block.

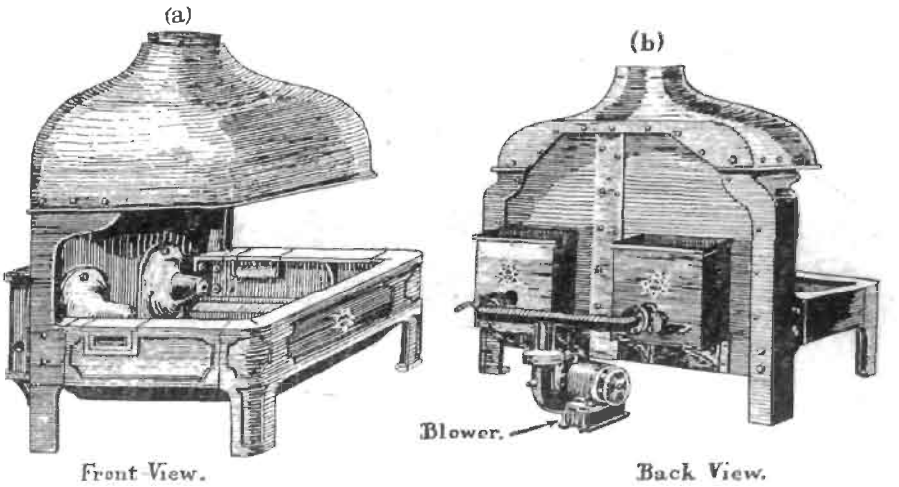


Fig. Smith's Hearth

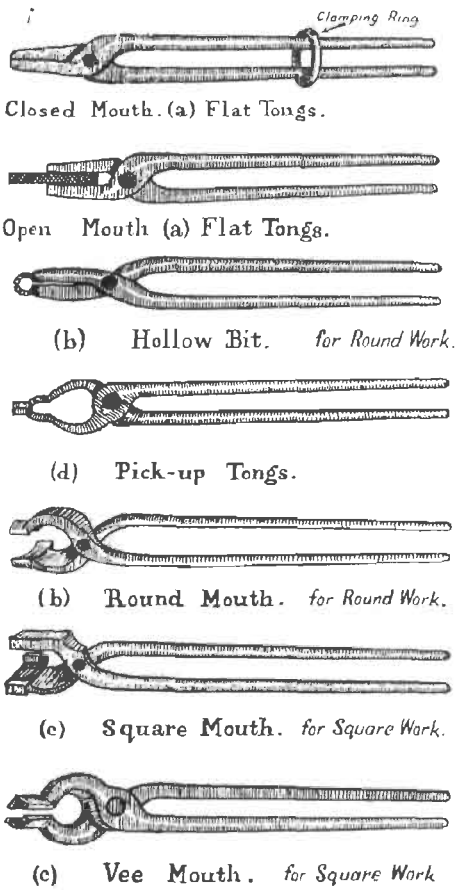


Fig. Blacksmith's Tongs.

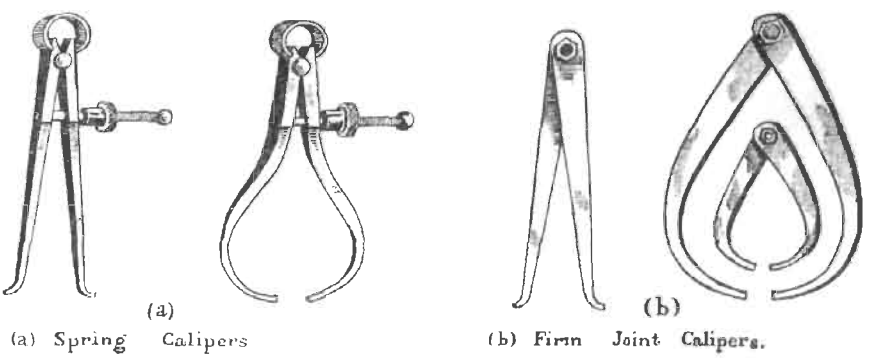


Fig. Calipers.

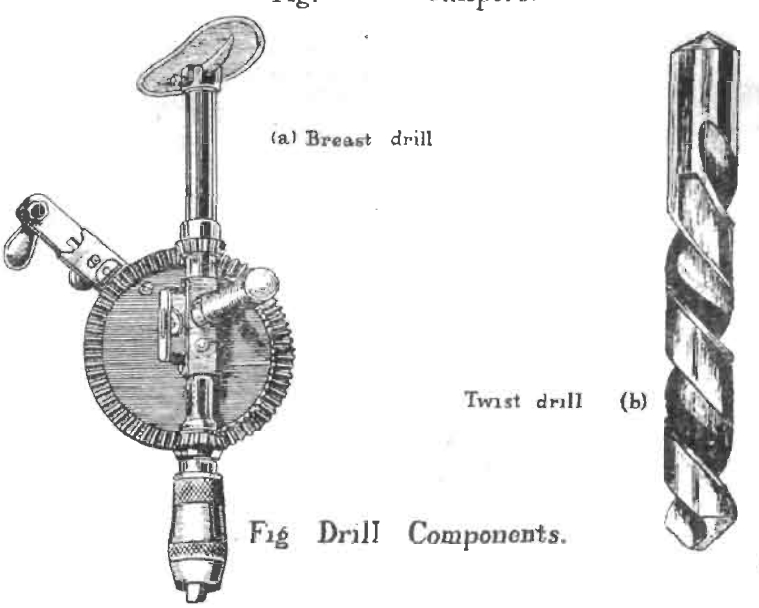


Fig Drill Components.

Improved handtools now being introduced

simple. A pair of sledge hammers, an anvil, a pair or two of tongs, a few small hammers, cold chisels, files, a furnace with bellows are all that they need for their working:

Furnace (bhatti)—For heating the metal a furnace locally known as 'bhatti' is necessary. For small works the blacksmith's furnace is commonly used and consists of a hearth for holding the combustible coke, an opening for leading forced air into the fire and a blower or bellow either of wood or leather for supplying air under slight pressure through the mouth of the bhatti. Generally the blacksmiths of the State are seen using locally made wooden or leather blowers.

Anvil (nia)—The anvil (nia) is meant for supporting the work or object while it is struck with the hammer as well as providing space for other forging operations. The weight of anvils may go up to about 3 cwt. and should stand with the top face about 2 feet from the ground. The height may be attained by resting the anvil on a cast iron or wooden base.

Hammer (haturi)- Two kinds of hammers (locally known as haturi) are generally used in hand forging:

(a) the hand hammer used by the smith himself, and

(b) the sledge hammer used by the striker, the smith's assistant. For roughly forming the metal, direct blows from the hammer alone may be used but for cutting off, forming and other finishing operations, various hand tools are necessary.

Chisels (cheni)- These are used for cutting metals. They may be hot or cold depending on whether the metal to be cut is hot or cold, and the main difference between the two is in the edge. The cold chisel has its edge hardened and tempered with an angle of about 60° while the angle of the hot chisel is 30° and hardening is not necessary since in any case the hot metal would re-soften it.

Tongs (chimta)- The village smith requires a good selection of tongs with which to hold his work during various operations.

Files (reth)—The chief scraping or cutting tool used by the blacksmiths now-a-days is the file locally known as 'reth'. A few types of files commonly used by the artisans are (a) flat file, (b) hand file, (c) square file, (d) round file, (e) half-round file, (f) triangular file, etc. These are generally imported from the outside market.

Processes of manufacture: In the fabrication of diverse articles, the ironsmith does not necessarily devise shapes and sizes himself. Very often they are made according to the pattern and specifications given or ordered by the customer.

The different stages of production of iron articles commonly met with are described below:

Shearing: The first stage of manufacture commences with the shearing of scrap iron or steel spring into desired sizes. This operation is performed with the help of a chisel locally known as 'cheni' and hammer. Before being sheared the scrap iron is to be heated into a red-hot condition.

Forging and shaping: The iron pieces sheared are then kept into the burning furnace locally known as 'bhatti' so as to make the scrap pieces malleable through heat and can always be hammered into the desired shapes. As soon as the iron piece becomes red hot, it is picked up with the help of a tong and placed on the surface of the anvil locally known as 'nia' and then hammered to form into a particular shape. This process of forging is repeated until the scrap piece takes the final shape required.

Grinding: In rural areas of the State, the usual practice is to sharpen the iron articles by grinding them on smooth slabs of stone. Often the sharpness of the tools and the implements is obtained by polishing them with sand or powdered brick.

At Karanga village, however, it is seen that the workers of the blacksmiths' cooperative society perform this operation with the help of power-driven grinding stones. After the iron pieces are formed into different shapes, they are brought to the factory for machine processing, namely, (a) rough grinding, where the uneven surface of the articles caused by hammering during the operation of forging is reduced to tolerably plain surfaces and formation of the object, and (b) when it is again subjected to a smooth grinding operation whereupon the surfaces are almost made smooth.

Filing: Then the objects are sent to the filing section, where they are finally sharpened and shaped with the help of hand files. The object at this stage is tightly held on a table vice for smooth and efficient working.

Tempering: The next is to temper the articles in order to impart a certain amount of strength to the tools and the implements to be manufactured. Only the skilled workers can do the job of tempering. Sharp-edged weapons or other objects require to be tempered for their appropriate use. Weapons and tools made of iron gets rusty and blunt or break with use, if untempered. Tempering is done to prevent rusting, impart strength and sustain the sharpness of implements. Implements meant for woodwork and similar materials are subjected to a mild temper, while those meant for working on stone or metal are given a hard temper. Briefly stated, anything that is to be tempered is in the first place heated to a pinkish red and then dipped into a solution of oil or water which may contain some chemicals. A mere knowledge of the theory or reading of books on the subject is not enough for mastering the art of tempering iron; it calls for years of rigorous practical training. In the Karanga Cooperative Society, it is also observed that after tempering is over, the products are stamped with the trade mark 'KARANGA'.

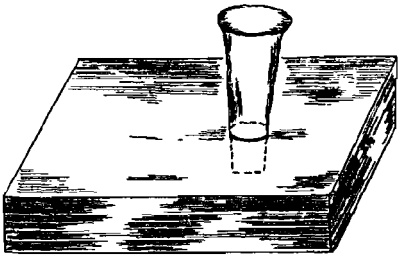
Fixing of wooden handles: There are some particular products such as daos, pruning knives, etc., which require either round or flat wooden handles properly fixed in the iron portion of the tool itself. At the Karanga factory these wooden handles are prepared with the help of a wood-working lathe and sometimes with hand-saws, chisels and files. Generally the village blacksmiths prepare these handles with saw, chisel and knife. After tempering is over, wooden handles prepared as such are fixed at one edge of the tool.

Polishing: The final operation is polishing the articles. The usual practice is to polish them on smooth slabs of stone, while occasionally this is done with sand or powdered brick. At the Karanga factory, it is observed that at this stage, the articles are subjected to a high friction with the power driven polishing bobs. In this operation, the scratches caused on the surface of the objects during grinding operations are also polished to a certain degree of fineness which gives a glazing and shiny effect to the finished products. The above process of manufacture are followed by indigenous blacksmithy all over the state with slight deviation here and there, in order to suit their individual convenience. Most of the artisans in Community Development Block areas are found using implements which are being encouraged by the Community Development personnel.

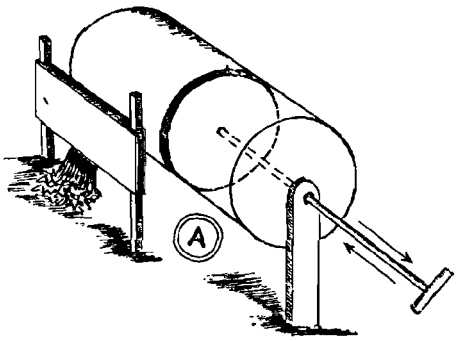
Finished products: Some of the products of blacksmithy all over the State and their uses are described below:

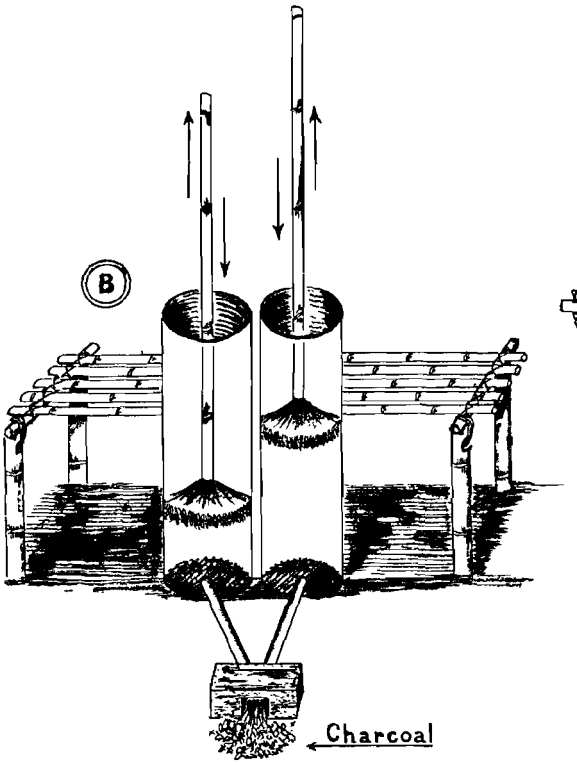
Products: 1. *Dao (chopper)*—Dao is a kind of chopper. It is the most useful tool to the householder and is used for a host of purposes in daily life. It is one of the stock tools to the bamboo and cane workers.

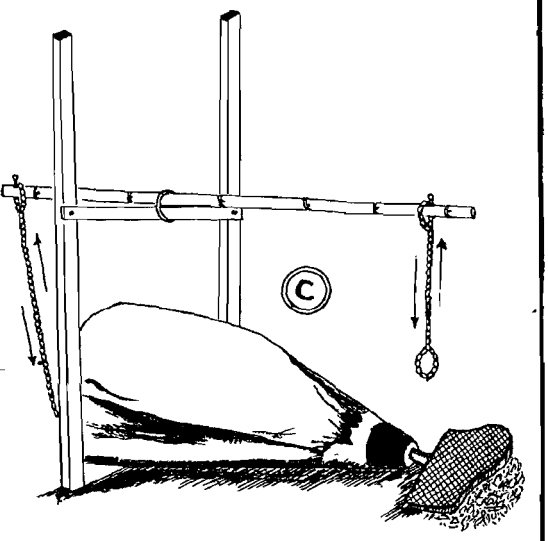
2. *Khurpi (spud)*—A tool with a broad flat head for earth digging especially prior to planting saplings and also for doing odd jobs in the field. The length of the tool is 4″ to 5″.



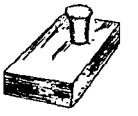
Anvil, fitted in wooden stand







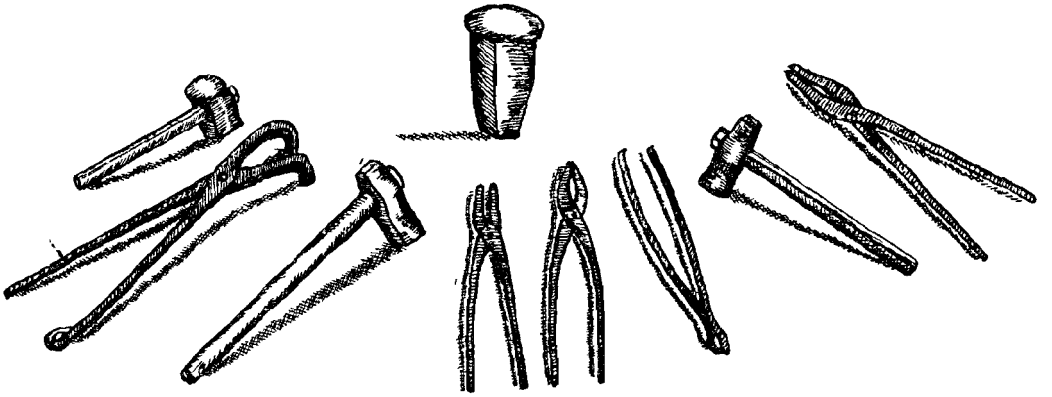




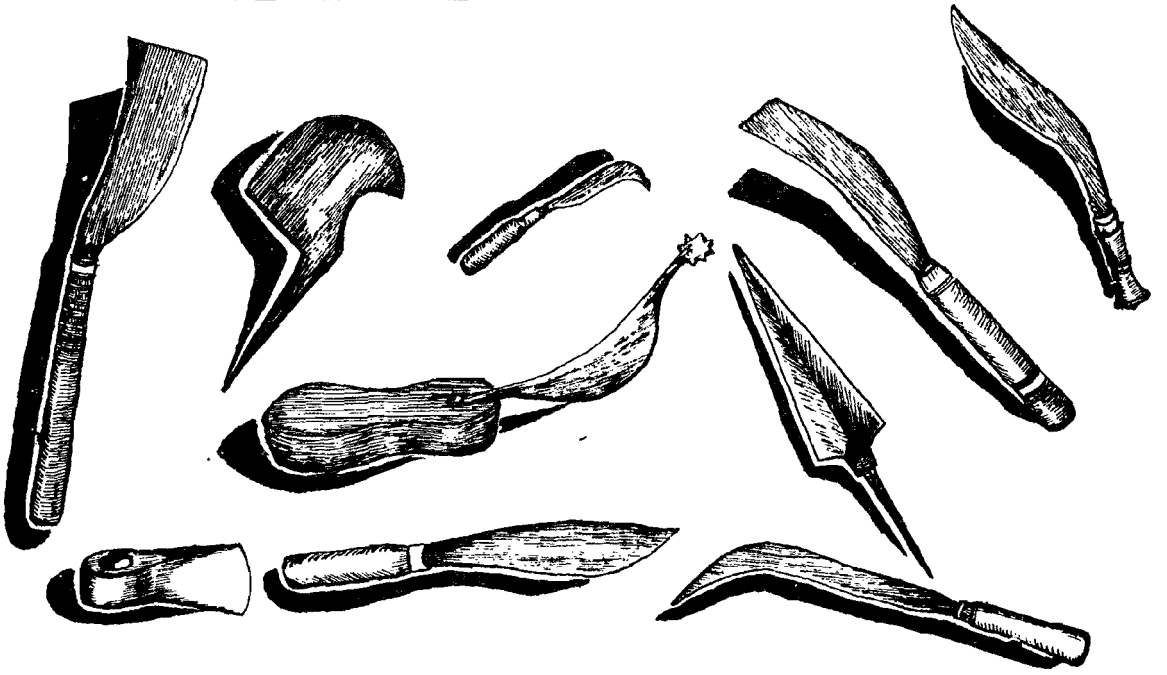
A.B.C. DIFFERENT TYPES OF COUNTRY BLOWER.

Ttinga P.c.
15/9/84

COMMON HAND TOOLS USED BY COUNTRY BLACK-SMITH



SOME SPECIAL IRON PRODUCTS OF ASSAM



3. *Kuthar (axe)*—Used for felling trees, chopping wood and other purposes. The handle is made of strong wood or bamboo. The blade of the axe and the ring for the handle are all of a piece. The blade is 2″ to 3″ in width and 3″ to 6″ in length.

4. *Kudali (spade)* — An agricultural implement used for earth cutting and other odd jobs. It is fitted with a sturdy handle of wood or bamboo.

5. *Kanchi or kaste (sickle)*—An indispensable tool to the cultivator used for harvesting paddy, etc. The blade is shaped like a crescent and is thin, sharp and has slanting teeth. It is generally mounted on a wooden handle. The blade is 1/16″ in thickness.

6. *Katari (knife)*—Mostly used in rural areas for bamboo work, cane weaving, betelnut cutting, for sharpening pencils and sundry other purposes. There are varieties of katari viz., hatnalia katari, nalia katari, jathi katari, tamul katari and dab katari. It is made of scrap file. These various knives are some of the many essential items required on occasions of marriage and other customs and rituals.

7. *Bathi or Maida (fish knife)*—A sort of knife on an iron or wooden stand and is used for cutting vegetables fruits, fishes, etc. It is usually made of good quality medium carbon steel.

8. *Kalam katari (pruning knife)*—Kalam katari is essentially used in tea gardens for proper pruning of tea bushes. The different sizes of the blades of the pruning knives vary from 3″ to 9″ in length.

3. *phal (ploughshare)*—*A small cone-shaped* blade fitted to a wooden plough used for agricultural purposes, particularly in tilling the soil.

Besides these, a few articles like 'itarkhuty' (hooked blade), prunning saw, weeding fork, shovel, cheel hoe, etc., are also manufactured by the local artisans.

The following table gives the number of persons engaged in blacksmithy as whole-time workers in different districts of Assam.

Table No. 1

THE NUMBER OF BLACKSMITHS, HAMMERSMITHS, FORGEMEN ENGAGED IN THE CRAFTS THROUGHOUT THE STATE IN 1961

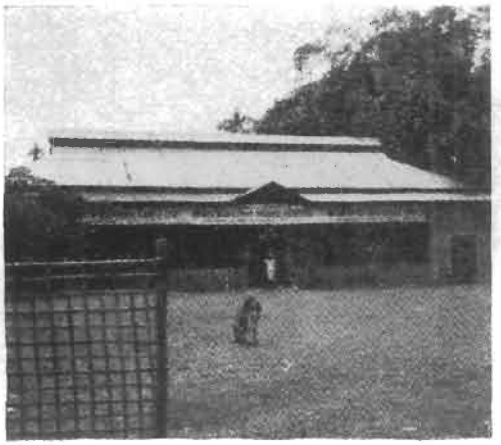
(1961 CENSUS – CODE NO. 733 OF NATIONAL CLASSIFICATION OF OCCUPATIONS)

Districts	Persons	Males	Females
Goalpara	548	543	5
Kamrup	806	805	1
Darrang	255	252	3
Lakhimpur	1,573	1,571	2
Nowgong	342	342	—
Sibsagar	321	320	1
Cachar	367	367	—
Garo Hills	73	68	5
United K. & J. Hills	228	225	3
United M. & N. C. Hills	196	195	1
Mizo Hills	104	104	—
Total:	4,813	4,792	21

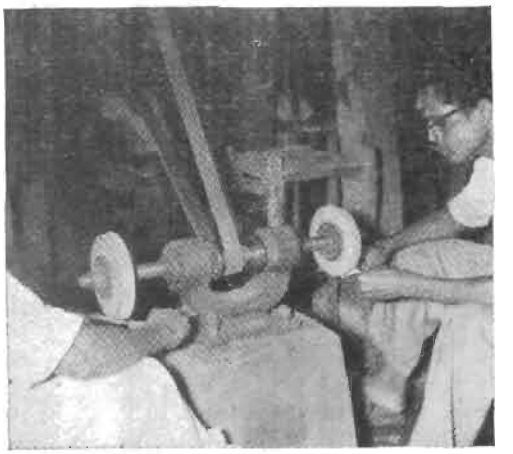
ECONOMICS OF THE INDUSTRY

Economics of production.: Blacksmithy is essentially a household industry. The artisans work in their homes. The hours of work are not fixed and fluctuate according to demand of finished goods. As stated earlier, the rural establishments with the exception of those of Karanga village provide only seasonal and subsidiary occupation to the workers. Purely seasonal establishments owned by the blacksmiths from Bihar are also met with. The working in these establishments usually starts in September-October and continues throughout the busy season till February-March when the slack season of the industry sets in and the blacksmiths close down their establishments and leave for their home in Bihar where, besides working in their own land, they also carry on blacksmithy throughout the agricultural busy season. But as soon as the rainy season is over and busy season of the industry sets in, they return to their places of work in Assam to resume their work in blacksmithy. The workers of Assam who carry on blacksmithy as their subsidiary and seasonal occupation never close down their establishments and do attend to their work during the agricultural busy season also according to their convenience and demand of the finished products. Only demand cannot induce them to work in the 'Kamarshala' but they attend only when there is time after working in their own land. As regards the artisans of the Karanga village, the majority of them carry on blacksmithy as their principal and whole-time occupation and a few of them have cultivation as their subsidiary occupation They generally do not cultivate their own land but give it tc others for cultivation for payment in money, kind or share.

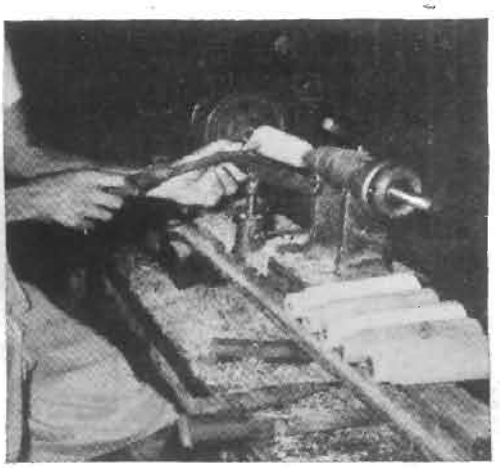
Usually iron products are sold by pieces, each piece having its own price. In some cases, however, the price is based on the making charge of the product, that is, on the amount of labour utilised in making it, calculated at the usual market rate of wages. The cooperatives pay the wages of the factory workers at daily rates which are often calculated down to a fraction of a day or even by the hour. Piece rates are also in vogue in many cases, while sometimes the worker is paid a part or whole of his wages in advance by the merchant or middleman. The daily earning of an ironsmith in this State varies from Rs. 2.00 to Rs. 4.00 depending on the skill and efficiency of the artisan concerned. An attempt has been made to determine the cost of production of a few iron articles in the State. It is calculated on the basis of raw materials (i.e. iron, steel and coal) consumed plus labour charges. The following table shows this attempt.



Front view of
Karanga Blacksmiths' Factory



Polishing operation at Karanga



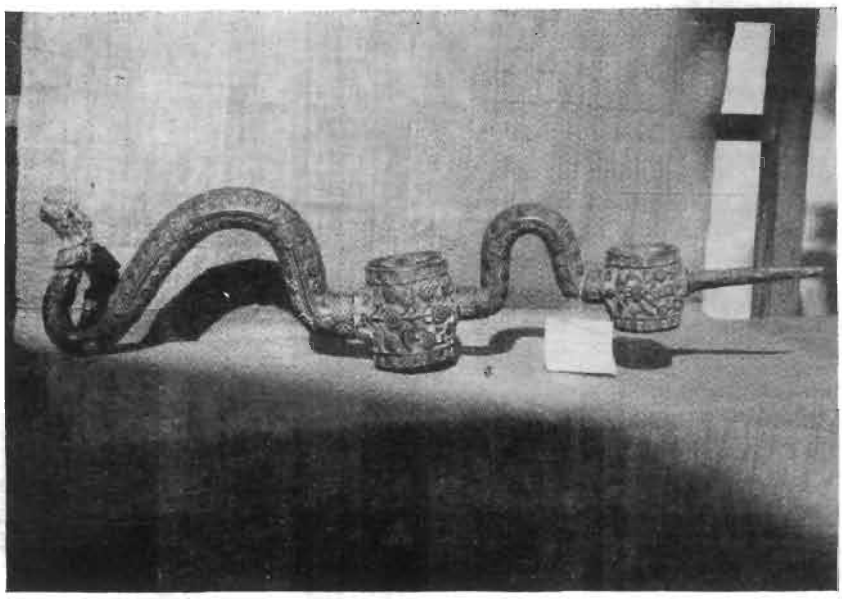
Preparing of wooden handles for 'daos', etc.



Drilling holes on the same handles



A Khasi blacksmith working
with his mechanical hand blower



A beautifully designed iron product of the past



Some ancient products of Assam

Table 2

Table showing the approximate cost of iron articles and their sale prices per dozen

Source :—Field Investigation.

Size of the products (Blade & handle)	Cost of raw materials including shearing & shaping charge	Rough grinding charge	Smooth grinding charge	Filing & Sharpening charge	Tempering & Sharpening charge	Handle preparing, fixing including cost of timber	Polishing charge	Storing and greasing charges	Establishment charge	Cost of production	Sale price per dozen with profit
1	2	3	4	5	6	7	8	9	10	11	12
1. Pruning knife 3'' 9''	12.00	1.50	1.00	0.75	1.50	4.00	2.00	2.00	5.25	30.00	42.00
2. Slashing knife 9'' 7''	30.00	3.00	2.00	1.25	2.00	5.50	6.00	4.00	3.25	57.00	89.00
3. Habi kata dao 1''×2½''	25.00	..	..	3.00	..	3.00	..	2.00	3.00	36.00	48.00
4. Kuthar (Axe)	36.00	3.00	2.00	..	3.00	.	3.00	..	12.50 + 3.00 (painting)	52.50	84.00
5. Dab-katari 2'' 2½''	2.00	0.75	0.50	0.25	0.25	5.00 (with brass handle)	1.00	0.25	2.50	12.50	18.00
6. Naga dao (wrapped with tin strip)	36.00	4.50	3.00	2.00	3.00	24.00	2.00	2.00	20.00	96.50	132.00

Marketing: It is found that the bulk of the products everywhere are sold in the local markets. The blacksmiths carry their products to the local or nearby markets or 'hats' for sale. A large quantity of the products of this industry is absorbed in agricultural operations and as such the rural 'hats' provide a good market to the industry. The hats in tea garden areas of Lakhimpur, Sibsagar, Nowgong, Darrang and Cachar districts are also found to be good markets for the products of this industry.

Development and assistance programmes: In the field of development of the blacksmithy industry, the State Government has chalked out various programmes in collaboration with different all-India boards. It is heartening to note that such elaborate programmes include training facilities in the improved technique for traditional artisans, popularising the use of improved tools and implements among rural artisans, arranging supply of raw materials viz., iron, steel and coal, to artisans at reasonable prices, standardisation of products and even creating markets for some particular products outside the State.

To meet the immediate financial need of the artisans, the State Industries Directorate is advancing loans upto Rs. 2,500.00 on personal surety basis to deserving blacksmiths of the State. Improved tools and implements are also supplied to rural artisans at 50 per cent subsidised rates or as grants-in-aid in kind either through District Industries Officers or through Community Development Blocks. The artisans in general are greatly benefited by these schemes. In some places of the State, a few co-operative societies have also been organised and registered to strengthen and revitalize the industry on a sound commercial line.

Cooperative Society: The first blacksmiths' cooperative society of the State was set up in 1954 and started functioning in the name of 'Karanga Kamar Silpa Samabay Samity'. The initiative came from the blacksmiths of the Karanga village. The society mainly confined its activities to owning a factory of its own with

improved machines and tools and employing the artisans on wage basis. A list of tools and implements manufactured by the society along with their prices is given in Appendix A. At the commencement, the total members enrolled were 50. By the end of 1962 it went upto 98. Out of the 98 members, 10 are ordinary and remaining 88 are all working members. Items of the manufacture undertaken by the society includes domestic tools, agricultural implements and tea garden tools. A detailed list of tools and implements manufactured by the society along with their prices is shown in Appendix B.

The paid up share capital of the society up-to-date stands at Rs. 2,375.00. Grants-in-aid received by the society from the Cottage Industries Department, Government of Assam, till the end of 1962-63 amount to Rs. 83,000.00 which has been utilised in the construction of buildings, installation of machines and purchase of hand tools. Besides these, two separate working capital loans of Rs. 20,000.00 and Rs. 10,000.00 were obtained by this society from the Cottage Industries Department and the Cooperative Department respectively. Against these loans a repayment of Rs. 3,000.00 with interest has already been made to the Cooperative Department.

The economic condition of the society at present is in a deplorable state. An examination of the business carried on it reveals that the society incurred a loss of Rs. 5,723.49 in 1960-61 and a further loss to the tune of Rs. 10,840.58 P. in 1961-62. This picture is not at all encouraging. In order to improve the condition of the society, the then Managing Committee of the society was dissolved by the Government in the year 1959-60, and to ensure a better management, an Executive Officer was posted there from the Government side. In spite of that, the condition of the society has been deteriorating further. The Government is now taking further steps to reorganise the society in the correct line. Of late, a few more cooperative societies have been set up in different districts with the avowed object of improving the blacksmithy and economic condition of the artisans. The following statement gives the number of societies functioning in the different districts of Assam.

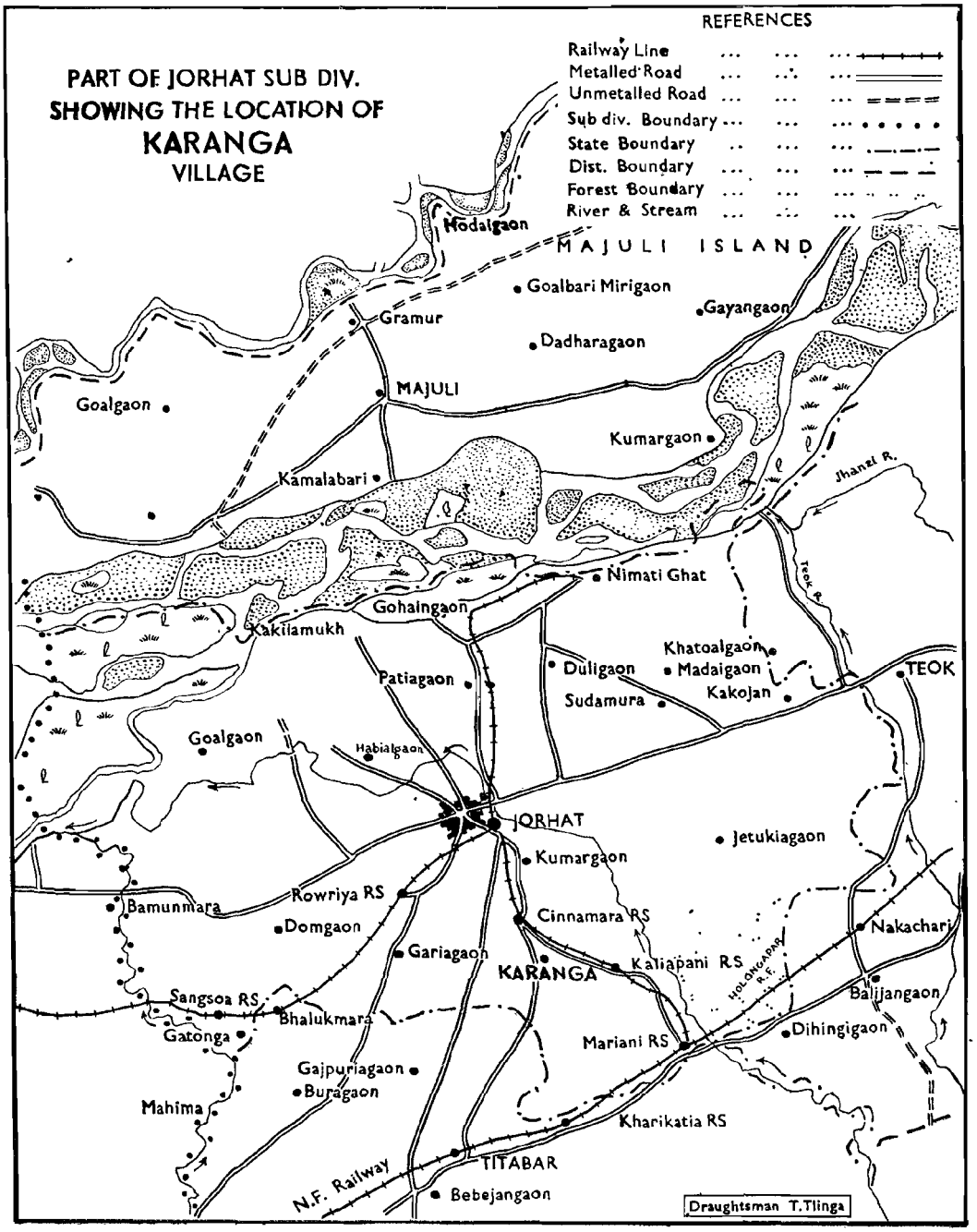
STATEMENT OF BLACKSMITHY COOPERATIVES IN ASSAM

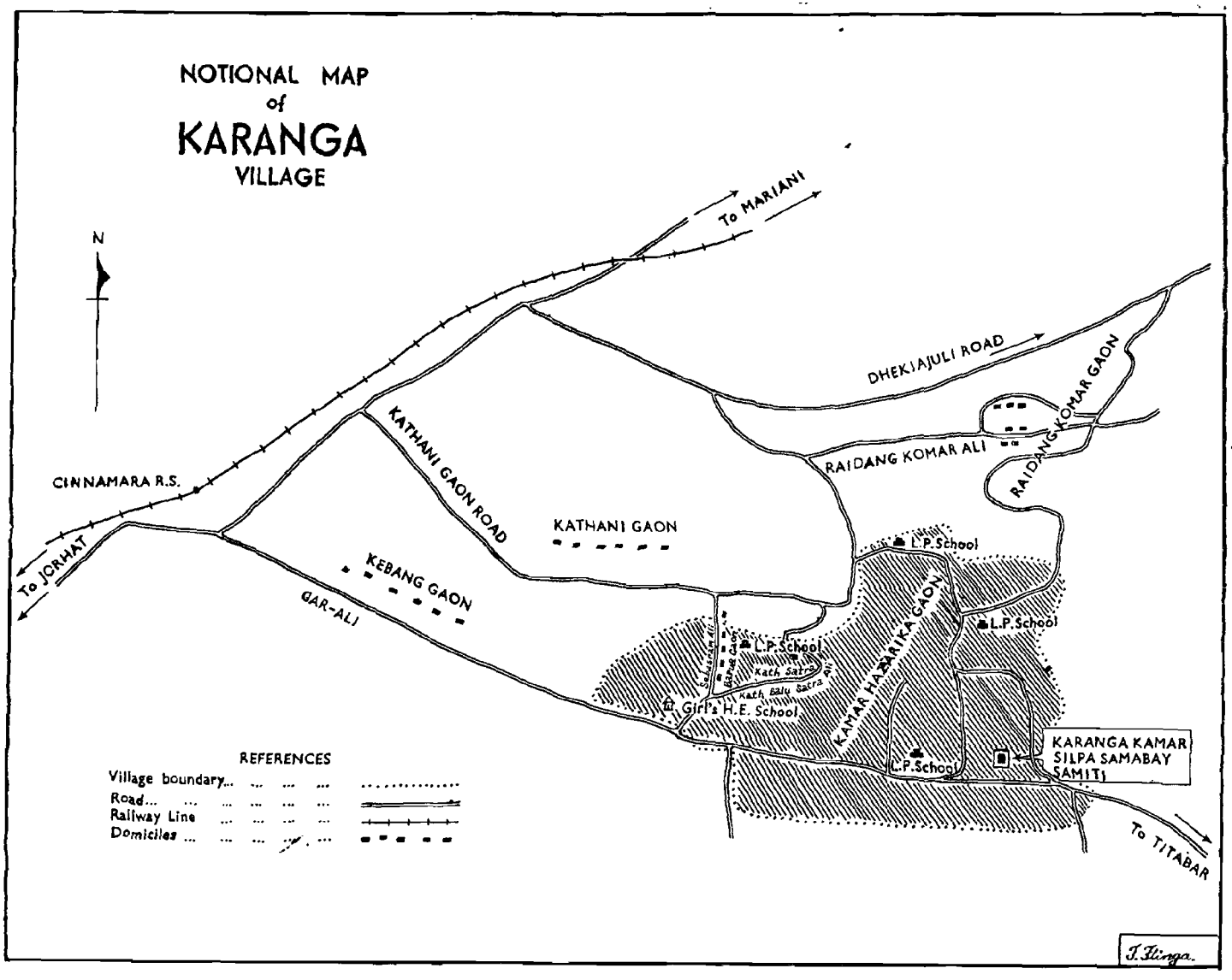
Name of the district	Number of cooperative societies	Number of shares	Amount of paid-up share capital	Total No. of workers	Extent of business done during 1961-62	Amount of loans and grants received from different sources
1	2	3	4	5	6	7
1. Goalpara	Nil	..	..	..	..	..
2. Kamrup	2	31	425.00	25	N.A.	N.A.
3. Darrang	Nil	..	..	..	..	..
4. Sibsagar	3	133	2,660.00	65	*56,210.70	82,000.00
5. Nowgong	Nil	..	..	..	..	..
6. Lakhimpur	Nil	..	..	..	..	..
7. Cachar	2	41	460.00	41	N.A.	1,000.00
8. United Khasi & Jaintia Hills	3	61	7,337.00	39	6,000.00	6,545.00
9. United Mikir & North Cachar Hills	Nil	..	..	..	..	..
10. Garo Hills	1	16	260.00	16	152.75	1,023.92
11. Mizo Hills	Nil	..	..	..	..	..
Total of all districts	11	282	11,142.00	186	62,363.45	90,568.92

N.A. — Stands for figures 'Not available'.

*Extent of business done by one society only.

Source : Registrar of Co-operative Societies, Government of Assam.


PART OF JORHAT SUB DIV.
SHOWING THE LOCATION OF
KARANGA
VILLAGE
REFERENCES
Railway Line
Metalled Road
Unmetalled Road
Sub div. Boundary
State Boundary
Dist. Boundary
Forest Boundary
River & Stream
Hodalgaon
MAJULI ISLAND
Goalbari Mirigaon
Gayangaon
Gramur
Dadharagaon
MAJULI
Goalgaon
Kumargaon
Kamalabari
Jhanzi R.
Nimati Ghat
Teok R.
Gohaingaon
Kakilamukh
Khatoalgaon
Madaigaon
Kakojan
TEOK
Duligaon
Sudamura
Patiagaon
Goalgaon
Habialgaon
JORHAT
Jetukiagaon
Kumargaon
Bamunmara
Rowriya RS
Cinnamara RS
Nakachari
Domgaon
Gariagaon
KARANGA
Kaliapani RS
HOLONGAPAR R.F.
Balijangaon
Sangsoa RS
Bhalukmara
Gatonga
Mariani RS
Dihingigaon
Gajpuriagaon
Buragaon
Mahima
Kharikatia RS
TITABAR
N.F. Railway
Bebejangaon
Draughtsman T.Tlinga


NOTIONAL MAP
of
KARANGA
VILLAGE
N
To MARIANI
DHEKIAJULI ROAD
RAIDANG KOMAR GAON
RAIDANG KOMAR ALI
KATHANI GAON ROAD
CINNAMARA R.S.
To JORHAT
KATHANI GAON
KEBANG GAON
GAR-ALI
L.P.School
L.P.School
L.P.School
Girl's H.E. School
KAMAR HAZARIKA GAON
L.P.School
KARANGA KAMAR SILPA SAMABAY SAMITI
To TITABAR
REFERENCES
Village boundary...
Road...
Railway Line
Domiciles ...
T. Hazarika.

The extent of help so far rendered by the different cooperative societies to the craftsmen's families under them has yet to be evaluated. With the object of improving the industry and economic condition of the artisans, the societies have started functioning under the guidance of the Government officials. They raise money from the artisans by issue of shares, by borrowing in the form of deposits. They purchase raw materials and appliance and sell the same to members either for cash or credit. The raw materials are advanced to the 'Kamars' who convert them into finished goods and deliver them to the societies which pay wages for the finished goods. They also act as agents to the members and for the sale of finished goods. The future growth and development of the industry will largely depend on the functioning of the co-operative societies organised so far.

Conclusion: The scope for employing improved tools and machineries in the industry is great but the artisans will have to be trained in the modern techniques of production as well as in the field of manufacture of new products such as cutleries, tea garden equipments, improved agricultural implements, builders' hardware, etc. without which the artisans in the present set up of the industry can hardly be expected to improve their technique of production. The artisans can improve their lot only in an organised way through cooperative societies. The cooperative societies in their turn should be well managed and devoted to their avowed task of bringing all round improvement to the lot of the artisans by following the right path.

APPENDIX A

A STATEMENT SHOWING THE PRICES IN 1961 OF A FEW INDIGENOUS IRON PRODUCTS OF THE KARANGA KUMAR SILPI SAMABAY SAMITY

Name and description of the products	Price (each) in Rs.	Remarks
1	2	3
1. Naga-dao	6.00	Used by local Nagas
2. Kirich	7.00	Sword
3. Mit-dao	4.50	Used by Miri, Mikir tribals
4. Tathi	4.00	A chopper used by Nagas
5. Kupuri	10.00	Nepali Kukri
6. Digal-dao	4.50	A chopper used by bamboo workers
7. Pholi-kotari	2.00	Kitchen knife
8. Moi-dao	3.12	Used for kitchen purposes (chopper)
9. Hat-karat	2.12	Hand saw
10. Gathi Kotari	2.50	A cutting tool used for domestic purposes
11. Hachua (9″ × 7″)	8.00	Slashing knife used in tea gardens
12. Kalam katari (3″ × 9″)	3.50	Pruning knife
13. Ita khuli	8.00	Hooked blade weeder
14. Habikota dao	3.12	Jungle cutting chopper
15. Dab-kotari	1.00	An instrument used for cutting coconut kernel
16. Kuthar	9.00	Axe
17. Kopi-dao	4.50	x
18. Hat nalia kotari	2.00	Used for making betelnut chips
19. Narikal rokani with fine wooden frame	8.00	Used for making chips out of coconut
20. Ploughshare (phul)	1.25	Used for agricultural tilling
21. Kanchi	1·50	Sickle

Source : Field Investigation.

APPENDIX B

TREND OF PRICES OF BLACKSMITHY PRODUCTS

Name of the district	Products	Unit	Price during (in Rupees) 1939 Urban	1939 Rural	1953 Urban	1953 Rural	1954 Urban	1954 Rural	1961 Urban	1961 Rural
1	2	3	4	5	6	7	8	9	10	11
			Rs-as-p	Rs-as-p	Rs-as-p	Rs-as-p	Rs-as-p	Rs-as-p	Rs.nP	Rs.nP
1. Goalpara	(i) Dao	No	0-10-0	0-10-0	.	1-8-0	..	1-8-0	2.50	2.50
	(ii) Sickle	„	0-4-0	0-4-0	..	0-8-0	..	0-8-0	1.00	1.00
2. Kamrup	(i) Dao	„	0-8-0	..	1-12-0	.	1-12-0	..	2·50	..
	(ii) Sickle	„	..	..	..	..	..	..	..	.
3. Darrang	(i) Dao	„	0-12-0	0-8-0	2-0-0	2-0-0	2-0-0	2-0-0	3.25	3.50
	(ii) Sickle	„	..	.	..	..	.	..	1.00	1.25
4. Nowgong	(i) Dao	„	0-6-0	0-6-0	1-8-0	1-8-0	1-8-0	1-8-0	2.50	2.50
	(ii) Sickle	„	.	.	.	.	.	..	..	0.75
5. Cachar	(i) Dao	„	0-8-0	0-8-0	2-0-0	2-0-0	2-0-0	2-0-0	2.00	2.00
	(ii) Sickle	„	.	0-5-0	.	0-7-0	..	0-7-0	0.75	0.75
6. Sibsagar	(i) Dao	„	0-14-0	0-14-0	2-0-0	2-0-0	2-0-0	2-0-0	.	..
	(ii) Sickle	„	0-3-0	0-3-0	0-12-0	0-12-0	0-12-0	0-12-0	..	..
7. Lakhimpur	(i) Dao	„	0-8-0	0-8-0	1-4-0	1-8-0	1-2-0	1-8-0	2·00	2.50
	(ii) Plough-share	„	.	..	..	..	..	.	..	3.00
8. Garo Hills	(i) Dao	„	..	0-10-0	..	0-14-0	..	1-0-0	2.25	2.00
	(ii) Sickle	„	..	..	..	..	.	..	1.00	1.00
9. United Khasi & Jaintia Hills	(i) Dao	„	0-4-0	0-6-0	1-0-0	1-8-0	1-0-0	1-8-0	5.00	6.00
	(ii) Knife	„	..	..	..	..	..	..	3.00	3.50
10. Mizo Hills	..	..	..	..	..	..	..	..	.	..
11. United Mikir & Noth Cachar Hills	(i) Dao	„	..	0-10-0	..	1-8-0	..	1-8-0	..	..

Source : Director of Industries and Director of Statistics and Economics, Government of Assam

APPENDIX C

TREND OF PRICES OF RAW MATERIALS IN ASSAM

Name of the district	Description of raw materials	Unit	Prices during (in rupees) 1939	1953	1954	1961
1. Goalpara	Iron	Maund	10-0-0	25-0-0	25-0-0	..
2. Kamrup	Scrap iron	„	4-0-0	11-8-0	11-8-0	12.00
3. Darrang	Iron	„	6-0-0	23-0-0	23-0-0	.
4. Nowgong	Iron	„	7-0-0	25-0-0	25-0-0	28.00
5. Cachar	Scrap iron	„	.	..	16-0-0	16.00
6. Sibsagar	Iron	„	10-0-0	20-0-0	20-0-0	..
7. Lakhimpur	Iron	„	8-0-0	29-0-0	29-0-0	36.00
8. Garo Hills	Scrap iron	„	..	.	5-0-0	10.00
9. United Khasi-Jaintia Hills	Iron	„	8-0-0	30-0-0	30-0-0	35.00
10. United Mikir & North Cachar Hills	Scrap iron	„	6-0-0	15-0-0	20-0-0	..
11. Mizo Hills	Iron	„	.	..	..	..

Source : Department of Industries and Department of Economics and Statistics, Government of Assam.

TERMINOLOGY

English term	Local term (Assamese)	Local term (Bengali)	Local term (Khasi)
1. Blacksmith	Kamar	Kamar	Kamar
2. Furnace	Bhatti	Bhatti	Shlem
3. Charcoal	Kath koila	Kath koyla	Rnga
4. Hammer	Hatudi	Hatudi or Hatura	Tyrnem
5. Tongs	'Sadash	Sadashi	Nap
6. Chisel	Cheni	Batail or Tangi	Shniah
7. Punch	Janti	Goxail	Nar kbong thliew
8. Anvil	Nchi	Nia	Rniang
9. File	Reth	Reth	Syrti
10. Forge	Kamarshal	Kamarshala	Pajwa
11. Bellows	Hapad	Hapad	Pyrsut

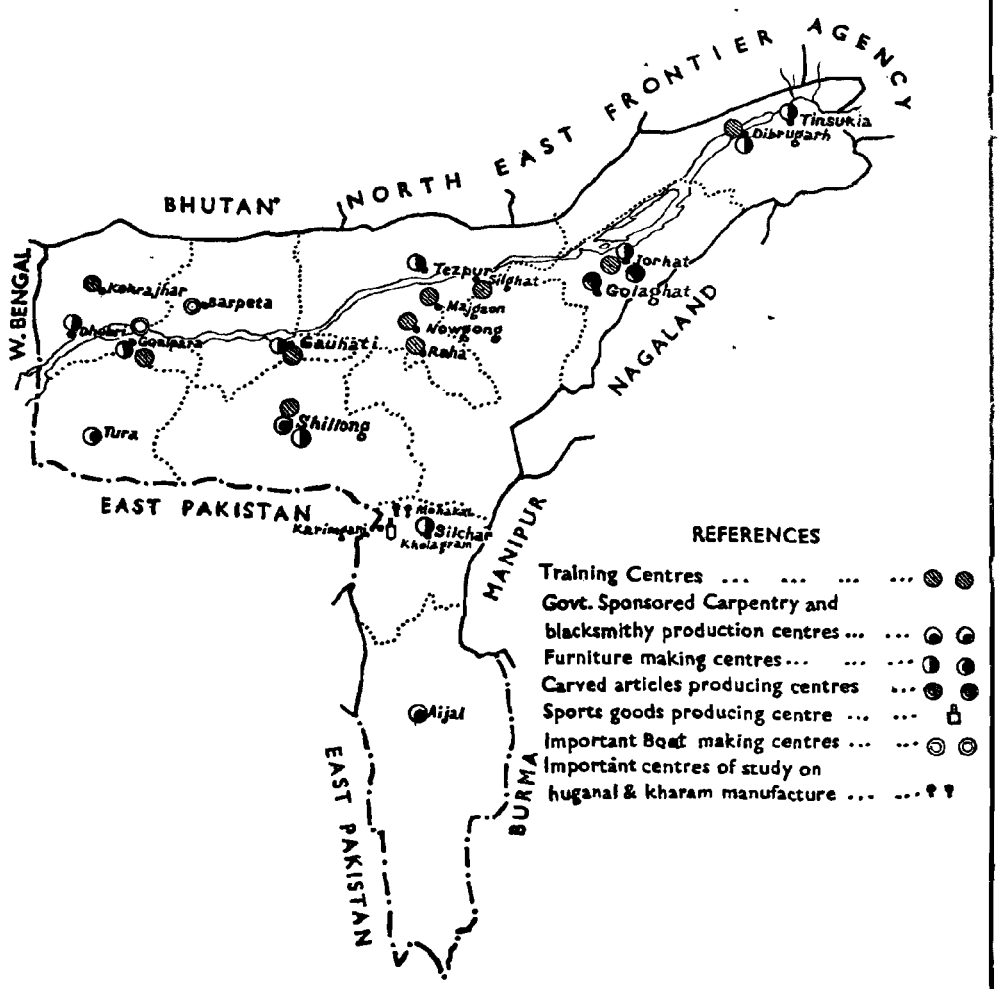
BIBLIOGRAPHY

1. Mineral Resources of Assam by the Publicity Department, Government of Assam, 1962.
2. A review of Industrial and Economic Situation of Assam by the Industries Department, Government of Assam, 1958.
3. A Monograph on the Indigenous Smelting of Iron by Shri M. Ahmed, M.A., I.A.S., Superintendent of Census Operations for Orissa, 1961.
4. History of Civilization of Assam by P.C. Choudhury, Ph. D.
5. Census of India, 1951, by A. Mitra, I.C.S., Vol. VI.
6. A Monograph on Iron & Steel Works by W.E.J. Dobles, 1897.
7. Report on the Survey of Cottage Industries in Assam by the Department of Economics & Statistics, Assam, Vol. I & II, 1958.

ASSAM

SHOWING THE LOCATION OF TRAINING CENTRES IN CARPENTRY AND WOODWORKS AND OTHER CENTRES OF TRADE AND COMMERCE

(NOTIONAL)



CHAPTER VI

Woodworks and Carpentry

Introduction : Making of wooden articles is a common industry pursued by the artisans both as a primary and subsidiary occupation throughout the State. The growth of the industry has been favoured by the availability of the main raw material, timber. Assam's forests are full of valuable trees and as such timber is abundantly available throughout the State.

In the plains districts of Assam, the traditional carpenters who have been the important elements in a village society belong to the community 'Sutradhar' and there is mention of this caste in the 'Vedas.' Since times immemorial their forefathers, from whom the mantle has fallen on them, have been working on wood. Generally, a carpenter earns his living by building houses, manufacturing carts, ploughs, looms, furniture, icons and boats. Occasionally he manufactures and repairs wooden structures such as doors, windows, battens, etc. for construction of a house or a bridge.

Though the main raw material is available in plenty throughout the State, there has been a dearth of indigenous carpenters for reasons unknown except in the United Khasi and Jaintia Hills. It is found that many of the big establishments in the urban areas of the plains which produce finer articles are run by the carpenters who migrated from East Pakistan, and to some extent, from Punjab and West Bengal. These artisans from outside the State are highly skilled and possess business ability of a high order. They carry on the industry on a comparatively large scale and articles produced by them are also attractive due to the finishing which is of very high quality. Unlike many indigenous artisans, they have expert knowledge of various paints. Such a knowledge enables them to impart an elegant look to their articles and make them more attractive. They have also been responsible for introducing new articles suited to the modern tastes and needs. The majority of indigenous carpenters are lagging behind their fellow carpenters from outside the State. In the Khasi Hills, however, Khasi carpenters are experts in house construction, and they are also capable of making very good furnitures.

It may be stated that, generally, Assam does not enjoy much reputation in this craft in comparison with other States of India, yet it is true that the industry has been carried on by local carpenters since the misty past. The investigation which has been conducted recently in this respect, reveals that this industry does not employ much labour and capital. But usually in any country which has timber that can be utilised for this trade and has progressed to the extent of possessing a few cutting tools, there are woodworks of some form or other. Nevertheless it can be said that there exist wide scopes for the growth and develop-

ment of this craft in the years to come having in view the fact that the State is very rich in forest resources.

History and origin : *"The art of wood carving is proved by Bana, who writes that the presents from Bhaskara to Harsavardhana included 'carved boxes with panels'. The Tezpur grant indicates that a large number of boats in the Brahmaputra were carved with beautiful designs and decorated with ornaments. Wood was used for the making of icons, as proved by an icon of Jagannatha in Ksetri in modern Kamarup. Speaking of the various articles of wood, a later source *Fathiyah-i-Ibriyah* enumerates wooden boxes, stools, trays and chairs which were made from a single piece of wood.

The Forests of Assam were noted for their valuable woods. Epigraphs mention a few of them, viz., sandal wood and agaru, besides others like 'Vata' (ficus indica), 'Asvattha' (ficus religiosa), 'Madhurasvattha', 'Salmali', 'Khadira' (acacia catechu), etc. These were used for both domestic and religious purposes. Classical writers make important mention of aloe and musk from Assam. We have reference to sandal wood and aloe from Kamarupa as early as the *Epics*. Kautilya in his 'Arthasastra' refers to some best varieties like *Jongaka* (from Jonga), *Dongaka* (donga) —both black aloe wood—*Grameruka* (grameru), *Aupaka* or *Japaka* (Japa), and *Taurupa*, all of which come from Kamarupa".

Till today, it can be seen in almost every district of the Assam Valley that a few 'namghars' or 'kirtanghars' (a house where congregational prayers are held by the Hindus) are decorated with fascinating carvings of wood. 'Kamalabari Satra' in Jorhat subdivision is noted for its exquisite carvings on wood. On a visit to a 'namghar' the first thing which attracts the attention of a visitor is its 'singhashun' which is placed at the farthest extremity of the house opposite the main entrance. The mythical animals like 'garoor, hanuman, nagar and ananta', also find prominent places inside a *namghar*. Besides these mythical figures, there can also be found the wood-carvings depicting lions, horses, men, birds, tigers, monkeys, etc. It is also not uncommon to witness a few carvings on the beams and cross beams of *namghars* which are generally covered with figures of deities and conventional flowers and creepers.

In the Assam Valley, carving was not confined to things in the religious sphere. Carved articles of furniture were also common in the olden days such as *paleng*, the *salpira*, the *barpira* and the *karoni*. Local craftsmen can also produce carved benches, chairs, thagi or bookrest, stools etc. The finer sense of artistic beauty among the wood workers can also be traced from the manufacture of minor weaving implements like *makoo, karhoni, durpati, nasoni*, etc., which were generally ornamented with the carving of parrots, peacocks, magars, monkeys and other floral designs. Carved figures of peacocks, parrots, lions and magars are often seen on the 'dolah' or palki. Boats were also ornamented with carved floral designs and figures of birds, animals and fishes. Amongst minor bits or carvings are those used in *kharams* (wooden sandal), *gosa* or lampstands and walking sticks.

The equivalent term for wood carver in the local language is 'khonikar'. In the sadar sub-division of the district of Sibsagar there is an old village of khonikars known as Khonikargaon. It is reported that during the reign of the Ahom kings, the khonikars of that village rendered services to the Rajahs. In those days the Ahom kings greatly patronised and encouraged the khonikars and their craft in all possible manners.

* An abstract from "The History of Civilization of the People of Assam" by Shri P. C. Choudhury, 1959, pp. 374-375.

Present set-up of the industry : The set up of woodworks in Assam is not yet organised and oriented on the lines of technical improvements. A very negligible proportion of the total population of Assam are found to be employed in this trade.

In the rural areas, most of the craftsmen carry on this industry as a subsidiary occupation. Due to their lack of improved technical knowledge, they cannot utilise their trade to fetch attractive dividends. They also experience financial difficulties which stand as a great obstacle on their way to development. On the initiative of the State Cottage Industries Department, a few co-operative societies have been organised among the rural carpenters and arrangements have been made to provide them with all types of financial and technical assistance to improve their lot. Presently, the village carpenters are found to do works on servicing and repairing of agricultural implements and manufacturing of other wooden articles on receiving orders.

The picture in the urban areas, on the other hand, is quite different and rather a bit encouraging. Though less in number, the carpenters living in the urban areas are more or less organised and work under contractors in bridge and building construction works on daily wage basis ranging from Rs. 4/- to Rs. 6/-. In almost all the towns and important places of the State can be found a good many units engaged in cabinet manufacture. These establishments are generally run by persons who are found to be experts in this trade. The carpenters usually work in these units on daily wage basis and manufacture the articles under the direction and guidance of the experts. The finished products are kept in the shop for display to attract the prospective buyers. These units also manufacture all varieties of articles as per plan and specification of the customers.

Workshed : In the rural areas there is no workshed for carrying on the jobs of woodworks and the artisans generally do the jobs in some open spaces near their dwelling houses in sharp contrast with the urban areas where the cabinet manufacturing establishments have got worksheds, godowns and selling depots.

TECHNIQUE OF PRODUCTION

Raw materials : The main raw material for woodworks is timber, both hard wood and soft wood. The forest resources in Assam are unique and there are more than a hundred species of timber that grow in its forests. There are at present about 144 saw mills employing nearly 2,327 workers and sawing a huge amount of timber sufficient enough to meet the local requirements. The woodworkers of the State generally procure sawn timbers of different sizes either from the mill or from the local dealers according to requirements.

The timbers generally used for woodworks are Cham (Anto Carpus Chaplasha), Sundi, Segoon, Bonsum, Titalsoom, Palash, Sal (Shorea Robusta), Nim wood (melia azadirachta), Jackwood (artocarpus integrifolia), Saral (henus Khasiya), Nahor or Nageswar (mesua ferra) Sissu (dalbargia sisso), Mango, Bael (argle marmeloo) Gomari (gmelina arborea), etc., to name a few. The woods especially selected are those which are easily worked, close grained, free from knot and are little liable to splinter. In the Brahmaputra Valley, the bael and mango woods are frequently used for making the images of gods.

Other subsidiary raw materials required for woodworks and carpentry are reported to be available from the local markets. These are glue for joining work, sand paper for polishing, and lac, resin, methylated spirit, rabi mustafi and a variety of indigo powder for varnishing, and some variety of ready-made paints for colouring.

* **Timber yield and its uses :** It is recorded that in the consumption of timber for general purposes, the following six species alone represent more than 75 per cent of the total supply.

1. Sal (Shorea robusta)	...	32%
2. Hollong (dipterocarpus)	...	20%
3. Hollock (terminalia myriocarpa)	...	9%
4. Simul (bambax malabaricum)	...	6%
5. Cham (artocarpus chaplasha)	...	5%
6. Makai (shorea assamica)	...	3.5%
		75.5%

The average annual recorded production of timber from the reserved, unclassed State and private forests is about 145,000 tons, yielding a revenue of Rs. 4,522,016 to the State.

The break-up of timber put to main industrial use is as follows. This excludes domestic and agricultural requirements.

1. Match industry	...	10,000 tons.
2. Plywood	...	24,000 tons.
3. Railway sleepers	...	6,000 tons.
4. Furnitures	...	2,000 tons.
5. Miscellaneous (small industry and wood carving, etc.)	...	3,500 tons.
		45,500 tons.

Nature of Establishment : The industry is found in both rural and urban centres. The urban units are comparatively well organised than the rural units and provide whole-time and principal occupation while many of the rural units provide only part-time and subsidiary occupation. The urban units generally produce costlier articles of finer varieties suited to modern tastes. The articles produced by the rural establishments are simple and intended to meet the demand of the village folk who are generally not interested in costly articles.

Most of the carpenters of the Brahmaputra Valley are from Bihar, Punjab, West Bengal and displaced persons from East Pakistan. While the carpenters from Bihar are mostly confined to manufacture cartwheels, the other artisans produce different types of cabinet and furnitures, boats, looms and their accessories.

In the district of Sibsagar, some special varieties of wood-works are done by the local village carpenters of which the following deserve special mention.

1. *Simhasan* :—A structure generally supported by 4 elephants, 24 lions and 7 tigers, kept in a Satra, Namghar or temple for religious purposes.

2. *Kharai* or book rest. It is also known as *thagi*.

3. Images of a few deities and figures of *magar, garoor, hanuman,* etc.

4. Carvings of conventional flowers, creepers, etc., on doors, windows and beams of *namghars* and *satras* can also be found.

It is gathered that generally these products are worked by the permanent residents of the satra who are known as 'bhakat'. On the basis of religious ideology, these 'bhakats' try to be self-sufficient in all respects. They are all Vaishnavites. In the manufacture of all these products mentioned above, the 'bhakats' gene-

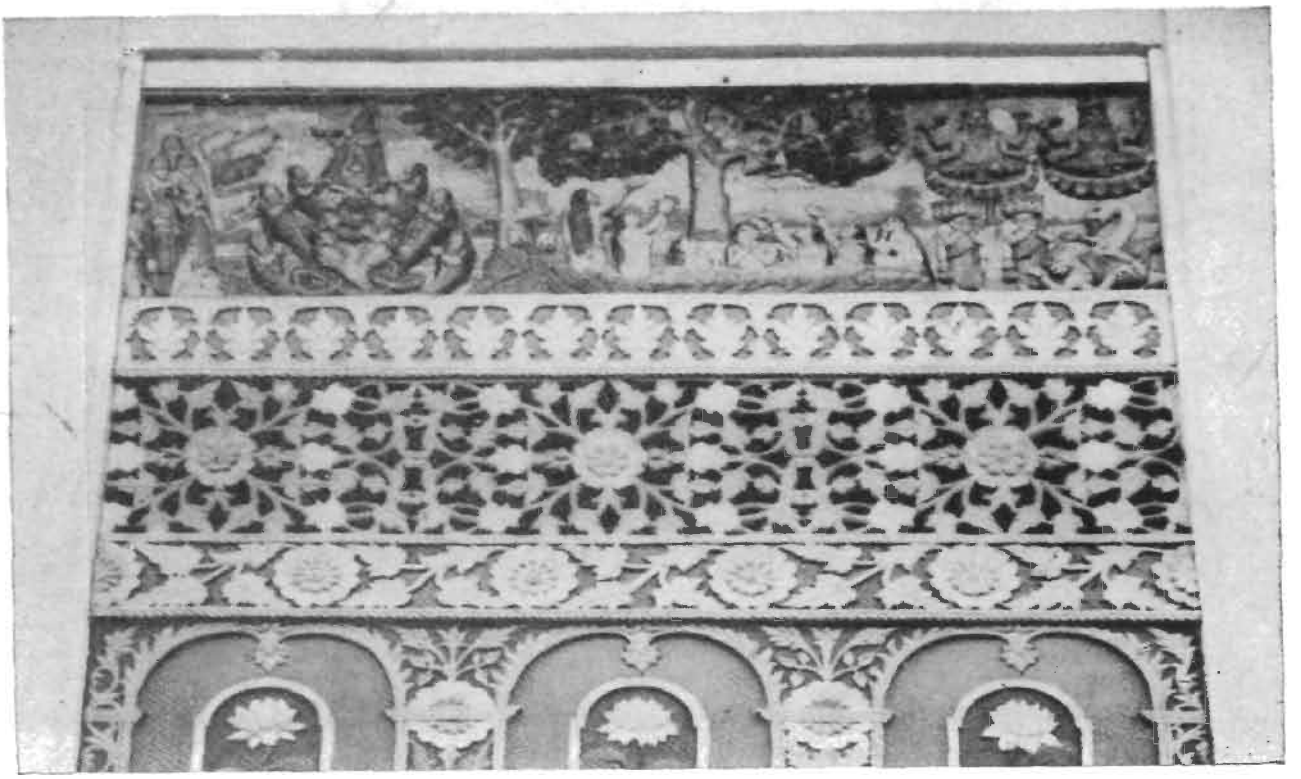
* Extract from 'A Review of the Industrial and Economic situation of Assam' July, 1958, by the Directorate of Industries, Government of Assam, Shillong.



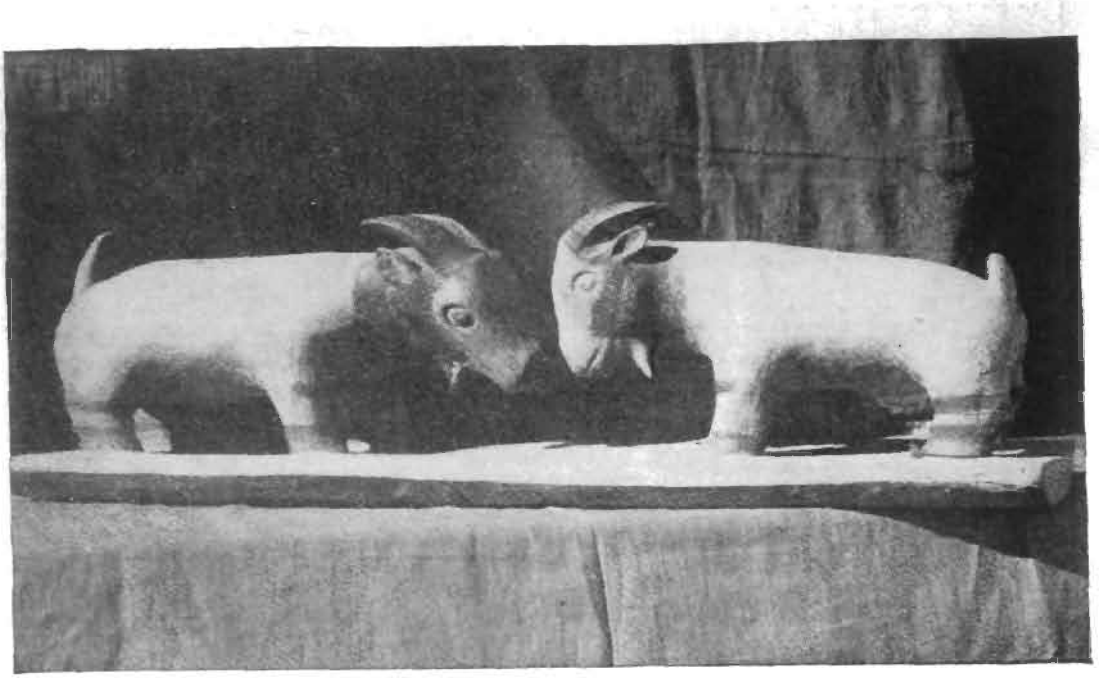
An exquisitely beautiful specimen of wood carving on a 'Singhasan'



Carved figures of various deities in wooden frames



Masterpiece of wood carving on the wall of a 'Satra'



Two fighting goats carved out of the same piece of wood

rally prefer to use wood of jack, bael and mango trees.

In the Cachar district, the craft is practised by the local carpenters and carpenters who migrated from East Pakistan. It is found that in rural areas, the work is taken up as a subsidiary means of occupation. They generally manufacture and repair agricultural implements, a few simple-designed furnitures like tables, chairs, bedsteads, etc., and occasionally construct wooden structures of bridges and buildings. Some of the rural craftsmen are also found to be proficient in the part of making fishing boats, ferry boats, mar-boats, etc. Further it is gathered that a few of the village carpenters can manufacture smoking pipe (hukanal)—mainly in the village of Kholagram under the Badarpur Police Station — and 'kharam' (wooden sandals). The rural carpenters generally do their work in some open spaces. A few of the timbers used by them are, in local language, cham, sundi, gamarui, pooma, katral and tula. The local carpenters can also manufacture bee-boxes and looms and their accessories.

In order to meet the demand for up-to-date furnitures a good many small factories have sprung up in urban areas. The proprietors of these workshops hire some skilled and semi-skilled workers on daily wage basis. The bulk of carpenters do not work in factories and prefer to work in private houses on daily wage basis. They generally do repair works of old furniture, doors, windows, etc. and construct the timber structure of buildings. They also manufacture cartwheels and a few small wooden articles as and when ordered for. In urban areas the daily wages vary from Rs. 4.00 to Rs. 6.00.

During the field survey a person named Shri Sadhan Chandra Das of village Kakalash under the Badarpur Police Station is found carrying on the manufacture of photo-frames of various designs. It is also known that he supplies his products throughout the Cachar district and his total annual production is sufficient enough to meet the local demand. The average cost of photo-frame per running foot ranges from 6 paise to 19 paise according to quality and design. Numerous varieties of soft woods, all locally available, are used for this purpose.

It is interesting to observe that one unit manufacturing sports goods, mostly wooden, has been functioning since 1956-57 in Karimganj which is rather a unique feature in the whole of the State. Though the unit is still in its infancy, has a promising prospect because sport is increasing in popularity in every nook and corner of the State. But they must compete with imported sports goods of better finish and durability at competitive rates. The said unit is registered with the Director of Industries, Assam, by the name *'Joyguru Sports Industries'*. It is gathered that the proprietor of the factory, Shri Ganga Charan Roy, acquired some skill in the manufacture of sports goods from a renowned factory at Jullundur City. On the initiative of some of his relatives, he came down to Karimganj and started his existing unit on a very small scale. At present seven workers, both skilled and unskilled, are engaged in this unit on monthly wage basis and are working under the direct control and supervision of the proprietor. The monthly wages of workers vary from Rs. 50.00 to Rs. 100.00 according to their respective skill. The factory is engaged in the production of the following sports goods:—

Name of products	Main raw materials used	Remarks
1. Carrom Board	Plywood and cham or sundi timber	Plywood is imported from outside the district
2. Chinese Checker	Plywood	—do—
3. Bagatelle Board	—do—	—do—
4. Chess Board	—do—	—do—
5. Table-Tennis Bat	—do— & rubber	—do—
6. Tea Tray	Plywood	—do—
7. Black Board	Seasoned pooma wood	Locally available
8. Wooden Slate	—do—	—do—

It is reported that the market for the products of this unit is spread throughout the State. But due to lack of funds the unit cannot cope with the increasing demand. The plywood used in this factory is stated to be imported from Margherita and Mariani. Given below is a price list for different products of this unit:—

1. Carrom Boards @ Rs. 6.00 to Rs. 27.00 as per size and quality.
2. Chinese Checkers @ Rs. 5.50 each.
3. Bagatelle Board @ 2.50 to Rs. 8.50 as per size and quality.
4. Wooden School Slate @ Rs. 5.50 to Rs. 6.00 per doz. and quality.
5. Black Boards @ Rs. 12.00 to Rs. 35.00 as per size and quality.
6. Chess Board @ Rs. 5.50 each.
7. Table-Tennis Bat @ Rs. 16.00 & Rs. 24.00 per doz. as per quality.
8. Tea Tray @ Rs. 24.00 per dozen.

Though timbers of various qualities are found in abundance in the Mizo Hills, the woodworks of this region of the State have never attained any predominence. But all the works including construction of bridges and buildings are being done by the Mizo carpenters and no carpenters from other districts can be found working there. The local carpenters manufacture a few simple-designed furnitures and a few wooden articles required for domestic use. A list of the wooden articles manufactured in this district are given below:—

1. *Sum suk—pestle and mortar*:—This is manufactured by local carpenters from ordinary jungle wood for husking of paddy.

2. *Chawthleng—rice plate*—In the interior parts of the district the rice plate is generally made out of ordinary wood so that 2 to 4 persons can take rice from the same plate. The stand to support the plate is also carved out of the same piece of wood.

3. *Thingthutlang — stool*—This is also made of wood and used for sitting purpose.

4. *Dawkhan—table*—This is made of wood of good quality.

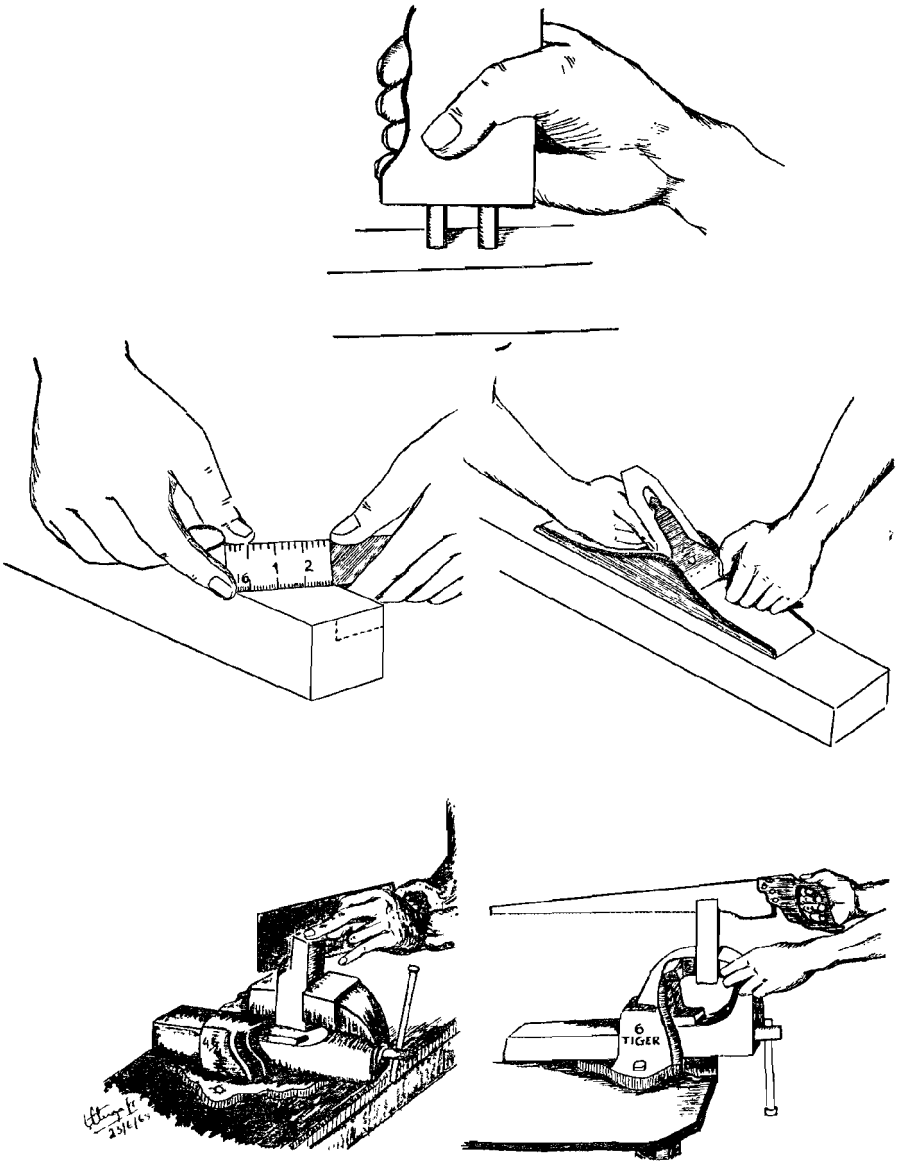
4. *Puanbu—loinloom set*—Manufactured by local carpenters.

6. *Hmui*—Traditional spinning charkha.

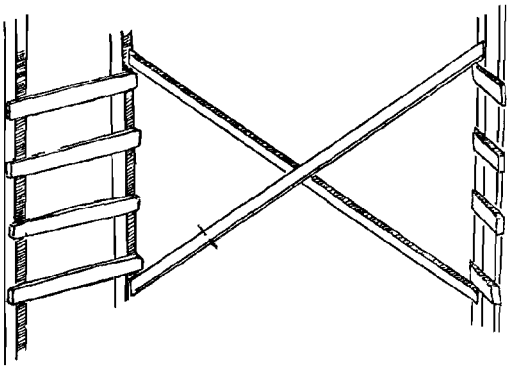
7. *Herawt—ginning machine*—This simple wooden tool is used for cleaning cotton.

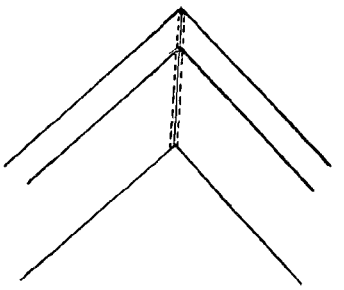
In the United Mikir & North Cachar Hills and Garo Hills districts, woodworks have not been developed to any appreciable extent and do not deserve any mention.

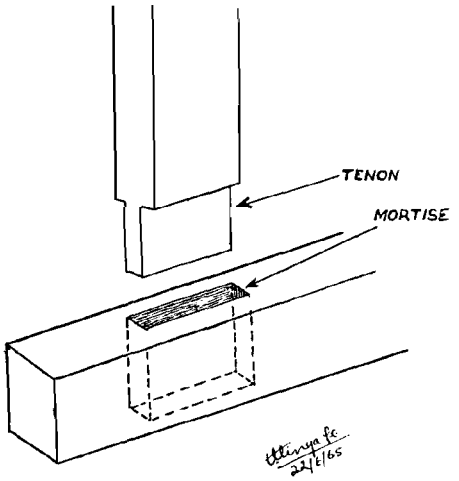
The picture is quite different in the United Khasi and Jaintia Hills, where local Khasi carpenters are found in plenty throughout the district and are famous for their wooden structural works and manufacture of up-to-date furnitures. The main timber used in all these works is pinewood which species of timber has a luxurious growth in this district. The

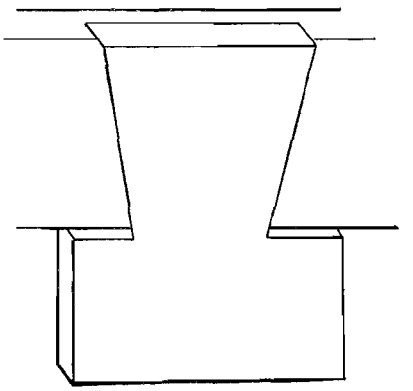


Top : Double-blind dowel joint;
Centre : Left-Measuring thickness of timber, Right-Jack plane in use ;
Bottom : Left-Tenon Saw in use, Right-Hand Saw in use









Top : Left—Butt joints, Right—Mitre angle joints
Bottom : Left—Mortise and tenon joints, Right—Dovetail joint

majority of carpenters are engaged in wooden structural works, motor-truck and bus-body making, while a few work in the manufacturing of furniture of up-to-date designs and fashions. The names of a few local furniture and wooden products in the rural areas are described below:—

1. (a) *Thlong—mortar—* } used for rice husking.
   (b) *Synrei — pestle —* }
2. *Pliang dieng—wooden plate*—used by rural people for taking foodstuff.
3. *Dieng kor—wooden chukki*—used for processing rice and other cereals.
4. *Lyngknot—a kind of stool*—used for sitting purpose.
5. *Khoh kit-dieng—A wooden carrier*—used by labourers, particularly for carrying firewood.
6. *Pham*—It is carved out of a round log of timber, preferably from the bottom of big trees, and is used for storing water.
7. *Diengshat pylleng*—A wooden plate with a handle specially used by non-Christian Khasi people for divining with an egg in order to know the cause of diseases etc.
8. *Dongsniang*—A wooden trough used for giving meals to cattle and pigs.
9. *Dong-ngab*—A typical type of wooden bee-box used by the Khasis for rearing bee.

Tools and implements : The industry requires various types of hand tools and implements. Some of these are saw, chisel, hammer, plane axe, adze, brace, vice, drill-machine, file, auger, curved chisel, screw drivers, pliers, scale, templets, squares, mallets, levels and clamp, gauges, etc. Tools are brought from the neighbouring town markets at reasonable prices.

Many of the establishments are very small, mostly one-man shows, particularly in the rural areas of the State, and carry on their business with inadequate tools. Even bigger establishments are lagging behind in the adoption of improved machines, tools and implements. Processes like planing, drilling and sawing can be done more efficiently and speedily with the employment of some power-driven machines designed for these purposes, but carpenters of such establishments have either no knowledge of these improved techniques of production or they have no resources for the adoption of such techniques. or there is no power supply in many areas.

Selection of tools : To advise on the purchase of improved tools is rather a difficult matter. The suitability of a particular tool for use varies from person to person. It is, in most circumstances, more convenient to purchase a set of carpenter's tools which, if kept in good condition, are better than a host of cheap implements. The care of tools is all the more important as their condition will be reflected in the work that is produced. Adequate provision in the shape of a good tool box is necessary for their storage. Tools left about in a haphazard way will undoubtedly be misused, thus chisels are used as screw drivers, hammers for breaking coal, and so on and so forth.

Essential implements and their description : The tools that are considered essential in modern carpentry works are two or three good saws, a couple of planes, two or three chisels, a rule and square, hammers, pincers, screw drivers, a brace with bits or hand drill. With these, a really skilled man can do a great deal, but at the same time, better and quicker works can be done with a more comprehensive range of well-chosen tools, remembering always that what matters most is the ability to use them. The following discussion on wood-working tools and their uses will serve as a pointer to their selection.

Hand saw—Hand saws for general purpose vary in length from 10″ to 28″. These are variously shaped and have different sized teeth

according to their particular use. The hand saw is used for cutting sized timbers across the grain or in the direction of the grain.

Tenon saw—These have a brass or steel stiffener along the back to keep the blade firm. They vary from 8″ to 18″ in length. A useful all-round size has a 12″ blade and 12 teeth per inch. The tenon saw is mostly used for cutting small materials and for working on the bench. The saw is normally used in a horizontal direction.

Key hole saw—Sometimes called a pad-saw. The blades range from about 6″ to 14″ in length. It is useful in cutting key-hole slots, internal work and often comes to great help for use in restricted spaces.

Jack plane—These planes are 14″ or 15″ in length and their cutters are about 2″ in width, according to their pattern. They are available both in wood and metal. The Jack plane is used for all planing except for finishing prior to scraping and glass papering.

Smooth plane—The smooth plane is of similar pattern to the jack plane but is smaller in dimensions. It is used for finer finishing or for small jobs where the jack plane would be too big. A few special planes are also available for working grooves, chambers, inlay works, etc.

Chisels—There are a few varieties of chisels most commonly used, viz., firmer chisel, bevel-edge chisel, mortise chisel, solid socket chisel. Among these, the most useful for general work are firmer chisels. The different sizes available are 1/4″, 3/8″, 5/8″and 1″.

Mortise chisels are made exceptionally strong to resist the lever action and heavy blows which they are subjected to.

Gauges—These are used for shaping hollow or rounded work.

Brace and bits—A brace, with a set of auger bits and a few centre bits and a counter sink are needed for drilling. Some workers are also found to use hand drills or straight-fluted drills for drilling holes from about 3/8″ diameter downwards.

Joiner's sash clamps—This device is made of either iron or any other metal. A five-foot long joiner's clamp is very useful for the majority of the carpenters. This implement helps the carpenter in making a compact joining of timbers possible.

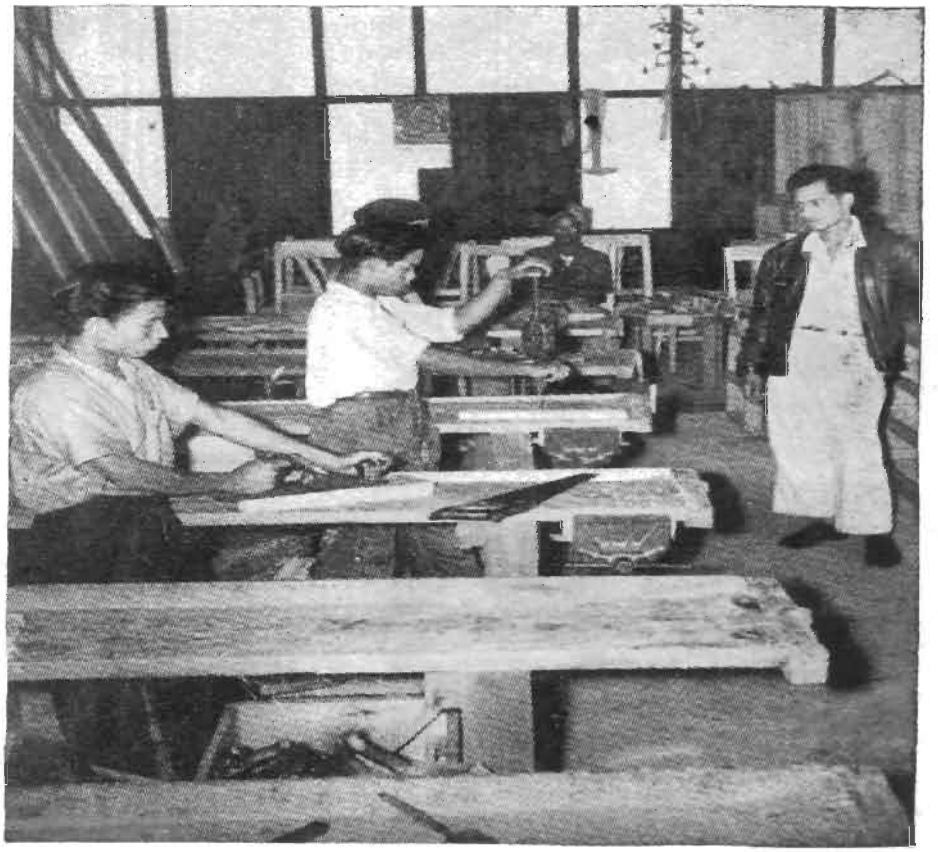
Miscellaneous tools—Little need be mentioned about hammers. A heavier claw hammer and a light ball hammer will meet most of the requirements. A bradawl is invaluable for boring holes and a spoke shave can give a finish to curved surfaces. Marking gauges and squares for setting out the work are most essential. Other desirable tools are a pair of pincers, a pair of long-nose pliers, one or two screw drivers and a scraper. One or two files, a cabinet rasp, a grinding stone, an oil stone, a spirit level, mallet and strong pair of compasses are most desirable. A working bench and vice will be of much help to a carpenter to carry out his job efficiently and smoothly.

Process of manufacture : Accurate measurements and their exact transference to the actual work lead to good craftsmenship and economy of time and material. It is observed that most of the rural carpenters are accustomed to use only a foot rule (sometimes locally made bamboo measuring device) and a plain wooden marking gauge for the purposes of marking and measuring. On the other hand, in the urban areas a few up-to-date measuring implements like mortise, gauge, try squares, steel squares, templates, bevels, plumb rule and spirit level, are found to have become popular with the skilled carpenters.

After the process of marking and measuring a sawn timber for a particular job is over, the carpenter is to use his saw for cutting the timber as per measurement or marking. In order to commence the saw cut, the carpenter is to rest the saw tightly on the work and the blade is to be guided by the left thumb. First, the saw is pulled backward and then pushed forward with long, regular, easy and steady



A Mizo carpenter weaving plastic cane seat of a wooden chair



Inside view of a carpentry production centre in the United Khasi-Jaintia Hills



Wooden furniture made in the centre

strokes. It is also noted that in this process about 60 strokes per minute is the maximum. The time required for sawing depends on the length or breadth of the object to be cut. Generally, a circular cutting is done by using a compass saw or a key-hole saw. To keep the blade of the saw clean and bright, the carpenters are found to rub it over with an oily rag. The village carpenters are also found to be very skilful in the art of sharpening the teeth of the blade by using files of different shapes and sizes.

Next to sawing comes the operation of planing. The purpose of planing is to remove the rough surface of the wood, to make it flat, smooth and straight and also to reduce the wood to exact dimension. This can only be done to perfection with years of experience and a high degree of skill. Village artisans perform this job with an age-old implement, that is, the adze. With some different strokes of the adze, they skilfully reduce the wood to exact dimension. Wooden or iron planes are found in use for this purpose. The handle at the back of the plane is firmly grasped by the carpenter's right hand to propel the plane forwards, while the knob at the front is used by the left hand to steady the front. This process of planing the surface of wood is continued till the wood is reduced to exact dimensions.

The next stage of operation is on joints. This work can be performed either by means of a saw or a chisel. There are several types of joints that are worked out by the carpenters. A few among them are known as bull joints, dowelled joints, cleated joints, mortise and tenon joints, devetail joints, etc. In the urban areas of the State, woodwork joints are generally secured with glue, and are cramped up by means of a joiner's cramp to ensure that the joint faces are in close contact and to prevent any strain while the glue is setting.

The different pieces of works are then fastened together to make the work complete. Fastening is generally performed with the help of glue, screws, nails, hinges, nuts and bolts as the case may be. Glue is generally applied to the job after heating.

Wood carving : Carved works on timber is found to be carried on by the local carpenters by using a chisel, a knife and a gauge. The quality of the carved work depends much on the skill and craftsmanship of the worker concerned. Regarding carved works, it will be well to keep in mind that from intelligent observation comes knowledge and with knowledge the much greater pleasure of designing and creating pieces of one's conception. The technique of carving also varies from person to person. It is also observed that a wood-worker of this State knows quite well only by experiences and not by wood technology or mensuration regarding the quality and quantity of timber required for his job.

Painting and varnishing : The carved objects like singhasan, icons, etc., are generally painted with different matching colours. Paints are purchased from the local markets. In the case of varnishing of wooden furniture, the artisans themselves prepare a solution out of the following ingredients :—

(i) Kalabati chach or lac.
(ii) Resin.
(iii) Methylated spirit.
(iv) Rabi mustafi and
(v) Some variety of indigo powder.

After the painting or varnishing operation is over, the products are kept in the sun or in an airy shade for drying, and when the varnish or paint is completely dried up and the colour becomes bright, the objects are ready for sale.

Cabinet and furniture making : Cabinet and furniture making is seasonal to some extent, but in urban areas, it continues throughout the year. In rural areas the busy season extends from the month of November to Febru-

ary. During this period the availability of timber as well as demand for the finished goods have been found to be more than at other periods of the year. Moreover, in rural areas many carpenters are cultivators at the same time. These carpenters-cum-cultivators pursue the industry when they are off from agricultural operations.

Hukanal and Kharam manufacture : The origin of the craft is, according to the reports of the villagers of Kholagram, that as the raw material for making the smoking pipe (naircha) that is 'polan', a locally available soft timber species, is abundantly available in the village, an artisan from Comilla (now in East Pakistan) came down to Kholagram some 80 years ago and taught the local villagers the delicate art of 'hukanal' manufacture. Since then the craft has been practised by the local people on commercial basis as a source of part-time employment. The artisans get the chief raw material 'polan' from the land owner at a very nominal price. They are also reported to grow the same timber in their own land.

The first process commences with the felling of 'polan' trees which are cut with the help of 'ari' or big saw from nearby jungles, collected by the artisans themselves and brought to the workshop. The 'polan' timber is then cut into different sizes of 24", 21", 18" or 9" in length with about 1" diameter as the case may be.

A rough shape of the 'naircha' is then produced on the small piece of timber by scraping it with the help of a 'dao' or 'chopper'. These roughshaped timbers are worked upon finally when they are fixed on the 'kund' or 'hal' (see photo) which is manipulated on the same principle as that of a wood working lathe. Two persons are required for this operation—one to pull both the ends of the string fixed with the object and thereby causing high speed rotation of the object, while the other to do the chiselling work with a high degree of skill to produce designs on the 'naircha' (see photo). To make the product finally ready for the market a hole is bored on the 'naircha' lengthwise with 'barma' or drill by rotating it on the same principle as that of the previous one. The hole is bored in order to facilitate passing of air or smoke through the 'naircha' or 'hukanal'.

It is gathered that the production capacity of a unit manufacturing hukanal is 20 per hour, if sized timbers are available. The 'nairchas' are sold unvarnished to the local merchants. It is also reported that no problems are faced by the artisans regarding procurement of raw material and the sale of finished products.

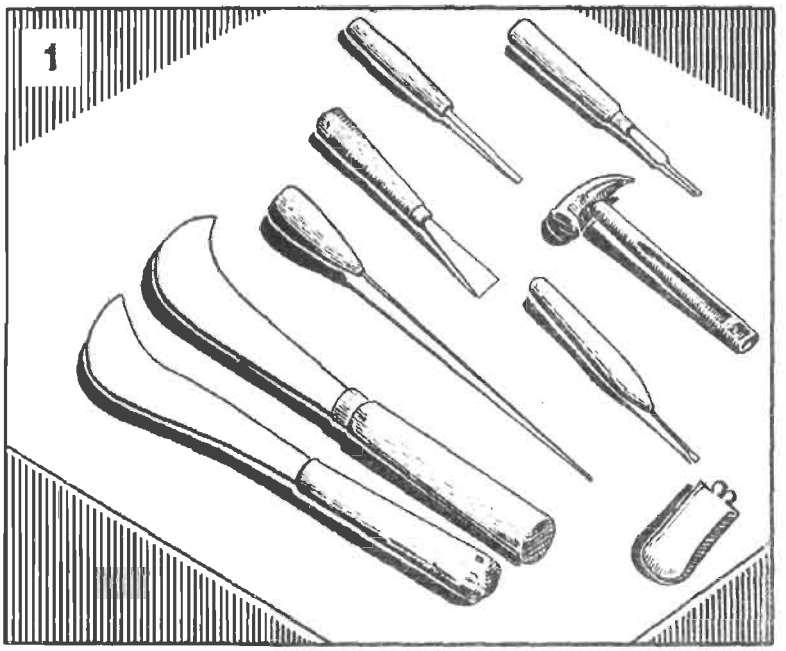
The sale price of different sizes of hukanals is recorded below:

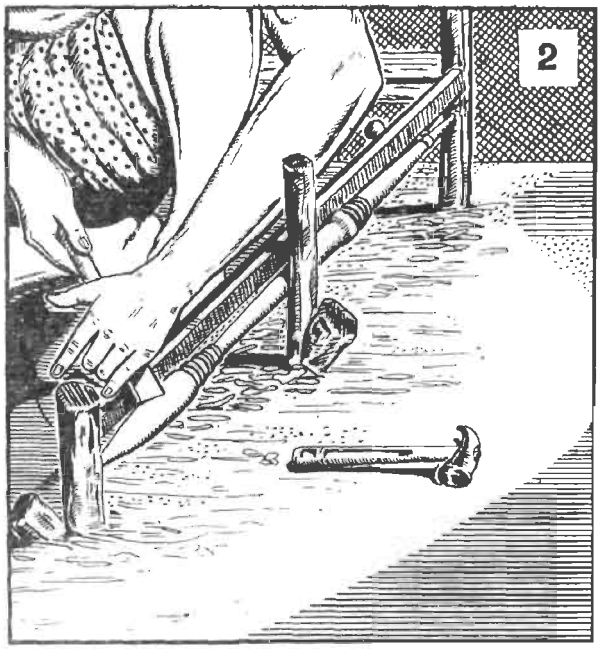
	Size	Price	Nos.
(i)	$13\frac{1}{2}$"	Rs. 5.00	per 100
(ii)	18"	Rs. 8.75	—do—
(iii)	$22\frac{1}{2}$"	Rs. 11.25	—do—
(iv)	24"	Rs. 12.50	—do—

It is estimated that by carrying on this craft, a full time worker can earn Rs. 1.75 to Rs. 2 - per day.

The mode of smoking is, however changing fast, huka-smoking being replaced by bidi and cigarettes. So, this industry may die down gradually.

Wooden sandals : Throughout the plains districts of Assam, the rural people use sandals of wood instead of leather. This is very economic and useful too. The timber used for the manufacture of wooden sandals comprises such soft timber species such as jarail, poma, rata, polan etc. To manufacture wooden sandals, timber logs are cut into pieces by village artisans with the help of a big saw These pieces are then shaped into different flat sizes with the help of axe, adze, wooden planer, etc.









1. Handtools used in 'Huganal' making.
2. Chiselling of huganal in the 'kund'.
3. Boring of 'Huganal'.
4. Making of 'Kharam' or wooden-sandal.



Boat making on the bank of Brahmaputra near Goalpara Town

The design on these flat pieces are sketched from a wooden mould with a pencil which are chiselled into proper shapes later on. Finally a wooden hook is fixed at the top of the sandal to make it complete. In the case of flat wooden sandals a strip of rubber is fixed at the top.

Different varieties of wooden sandals can be seen in the market and the bulk of them are being produced in the Cachar district The cost of these sandals per dozen range from Rs. 3/- to Rs. 16/- depending on the designs and sizes. Generally a person can produce 6 to 12 pairs per day.

Wooden sandals are also slowly being replaced by leather or rubber-soled slippers and sandals with the possibility of this petty trade going down as people become more literate and richer.

Boat manufacture : The boat plays an important role in the internal communications of the State. The establishments in this industry are scattered and no concentration has been observed. In all the plain districts boat-making is practised. As navigation is hardly possible in most of the rivers in the hills, boat-making establishments do not exist in the hill districts. The demand for boats is uncertain and low, resulting in a small scale of production. Further, it is reported that many families make boats not for sale but for their own uses. As such, the commercial prospect of the industry does not seem to be bright.

The process of production such as sawing, shaping, joining, colouring, etc. are carried on outdoors. Tools and implements employed include the axe, adze, hammer, saw, brace, etc. Suitable timber for construction of boats is available in plenty in districts in the local markets. Most of the boat makers collect the raw material from the driftwood carried by the rivers during floods. For such collections they are, however, required to pay a nominal royalty to the Forest Department.

Cart wheel manufacture — its occupational trend : Making of wooden wheels with iron rims for bullock and buffalo carts in Assam is practically a monopoly of a section of blacksmiths from Bihar. Workers in this industry have to be proficient and attain skill in two different branches, blacksmithy and carpentry. In most of the rural areas of the State, bullock carts constitute a major means of transport, and this mode of transport is likely to remain for still many years ahead. But a discouraging fact is that the indigenous people have not taken to this industry. One of the main reasons may perhaps be the prejudices arising out of caste, and some other reasons and social barriers may also prevent the growth of the combined skill in both blacksmithy and carpentry. As a result, the necessary skill for manufacturing cart wheels is a rare phenomenon amongst the indigenous people and the artisans from Bihar enjoy unhindered monopoly in this field.

The industry is purely seasonal. The busy season is very short and lasts for over 5 months only, from late October to early April. During these months the roads become dry and most of the agricultural produces are harvested. So the need for transport of these produces reaches the maximum during this period of the year. It is observed that the industry has a tendency to grow in urban and semi-urban areas.

Seasoned trunk of hardwood is ideal for the cart wheel. The usual practice is to purchase logs or whole trees which are sawn off into pieces and left in the open for a considerable period so that the wood may be sufficiently seasoned. A carpenter does not depend on anybody for bringing to shape the cart wheel. With his age-old tools like hammer, chisel, saw, mallet, drill and adze of different sizes, he himself completes the whole work. He even makes the iron parts sometimes, using a forge of the fellow blacksmiths. A pair of cart wheels fetches him Rs. 100.00 to Rs. 120.00.

Apart from manufacturing new wheels the cart wheel makers also undertake servicing works like repair of old and worn out wheels and overhauling of carts. In fact, a substantial part of their income is derived from such servicing activities.

ECONOMICS OF THE INDUSTRY

Economics of production : It is very difficult to discuss elaborately the various economic aspects of the industry in a general way. The cost of production of a particular item differs from district to district. The price and quality of timbers used for particular item are also not uniform in all the districts. The principal factors which determine the cost of production include cost of timber, nails, screws, glue, varnish, etc., plus wages and a nominal amount of depreciation charge for tools, whereas the cost of carved wooden articles are determined by the amount of skill applied by the craftsmen concerned. It is gathered that the average monthly income of full-time wood workers or carpenters vary from Rs. 80/- to Rs. 140/- according to their respective skill and craftsmanship.

The following table gives the number of persons engaged in carpentry, joining and pattern-making (wood) in the different districts of the State according to the 1961 Census.

Table No. 1

Name of the district	Total No. of persons	Males	Females
Goalpara	2,574	2,574	—
Kamrup	4,247	4,247	—
Darrang	2,545	2,545	—
Lakhimpur	3,417	3,417	—
Nowgong	1,472	1,472	—
Sibsagar	3,671	3,671	—
Cachar	1,962	1,962	—
Garo Hills	229	229	—
United K. & J. Hills	1,342	1,342	—
United M. & N.C. Hills	214	207	7
Mizo Hills	324	324	—
	21,997	21,990	7

Problems in stock piling & products : It is observed that in different parts of the State wood workers and carpenters produce only such quantities as are likely to be sold out immediately. No production for stock piling is undertaken. This is due to several reasons. Firstly, the carpenters lack adequate finance to enable them to block huge sums of money in the form of finished products for a long period of time. Secondly lack of proper storage facilities force the carpenters to produce only such quantity of output as can be disposed of within a short period of time. In addition, the carpenters do not have proper marketing organisation to facilitate the disposal of a large number of articles. In fact many of the carpenters produce only those articles which are ordered by customers. Only the biggest few, and that too in urban centres, have provisions for advertisements and canvassing for the articles. As a result, production is intended to meet only local demands. The absence of modern advertising media like showrooms is keenly felt. As far the majority of wood workers and carpenters, there is almost no scope for the display of particular skills to the customers. It is also observed that generally the role of the artisans and producers in the matter of marketing is limited mainly to local markets only. Most of the rural artisans are quite unaware of markets for their products in different parts of the State. Manufacturers' association or organisation for the marketing of products is still a rare phenomenon.

Government assistance : In order to encourage and develop wood works in the State, the Directorate of Industries has formulated various schemes. Deserving carpenters of both rural and urban areas are said to have been provided with financial, technical and such other facilities as are likely to promote the growth of the industry. The State Industries Department is also trying to popularise improved carpentry tools among the rural artisans by way of distributing them at subsidised rates. Some

of the skilled carpenters are advanced with loans on easy terms and conditions. Willing carpenters are also trained in the application of modern techniques of production and use of improved tools for better and efficient working. In the Community Development Block areas, there are schemes for promoting the growth of rural industries by distributing improved tools at subsidised rates and financing cooperative societies organised among the carpenters for the betterment of rural artisans engaged in and promoting of the crafts. Consequently carpenters' cooperative societies have been on the increase in rural areas a list of which with the available details is given in the appendix. Government has also started a few production centres equipped with up-to-date machinery in some of the Hill districts. In such centres, local skilled carpenters find scope of employment. A good number of carpenters hailing from within the State itself are also getting scope in the manufacture of bee-boxes and parts of ambar-charkha under the schemes put into operation by the Assam State Khadi and Village Industries Board. This Board, it is reported, is also financing and patronising a few carpentry workshops which are engaged in the manufacture of looms and their accessories, bee-boxes, ambar-charkha, etc.

Scope for development : It has been noted that the inadequacy of financial resources has prevented the industry from adopting machines and tools. As a result, the technique of production has not improved and the productivity has remained still very low. There is considerable scope for the improvement of the quality and increase of the quantity of products. It has been found that the products manufactured by the skilled artisans coming from outside the State are much more elegant than those of the local artisans. The finishing of these products is very good and it gives added attraction to them and therefore whenever such highly skilled artisans established their shops, the indigenous producers have been placed at a disadvantage and cannot compete successfully with the former. So, measures should immediately be taken to train the young indigenous artisans in the modern methods of production so as to enable them to compete successfully with the immigrant artisans. Moreover, the local woodworkers should be trained to produce articles to suit modern changing tastes and needs of the customers. The carpenters should also be trained to handle modern machines and tools employed in this industry. It is heartening to find that a right move in this direction has already been taken by the State Government by launching various schemes for the promotion of this prospective trade although the results thereof are yet to be seen.

CONCLUSION

It is rather astonishing to note that though Assam is very rich in forest resources, there is an import of wooden articles. A few articles are made only to meet the local demands which fluctuate from time to time. Carved articles are made on order only. Save and except carved wooden icons of Hindu gods and goddesses, many wooden articles are being gradually ousted by more elegant and durable articles imported into the State. There is also a constant dearth of carpenters in this State due to the fact that manual labour is not viewed with favour. For this reason, many carpenters have to be imported regularly from other parts of India. It has also been noticed that in the construction of private houses, people's modern taste is to have plain and simple carpentry and this tendency has led to the disuse of carved architecture in construction work. The wind of change cannot be stopped, but with improvement in economic conditions, more houses and furniture will be required, and so there will be demand for more carpenters with skill to adapt the modern tastes.

With the opening of more and more technical schools throughout the State, it is hoped that the elements of wood carving and manufacture of fancy wooden articles will be included in the carpentry curriculum, which course will not fail to be a source of pleasure and profit to the enterprising carpenter.

The ultimate object of the village and rural industries programme is to increase the per capita income of the village artisans and also to increase the national income by expanding production through village industries. It is already an established fact that through village industries alone it is possible to find congenial work for the people near their homes, part-time throughout the year or full-time during off-seasons. This in turn will help in keeping the locally available raw materials worked up to the stage of manufacture much to the enrichment of the rural economy.

It is now recognised that while it is essential that village industries should receive the maximum support in terms of State policy and specific Government assistance, these measures cannot have more than short term benefit, unless the techniques of cottage industries improve rapidly. The Planning Commission, therefore, rightly attaches utmost importance to arrangements for research and training in village industries. With the prospect of rural electrification, the present rudimentary character of village industries is likely to be gradually transformed. The elimination of the middleman and the substitution of the system of co-operative finance for him will go a long way in re-establishing many of the decadent village industries.

APPENDIX A

STATEMENT ON WOODWORKS (CARPENTRY) CO-OPERATIVES OF ASSAM

Name of the district	No. of Co-operative Societies	No. of paid-up shares	Amount of paid-up share capital (Rs.)	Total No. of workers	Extent of business done during 1961-62 (Rs.)	Amount of loans and grants received from different sources (Rs.)
1	2	3	4	5	6	7
1. Goalpara	4	25	967.00	25	* 183.00	1,250.00
2. Kamrup	2	24	285.00	24	N.A.	N.A.
3. Darrang	Nil	..	..	..	..	..
4. Sibsagar	2	43	1,050.00	30	* 35,000.00	34,140.00
5. Nowgong	2	35	310.00	30	* 5,000.00	Nil
6. Lakhimpur	2	40	450.00	35	N.A.	N.A.
7. Cachar	Nil	..	..	..	..	..
8. U. K. & J. Hills	3	101	2,810.00	34	* 3,820.00	1,950.00
9. U. M. & N.C. Hills	Nil	..	..	..	..	..
10. Garo Hills	1	22	230.00	22	N.A.	N.A.
11. Mizo Hills						
Total of all districts	16	290	6,092.00	200	44,003.00	37,340.00

* Extent of business done by only one society. All figures in different columns exclude 2 societies.
N.A. : Not Available.
Source : Registrar, Co-operative Societies, Assam, Shillong.

APPENDIX B

TREND OF PRICES OF CABINET AND FURNITURE PRODUCTS

Name of the district	Products	Unit	Prices during (in rupees)							
			1939		1953		1954		1961	
			Urban	Rural	Urban	Rural	Urban	Rural	Urban	Rural
1. Goalpara	(i) Chair	One	..	..	..	5-0-0	.	5-0-0	11.00	.
	(ii) Cartwheel	Pair	..	30-0-0		65-0-0	64-0-0	64-0-0	45·00	45.00
2. Kamrup	(i) Chair	One	3-8-0	..	10-8-0	..	10-8-0	..	12.00	..
	(ii) Cartwheel	Pair	..	25-0-0	..	75-0-0	..	80-0-0	..	..
3. Darrang	(i) Chair	One	3-0-0	4-0-0	12-0-0	14-0-0	12-0-0	15-0-0	10.00	12.00
	(ii) Table	,,	.	..	..	..	.	..	10.00	12.00
4. Nowgong	(i) Chair	,,	7-0-0	15-0-0	15-0-0	..	15-0-0	..	15.00	15.00
	(ii) Bench	,,	..	..	.	.	..	..	17.00	17.00
5. Cachar	(i) Chair	,,	5-0-0	..	15-0-0	..	15-0-0	.	12.50	12.50
	(ii) Table	,,	..	.	..	..	..	.	15.00	15.00
6. Sibsagar	(i) Chair	,,	2-10-0	2-10-0	13-0-0	13-0-0	13-0-0	13-0-0	.	..
	(ii) Table	,,	4-8-0	4-8-0	17-0-0	17-0-0	17-0-0	17-0-0	.	..
7. Lakhimpur	(i) Chair	,,	4-0-0	4-8-0	15-0-0	15-0-0	15-0-0	15-0-0	20.00	22.00
	(ii) Cartwheel	Pair	..	35-0-0	..	80-0-0	..	80-0-0	..	120.00
8. Garo Hills	(i) Wooden Chair	One	..	..	.	..	20-0-0	18-0-0	34.00	22.00
	(ii) Table	,,	..	..	..	..	40-0-0	35-0-0	48.00	45.00
9. U.K. & J.Hills	(i) Chair	,,	1-4-0	..	6-0-0	..	6-0-0	..	12.40	10.00
	(ii) Table	,,	..	..	..	..	..	..	40.00	35.00
10. Mizo Hills	(i) Chair	,,	.	..	.	.	.	..	20.00	..
	(ii) Table	,,	..	..	.	.	..	..	35.00	..
11. U.M. & N.C. Hills	(i) Chair	,,	..	.	..	12-0-0	..	12-0-0	..	..

Sources : Department of Economics & Statistics and Department of Industries, Assam

APPENDIX C

TREND OF PRICES OF RAW MATERIALS IN ASSAM

Name of the district	Description of raw materials	Unit	Prices during			
			1939	1953	1954	1961
1	2	3	4	5	6	7
1. Goalpara	Timber	Cft.	2-8-0	5-0-0	5-0-0	Rs. 6.00
2. Kamrup	,,	,,	2-4-0	6-12-0	6-8-0	Rs. 7.00
3. Darrang	,,	,,	2-0-0	7-8-0	7-8-0	Rs. 7.50
4. Nowgong	,,	,,	3-0-0	8-0-0	8-0-0	Rs.10.00
5. Cachar	,,	,,	2-4-0	6-4-0	6-4-0	Rs. 8.00
6. Sibsagar	,,	,,	2-8-0	7-8-0	7-8-0	..
7. Lakhimpur	,,	,,	2-0-0	6-0-0	5-4-0	Rs. 7.50
8. U K & J. Hills	Pinewood	100 Sq.ft.	3-0-0 to 15-0-0	15-0-0 to 60-0-0	15-0-0 to 60-0-0	Rs.25.00 to Rs.45.00
9. Garo Hills	Timber	Cft.	..	..	8-0-0	Rs.12.00
10. United Mikir & N.C. Hills	,,	,,	..	..	..	..
11. Mizo Hills	,,	,,	..	.	.	Rs. 5.00

Sources : Department of Economics & Statistics and
Department of Industries, Assam.

APPENDIX D

BOTANICAL NAMES OF A FEW VALUABLE TIMBERS OF ASSAM AND THEIR USES

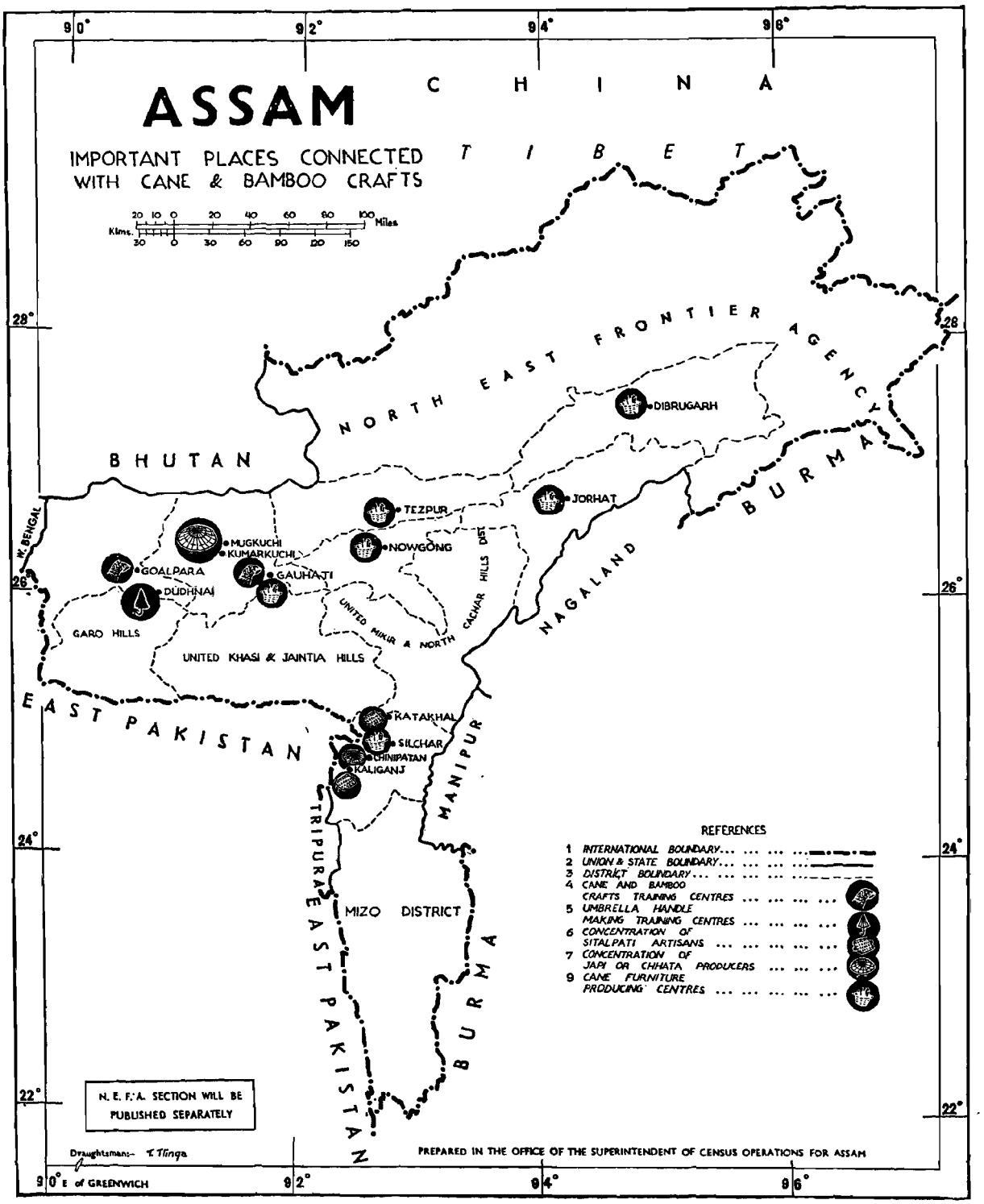
	Local term	Botanical equivalent	Remarks/uses
1.	Am	Mangifera Indica	Used as ply-wood
2.	Cham, Kathal	Artocarpus Integrifolia	Furniture, dug-out canoes and boats
3.	Cham or Sam	Artocarpus Chaplasha	Used for beams, dug-out canoes and boats
4.	Gomati	Gmelina Arborea	Used for house posts and furniture
5.	Hortuki or harra	Terminalis Citrina	Plywood
6.	Jam	Eugenis Species	Used for furniture and agricultural implements
7.	Jarul or Ajhar	Lagerstroemia Flos Reginae	Used for house posts and boats
8.	Jhalna	Terminalia Bicolorata	Used for beams
9.	Kadam	Anthocephalus Cadamba	Ply-wood
10.	Kurta	Isonandra Polyantha	Used for beams
11.	Nagasvar	Mesua Ferrea	Used for house posts, railway sleepers and boats
12.	Odal or Udal	Sterculia Villosa	Ply-wood
13.	Poma	Cedrela Toona	Used for beams and dug-out canoes and boats
14.	Ramdaladewa	Artocarpus Lakoocha	Used for beams
15.	Rata	Dysoxylum Binectarfiferum	Used for beams
16.	Poma, Gundroi or Gunserai	Cinnamomum Glanduliferum	Used for furniture
17.	Sal	Shorea Robusta	Used for posts and railway sleepers
18.	Sida	Lagerstroemia Parviflora	Beams and railway sleepers
19.	Simul	Bombax Malabaricum	Ply-wood
20.	Sonaru	Cassia Fistula	Used for construction work and furniture
21.	Sundi, Ramdala	Duabanga Sonceratioides	Used for house posts
22.	Tailo	Castanopsis Indica	Used for beams
23.	Titasopa	Michelia Champaka	Used for furniture
24.	Tula, Dhumboil	Beilschmiedia Sikkimensis	Ply-wood

TERMINOLOGY

Description in English	Local term in Assamese	Local term in Bengali	Local term in Khasi
1. Adze	Tangi	Tangi	Sdie-puh-dieng
2. Bamboo or wooden string	Karhoni	.	Saisiej, Saiuieng
3. Bed stead (designed)	Salpira or paleng	Palong	Jingthiah, Palong
4. Boat	Nau	Nau or Nauka	Lieng
5. Book-rest	Thagi or Kharai	..	Jingbuh kot
6. Brace & bit	Agor	Agor	Budu
7. Chisel	Batali	Batali	Shniah or Paintali
8. Dril	Girmit Buddu	..	Budu
9. File	Reth	Reth	Syrti
10. Hammer	Hatudi	Hatudi	Tyrnem
11. Jack-plane	Randa	Randa	Rynda
12. Joiner's-sashclamp	..	Serganja	Klam
13. Lamp stand	Gasa	Botni or Gasa	Ka jingpynieng sharak
14. Palanquin	Dolah	Palki	Palki
15. Picker	Nasoni	..	
16. Pincer	Peskos	Jambura	Halai
17. Saw	Korath	Korath	Kurat
18. Shuttle	Mako	Makoo	Tadonghashei
19. Sley	Durpati	.	.
20. Smoking pipe	Hukanal	Huka or Naircha	Tangeuma
21. Wood-carver	Khonikar	..	Nongoh-dur
22. Wooden device for pressing of paddy	Dheki	Dhenki	Thlong
23. Wooden sandal	Kharam	Kharam	Slipar dieng
24. Wooden seat	Borpira	Pira	Lyngknot

BIBLIOGRAPHY

1. The History of Civilization of the People of Assam, by Shri P.C. Chowdhury, Ph. D., 1959.
2. Wood Carving in Assam, by A. Mazid, 1915.
3. A Review of the Industrial and Economic Situation of Assam, 1958, by the Industries Department, Government of Assam, Shillong, 1960.
4. Carpenter's Practical Guide by E.W. Hobbs, 1953.
5. Report on the Survey of Cottage Industries in Assam, Vol. I & II, by the Department of Economics and Statistics, Assam, Shillong, 1958.


ASSAM
IMPORTANT PLACES CONNECTED WITH CANE & BAMBOO CRAFTS
20 10 0 20 40 60 80 100 Miles
Kms. 30 0 30 60 90 120 150
90°
92°
94°
96°
28°
26°
24°
22°
CHINA
TIBET
NORTH EAST FRONTIER AGENCY
BHUTAN
BURMA
W. BENGAL
NAGALAND
EAST PAKISTAN
MANIPUR
TRIPURA
EAST PAKISTAN
BURMA
DIBRUGARH
JORHAT
TEZPUR
NOWGONG
MUGKUCHI
KUMARKUCHI
GOALPARA
GAUHATI
DUDHNAI
GARO HILLS
UNITED KHASI & JAINTIA HILLS
UNITED MIKIR & NORTH CACHAR HILLS DIST
RATAKHAL
SILCHAR
CHINIPATAN
KALIGANJ
MIZO DISTRICT
REFERENCES
1 INTERNATIONAL BOUNDARY
2 UNION & STATE BOUNDARY
3 DISTRICT BOUNDARY
4 CANE AND BAMBOO CRAFTS TRAINING CENTRES
5 UMBRELLA HANDLE MAKING TRAINING CENTRES
6 CONCENTRATION OF SITALPATI ARTISANS
7 CONCENTRATION OF JAPI OR CHHATA PRODUCERS
9 CANE FURNITURE PRODUCING CENTRES
N. E. F. A. SECTION WILL BE PUBLISHED SEPARATELY
Draughtsman:- T. Tinga
PREPARED IN THE OFFICE OF THE SUPERINTENDENT OF CENSUS OPERATIONS FOR ASSAM
90° E of GREENWICH

CHAPTER VII

Cane and Bamboo Works

Introduction : Assam is rich in sylvan resources and most of its forests are richly stocked with bamboos and canes of various species. Special mention may, however, be made of the forests of Mizo Hills, Cachar, United Mikir and North Cachar Hills, Nowgong and Lakhimpur districts. While touring in the hill districts, one sometimes wonders at the vast expanses of bamboos and canes. The important species of bamboos of economic value are the Muli (melocanna bambusoides), Dalu (Teinostachyum dalloa), Khang (Dendrocalmus longispatnus), Kaligoda (Oxytenanthera nigrociliata) and Pecha (Dendrocalamus Hamilton-ii). The Muli and the Dalu have great commercial importance, the former for pulping, constructional and fencing purposes, and the latter for the mat and basket industry.

The making of bamboo and cane products is perhaps the most universal of all the crafts practised by a large number of artisans scattered throughout the State. It is practised as a household industry and no mechanical device is used. Cane and bamboo products have a wide range of purposes and are extensively used in every household. This industry has carved for itself an important place among the handicrafts of the State. It provides part time employment to the cultivators in their spare time, and full time employment to the few highly skilled artisans who produce only fine decorative baskets, furniture and mats on a commercial basis.

History and origin : No definite records are available to establish the antiquity, history and origin of this craft in Assam. But it can be safely assumed that the craft was practised since the misty past with the very dawn of civilization.

An idea about the flourishing state of cane and bamboo products of Assam even during the time of Bhaskara Varman (early part of the 7th century A.D.), the king of Assam, may be had from the following extract.

*"Early literature refers to the well-decorated and coloured *sital patis* (cool mats) used by the rich people. Mats were usually made of cane. The abundance of cane in the forests of Assam is testified by the classical writers. Ptolemy, for instance, states that to the east of Serica, which we have identified with Assam, there were hills and marshes where canes were grown and used as bridges. Evidence of the production of other cane articles is also supplied by the '*Harsacarita*', which mentions stools of cane. The cultivation of bamboo and its use for various purposes are well known. Bana again testifies to this highly developed craft. He states that Bhaskara sent to Harsha 'baskets of variously coloured reeds', 'thick bamboo tubes' and various birds in 'bamboo cages'. All these prove that various industrial arts were developed in Assam at an early period and were continued to be practised till recent times, based on that traditions like those of the craftsmen of other parts of India...".

*An extract from "The History of Civilization of the People of Assam" by Dr. P. C. Choudhury, (pp. 378)

Distribution according to caste or community : It is observed that there is no particular caste or community in the Assam Valley who are exclusively connected with this traditional craft. It is generally carried on by all, particularly the peasants irrespective of caste, community or creed. But the bulk of the artisans practising this craft are found among certain castes like the Kalitas, the Katonis, the Ahoms and the Namasudras of Cachar. In the hill areas of the State, however, everybody can take up this craft.

Kalitas **:** The Kalitas occupy a high position among the indigenous castes of the Brahmaputra Valley, and rank next to the Brahmans and Ganaks. There are several functional sub-divisions of the Kalita caste, the true position of which is doubtful. These are Kumar Kalitas, Mali Kalitas, Nath Kalitas, Bez Kalitas and so on. Some assert that these are not Kalitas at all, and have no title to describe themselves as such ; while others maintain that they are true Kalitas who have degraded themselves by taking to the occupations which their names denote. On the whole, the evidence is in favour of their being Kalitas, but it seems not unlikely that persons of other castes have also gained admission to these lower grades. These different artisan sections seem to be much on the same level, and intermarriage is said to be permitted amongst them. They often describe themselves as 'Saru Kalita' instead of by their occupational name, and are thus distinguished from the better class Kalitas or 'Bar-Kalita'.

Katonis or Jugis **:** It is gathered that some Jugis or Katonis of Cachar claim to be descended from 'Lord Siva' while others trace their descent to the amours of Brahman widows and ascetics. Many of them in consequence take the sacred thread, and refuse to take food or drink in the house of people belonging to other castes. They select their priests from their own caste. But their position in the Brahmaputra Valley is some what different because the caste originally contained five principal sub-divisions, viz. the 'Sapmelas' or snake-Charmers, the 'Polupohas' or rearers of a particular kind of silkworm, the 'Duliyas' or palki bearers, the 'Katonis' or weavers and the 'Thiyapotas' who were so-called because they buried their dead in a standing position. Of these, the 'Katonis' and 'Duliyas' are the chief sub-divisions now remaining.

Ahoms **:** It is generally believed that the Ahoms are descendants of 'Mau Shans', a sub-division of the 'Thai' race. Their social status in the village community is not very high. They are an independent and self-contained group of people. Ahoms do not discourage any member of their community to take up whatever occupation he chooses.

Namasudra **:** In the Cachar district this craft is found to be mostly confined to a particular caste known as 'Namasudra' who claim it to be one of their traditional occupations. Namasudra is a Scheduled Caste of Assam. They are generally diligent, painstaking and business minded. The principal occupation of the people belonging to this caste is fishing, the selling of which necessitates the manufacture of fishing contraptions. Apart from nets, most of the other fishing implements are made of bamboo and cane.

TECHNIQUE OF PRODUCTION

Raw materials and their uses : *Assam has probably the maximum resources for the cane and bamboo industry in India, comparable with Canada and the Scandinavian countries- -Sweden, Norway and Finland. In bamboo and reeds, Assam has the most concentrated forests in the whole of India.

*Ref.: 'A Review of the Industrial and Economic Situation of Assam' by the Deptt. of Industries, Assam, Shillong, 1958

As many as 51 species of bamboo grow in Assam and they are being used for diverse purposes, mainly for buildings, furniture and diverse contraptions. Studies are being conducted in some research centres in India as to the suitability of bamboo being used as a reinforcement to replace mild steel bars in light concrete structures. Bamboo is also used for umbrella handles, walking sticks, tool handles, fishing rods, tent poles, cordage, ladders, yokes, baskets, toys, handfans and various other domestic and agricultural implements. All these articles can be produced on a cottage and small scale basis with small machineries. A team of Japanese experts who visited the state during 1955-56 made recommendations for the establishment of what are called bamboo mills, in addition to cane mills and furniture mills, to exploit the forest resources of Assam in the small scale industries sector. One such mill has been set up in Gauhati.

Bamboo : At present only 50,000 tons of bamboo are extracted while the potential is of the order of 1,250,000 tons. Such a huge surplus of bamboos is not to be found elsewhere in India Therefore the prospects of establishment of several pulp and paper plants, big and small, based on bamboo alone, is tremendous. The important species mentioned below are available in commercial quantities.

1. Muli (Malocanna Bambusoides).
2. Hill Jati (Oxytenanthera Parvifola).
3. Kako (Dendrocalamus Hamiltoni).
4. Dalu(Teinostachyum Dalloa).

Apart from forests, bamboo is also plentiful in the villages throughout the State. Some of the rural establishments are found to have their own groves. Such establishments generally meet their requirements partly or mainly from their own groves. Others obtain their requirements through 'mahajans' or middlemen as advance against the supply of finished products. The urban establishments on the other hand are found to meet their requirements mainly or wholly by purchase from the merchants. A statement showing the areas and quantities of bamboo, rates, etc. in different forest divisions of Assam is presented below :—

*TABLE I

Name of division	Total bamboo area (in acres)	Annual yield of Govt. forests (in tons)	Annual yield of forests under District Councils (in tons)	Rate per ton	Remarks
1	2	3	4	5	6
Lakhimpur	2[illegible]300	34,000	..	Rs. 40- to 45/- at railhead.	Mainly Kako.
Sibsagar	12,800	14,000	..	Rs. 50/- at Ledo, Joypore.	,, Waithai & Dalu.
Dhansiri	3,500	3,000	..	Rs. 45/- at railhead.	,, Hill Jati & Muli
Nowgong	44,000	68,000	..	Rs. 45/- at Lumding	,, ,,
Cachar	148,000	497,000	..	Rs. 45/- at Karimganj, Badarpur	Muli.
Kamrup	7 200	11 400	..	Rs. 52/- to 70/- at Gauhati.	,, Kako & Muli.
Garo Hills	7[illegible],000	[illegible] 500	25,000	Rs. 70/- to 78/- at Gauhati	,, Terai
K. & J. Hills	28,000	9,400	8,500	Rs. 50/- to 60/- at Gauhati	,, Kako & Muli.
Mizo Hills	192,000	200,000	600,000	Rs. 60/- to 65 at Karimganj	,, Muli.
Mikir & N.C. Hills	67,000	N.A.	50,000	Rs. 40/- to 50/- at railhead.	,, Hill Jati & Muli
	549,800	839,300	683,500		

*Total of column n and 4 1,522,800.

Source :- 'A Review the Industrial and Economic Situation of Assam', Government of Assam, Shillong.

TABLE 1 (contd.)

Average air dry weight of bamboos per ton-

(i)	Full length air dry	..	100 Kako to a ton
(ii)	,, ,, ,,		400 Muli ,, ,,
(iii)	,, ,, ,,	..	400 Dalu ,, ,,
(iv)	,, ,, ,,	150 Hill Jati	,, ,,
(v)	,, ,, ,,	200 Waithai	,, ,,
(vi)	,, ,, ,,	500 to 600 Torai	,, ,,

Cane : Various cottage and small-scale industries in Assam are dependent on the supply of different kinds of cane and reeds. It is observed that generally 3 species of cane are exploited inc ommercial quantities—Jati (Calamus tenuis), Tita (Calamus leptesadix) and Lejai (Calamus floribundus). Some less important qualities like Sundi (Calamus garuba) and Raidang (Calamus flagellum) are also extracted.

Cane is also found in abundance almost throughout the state. Generally, the artisans obtain their supply of cane from the merchants or 'mahaldars'[1] except in the case of big firms which are occasionally found to be also 'mahaldars' of cane. Thus some of the bigger firms obtain their supply of raw materials at much lower costs than the others.

The following table shows the average annual recorded production of cane in different forest divisions of Assam.

*TABLE 2

Name of division	Annual outturn in bundles	Remarks
1. Lakhimpur	61,000	(i) Average length of a bundle is 3,200 Rft.
2. Sibsagar	26,200	
3. Nowgong	43,000	(ii) Mainly tita, jati & lejai species.
4. Darrang	69,000	
5. Dhansiri Valley	21,200	
6. Cachar	3,500	
	223,900	

[1]The rights to exploit the forest cane'mahals' are auctioned by the Forest Deptt. and a person holding such right is known as a 'mahaldar' of cane.

*(Source :- 'A Review of Industrial and Economic Situation of Assam'—published by the Directorate of Industries, Government of Assam, Shillong).

There are a few more varieties of bamboo and cane used for manufacturing of different products. A kind of muli bamboo locally known as 'muli bazail' is used for making umbrella handles. Two other varieties of bamboo locally known as 'Mrithinga' and 'Bethua' and different varieties of canes locally known as 'sundi', 'barjali'; 'rangajali', 'harua', 'golla' etc. are required for making furniture and baskets.

'Murta' (Clinogyne Dichotoma), a plant of the reed family or patidai, is required for making the famous "Sital Pati" (cool-mat).

For manufacture of 'japis' (umbrella), a type of palm leaves locally known as "tokow pat" are used. A 'fulam japi' (decorated bamboo umbrella) requires, besides palm leaves, coloured wool, cotton, dyed yarn, mica, etc. For colouring and varnishing of their products, the artisans use the following materials—'bhatar phen' (boiled rice juice), 'amrapata' (Hibicus Subdariffa), 'tamarind leaves', mezenta (a kind of chemical dye staff), kalabati chach (lac) resin, methylated spirit, rabi mustafi, etc.

Tools and implements : In the sphere of bamboo and cane works simple and inexpensive tools and implements are extensively used throughout the state. The essential tools required for bamboo craft consist of a 'dao' (bill-hook), a knife and a "jak" ('v' shaped wooden frame). In manufacturing cane products also, 'daos' and knives are mainly used, and only the furniture making establishments use a few saws, hammers, pliers and pincers in addition to daos and knives. A few more hand tools are found to be employed in the manufacture of bamboo umbrella handles. The tools and implements used in this craft are 'bakai kol' (bending frame), 'narum' (sharp and pointed carving blade), files, saws, knives, blow pipes, tongs, oven, etc. It was

reported that spotting equipments with carbide flame arrangement designed to step up production are available for use in the craft. But it was also reported that these could not be purchased for want of funds.

Workshed : The manufacture of cane and bamboo products are mostly carried on outdoors and the craft does not require any housing arrangement particularly in rural areas. It the urban areas, however, it is found that cane products are manufactured in workshops and the majority of the establishments are housed in rented buildings.

Products and process of manufacture : Various types of cane and bamboo products are found in this state. The people of the plains and hills districts of Assam have their own bamboo and cane products with distinctive features and typical designs. The products of the plains districts differ from that of the hills districts in use, shape and design.

It is found that a variety of products like bamboo mats, sital pati, baskets of various sizes and shapes, winnowing trays, sieves, japi or chatta, various types of fishing implements, etc. are manufactured in large numbers in the plains districts of the state. The cane and bamboo products used for domestic purposes are prepared in every nook and corner of the state out of split bamboo and fine flexible canestrips.

A brief description of some cane and bamboo products manufactured for use in domestic purposes is given below :

1. **Chalani (sieve) :** It is woven with fine bamboo slips in a criss-cross way, keeping some open spaces between the different slips as required for different purposes. The 'chalani' is a round-shaped disc-like object and its diameter varies from $1\frac{1}{2}'$ to $3\frac{1}{2}'$. It is used for sieving rice, paddy, tea leaves, etc. and also for washing fish. The cost of one sieve varies from Re. 0.75 P. to Rs. 1.50 P.

2. **Kula (winnowing fan) :** It is prepared out of flat bamboo slips for winnowing purposes in different sizes and shapes. Twilled design is used for a 'kula'. The edge of the 'kula' is made strong by fixing of two sets of one inch wide bamboo pieces wrapped up in flexible cane strips.

3. **Khorahi (small basket) :** Khorahi is made of fine bamboo splits for washing rice, vegetables, fish, etc. It is a small basket-like thing with provision to allow water and dirt to pass out. In the Bengali language, it is termed as 'dhusain' or 'patangi'. The khorahi is woven in plain and square form but is gradually bent in a round form at the time of final stitching by flexible cane slips. The cost of each khorahi ranges from 0.37 P. to Re.1.00.

4. **Dukula** (Assamese), **Tukuri** (Bengali) **Big basket :** The shape of a dukula is exactly the same as that of the khorahi, but the size and process of preparation is a bit different. The required shape of a 'tukuri' is made by bending the bamboo splits forming the warp gradually when the process of weaving with the weft is in operation. The edge is strengthened by fixing two or four flat bamboo strips. The last stage is to stitch the edge along with those flat bamboo strips with some flexible cane slips. The size of a 'dukula' or a 'tukuri' is much bigger than that of the khorahi and is used for carrying as well as keeping paddy, rice, etc. The cost of each 'tukuri' varies from 0.75 P. to Rs. 2.00 according to size and quality.

5. **Dala (bamboo tray) :** Dala is prepared out of flexible bamboo slips in twilled design. The shape of a dala is exactly like a disc with

various sizes for different purposes. The edge around the dala is stitched in the same way as that of the edge of a tukuri or dukula, but the bamboo rims used in the edge of the dala is about 1½". The dala is used specially for rearing silk worms and for winnowing in addition to other domestic purposes. The cost of a dala varies from Re.1.00 to Rs. 3.00 according to various sizes.

6. **Duli (Assamese) : Tali (Bengali)- big basket** : The 'duli' or 'tali' is used for preserving paddy. The process of weaving is almost the same as that of a tukuri but the size of bamboo slips used are more flat and flexible. The dulis are much bigger than the 'tukuri' and the shape is also a bit different. The cost of a duli varies from Rs.4.00 to Rs. 7.00 according to different sizes.

7. **Doon (Assamese), Kathi (Bengali)- measure :** It is prepared in an almost conical shape with fine bamboo strips for measuring rice or paddy. Its holding capacity varies from 2 seers to 3½ seers from place to place. A ring is attached at the bottom to enable it to stand on the ground. These are sold in the local markets @ Re.1.00 to Rs.2.00 but in the Brahmaputra Valley these are not generally sold.

8. **Dhol (big measure) :** The process of preparation of 'dhol' is just like that of the doon. But it is much bigger in size. It is used for measuring paddy only. In the Cachar district it is known as 'pura'. This is not generally bought or sold in the markets.

Fishing implements : There are various kinds of fishing contraptions prepared out of cane and bamboo in different parts of the state. A brief description of some of the fishing implements widely used like polo, Jakai, khalai, dori, chepa, paran, jhuti, hogra, etc. is given below :

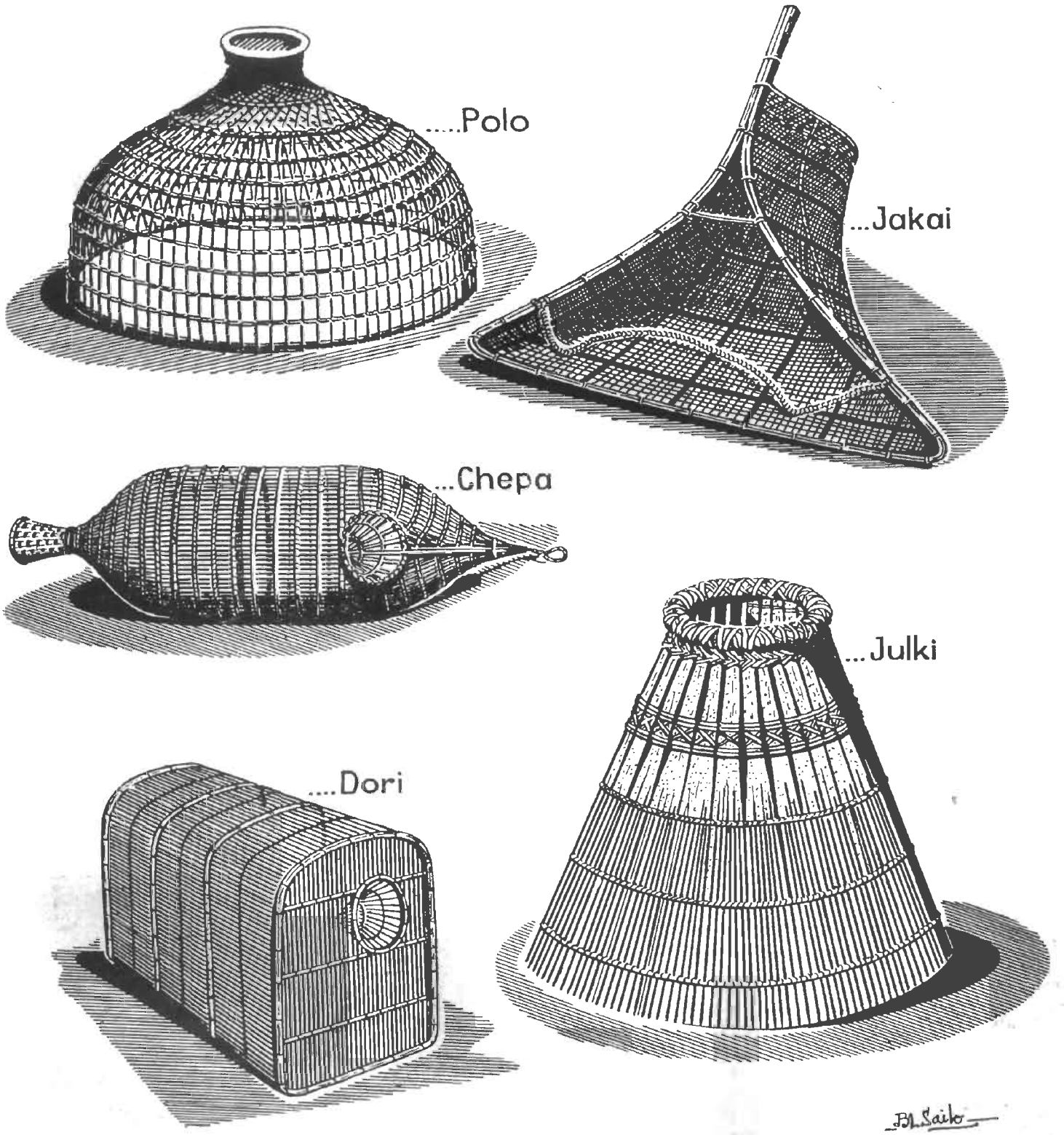
1. **Polo :**—It resembles the shape of a dome with short stem of about 6 ' diameter open at the top. The diameter at the bottom varies from 2 ft. to 3½ ft. and even upto 4 ft. and the height varies from 2 ft. to 3 ft It is prepared out of small bamboo strips fastened with fine and flexible cane slips. Polo is used for fishing in shallow water. The man who uses it holds it by the side of the stem, presses its rim on the mud, then pulls it back and lifts above or up to the level of water and again presses it as before while moving on through water. Whenever any fish is caught, he puts his hand inside through the stem to catch hold of the fish.

The cost of each polo varies from Rs. 3.00 to Rs. 5.00 according to size and quality. 'Julki' is a small polo prepared in the same fashion.

2. **Jakai :-** The 'jakai' is a species of wicker work shovel which is either dragged along the bottom or placed on the water bed to catch the small fishes which take refuge in it when the weed is trampled. It is prepared with bamboo slips which are locally known as 'dai'. 'Jati' bamboo is specially used for making this particular implement.

3. **Khalai :-** The 'khalai' is also prepared with bamboo strips. The strips required for the weft are very long, while those for the warp are short. The 'khalai' is woven in the shape of an earthen 'kalasi' or pitcher. This is used for temporary keeping of fishes during hand-net fishing.

Chepa :- The Chepa is made of some prepared bamboo rods according to required size. These are woven in a roundish fashion with jute strings or soft cane slips. A bamboo-made valve locally known as 'par' (Bengali) and 'kal' (Assamese) is fitted in the



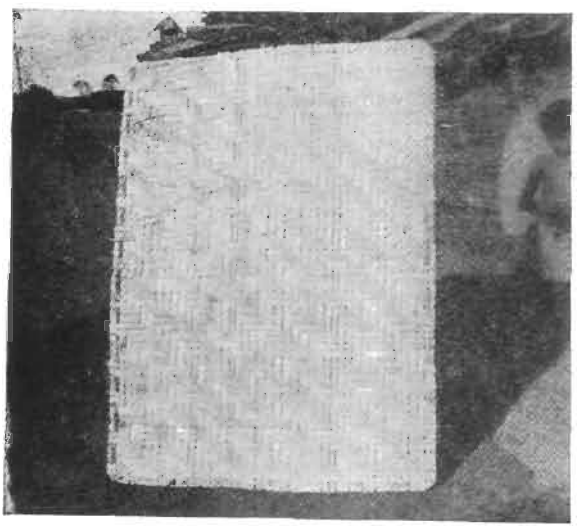
Some fishing implements



Splitting of bamboo for mat making



An old lady engaged in mat weaving



A finished 'dhara' or bamboo mat



Making of bamboo umbrella handle

middle of the chepa to allow the fishes to enter inside with no scope for going out. The cost of a chepa varies from Re. 1.00 to Rs. 2.00 according to size and quality.

Dori :—Dories of various types are manufactured by fishermen in the rural areas of the state. A 'dori' is generally rectangular in shape. It is prepared out of small bamboo strips woven with flexible cane slips. A trap is fitted with a 'dori' in such a way that a bamboo-made screen is prolonged inwards from either side of the oval mouth and the pointed splints of the two sides interlock together. The cost of a 'dori' varies from Re. 1/ to Rs. 6/- according to sizes.

Parans :—These are various cages or basket traps made of bamboo splits used for catching fishes. There are two kinds of 'parans', namely (i) 'uba paran' (vertical cage) and (ii) 'pora paran' (horizontal cage). These are provided with one or two valves or trap-doors through which fishes can be easily trapped.

All these implements mentioned above are generally used in catching fishes in shallow water. Apart from these contrivances, there are some other implements made of bamboo and cane and are used for catching fishes in deep waters. These are locally known as gui, jhuti, dingaru, thupa, hogra, etc.

BAMBOO MATS

Bamboo mats of various types are manufactured on a commercial basis in the districts of Nowgong, Darrang and Cachar. Large scale commercial production is reported from Karimganj sub-division of the Cachar district where mats are locally known as 'dhara', 'jharia' or 'darma', and thousands of people are engaged in this craft. In the districts of Darrang and Nowgong such mats are produced from the dried stalks of various kinds of marshy plants and weeds, while in the district of Cachar, it is produced out of bamboo slips.

Observation on the manufacturing processes as recorded here is taken at a village of Cachar. To start with, long bamboos are cut into several parts according to the desired length of the mats to be woven. Then each part is splitted into several thin pieces, the breadth of such pieces being about 1/8" to 1/16". Splitting is generally done by a 'hatu dao' (small bill hook), which is fixed on a 'jak' ('v' shaped wooden frame). Then the soft portion of such bamboo split is removed with a 'dao', whereupon the flat flexible bamboo strips are obtained for manufacturing mats. Thus with the bamboo slips ready, the artisans commences actual weaving. In weaving bamboo mats, generally the twilled pattern is followed wherein three slips are taken at a time and woven breadthwise one after another and the same process is repeated. As soon as weaving is complete, all the four sides of the mat are twisted a little and tide up with a long bamboo slip in order to frame the outer edges which keeps the woven slips compact.

This craft is found to be carried on by all the members of the artisans—family/men preparing the bamboo slips and women and children weaving the mats. Bamboo mats are extensively used for various purposes like construction of temporary walls and sheds, big pandals, roofing of country boats, dwelling houses, etc. Besides domestic uses, mats are also required by big mills and factories for various uses.

Umbrella handle :—It is only in the district of Cachar where this craft is prac-

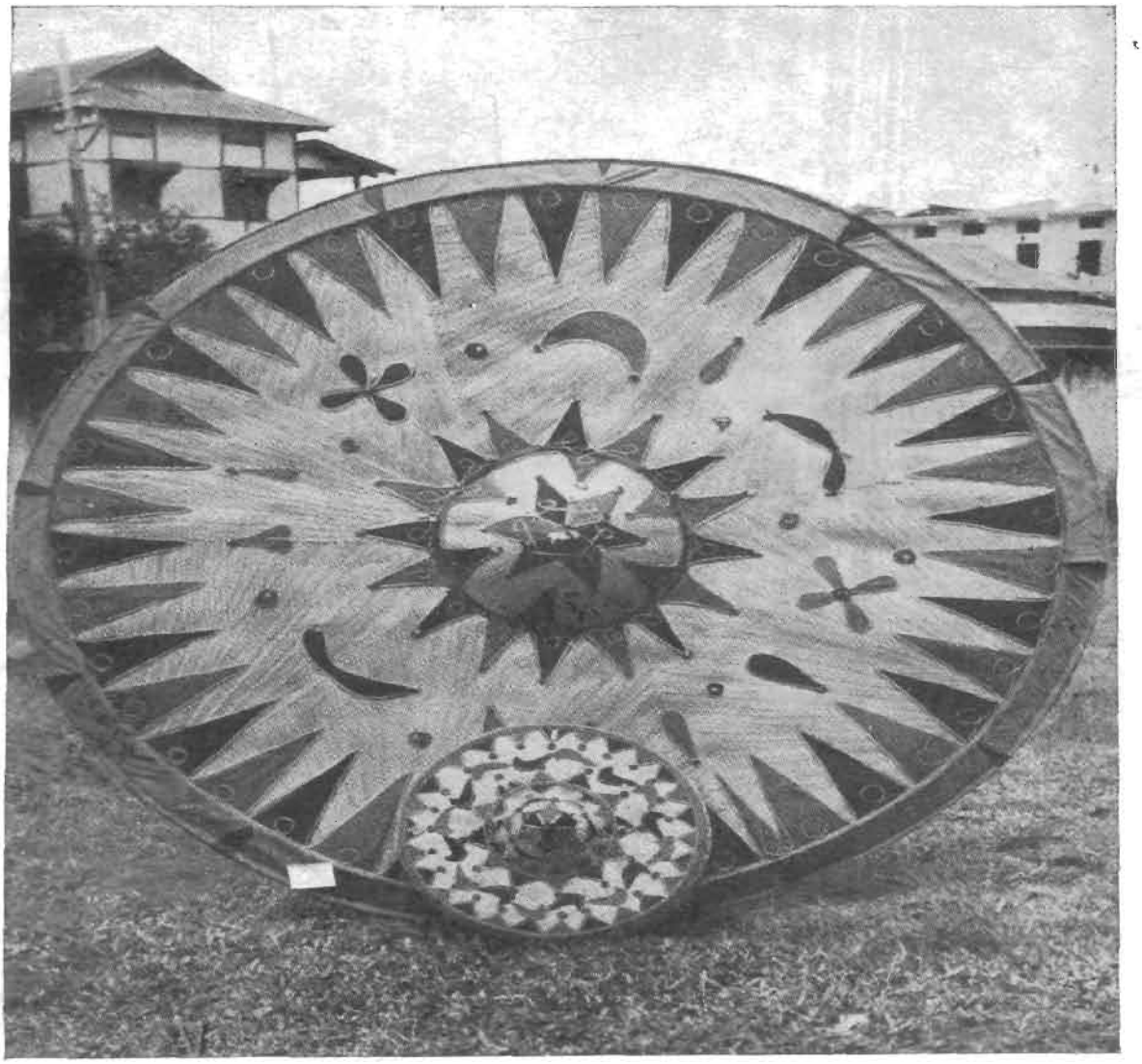
tised on a commercial basis. It is reported that all the craftsmen engaged in manufacturing of umbrella handles are immigrants from East Pakistan after the Partition of India. This craft did not exist in this State in the pre-partition days and most of the bamboo used to be exported to Calcutta. It is, therefore, a new addition to the list of other bamboo products of the State. An umbrella handle-making cooperative society is functioning in the village Paldahar under Ratabari Police Station of the Cachar district. The State Government has recently opened a training centre for imparting training to the artisans on handle-making at Dudhnai in the Goalpara district.

A special variety of muli bamboo locally known as muli-bazail is used for umbrella handles. There are different processes involved in manufacturing umbrella handles. First, the uppermost chamber of the bamboo pieces, sorted out for use, is filled with sand so that it may not crack at the time of bending. The mouth of the top chamber so filled up is then sealed with cowdung. Next comes the bending operation which is generally done with the help of a bending hand-tool locally known as 'bakai-kol'. The upper portion of the bamboo is pressed into the 'bakai-kol' after getting it red-hot while holding tightly the bamboo with a pair of tongs. After a few minutes the bamboo is removed from the 'bakai-kol' and the bent portion of the umbrella handle is tied with a rope and kept in shade for sometime to allow it to cool down. The edge of the 'U' bend is then cut in a uniform manner and the opening is plugged with a thin piece of reed locally known as 'gollar doga' to prevent the filled up sand coming out of the handle. The sand thus remained minimizes the chance of cracking or splitting of the bend. The plugged portion is smoothened with an iron file. Now the rough surface of the umbrella handle is smoothened with a sharp knife, locally known as 'chacha', followed by carving out or scratching various designs on the handle. These are scooped out by swift and deft strokes of a sharp and pointed blade locally known as 'narain' or in some, cases, with a hand tool known as 'rally machine'. A few of the vaious designs depicted are of flowers, creepers, leaves, plants, ring, etc. It is observed that the design work is exclusively performed by the female members of the artisan families. As soon as the design work is over, the surface of the umbrella handle is polished with 'sirish' 'glass paper' followed by the application of varnish and spotting the handles with castor oil flames blown with the help of blow pipes. The handles made as such are then ready to be sent to the market packed up in bundles of grosses.

'Japi or Chhata' (Bamboo leaf head-gear).

Bamboo and leaf headgear is the most indispensable item of the open air workers Such ordinary headgears are produced throughout the State. 'Chhatas' on commercial basis are largely produced in some villages (e. g. Rangpur, Chinipa an, etc) of Cachar district and supplied mainly to neighbouring tea-gardens and some parts of Nowgong district. The products are generally carried on shoulder loads to the nearby 'hats' and sold to consumers in retail or to middle men in bulk.

Many varieties of 'Japis' such as 'halua japi, 'pitha japi', 'sorudoiya japi', 'bordoiya japi' 'cap japi' etc. are produced in the districts of Kamrup, Nowgong. Darrang, Sibsagar and Lakhimpur. Nalbari and its neighbouring villages (such as Kamarkuchi, Mughkuchi, etc.) of Kamrup district deserve special mention in respect of manufacturing of

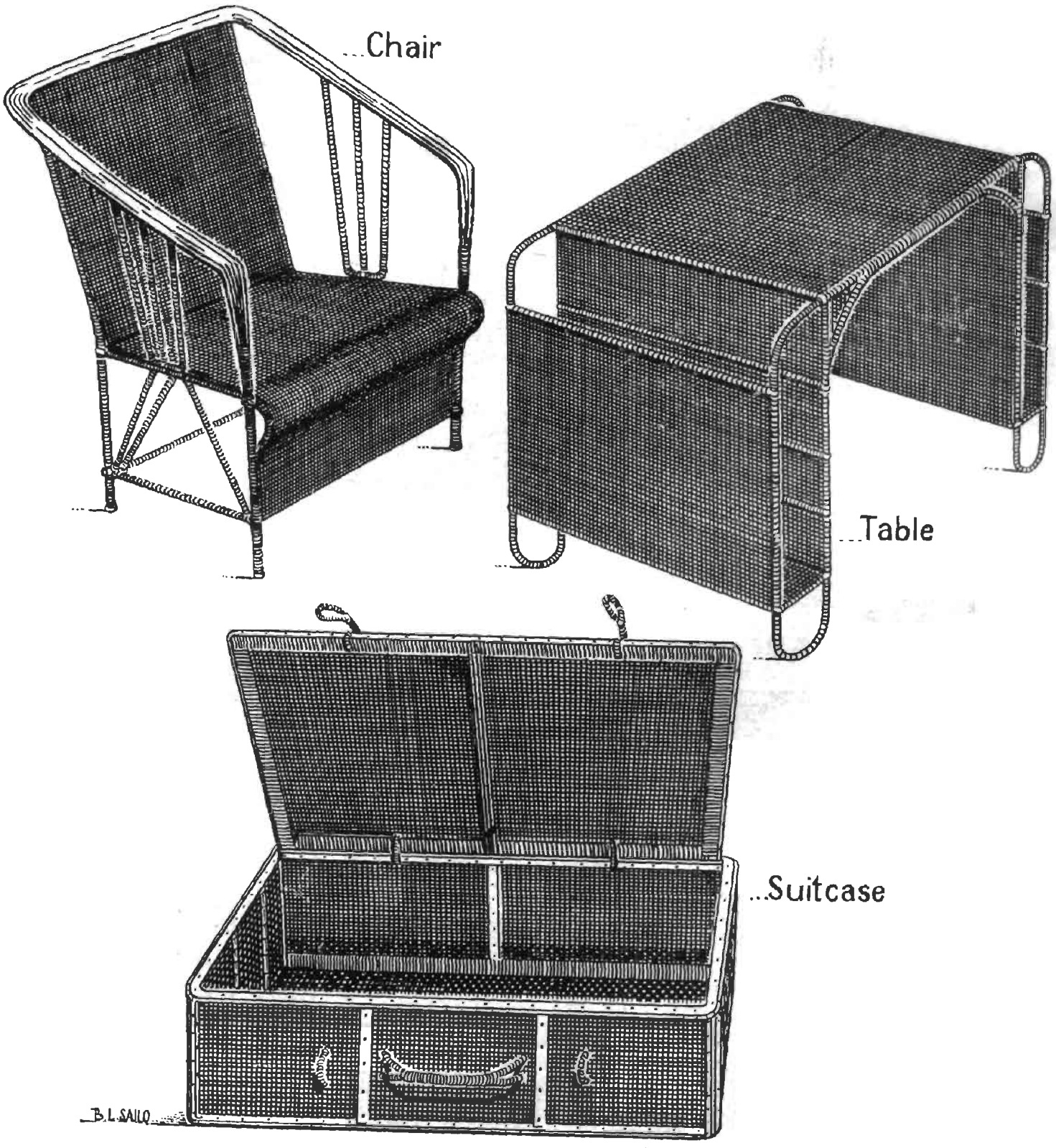


'Fulam Japis' or decorated headgear



Cane furniture makers at work inside an urban factory

CANE PRODUCTS *of* ASSAM



'fulam japis' (decorated bamboo umbrellas). In olden days, this particular type of 'japis' served as headgears for the females of noble and rich families, but now it has become outdated. Production of 'fulam japis' are now only intended to serve as items of drawing-room decorations.

This traditional headgear (japi) is made of strips of bamboo and a special kind of dried palm leaves locally known as 'tokow pat'. The manufacture of ordinary 'japi' does not require any special skill. First of all, the selected bamboos are split into small strips of required sizes. Then the strips are woven in open hexagonal design into a circular disc with a dome in the centre for the head to fit in, putting a few dried 'tokow' leaves (previously cut into required sizes) in between two such discs and finally sewing them securely with yarn and fine strands of cane. And thus the manufacture of ordinary 'japi' is completed.

A 'japi' is more advantageous to the cultivators and other open air workers than the conventional umbrella, because the cultivator after putting it on can tie the strings around his chin leaving his hands free to work in any position—standing, squatting or stooping. 'Japi' can also be called a poor man's umbrella, because of its cheap price. The price of an ordinary 'japi' varies from Re. 1.00 to Rs. 2.00 according to size and quality while ordinary 'chhatas' are sold in between 0.50 to 0.75 paise each in the district of Cachar. The price of 'Fulam japi' (decorated umbrellas) ranges from Rs. 3.00 to Rs. 15.00 according to size and decoration.

Cane furniture

The manufacture of cane furniture, however, calls for a high degree of skill on the part of the workers. Such skill is found to be traditional. The cane furniture makers are mostly displaced persons from East Pakistan. In the manufacture of cane furniture, Cachar district enjoys a special advantage over the other districts of the State in so far as skilled artisans (East Pakistan refugees and local muslims) are concerned. The craft has commercial production in almost all the important urban areas of the State.

The manufacture of cane furniture starts with the preparation of requisite amount of bamboo slips. Canes of various diameters are also reduced into slips of various sizes according to adaptability. The artisans then prepare a rough structure of the furniture by joining the different bamboo parts (previously sized) with the help of nails. In case of round-cane furniture, thin iron rods are used to get the round cane bend to the required shape. The actual weaving or coiling of the structure so made is done with fine slips of flexible cane. The more skilful an artisan is, the finer slips of cane he can use in coiling and plaiting.

To meet the growing needs of the consumers, urban establishments are found to be engaged in the manufacture of various types of cane furniture and other sundry articles, like boxes, murrahs, cradles, office trays, bottle carriers, tiffin baskets, bicycle baskets, waste paper baskets, etc. The cost of such products is higher than other common bamboo products. The cost of one complete sofa set made of cane ranges from Rs. 75.00 to Rs. 175.00 or more according to quality and workmanship.

Cane baskets

Cane baskets of various types are produce in different parts of the State.

Baskets are used for various purposes. These are used mainly for carrying goods, storing grains and keeping valuables. The Kukis, Mikirs and Mizos specially use the baskets for keeping ornaments and clothes with locking arrangements. In the plain districts also a few persons keep their clothes, etc. in cane suitcases. Extensive manufacture of 'plucking baskets' are found in all the plain districts on a commercial basis. Generally the tea planters purchase these baskets in big lots from time to time. So the manufacture of plucking baskets is a monopoly of a few big firms with substantial financial backing. These firms also manufacture various types of baskets used in the carrying of earth, coal etc. It is also reported that these bigger firms obtain their supply of raw materials at much lower costs than the others.

In Assam, baskets are prepared in different designs and by different methods. They may be prepared of both bamboo and cane or cane alone. The different methods of production is confined to the following types, e.g. (1) Plaited or woven work, (2) Wicker work, and (3) Coiled basketry.

(1) Plaited or woven baskets :— Plaited basketry consists essentially of two sets of elements (warp and weft) crossing each other. The plaited baskets are prepared in different designs such as check, twilled, twined, wrapped and hexagonal. Baskets used for keeping clothes and ornaments, cane suitcases, etc. are generally prepared in this method.

(2) Wicker work :—In wicker work, the warp is not pliable, but the weft is pliable and passed alternately over and under the warp. In this method, the warp is kept in a lesser rigidity. Plucking baskets are prepared in this method.

(3) Coiled basketry :- The warp is arranged by cane of sufficient length Before arrangement, such cane is soaked in water for sometime to give it a flexible character. The shape of the basket is preserved by simply binding coiled cane while the process of weaving is in operation. Finally, the edge of the basket is stitched with a thin and flexible cane slip. Plucking baskets, ration baskets, baskets used for carrying earth, stone-chips, coal etc. are manufactured in this method, technically known as 'bee-skip' design.

Sital pati (Cool mat)

The production of sital pati is purely a household industry. Generally, men prepare the cane slips, while women do the weaving work with few exceptions here and there. The highest concentration of artisans pursuing this craft is noticed only in the district of Cachar. The patikars (mat makers) mostly belong to the Muslim community and also to Hindu refugees from East Pakistan belonging to the Mahesya Das, Mali, Nath and Dhobi castes. The villages renowned for the production of sital pati in Cachar are Katakhal, Kaliganj, Karimpur, Bashigram and Sridurgapur. Three co-operative societies are also reported to be functioning in this district by the end of 1961

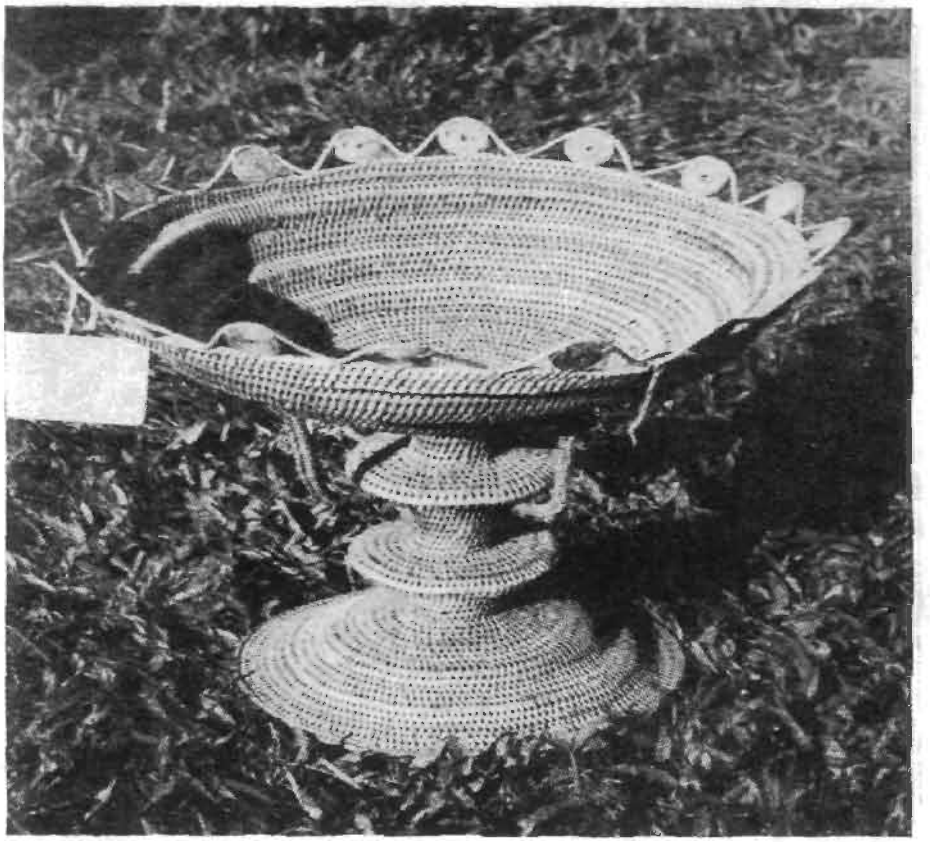
Pati is made from 'murta' (clinogyne dichotoma) a plant of the reed family. Unlike reeds of the 'khag' variety, it has no joints. Like other reeds, it grows on marshy and water logged areas and is found in abundance in choked up tanks and damp hill slopes. It is a periodical crop and yields good 'murta' for 10 to 12 years consecutively. The raw material for this craft is available in different forests of Assam, but it is abundantly available in different forests of the Cachar district.



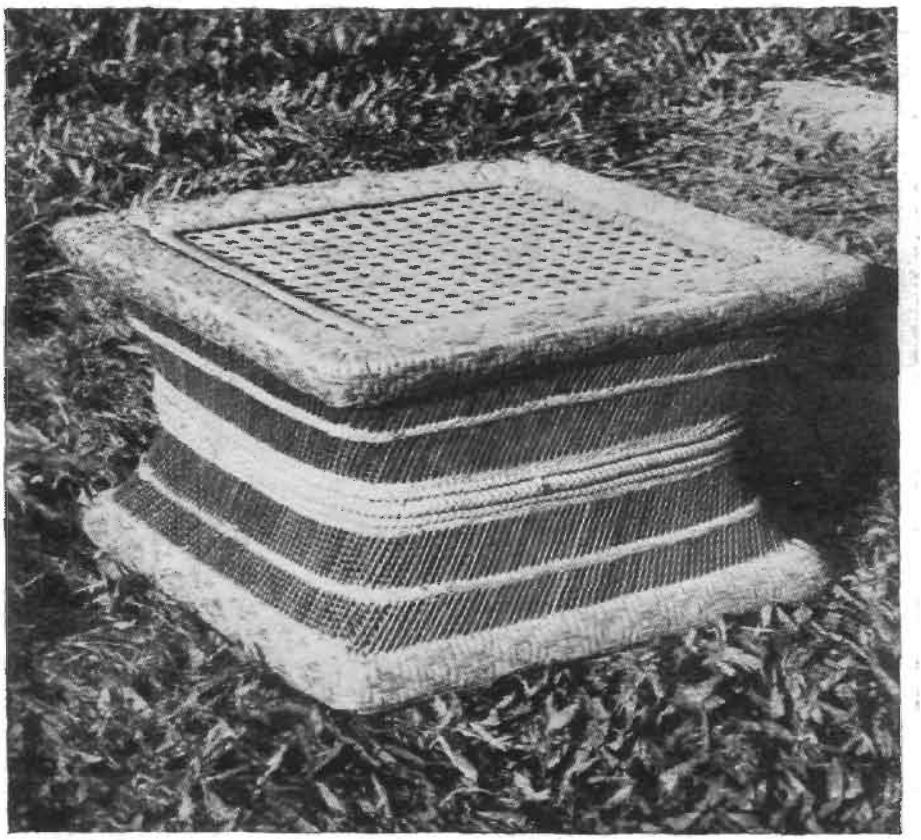
A tea-tray



A fine basket



A 'man-bata' or 'pan-bata'



A 'murrah'



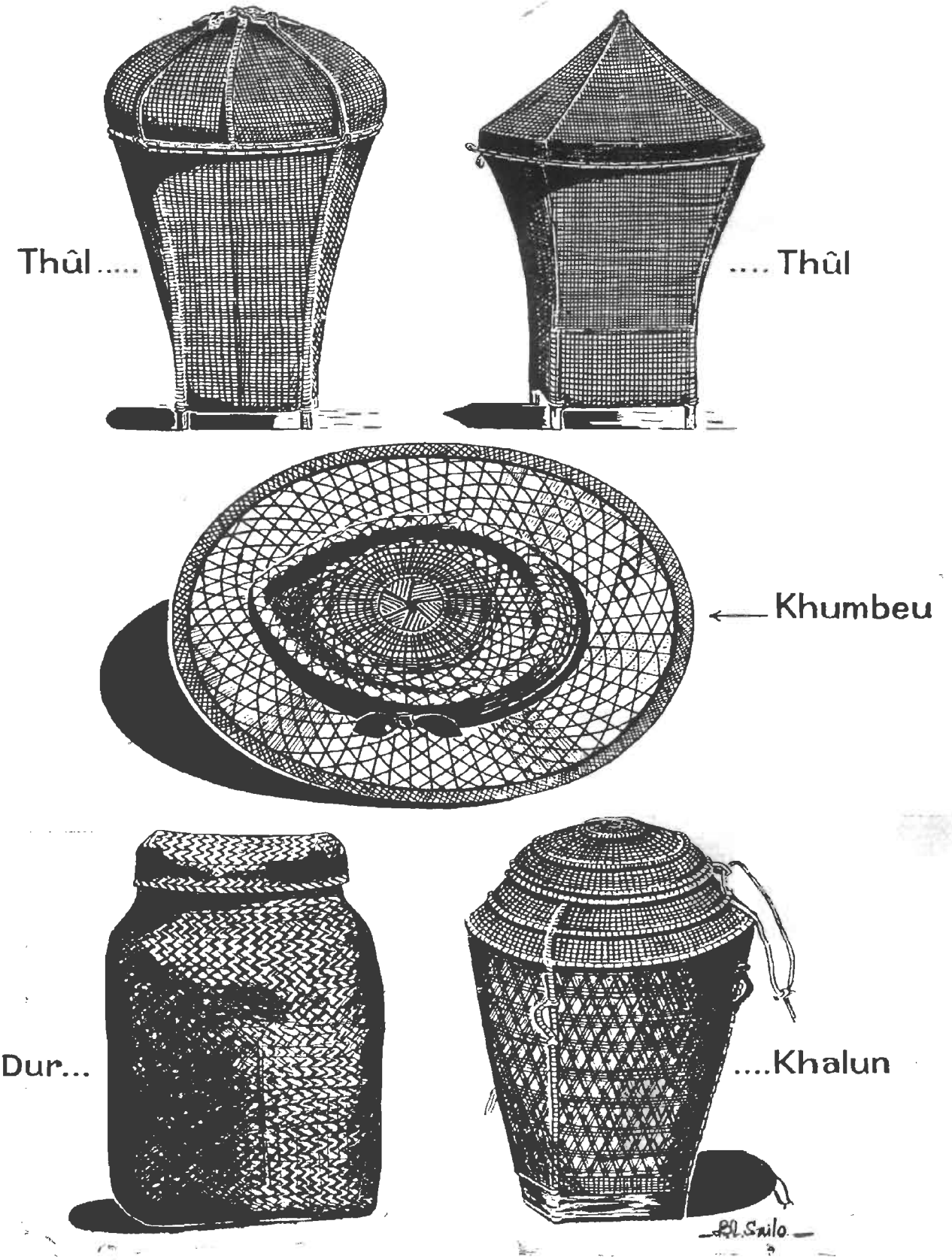
Splitting of cane for 'pati' making



Weaving of cane 'pati' in progress



Cutting the edges of 'pati'—a finishing touch



Objects of interest from the Hills of Assam

The manufacture of 'sital pati' involves many complicated processes. The most difficult part of it is, the preparation of fine cane slips fit for 'pati' weaving. First of all, the canes to be used for obtaining requisite slips are washed in water mixed with some amount of washing soda. After washing, these canes are kept in the open sun for drying. These are then divided into equal halves lengthwise, with the help of a bill-hook. These divided halves are again sub-divided into four splits of equal length and breadth, and the 'boka' (soft inside portion) from slips is chopped out with the help of a 'chip' (chopping tool). This process is locally known as 'aushani' (planing). The next operation known as 'nawkhani' (sizing) in which all the pieces of the splits are sized to equal breadths throughout the whole of their length. This operation is also done with the 'chip'. With the sizing operation complete, the requisite splits for 'sitalpati' weaving is obtained. The splits so prepared are then boiled in water for about 3 to 4 hours. During this process of boiling the colour of the splits undergoes a change and becomes brownish. After boiling, the splits are washed in clean water and left to the open for drying. When the splits are properly dried up, they are kept in stock for somedays. The splits kept as such may be used directly for 'sital pati' weaving or after colouring them, if need be. Weaving of 'pati' is generally done in twill or check pattern with slight variations here and there.

Colouring of the splits is done by indigenous methods. White (ivory) colour is obtained by boiling the splits in water, wherein other ingredients, such as 'bhatar phen' (boiled rice juice), 'amrapata' (hibicus safdariffa) and tamarind leaves are mixed. For black colour, the splits packed into bundles are wrapped up with mango barks and kept under the mud for about 7 days. In order to obtain red colour, the cane splits are boiled in water mixed with 'mezenta' (a kind of chemical dye-stuff).

The different designs produced by the 'patikars' of Assam are the following (local nomenclatures) : 1. Fulpata (flower leaves with creepeers), 2. Dalani, 3. Cup-plate, 4. Taj Mahal, 5. Aeroplane, 6. Tree, 7. Birds, etc.

1. **A 'pati' of ordinary quality is used as:-**

(a) A covering for bedstead, (b) A convenient 'camp chair' in villages to be spread out when there are guests to be received.

2. 'shap' or long 'pati' :-This is used for ceremonial occasions, meetings and musical performances. Sizes vary from 12 x 3, 2¼ x 1½ to 2½ x 1¾ feet.

3. Sital patis :- Very fine pieces are spread out on the bed during summer and form luxuriously cool coverings that easily induce sleep. These are real works of art and command a very high price.

4. Floor mats :- Most of the people use it as a floor covering. For this purpose furniture fitters cut up a long 'pati' and weave the sides into perfectly whole joints, giving the appearance of one single 'pati' exactly fitting the room.

ECONOMICS OF THE INDUSTRY

Cost of Production : It is not possible to cover the whole range of production for analysing the cost of production of various bamboo and cane products and hence a few of the items are discussed below :

TABLE No. 3

STATEMENT SHOWING THE COST OF PRODUCTION OF BAMBOO MATS (DHARA) AS IN 1961

Sl. No.	Size of the bamboo mats	Cost of raw materials per 100 pieces (in Rs.)	Labour cost including weaving and preparing of bamboo slips per 100 nos. (in Rs.)	Cost of production per 100 pieces (in Rs.)	Sale price of 100 pieces at the producing centre	Profit earned by the local agents on 100 pieces (in Rs.)	Transportation costs of 100 pieces of mats to the main marketing centres (in Rs.)	Sale price of 100 pieces at main market (in Rs.)	Remarks
1	2	3	4	5	6	7	8	9	10
1.	4½′ 3′	7.50	10.50	18.00	18.00	4.25	0.75	23.00	Rarely the artisans themselves sell the products in local market except through co-operative.
2.	4′×2½′	7.00	9.00	16.00	16.00	4.00	0.75	20.75	
3.	3¾′ 2¼′	6.50	8.50	15.00	15.00	3.25	0.75	19.00	
4.	6′×4′	11.00	16.00	27.00	27.00	6.00	1.50	34.50	
5.	9′×6′ (dam)	20.00	55·00	75.00	75.00	..	.	75.00	This type of mats is generally sold in local market.

Source : Field Investigation

TABLE No. 4

STATEMENT SHOWING THE COST OF PRODUCTION OF BAMBOO UMBRELLA HANDLES

(figs. in Rs.)

Cost and collection charge of bamboo sticks (per 1,000 pieces)	Cost of subsidiary materials such as varnish, charcoal, sand etc. (for 1,000 pieces)	Labour charges for 1,000 pieces covering the different stages of production	Cost price of 1,000 pieces at producing centre
1	2	3	4
30.00	37.00	73.00	140.00

Source : Field Investigation

TABLE No. 5

SITAL PATIS- -PLAIN DESIGN (WHITE)

STATEMENT RELATING TO STANDARD PRICE OF VARIOUS SITAL PATIS AS PER QUALITY AND DESIGN

Qualities of sital patis	Specification of the size of the split in breadth	Standard size of sital patis	Average cost of raw materials (approx.) including the main basic raw materials, murta plants, @ Rs.2/- per 'pan' (80 nos.)	Labour charges (approx.) per day @ 8 hours work per day	Time required for preparation (approx.)	Actual manufacturing cost (col.5 6- -4)	Percentage of profit @Rs.6¼ %	Actual purchasing cost from the producers (total of cols. 7 & 8)	Remarks
1	2	3	4	5	6	7	8	9	10
Superfine	1.25 mm.	5 3½ cubits (2 5146 6002 meters)	8.00	2.00	20.0 days	48·00	3.00	51.00	of the same size
Fine	1.5 mm.		7.00	2.00	8.5 days	24.00	1 50	25.50	
Medium	1.75 mm.		7.00	2.00	5.0 days	17.00	1.06	18.06	
Ordinary 1	12 mm.		6.00	2.00	4.5 days	15.00	0.94	15.94	
-do- 2	12 mm.		5.00	2.00	2.5 days	10.00	0.62	10.62	
Rough	3 mm. or above		3.00	2.00	1 5 days	6.00	0.37	6.37	
Superfine	1.25 mm.	5 3 cubits (2.2860 1.3716 meters)	7.00	2.00	15.5 days	38.00	2.37	40.37	of the same size
Fine	1 mm.		6.00	2.00	6.5 days	19.00	1.19	20.19	
Medium	1.75 mm.		6.00	2.00	4 days	14.00	0.87	14.87	
Ordinary 1	2 mm.		5.00	2.00	3 days	11.00	0.69	11.69	
-do- 2	2.5 mm.		4.00	2.00	2 days	8.00	0.50	8.50	
Rough	3 mm.		2.50	2.00	1 day	4.50	0.28	4.78	
Superfine	1 mm.	4½ 2½ cubits (1.9812 1.2573 meters)	6.00	2.00	13 days	32.00	2.00	34.00	of the same size
Fine	1 5 mm.		5.00	2.00	5.5 days	16.00	1.00	17.00	
Medium	1 75 mm.		5.00	2.00	3.5 days	12.00	0.75	12.75	
Ordinary 1	2 mm.		4.00	2.00	2.5 days	9.00	0.56	9.56	
-do- 2	2.5 mm.		3.00	2.00	2 days	7.00	0.44	7.44	
Rough	3 mm. or above		1.75	2.00	1 day	3.75	0.25	4.00	

Source : Office of the Sital pati Instructor, Karimganj, Cachar.

TABLE No. 6

SITAL PATIS—ORNAMENTAL DESIGN (WHITE)

STATEMENT RELATING TO STANDARD PRICE OF VARIOUS SITAL PATIS AS PER QUALITY AND DESIGN

Qualities of sital patis	Specification of the size of the split in breadth	Standard size of sital patis	Average cost of raw materials (approx.) including the main basic raw materials, murta plants @ Rs.3.00 per 'pan' (80 nos.)	Labour charge (approx.) per day@8 hours work per day	Time required for preparation (approx.)	Actual Manufacturing cost (cols. 5×6÷4)	Percentage of profit @Rs.6.00	Actual purchasing cost from the producers (total of cols. 7 & 8)	Remarks
1	2	3	4	5	6	7	8	9	10
Superfine	1.25 mm.	5½×3½ cubits (2.5146×1.6002 Meters)	8.00	2.00	23.5 days	55.00	3.44	58.44	
Fine	1.5 mm		7.00	2.00	12 days	31.00	1.94	32.94	
Medium	1.75 mm		7.00	2.00	6.5 days	20.00	1.25	21.25	
Ordinary 1	2 mm		6.50	2.00	5.5 days	17.50	1.12	18.62	
-do- 2	2.5 mm		5.00	2.00	3.5 days	12.00	0.75	12.75	
Rough	3 mm.or above		5.00	2.00	3.0 days	11.00	0.59	11.59	

Source : Office of the Sital pati Instructor, Karimganj.

From the above tables, it may be seen that labour charges constitute the major portion in the cost of production of cane and bamboo articles. The proportion of the cost of raw materials to the labour charges for the manufacture of 100 pieces of bamboo mats (4½' x 3') is found to be 1 : 1. 4. Similarly for 1,000 pieces of bamboo umbrella handles, the proportion is 1: 1. 09. In the manufacture of ornamental superfine sital pati, the labour charge is still higher and the average proportion of the cost of the raw materials and that of the labour charges comes up to 1: 6.9.

Demand and sale :—It is not possible to give here a complete picture about the demand and sale position of various items of cane and bamboo products of the State. Let us however try to ascertain it to a limited extent by taking into account a single craft, namely, 'sital pati'. The following trading account of the Kaliganj Patikar Co-operative Society Ltd., available at the moment, will give us an impression about the increasing volume of business done in such products every year.

TABLE No. 7

PRODUCTION AND SALE OF SITAL PATIS

Year	Total value of production	Sale proceeds	Stock value of articles at the close of the year	Gross profit
1	2	3	4	5
1958-59	Rs. 647.19	Rs. 298.62	Rs. 370.56	Rs. 21.99
1959-60	Rs. 8,366.02	Rs. 7,437.55	Rs. 1,538.94	Rs. 239.91
1960-61	Rs. 10,238.68	Rs. 10,374.56	Rs. 1,903.79	Rs. 500.73

Source—Field Investigation

Marketing :—A variety of bamboo products like mats for ceilings, baskets of various sizes and shapes, winnowing trays, sieves, various types of fishing contraptions, etc. are manufactured by the artisans in this industry. The products are mainly sold in the local markets. Sale of bamboo products in markets outside the State was reported only from the district of Cachar. It is found that large quantities of bamboo mats are exported from the district of Cachar to Calcutta. In other districts the scale of production is very low and the sale of products is invariably confined to local markets. It is also found that contractors or middlemen play a vital part

in the marketing of bamboo mats and thereby deprive the producers of a certain amount of profit.

It is also reported that the sale of cane products is mostly direct from the producers to the consumers and middlemen known as contractors appointed by the tea gardens or big firms are only occasionally found to be engaged in purchasing the articles like baskets from the producers and supplying the same to the consumers. The producers have no marketing organisation. As such the sale of products is mostly confined to local markets and at best to markets within the State. Some of the bigger units are however found to advertise their products and push the sales through canvassing and such establishments are able to sell their products even in markets outside the State. A large quantity of 'sital pati' is also exported to the Calcutta market from the district of Cachar. State Government Emporia are also trying to popularise a few artistic and decorative cane and bamboo products outside the State.

Working force :—From the table in Appendix A, it may be seen that the total number of bamboo and cane workers in the plains districts of Assam comes to 7,922, with highest number of artisans, i. e., 2,478 in the district of Cachar against 1,005 persons in the hill districts.

Trend of prices of products :—It appears from the table in Appendix D that the prices of all the products of bamboo and cane have registered increase over prices prevailing in 1939. Almost all the districts have shown an increase in the price level of the products ranging from 200 to 400 per cent over 1939 prices. It is also seen that while most of the products have recorded an increase over pre-war prices, the prices remained more or less stationary over 1953 and 1954. Only in a few cases, slight fall in prices have been registered. One of the main reasons for difference of prices in rural and urban areas is known to be the lack of transport facilities in the producing centres in the interior of the State.

Assistance programme :—In order to encourage the growth and development of this widely practised craft, the State Cottage Industries Department has taken various steps. During the Second Plan period, the department has opened several training centres in different districts to impart training to cane and bamboo workers of the State in the manufacture of artistic cane and bamboo products. During the said period, Government have also started two cane and bamboo mills, one at Gauhati and the other at Karimganj, to increase production. Government is also rendering financial assistance to deserving artisans and co-operative societies in the shape of loans and subsidies.

Profit : In order to get a rough idea about the percentage of profit earned by producers engaged in this craft, an example of bamboo mats, according to Karimganj market of Cachar district is given below :

TABLE No. 8

Price of mats purchased by dealers at Karimganj	Distribution of the sum of Rs.100.00 (Sale price)			
	Cost of materials	Wage of producers	Profit of agents	Profit of main dealers
1	2	3	4	5
Rs. 100.00 (100%)	Rs. 58.00 (58%)	Rs. 21.00 (21%)	Rs. 10.50 (10.5%)	Rs. 10.50 (10.5%)

Source : Field Investigation

Scope for development :— The main raw materials for the industry, i.e. bamboo and cane of various types are abundantly available throughout the State. It requires little or no investment and can be persued

by anybody and everybody of a household as a subsidiary occupation. As such, the industry has considerable scope for development and various new products suited to modern tastes can be manufactured out of bamboo and cane. With a little training in artistic designs and the use of colours, the artisans will be able to manufacture products like cane and bamboo baskets for bicycles, bamboo screens, electric lamp shades, cradles, boxes, bottle carriers, tiffin baskets, various fancy products, etc. which are likely to find markets even outside the State.

APPENDIX A

STATEMENT SHOWING THE NUMBER OF CANE AND BAMBOO WORKERS OF THE STATE ENGAGED IN HOUSEHOLD AND NON-HOUSEHOLD INDUSTRY

Name of the district	Total	Males	Famales
1. Goalpara	1,301	906	395
2. Kamrup	1,475	1,153	322
3. Darrang	849	680	169
4. Lakhimpur	1,133	973	160
5. Nowgong	208	194	14
6. Sibsagar	478	411	67
7. Cachar	2,478	1,625	853
8. Garo Hills	11	10	1
9. United Khasi & Jaintia Hills	627	179	448
10. United Mikir & North Cachar Hills	222	154	68
11. Mizo Hills	145	145	.
	8,927	6,430	2,497

Source - Census 1961 - Code No. 228 of Indian Standard Industrial Classification.

APPENDIX B

STATEMENT OF CANE AND BAMBOO CO-OPERATIVES OF ASSAM

Name of the district	No. of Co-operative Societies	No. of paid-up shares	Amount of paid-up share capital (in rupees)	Total number of workers	Extent of business done during 1961-62 (in rupees)	Amount of loans and grants received from different sources (in rupees)	Remarks
1	2	3	4	5	6	7	8
1. Goalpara	1	19	210.00	19	8,554.00	2,500.00	
2. Kamrup	Ni	..	..	..		..	
3. Darrang	Nil	..	.	.	.	..	
4. Lakhimpur	6	200	7,383.00	140	N.A.	N.A.	
5. Nowgong	Nil	.		.		..	
6. Sibsagar	2	N.A.	N.A.	N.A.	N.A.	N.A.	
7. Cachar	10	376	2,820.00	256	*53,105.00	34,440.00	*The amount stated is the extent of business done by six societies only.
8. Garo Hills	2	39	330.00	39	864.00	1,700.00	
9. United Khasi & Jaintia Hills	2	69	2,325.00	69	N.A.	2,750.00	
10. United Mikir & North Cachar Hills	1	22	965.00	22	11,547.00	1,500.00	
11. Mizo Hills	1	15	150.00	15	350.00	Nil	
	25	740	14,183.00	560	74,420.00	42,890.00	

N.A.- Stands for figures not available.

* Source : Registrar of Co-operative Societies, Government of Assam.

APPENDIX C

TREND OF PRICES OF RAW MATERIALS IN ASSAM

Name of the district	Description of raw materials	Unit	Prices (in Rupees) during 1939	1953	1954	1961
1. Goalpara	Bamboo	one	0-3-0	0-8-0	0-8-0	0.65
2. Kamrup	Bamboo	One	0-2-0	0-6-0	0-7-0	0.75
3. Darrang	Bamboo Cane	One Kup	0-2-0 0-1-6	0-10-0 1-0-0	0-10-0 0-12-0	0.50 1.25
4. Nowgong	Bamboo	One	0-2-0	0-8-0	0-8-0	1.00
5. Cachar	Muli bamboo Cane	100 nos. Mura (84 nos.)	4-10-0 1-12-0	14-0-0 5-8-0	14-0-0 6-0-0	15.00 8.00
6. Lakhimpur	Bamboo Cane	One Kup	0-2-0 0-2-6	0-6-0 0-7-0	0-6-0 0-7-0	1.00 0.50
7. Sibsagar	Bamboo Cane	One Kup	0-3-0 0-1-6	0-8-0 0-6-0	0-8-0 0-6-0	N.A. N.A.
8. United Khasi & Jaintia Hills	Bamboo Cane	One Kup	0-2-0 0-4-0	0-8-0 2-0-0	0-8-0 2-0-0	1.00 2.00
9. United Mikir & North Cachar Hills	Beti (for pati)	Bundle	20-0-0	30-0-0	32-0-0	N.A.
10. Garo Hills	Bamboo	One	..	..	0-12-0 to 1-0-0	1.25 to 1.50
11. Mizo Hills	Bamboo	100 nos.	..	..	..	15.00

N.A. -Not Available.

Sources : Department of Economics and Statistics and Department of Industries, Government of Assam.

APPENDIX D

TREND OF PRICES OF CANE AND BAMBOO

Name of the district	Products	Unit	Prices (in rupees) during 1939		1953		1954		1961	
			Urban	Rural	Urban	Rural	Urban	Rural	Urban	Rural
1. Goalpara	'doli'	One	.	0-10-0	.	2-0-0	..	2-0-0	3.00	2.00
	Mat	,,	..	0-4-0	.	0-8-0	..	0-8-0	0.87	0.76
2. Kamrup	'japi'	,,	..	0-3-6	.	0-9-0	..	0-9-0	1.50	1.50
	Cane basket	,,	.	..	..	..	.	..	4.00	3.00
3. Darrang	Cane basket	,,		0-5-0	..	1-0-0	.	1-0-0	N.A.	N.A.
	Cane chair	,,	2-0-0	..	13-0-0	..	12-0-0	..	8.00	8.00
4. Nowgong	'ugha'	Set	.	0-8-0		2-8-0	..	2-8-0	..	2.50
	'japi'	One	.	0-10-0		1-0-0	..	1-0-0	2.00	1.00
5. Cachar	Mat	100 nos.	..	12-0-0	.	15-0-0	..	14-0-0	20.00	18.00
	Cane basket	One	..	1-0-0	..	2-4-0	..	2-4-0	3.00	2.75
6. Lakhimpur	Mat	,,	.	0-3-0	..	0-10-0	.	0-10-0	1.00	0.75
	Cane chair	,,	3-0-0	..	10-0-0	.	12-0-0	..	12.00	10.00
7. Sibsagar	Mat	,,	0-10-0	0-8-0	2-0-0	1-10-0	2-0-0	1-10-0	N.A.	N. A.
	Cane chair	,,	2-4-0	2-4-0	7-0-0	7-0-0	7-0-0	7-0-0	N.A.	N. A.
8. United Khasi & Jaintia Hills	Cane basket	,,	.	0-4-0	.	1-0-0	..	1-0-0	3.00	2.50
	Cane 'murrah'	,,		0-4-0	..	2-0-0	..	1-8-0	4.00	2.50
9. United Mikir & North Cachar Hills	Cane 'pati'	bundle	.	45-0-0	.	65-0-0	..	70-0-0	N.A.	N. A.
10. Garo Hills	Cane basket	One	..	.		..	2-0-0	1-12-0	2.50	2.00
	Bamboo 'dala'	,,	..	..		.	1-8-0	1-8-0	2-00	1.75
11. Mizo Hills	Cane chair	,,	..	..	.	13-0-0	.	13-0-0	15-00	13.00
	Cane basket (thul)	,,	..	..	..	15-0-0	..	15-0-0	15.00	10.00

Sources : Department of Economics and Statistics and Department of Industries, Government of Assam
Note :—N. A. stands for 'Not Available'.

GLOSSARY

Local term		Botanical or English equivalent
1. Muli (bamboo)	..	Malocanna bambusoides
2 Jati ,,	..	Oxytenanthera parvifola
3. Kako ,,	..	Dendrocalamus Hamilton-ii
4. Dalu ,,	.	Teinostachyum Dalloa
5. Jati (cane)		Calamus Tenius
6. Tita	..	Calamus Leptesadix
7. Lejai ,,	..	Calamus Floribundus
8. Sundi ,,	..	Calamus Garuba
9. Raidung,,		Calamus Flagellum
10. Murta ,,	..	Marantha or Clinogyne Dichotoma
11. Chalani	..	Sieve
12. Kula		Winnowing fan
13. Khorahi or dhusain	..	Small bamboo basket
14. Dukula or Tukuri	..	Big bamboo basket
15. Dala	..	Bamboo tray
16. Doon or kathi		Measure
17. Dhara, dharia or darma	.	Bamboo mat
18. Japi or chhata	.	Bamboo and leaf head-gear
19. Sital pati	.	Cool mat
20. Patikar	..	Cool mat maker

BIBLIOGRAPHY

1. History of Civilization of the People of Assam, by Dr. P. C. Choudhury, 1959
2. A review of the Industrial and Economic Situation of Assam, July, 1958- -Published by the Directorate of Industries, Govt. of Assam, Shillong.
3. Report on a Survey of Cottage Industries in Assam, 1958– Published by the Department of Economics and Statistics, Assam, Shillong.
4. Report on the Census of Assam, 1891 (pp.210-270)

APPENDIX

TRADE-WISE (CRAFTS) ANALYSIS AND NUMBER OF

SOURCE : DIRECTORATE OF NATIONAL EMPLOYMENT

Name of the Institution, address and year of foundation	Managing authority and type of institution Central/State/ Private/Aided/ Unaided	Duration of the course (years)		Requirements of Admission		Method of Admission by Competition, by Interview, by Marks obtained
		Institutional Training	Practical Training	Age Limit	Minimum educational requirements	
1	2	3	4	5	6	7
CARPENTRY AND WOODWORKS						
1. Industrial Training Institute, Jorhat, Sibsagar, (1947).	State	1½ years	6 months	16-25 yrs.	Class VIII	Interview
2. Don Bosco Technical School, Shillong, Assam (1922)	Private Unaided	5 years	Including	12 years	Class V	-do-
3. Silghat Industrial and Technical School, Silghat, Nowgong (1920)	-do-	3 years	-do-	14 years	M. E.	-do-
4. Dhireswar Dev Goswami Technical School, Majgaon, Chaklaghat, P. O. Nowgong (1948)	-do-	1 year	-do-	16-18 yrs.	Class VI	-do-
5. Industrial Training Institute Nowgong (1948)	State	1½ years	6 months	16-25 yrs.	Class VIII	Interview, Marks obtained
6. Carpentry and Bucket Making Training Centre, Raha, Nowgong	Kopili Development Block	2 years	Including	15 years	-do-	Interview
7. Cottage Industries Training Institute, Kalapahar, Gauhati (1954)	State	3 years	-do-	16-21 yrs.	-do-	-do-
8. Dibrugarh Chah Mazdoor Kalyan Samaj, Khalihamari, Dibrugarh (1956)	Private Aided	2 years	-do-	12-35 yrs.	L. P..	-do-
9. Cottage Industries Training-cum-Production Institute, Kokrajhar, Goalpara	State	1 year	-do-	18-28 yrs.	Class VI-X	-do-
10. Training-cum-Production Centre, Goalpara	-do-	3 years	-do-	-do-	-do-	-do-
11. Vocational Collegiate High School, Jorhat.	Private Aided	1 year	3 months	12 years	M. E.	Interview
12. Industrial Training Institute, Rehabari, Gauhati.	State	1½ years	6 months	16-25 yrs.	Under-Matric	Competitive Examination
BLACKSMITHY						
1. Industrial Training Institute, Nowgong, (1943)	State	1½ years	6 months	16-25 yrs.	Class VIII	-do-
2. Industrial Training Institute, Jorhat, (1947)	-do-	-do-	-do-	-do-	-do-	-do-
3. Industrial Training Institute, Rehabari, Gauhati	-do-	-do-	-do-	-do-	-do-	Competition
4. Dibrugarh Chah Mazdoor-Kalyan Samaj, Khalihamari, Dibrugarh (1956)	Private Aided	2 years	Including	12-35 yrs.	L. P.	Interview
5. Silghat Industrial and Technical Institute, Silghat, Nowgong (1920)	Private Unaided	3 years	-do-	14 years	M. E.	-do-

I

TRAINING INSTITUTES, INTAKE CAPACITY, ETC.

SERVICE, ASSAM, SHILLONG.

Nature and titles of the Award, Degree/ Diploma/Certificate	Whether Degree/ Certificate is recognised by the Government	Authority conferring Award	Tuition fee charged	Rate of stipend	Date/month of commencement of course	Sanctioned intake capacity for 1st year course session	Remarks
8	9	10	11	12	13	14	15
Certificate	Yes	S. C. V. T.	Nil	Rs. 35/-p.m. to 33⅓% of admitted students	February, August, November/May	176	**Abbreviation :-** S/C-Scheduled Caste,
-do-	No	School Authority	Rs. 10/	Nil	15th February	15	S/T-Scheduled Tribe
Diploma	Yes	Subdivisional Officer, Nowgong	Free	Nil	April	15	S. C. V. T.- State Council for Vocational Training
Certificate	No	Cottage Industries	Free	Nil	August	30	N. C. V. T.- National Council for Vocational Training
-do-	Yes	Director of Technical Education	Nil	Competitive 33⅓% of the admitted students	August/November	15	N. A.-Not Available
-do-	Yes	B. D. O., Kopili	Nil	Nil	October	10	
-do-	-do-	Director of Industries	Nil	Rs. 50/-for S/C & S/T, Rs. 42/- & Rs. 30/-for general	October	16	
-do-	-do-	President, Chah Mazdoor	Nil	Rs. 20/-for S/T & S/C	April	4	
-do-	-do-	Director of Cottage Industries, Assam	Nil	Rs. 30/-p. m.	Not fixed	N. A.	
-do-	-do-	-do-	Nil	Rs. 42/- p. m.	October	23	
Certificate	-do-	S. C. V. T	Rs. 5/	Rs. 10/- p. m.	January, April, July	13	
-do-	-do-	-do-	Free	Nil	Not fixed	16	
Certificate	-do-	Director of Technical Education, Assam	Nil	At 33⅓%	August/November	15	
-do-	-do-	S. C. V. T.	Nil	Rs. 25/- p. m. for 33⅓%	February August, November/May	176	
-do-	-do-	D. G. E. & T., Govt. of India	Nil	33⅓%	N. A.	16	
-do-	-do-	President of Chah Mazdoor	Nil	Rs. 20/- p. m. for S/T & S/C	April	4	
Diploma	-do-	S. D. O., Nowgong	Nil	Nil	-do-	5	

APPENDIX

TRADE-WISE (CRAFTS) ANALYSIS AND NUMBER OF

SOURCE :— DIRECTORATE OF NATIONAL EMPLOYMENT

Name of the Institution. address and year of foundation	Managing authority and type of institution Central/State/ Private/Aided/ Unaided	Duration of the course (years)		Requirements of Admission		Method of Admission by Competition, by Interview, by Marks obtained
		Institutional Training	Practical Training	Age Limit	Minimum educational requirements	
1	2	3	4	5	6	7
6. Cottage Industries Training Institute, Kalapahar, Gauhati, (1954)	State	2 years	Including	16-21 yrs.	Class VIII	-do-
7. Dhireswar Dev Goswami Technical School, Majgaon, Chaklaghat, Nowgong	Private Aided	1 year	-do-	16-18 yrs.	Class V	-do-
WEAVING						
1. Girls' Weaving School, Nong-Malki, Shillong (1958)	Private Unaided	N. A.	N. A.	10-15 yrs.	Class VI	-do-
2. Vocational Collegiate High School, Jorhat.	Private Aided	1 year	3 months	12 years	M. E.	-do-
3. Sarupathar Weaving Training Class, Sarupathar, Sibsagar.	State	1 year	Including	-do-	L. P.	-do-
4. Dibrugarh Chah Mazdoor Kalyan Samaj, Khalihamari, Dibrugarh (1956)	Private Aided	2 years	-do-	12-35 yrs.	-do-	-do-
5. Khagarijan Industrial Co-operative Handloom Weaving Training Centre, Nowgong	Private Unaided	1 year	6 months	14 years	M.E. & M.V.	-do-
6. Government Weaving Training Class, Raha, Nowgong (1957)	State	1 year	Including	15-25 yrs.	-do-	-do-
7. Industrial Training Institute Jorhat, (1947)	State	-do-	-do-	16-25 yrs.	Class VIII	-do-
SPINNING						
1. Girls' Weaving School, Nong-Malki, Shillong (1948)	Private Unaided	N. A.	N. A.	15-40 yrs.	Class VI	-do-
2. Government Weaving Training Class, Raha, Nowgong, (1957)	State	1 year	Including	15-25 yrs.	M. E.& M. V.	-do-
BLEACHING AND DYEING						
1. Industrial Training Institute Jorhat, (1947)	-do-	-do-	-do-	16-25 yrs.	Class VIII	-do-
DYEING AND PRINTING						
1. Girls' Weaving School, Nong-Malki, Shillong (1948)	Private Unaided	N. A.	N. A.	15-40 yrs.	Class VI	Interview
2. Government Weaving Training Class, Raha, Nowgong (1957)	State	1 year	Including	15-25 yrs.	M. E. & M. V	-do-
3. Khagarijan Industrial Co-operative Handloom Weaving Training Centre, Nowgong	Private Unaided	1 yrs.	6 months	14 years	-do-	-do-
4. Amguri Commercial Institute, Amguri (1955)	-do-	-do-	Including	16-25 yrs.	Matric	-do-

I (Contd.)

TRAINING INSTITUTES, INTAKE CAPACITY, ETC.

SERVICE, ASSAM, SHILLONG.

Nature and titles of the Award, Degree/ Diploma/Certificate	Whether Degree/ Certificate is recognised by the Government	Authority conferring Award	Tuition fee charged	Rate of stipend	Date, month of commencement of course	Sanctioned intake capacity for 1st year course session	Remarks
8	9	10	11	12	13	14	15
Certificate	-do-	Director of Cottage Industries	Nil	Rs. 50/-p. m. for S/T & S/C and Rs. 42/- & 30/- for general	October	16	
-do-	No	-do-	Nil	N. A.	August	30	
-do-	Yes	N. A.	N. A.	N. A.	March	30	
-do-	-do-	S. C. V. T.,Assam	Rs.5/-p.m	-do-	January/April/ July	13	
-do-	-do-	Director of Sericulture & Weaving Assam	Nil	Rs. 25/- p. m.	March	15	
-do-	-do-	President, Chah Mazdoor	Nil	Nil	April	4	
-do-	-do-	Secretary of the Society	Nil	Rs. 10/- to Rs.15/- p. m.	January	10	
-do-	-do-	Director of Weaving and Sericulture , Assam	Nil	Rs. 25/- p. m.	March	15	
-do-	-do-	N. C. V. T.,Assam	Nil	-do-	August	80	
-do-	-do-	N. A.	Nil	Nil	March	80	
-do-	-do-	Director of Sericulture & Weaving, Assam	Nil	Rs. 25/- p. m.	March	15	
-do-	-do-	N. C. V. T.,Assam	Nil	Rs. 25/- p. m.	August	80	
Certificate	-do-	N. A.	Nil	Rs. 25/- p. m.	August	80	
-do-	-do-	Director of Sericulture & Weaving, Assam	Nil	Rs. 25/- p. m.	March	15	
-do-	No	Secretary of the Centre	Nil	Rs. 10/- to 15/-p.m	January	20	
Diploma	-do-	Principal	Rs.8/- p. m.	Nil	May	50	

APPENDIX

TRADE-WISE (CRAFTS) ANALYSIS AND NUMBER OF

SOURCE :— DIRECTORATE OF NATIONAL EMPLOYMENT

Name of the Institution, address and year of foundation	Managing authority and type of institution Central/State/ Private/Aided/ Unaided	Duration of the course (years)		Requirements of Admission		Method of Admission by Competition, by Interview, by Marks obtained
		Institutional Training	Practical Training	Age Limit	Minimum educational requirements	
1	2	3	4	5	6	7
CANE AND BAMBOO						
1. Cottage Industries Training Institute, Kalapahar Gauhati (1954)	State	2 years	Including	16-21 yrs.	Class VIII	Interview
2. Training-cum-Production Centre, Goalpara.	-do-	1 year	-do-	18-28 yrs.	Class VI	-do-
UMBRELLA HANDLE MAKING						
1. Umbrella Handle Making Centre, Dudhnai, Goalpara	-do-	6 months	-do-	18-28 yrs.	-do-	-do-
POTTERIES						
1. Cottage Industries Training Centre, Kalapahar, Gauhati (1954)	-do-	2 years	-do-	16-21 yrs.	Class VIII	-do-
SERICULTURE TRAINING						
1. Sericulture Training Institute Titabar, Jorhat (1956)	State & Union	2 years (Diploma) 1 year (Certificate)	-do-	Below 25 years	I.Sc for Diploma Matric for Certificate	-do-
CUTTING AND TAILORING						
1. Girls' Weaving School, Nong Malki, Shillong (1948)	Private Unaided	N.A.	N.A.	15-40 yrs.	Class VI	-do-
2. Dhireswar Goswami Technical School, Majgaon, Nowgong (1948)	-do-	1 year	Including	16-18-yrs.	-do-	-do-
3. Government Aided Sibsagar Commercial Institute, Sibsagar (1947)	-do-	-do-	-do-	16 years	Class IV	Application & Marks
4. Vocational Collegiate High School, Jorhat	-do-	-do-	3 months	12 years	M.E. & M.V.	Interview
5. Dibrugarh Chah Mazdoor Kalyan Samaj, Khaliamari, Dibrugarh (1956)	-do-	2 years	Including	12-35 yrs.	L.P.	-do-
6. Tailoring Training Centre, Kopili, Nowgong (1959)	Kopili Development Block	1 year	-do-	15 years	Class VI & M.V.	-do-
7. Government Aided Progressive Commercial Institute, Lakhimpur, (1952)	Private Aided	-do-	-do-	18 years	M.E.	-do-
8. Amguri Commercial Institute Amguri, Sibsagar (1955)	Private Unaided	-do-	-do-	16-25 yrs.	Matric	-do-
9. Usha Tailoring and Embroidery School, Dibrugarh	-do-	-do-	-do-	Not fixed	Not fixed	-do-
10. Don-Bosco Training School, Shillong (1922)	-do-	5 years	-do-	12 years	Class V	-do-

I (Contd.)

TRAINING INSTITUTES, INTAKE CAPACITY, ETC.

SERVICE, ASSAM, SHILLONG.

Nature and titles of the Award, Degree/ Diploma/Certificate	Whether Degree/ Certificate is recognised by the Government	Authority conferring Award	Tuition fee charged	Rate of stipend	Date, month of commencement of course	Sanctioned intake capacity for 1st year course session	Remarks
8	9	10	11	12	13	14	15
Certificate	Yes	Director of Industries, Assam	Nil	Rs.1.50 per day	October	16	
-do-	-do-	-do-	Nil	Rs.30/-p.m.	N.A.	N.A.	
-do-	-do-	-do-	Nil	Rs.42/-p.m.	February	12	
-do-	-do-	-do-	Nil	Rs.1.50 per day	October	16	
Diploma/ Certificate	-do-	Director of Sericulture & Weaving, Assam	Nil	Rs.40-/p.m. for Diploma, Rs.30/-p.m. for Certificate	September	30	
Certificate	-do-	N.A.	Nil	Nil	March	30	
-do-	No	Director of Cottage Industries, Assam	Nil	Nil	August	30	
Diploma	Yes	Managing Committee	Rs.10/-	Nil	Not fixed	30	
Certificate	-do-	State Council of Vocational Training	Rs.5/-	Rs.10/-p.m.	January/April/ July	13	
-do-	-do-	President of Chah Mazdoor	Nil	Nil	April	4	
-do-	-do-	B.D.O., Dev. Block	Nil	Not known	March	10	
Diploma	-do-	Principal	Rs.10/-	Nil	Not fixed	30	
-do-	No	Principal	Rs.8/-	Nil	May	50	
-do-	-do-	M/s Joy Engineering Works, Calcutta	Rs.5/-	Nil	Not fixed	100	
-do-	-do-	School Authority	Rs.10/	Nil	February	8	

APPENDIX

TRADE-WISE (CRAFTS) ANALYSIS AND NUMBER OF

SOURCE :—DIRECTORATE OF NATIONAL EMPLOYMENT

Name of the Institution, address and year of foundation G	Managing authority and type of institution Central/State/ Private/Aided/ Unaided	Duration of the course (years)		Requirements of Admission		Method of admission by Competition by Interview, by Marks obtained
		Institutional Training	Practical Training	Age Limit	Minimum educational require-ments	
1	2	3	4	5	6	7
11. Industrial Training Institute, Jorhat	State	1 year	-do-	16-25 yrs.	Class VIII	-do-
PRINTING TECHNOLOGY						
1. Don Bosco Technical School, Shillong (1922)	Private Unaided	5 years	-do-	12 years	Class V	-do-
2. Karmabir Nabin Chandra Bordoloi Vocational Institute, Nowgong, (1957)	Private Aided	9 months	-do-	15-24 yrs.	M.E. & M.V.	-do-
3. Vocational Collegiate High School, Jorhat	-do-	1 year	3 months	12 years	-do-	-do-
LEATHER CRAFT						
1. Cottage Industries Training Centre, Kalapahar, Gauhati(1954)	State	2 years	Including	16-21 years	Class VIII	Interview
TOY AND DOLL MAKING						
1. Cottage Industries Training Centre, Kalapahar, Gauhati(1954)	-do-	-do-	-do-	-do-	-do-	-do-
2. Government Toy Making Factory, Diphu, United Mikir &North Cachar Hills (1959)	-do-	-do-	-do-	18 years	Class VI	Application
BUCKET MAKING						
1. Carpentry and Bucket Making Training-cum-Production Centre, Raha, Nowgong (1962)	-do-	1½ years	-do-	15 years	Class VIII	Interview
EMBROIDERY						
1. Dibrugarh Chah Mazdoor Kalyan Samaj, Khalihamari, Dibrugarh (1956)	Private Aided	1 year	-do-	12-35 years	Primary	-do-
KNITTING						
1. Dibrugarh Chah Mazdoor Kalyan Samaj, Khalihamari, Dibrugarh (1956)	-do-	2 years	-do-	-do-	-do-	-do-
PAPER MODELLING						
1. Dibrugarh Chah Mazdoor Kalyan Samaj, Khalihamari, Dibrugarh (1956)	-do-	6 months	-do-	-do-	-do-	-do-
BOOK BINDING						
1. Don Bosco Technical School, Shillong (1922)	Private Unaided	5 years	-do-	12 years	Class V	Application & Interview
NAIL MAKING						
1. Cottage Industries Training Centre, Kalapahar, Gauhati(1954)	State	6 months	-do-	16-21 years	Class VIII	Interview

I (Concld.)

TRAINING INSTITUTES, INTAKE CAPACITY, ETC.

SERVICE ASSAM, SHILLONG.

Nature and titles of the Award, Degree/ Diploma/Certificate	Whether Degree/ Certificate is recognised by the Government	Authority cenferring Award	Tuition fee charged	Rate of stipend	Date, month of commencement of course	Sanctioned intake capacity for 1st year course session	Remarks
8	9	10	11	12	13	14	15
Certificate	Yes	N.C.V.T.	Nil	Rs.25/-p.m.	August	80	
Diploma	No	School Authority	Rs.10/-	Nil	February	12	
Certificate	-do-	Governing Body	Rs.5/-	Nil	January/April/ July/October	16	
-do-	Yes	S.C.V.T.	Rs.5/-	Nil	January/April/ July	13	
Certificate	Yes	Director of Industries, Assam.	Nil	Rs.50/-p.m. for S/T &S/C& Rs.40/ and 30/-for general	October	16	
-do-	-do-	-do-	Nil	Rs.1.50 per day	October	16	
-do-	-do-	-do-	Nil	Rs.50/-p.m. for S/T&S/C& Rs.40/- & 30/-for general	October	16	
-do-	-do-	-do-	Nil	Rs.40/-p.m.	April	10	
-do-	-do-	President of Chah Mazdoor	Nil	Rs.20/p.m. for S/T and S/C	April	4	
-do-	-do-	-do-	Nil	-do-	April	4	
-do-	-do-	-do-	Nil	-do-	April	4	
Diploma	No	School Authority	N.A.	N.A.	February	N.A.	
Certificate	Yes	Director of Industries, Assam	Nil	Rs.50/-p.m. for S/T&S/C and Rs.40/-&30 for general	October	8	

APPENDIX II

CONSOLIDATED STATEMENT OF THE CRAFTS OF ASSAM

(Source : Table I (Crafts), Census 1961)

STATE/DIVISION/ DISTRICT	NAMES OF CRAFTS															
	WEAVING				SERICULTURE				CARPENTRY				POTTERY			
	Rural		Urban		Rural		Urban		Rural		Urban		Rural		Urban	
	X	P	Y	P	X	P	Y	P	X	P	Y	P	X	P	Y	P
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
1. Assam State	60	365	24	102	2	19	21	24	145	499	79	348	34	190	22	162
2. Assam Plains Division	55	355	21	97	..	..	1	2	133	479	73	326	30	179	20	147
3. Goalpara	13	52	2	23	..	..	..	..	12	51	3	24	9	74	6	58
4. Kamrup	10	65	5	13	..	..	1	2	10	74	13	44	4	12	4	14
5. Darrang	5	13	1	2	..	..	..	..	11	44	13	42	3	10	5	69
6. Lakhimpur	..	..	..	..	..	..	..	..	13	39	15	85	3	4	3	3
7. Nowgong	5	34	2	6	..	..	..	..	7	23	..	..	2	4	...	..
8. Sibsagar	8	22	3	17	..	..	..	..	73	214	18	96	2	16	..	..
9. Cachar	14	169	8	36	..	..	..	..	7	34	11	35	7	59	2	3
10. Assam Hills Division	5	10	3	5	2	19	20	22	12	20	6	22	4	11	2	15
11. Garo Hills	2	3	1	1	..	..	..	..	..	..	..	..	2	8	2	15
12. United Khasi & Jaintia Hills	1	4	2	4	..	..	..	..	1	2	4	13	1	1	..	..
13. United Mikir & North Cachar Hills	..	..	..	..	2	19	20	22	2	4	1	7	1	2	..	..
14. Mizo Hills	2	3	..	..	..	..	..	..	9	14	1	2	..	..	..	..

	NAMES OF CRAFTS																					
	BLACKSMITHY				CANE AND BAMBOO WORKS				GOLDSMITHY				TAILORING				TINSMITHY				SPIN-NING	
	Rural		Urban		Rural		Urban		Rural		Urban		Rural		Urban		Rural		Urban		Rural	
	X	P	Y	P	X	P	Y	P	X	P	Y	P	X	P	Y	P	X	P	Y	P	X	P
	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39
1.	379	1,129	140	523	6	15	3	6	117	423	90	428	94	382	60	418	4	10	9	27	1	16
2.	224	790	90	405	5	11	3	6	114	415	78	378	91	374	49	314	3	6	9	27	1	16
3.	24	111	17	97	..	..	..	..	21	114	18	89	11	81	7	74	..	..	..	..	..	..
4.	52	194	18	76	.	..	..	..	22	97	14	60	14	57	13	72	..	..	3	7	..	..
5.	26	90	17	66	..	..	2	4	24	85	10	40	23	54	9	22	1	1.	..	..	1	16
6.	25	70	10	33	..	..	..	..	7	22	13	36	9	23	7	24	.	..	..	..	..	..
7.	20	69	6	17	2	5	..	..	8	26	3	21	8	19	2	8	..	..	..	..	..	..
8.	60	197	14	53	1	2	..	..	16	36	8	32	11	31	5	24	..	..	4	12	..	..
9.	17	59	8	63	2	4	1	2	16	35	12	100	15	109	6	90	2	5	2	8	..	..
10.	155	339	50	118	1	4	..	..	3	8	12	50	3	8	11	104	1	4	..	..	..	..
11.	4	12	3	11	1	4	..	.	1	1	2	7	1	6	1	3	..	..	..	..	..	..
12.	14	41	3	4	..	..	..	..	1	5	8	37	..	..	8	87	..	..	..	..	..	..
13.	2	2	1	1	..	..	..	..	1	2	1	5	..	..	1	12	..	..	..	..	..	..
14.	135	284	43	102	..	..	..	..	..	..	1	1	2	2	1	2	1	4	..	..	..	..

Note :—
X denotes No. of Villages
Y denotes No. of Wards or Blocks
P denotes No. of Persons employed

APPENDIX II (Contd.)

CONSOLIDATED STATEMENT OF THE CRAFTS OF ASSAM

STATE/DIVISION/ DISTRICT	NAMES OF CRAFTS																	
	RICE & FLOUR MILLS				SHOE REPAIRING & LEATHER WORKS				BRICK MAKING		BAKERY				UMBRELLA MAKING			
	Rural		Urban		Rural		Urban		Rural		Rural		Urban		Rural		Urban	
	X	P	Y	P	X	P	Y	P	X	P	X	P	Y	P	X	P	Y	P
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
1. Assam State	295	858	65	224	34	128	31	87	10	30	84	396	72	362	3	5	1	3
2. Assam Plains Division	293	854	63	217	33	124	23	56	10	30	76	369	61	311	3	5	1	3
3. Goalpara	19	73	12	37	..	..	5	15	1	4	12	110	12	30	..	..	..	..
4. Kamrup	21	89	11	38	11	61	2	6	2	6	17	79	22	127	..	.	..	..
5. Darrang	30	88	6	17	7	15	2	8	2	4	13	42	10	41	.	..	..	..
6. Lakhimpur	31	72	11	43	3	6	5	10	1	5	10	37	7	27	1	1	..	..
7. Nowgong	39	119	8	33	4	11	3	6	1	3	6	37	1	4	..	..	..	..
8. Sibsagar	120	312	13	41	4	25	4	8	3	8	8	25	3	10	1	1	..	..
9. Cachar	33	101	2	8	4	6	2	3	..	..	10	39	6	72	1	3	1	3
10. Assam Hills Division	2	4	2	7	1	4	8	31	..	..	8	27	11	51	..	..	..	..
11. Garo Hills	..	..	..	..	1	4	1	4	..	..	3	9	1	9	..	..	..	..
12. United Khasi & Jaintia Hills	..	..	..	..	..	..	3	16	..	..	5	18	8	30	..	..	..	..
13. United Mikir & North Cachar Hills	2	4	2	7	..	..	2	6	..	..	..	..	2	12	..	..	..	..
14. Mizo Hills	..	..	..	..	..	..	2	5	..	..	..	..	..	..	..	..	..	..

	NAMES OF CRAFTS																	
	SAWING OF TIMBER				PRINTING PRESS				SOAP MAKING				AGARBATI MAKING		OILSEED PRESSING			
	Rural		Urban		Rural		Urban		Rural		Urban		Rural		Rural		Urban	
	X	P	Y	P	X	P	Y	P	X	P	Y	P	X	P	X	P	Y	P
	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37
1.	13	52	19	46	3	8	18	66	8	26	33	139	8	52	15	37	5	15
2.	13	52	19	46	3	8	14	54	8	26	31	135	8	52	14	35	4	12
3.	.	..	1	4	.	..	2	8	2	6	4	18	..	..	3	10	..	..
4.	2	9	2	7	3	8	2	8	1	9	6	23	..	..	8	14	1	3
5.	2	4	1	4	..	..	..	..	2	4	5	16	..	..	..	..	..	..
6.	2	10	3	10	..	..	1	2	1	2	6	15	..	..	1	4	..	..
7.	..	.	1	1	..	..	1	2	..	..	3	22	5	28	1	2	1	1
8.	4	15	2	4	..	..	1	3	2	5	3	13	3	24	1	5	2	8
9.	3	14	9	16	.	..	7	31	..	..	4	28	..	..	..	..	..	..
10.	..	..	..	..	..	..	4	12	..	..	2	4	.	..	1	2	1	3
11.	..	..	.	..	..	..	1	3	..	..	..	..	..	..	..	..	..	..
12.	..	..	..	..	..	..	3	9	..	..	2	4	..	..	..	..	..	..
13.	..	..	..	.	..	..	..	..	..	..	..	..	..	..	1	2	..	..
14.	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	1	3

APPENDIX II (Contd.)

CONSOLIDATED STATEMENT OF THE CRAFTS OF ASSAM

STATE/DIVISION/ DISTRICT	NAMES OF CRAFTS															
	DAIRY PRODUCTS				ELECTRICAL WORKSHOP				MOTOR REPAIRING				CYCLE & RICKSHAW REPAIRING WORKS			
	Rural		Urban		Rural		Urban		Rural		Urban		Rural		Urban	
	X	P	Y	P	X	P	Y	P	X	P	Y	P	X	P	Y	P
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
1. Assam State	1	4	3	7	16	36	14	60	29	96	95	433	179	436	108	484
2. Assam Plains Division	1	4	..	..	15	34	14	60	27	90	74	372	172	418	106	469
3. Goalpara	..	..	..	..	1	3	..	..	..	..	6	18	18	58	12	30
4. Kamrup	1	4	..	..	1	2	..	..	3	11	13	51	28	70	28	111
5. Darrang	..	..	..	..	6	13	2	5	7	27	7	24	31	76	14	49
6. Lakhimpur	..	..	..	..	2	5	7	31	6	21	20	123	22	52	24	112
7. Nowgong	..	..	..	..	..	..	..	..	3	6	6	33	13	22	3	11
8. Sibsagar	..	..	..	..	5	11	5	24	6	22	15	87	50	98	15	77
9. Cachar	...	..	..	..	..	..	..	..	2	3	7	36	10	42	10	79
10. Assam Hills Division	..	..	3	7	1	2	..	..	2	6	21	61	7	18	2	15
11. Garo Hills	..	..	..	..	..	..	..	..	..	..	2	6	2	5	1	3
12. United Khasi & Jaintia Hills	..	..	3	7	1	2	..	..	2	6	6	28	...	...	..	..
13. United Mikir & North Cachar Hills	..	..	..	..	..	..	..	..	..	..	1	3	5	13	1	12
14. Mizo Hills	..	..	..	..	..	..	..	..	..	..	12	24	..	..	..	..

	NAMES OF CRAFTS																							
	REPAIRING OF MUSICAL INSTRUMENTS				GUN REPAIRING				REPAIRING OF MISC. ARTICLES				RADIO REPAIRING				CARTWHEEL MAKING & REPAIRING				WATCH REPAIRING			
	Rural		Urban		Rural		Urban		Rural		Urban		Rural		Urban		Rural		Urban		Rural		Urban	
	X	P	Y	P	X	P	Y	P	X	P	Y	P	X	P	Y	P	X	P	Y	P	X	P	Y	P
	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41
1.	7	12	16	38	1	2	9	14	7	33	25	65	1	2	9	32	9	25	3	13	14	20	29	71
2.	6	11	14	35	1	2	6	10	4	15	17	43	1	2	8	31	9	25	2	11	14	20	25	64
3.	2	2	3	6	..	..	..	..	..	..	2	2	..	..	..	..	..	..	..	..	1	1	2	4
4.	1	2	4	8	..	..	2	5	..	..	4	11	..	..	4	13	..	..	1	8	1	1	5	12
5.	2	2	4	16	1	2	1	1	1	1	6	15	1	2	1	3	8	22	..	..	3	4	4	8
6.	..	..	..	..	..	..	..	..	1	10	..	..	..	..	1	8	..	..	..	..	1	2	2	6
7.	..	..	..	..	..	..	..	..	1	2	..	..	..	..	1	6	1	3	..	..	1	1	1	4
8.	..	..	..	..	..	..	2	2	1	2	4	14	..	..	..	..	..	..	1	3	4	4	5	15
9.	1	5	3	5	..	..	1	2	..	..	1	1	..	..	1	1	..	..	..	..	3	7	6	15
10.	1	1	2	3	..	..	3	4	3	18	8	22	..	..	1	1	..	..	1	2	..	..	4	7
11.	..	..	..	..	..	..	2	2	1	2	1	2	..	..	1	1	..	..	1	2	..	..	..	..
12.	1	1	2	3	..	..	1	2	2	16	4	10	..	..	..	..	..	..	..	..	..	..	3	4
13.	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	1	3
14.	..	..	..	..	..	..	..	..	..	..	3	10	..	..	..	..	..	..	..	..	..	..	..	..

APPENDIX II (Contd.)

CONSOLIDATED STATEMENT OF THE CRAFTS OF ASSAM

STATE/DIVISION/DISTRICT	NAMES OF CRAFTS													
	BELL METAL & BRASS MAKING INDUSTRY				FIRE-WORKS		BIDI MAKING				BOOK BINDING			
	Rural		Urban		Rural		Rural		Urban		Rural		Urban	
	X	P	Y	P	X	P	X	P	Y	P	X	P	Y	P
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1. Assam State	24	107	35	184	1	1	4	12	3	10	1	1	12	26
2. Assam Plains Division	24	107	35	184	1	1	4	12	3	10	1	1	10	23
3. Goalpara	4	10	12	85	..	..	2	3	1	2	..	..	..	..
4. Kamrup	6	52	4	39	..	..	1	6	1	4	..	..	6	14
5. Darrang	3	4	3	4	..	..	..	..	..	..	..	..	2	5
6. Lakhimpur	..	..	3	11	..	..	..	..	..	..	..	..	..	..
7. Nowgong	..	..	1	4	..	..	..	..	..	..	..	..	..	..
8. Sibsagar	5	15	6	15	..	..	..	..	1	4	..	..	..	..
9. Cachar	6	26	6	26	1	1	1	3	..	..	1	1	2	4
10. Assam Hills Division	..	..	..	..	..	..	..	..	..	..	..	..	2	3
11. Garo Hills	..	..	..	..	..	..	..	..	..	..	..	..	..	..
12. U.K. & J. Hills	..	..	..	..	..	..	..	..	..	..	..	..	2	3
13. U.M. & N.C. Hills	..	..	..	..	..	..	..	..	..	..	..	..	..	..
14. Mizo Hills	..	..	..	..	..	..	..	..	..	..	..	..	..	..

	NAMES OF CRAFTS													
	SWEETMEAT & CONFECTIONERY WORKS				MATCH MAKING		PHOTOGRAPHY & PHOTO BINDING		SPICES MAKING		WIRE MAKING		PAPER MAKING	
	Rural		Urban		Urban		Urban		Urban		Urban		Urban	
	X	P	Y	P	Y	P	Y	P	Y	P	Y	P	Y	P
	16	17	18	19	20	21	22	23	24	25	26	27	28	29
1.	4	12	19	109	1	1	15	48	1	3	1	2	1	4
2.	4	12	13	87	1	1	13	38	1	3	1	2	1	4
3.	..	..	2	9	..	..	1	2	..	..	..	..	..	..
4.	..	..	2	4	..	..	5	12	1	3	1	2	1	4
5.	1	3	2	15	..	..	1	1	..	..	..	..	..	..
6.	2	5	3	39	..	..	3	5	..	..	..	..	..	..
7.	1	4	2	15	..	..	..	..	..	..	..	..	..	..
8.	..	..	..	..	..	..	2	10	..	..	..	..	..	..
9.	..	..	2	5	1	1	1	8	..	..	..	..	..	..
10.	..	..	6	22	..	..	2	10	..	..	..	..	..	..
11.	..	..	1	1	..	..	..	..	..	..	..	..	..	..
12.	..	..	1	8	..	..	2	10	..	..	..	..	..	..
13.	..	..	4	13	..	..	..	..	..	..	..	..	..	..
14.	..	..	..	..	..	..	..	..	..	..	..	..	..	..

APPENDIX II (Contd.)

CONSOLIDATED STATEMENT OF THE CRAFTS OF ASSAM

STATE/DIVISION/ DISTRICT	NAMES OF CRAFTS									
	SODA WATER MAKING		CANDLE MAKING		ROPE MAKING		QUILT MAKING		DYEING	
	Urban		Urban		Rural		Urban		Urban	
	Y	P	Y	P	X	P	Y	P	Y	P
1	2	3	4	5	6	7	8	9	10	11
1. Assam State	4	10	2	5	1	3	5	32	1	4
2. Assam Plains Division	3	7	2	5	1	3	5	32	1	4
3. Goalpara	..	..	1	3	..	..	1	20	..	..
4. Kamrup	2	4	1	2	1	3	..	..	..	..
5. Darrang	..	..	..	..	..	..	3	9	..	..
6. Lakhimpur	..	..	..	..	..	..	..	..	..	..
7. Nowgong	..	..	..	..	..	..	..	..	..	..
8. Sibsagar	1	3	..	..	..	..	..	..	..	..
9. Cachar	..	..	..	..	..	..	1	3	1	4
10. Assam Hills Division	1	3	..	..	..	..	..	..	..	..
11. Garo Hills	..	..	..	..	..	..	..	..	..	..
12. U. K. & J. Hills	1	3	..	..	..	..	..	..	..	..
13. U. M. & N. C. Hills	..	..	..	..	..	..	..	..	..	..
14. Mizo Hills	..	..	..	..	..	..	..	..	..	..

	NAMES OF CRAFTS											
	MANUFACTURE OF MUSICAL INSTRUMENTS				MANUFACTURE OF SPORTS GOODS		TEA-CHEST MAKING		LOOKING GLASS MAKING		NET MAKING	
	Rural		Urban		Urban		Rural		Rural		Urban	
	X	P	Y	P	Y	P	X	P	X	P	Y	P
	12	13	14	15	16	17	18	19	20	21	22	23
1.	2	2	3	6	1	2	4	20	1	4	2	7
2.	2	2	3	6	1	2	4	20	1	4	2	7
3.	..	..	..	..	1	2	..	..	..	..	1	3
4.	..	..	..	..	..	..	..	..	..	..	..	..
5.	..	..	2	4	..	..	..	..	..	..	..	..
6.	..	..	..	..	..	..	1	3	..	..	..	..
7.	..	..	1	2	..	..	..	..	..	..	..	..
8.	2	2	..	..	..	..	..	..	..	..	1	4
9.	..	..	..	..	..	..	3	17	1	4	..	..
10.	..	..	..	..	..	..	..	..	..	..	..	..
11.	..	..	..	..	..	..	..	..	..	..	..	..
12.	..	..	..	..	..	..	..	..	..	..	..	..
13.	..	..	..	..	..	..	..	..	..	..	..	..
14.	..	..	..	..	..	..	..	..	..	..	..	..

APPENDIX II (Concld.)

CONSOLIDATED STATEMENT OF THE CRAFTS OF ASSAM

STATE/DIVISION/ DISTRICT	NAMES OF CRAFTS															
	PIPE MAKING		MANU-FACTURE OF ICE CREAM		TUBE-ROD MAKING		COUNTRY LIQUOR MAKING				IRON & STEEL FURNI-TURE		MANU-FACTURE OF PER-FUMES		FRUIT CANNING	
	Urban		Urban		Rural		Rural		Urban		Urban		Urban		Urban	
	Y	P	Y	P	X	P	X	P	Y	P	Y	P	Y	P	Y	P
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
1. Assam State	1	1	2	7	1	3	10	37	5	81	1	1	1	3	1	2
2. Assam Plains Division	1	1	2	7	1	3	..	..	..	..	1	1	1	3	..	..
3. Goalpara	..	..	1	3	..	..	..	..	..	..	..	..	..	..	..	..
4. Kamrup	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
5. Darrang	..	..	..	..	..	..	..	..	..	..	1	1	1	3	..	..
6. Lakhimpur	..	..	1	4	..	..	..	..	..	..	..	..	..	..	..	..
7. Nowgong	..	..	..	..	1	3	..	..	..	..	..	..	..	..	..	..
8. Sibsagar	1	1	..	..	..	..	..	..	..	..	..	..	..	..	..	..
9. Cachar	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
10. Assam Hills Division	..	..	..	..	..	..	10	37	5	81	..	..	..	..	1	2
11. Garo Hills	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
12. United Khasi & Jaintia Hills	..	..	..	..	..	..	9	29	5	81	..	..	..	..	1	2
13. United Mikir & North Cachar Hills	..	..	..	..	..	..	1	8	..	..	..	..	..	..	..	..
14. Mizo Hills	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	—

APPENDIX

POLICE STATION-WISE LIST OF CRAFTS AND

GOALPARA DISTRICT

Sl. No.	Names of Crafts	Gossain-gaon		Kokra-jhar		Sidli		Bijni		Golok-ganj		Bilasi-para		Dhubri		South-Salmara		Manka-char		North Salmara	
		X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
1	Weaving	..	..	..	..	2	6	1	4	1	3	..	..	..	..	2	8	..	..	..	..
2	Carpentry	1	1	3	11	1	2	1	1	1	6	..	..	..	..	..	..	..	..	..	..
3	Pottery	..	..	..	..	..	..	..	..	..	..	..	..	2	54	1	2	..	..	1	1
4	Blacksmithy	3	7	5	16	..	..	2	9	3	28	1	3	..	..	3	21	1	4	1	2
5	Goldsmithy	2	8	2	5	2	20	2	11	1	4	..	..	1	7	3	37	1	3	..	..
6	Tailoring	..	..	2	5	..	..	..	..	2	50	..	..	..	..	..	..	..	..	..	..
7	Rice & Flour Mills	2	12	2	7	2	9	3	17	5	15	..	..	..	..	..	..	..	..	2	5
8	Shoe Repairing & Leather Works	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
9	Brick Making	...	..	..	..	..	..	..	..	..	..	1	4	..	..	..	..	..	..	..	..
10	Bakery	1	2	2	28	1	12	..	..	3	33	..	..	..	..	2	25	..	..	1	3
11	Printing Press	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
12	Soap Making	..	..	..	..	2	6	..	..	..	..	..	..	..	..	..	..	..	..	..	..
13	Oilseed Pressing	..	..	..	..	1	2	..	..	2	8	..	..	..	..	..	..	..	..	..	..
14	Dairy Products	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
15	Electrical Workshop	1	3	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
16	Motor Repairing	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
17	Cycle & Rickshaw Repairing	1	2	2	3	1	10	2	6	2	10	..	..	..	..	3	4	..	..	..	..
18	Repairing of Musical Instruments	..	...	..	..	..	..	1	1	..	..	..	..	..	..	..	..	..	..	..	..
19	Watch Repairing	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
20	Bell metal & Brass Making Industry	1	4	..	..	..	..	1	4	1	1	..	..	..	..	..	..	..	..	..	..
21	Bidi Making	1	2	..	..	1	1	..	..	..	..	..	..	..	..	..	..	..	..	..	..

Note :—X denotes No. of Villages

P „ No. of Persons employed

VI (i)

NUMBER OF PERSONS EMPLOYED IN PRODUCTION

GOALPARA DIST. (Concld.)						KAMRUP DISTRICT																		Sl. No.
Lakhi-pur		Goal-para		Dudh-nai		Sorbhog		Patachar-kuchi		Barpeta		Bagh-bor		Tara-bari		Barama		Tamul-pur		Nal-bari		Rangia		Hajo		
X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	
23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	1
3	14	2	5	2	12	2	28	..	..	..	..	..	..	4	19	..	..	..	..	..	..	..	..	..	..	1
1	2	..	..	4	28	1	1	..	..	..	..	..	..	4	57	1	3	1	5	..	..	..	..	..	..	2
..	..	2	5	3	12	..	..	..	..	..	..	..	..	3	11	..	..	..	..	..	..	..	..	..	..	3
2	13	1	5	2	3	2	6	4	21	2	9	..	..	9	43	4	12	2	16	2	3	3	9	1	2	4
2	10	..	..	5	9	1	10	1	1	2	2	..	..	5	39	..	..	1	3	1	2	3	23	1	2	5
1	8	2	7	4	11	..	..	1	1	1	2	..	..	..	..	..	..	..	..	..	..	2	6	..	..	6
2	4	..	..	1	4	3	11	5	12	2	7	1	2	2	3	1	3	3	13	1	1	..	..	..	..	7
..	..	..	..	..	..	..	..	2	9	1	1	..	..	2	22	..	..	..	..	..	..	..	..	1	2	8
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	9
..	..	2	7	..	..	1	3	3	12	..	..	..	..	5	34	3	12	..	..	..	..	1	3	..	..	10
..	..	..	..	..	..	..	..	1	3	..	..	..	..	..	..	..	..	..	..	..	..	1	4	..	..	11
..	..	..	..	..		..	..	..	..	1	9	..	..	..	..	..	..	..	..	..	..	..	..	..	..	12
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	2	2	..	..	..	..	..	..	..	..	13
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	1	4	..	..	..	..	..	..	14
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	15
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	16
1	5	1	1	5	17	1	2	5	17	1	1	..	..	4	8	1	4	2	4	1	2	1	1	2	4	17
..	..	..	..	1	1	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	18
..	..	..	..	1	1	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	19
..	..	1	1	..	..	..	..	2	3	1	42	..	..	1	2	..	..	..	..	..	..	..	..	1	3	20
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	21

APPENDIX

POLICE STATION-WISE LIST OF CRAFTS AND

Sl. No.	Names of Crafts	KAMRUP DISTRICT—Concld.										DARRANG DISTRICT											
		Kamal-pur		Boko		Chhay-gaon		Palas-bari		Gau-hati		Paneri		Udal-guri		Kalai-gaon		Dal-gaon		Mangal-dai		Dhekia-juli	
		X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P
1	2	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
1	Weaving	..	..	..	..	..	..	1	13	2	5	..	..	..	..	..	..	..	..	2	3	1	3
2	Carpentry	..	..	1	2	..	..	1	3	1	3	..	..	1	11	..	..	..	..	2	4	2	8
3	Pottery	..	..	..	..	..	..	..	..	1	1	..	..	..	..	..	..	..	..	..	..	..	..
4	Blacksmithy	3	6	3	15	..	..	4	15	13	37	4	15	1	18	2	4	..	..	3	8	..	..
5	Goldsmithy	..	..	..	..	..	..	4	8	3	7	4	22	3	5	1	5	..	..	6	23	1	2
6	Tailoring	2	5	..	..	..	..	6	32	2	11	2	3	3	11	1	2	..	..	9	18	1	2
7	Tinsmithy	..	..	..	..	..	..	..	..	..	..	..	..	1	1	..	..	..	..	..	..	..	..
8	Spinning	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	1	16	..	..
9	Rice & Flour Mills	1	20	..	..	..	..	1	15	1	2	3	10	1	1	2	7	6	17	3	8	..	..
10	Shoe repairing & Leather works	..	..	1	2	..	..	3	17	1	8	1	1	2	3	1	1	..	..	2	8	..	..
11	Brick Making	1	3	..	..	..	..	1	3	..	..	..	..	1	2	..	..	..	..	..	..	..	..
12	Bakery	1	3	..	..	..	..	1	6	2	6	4	8	..	..	1	7	..	..	3	5	..	..
13	Umbrella making	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
14	Sawing of Timber	..	..	..	..	..	..	..	..	2	9	..	..	..	..	..	..	..	..	..	..	..	..
15	Printing Press	..	..	..	..	..	..	1	1	..	..	..	..	..	..	..	..	..	..	..	..	..	..
16	Soap making	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	1	1	..	..
17	Oil Seed pressing	1	4	1	1	..	..	..	..	4	7	..	..	..	..	..	..	..	..	..	..	..	..
18	Electrical Workshop	..	..	..	..	..	..	..	..	1	2	..	..	..	..	..	..	..	..	..	..	2	6
19	Motor Repairing	..	..	..	..	..	..	1	2	2	9	1	5	..	..	..	..	..	..	..	..	1	2
20	Cycle & Rickshaw Repairing	2	4	..	..	1	5	4	12	3	6	3	9	4	10	2	3	..	..	5	6	..	..
21	Repairing of Musical Instruments	..	..	..	..	..	..	..	..	1	2	..	..	..	..	..	..	..	..	..	..	..	..
22	Gun Repairing	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
23	Radio Repairing	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
24	Cart wheel making & repairing	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	2	5	3	7	3	10
25	Watch Repairing	..	..	..	..	..	..	1	1	..	..	..	..	1	1	..	..	..	..	..	..	..	..
26	Repairing of Misc. Articles	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	1	1	..	..
27	Bell metal and Brass making Industry	..	..	..	..	..	..	1	2	..	..	1	2	..	..	..	..	..	..	..	..	..	..
28	Bidi Making	..	..	..	..	..	..	..	..	1	6	..	..	..	..	..	..	..	..	..	..	..	..
29	Sweetmeat and Confectionery Works	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
30	Rope Making	..	..	1	3	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..

II (i) (Contd.)

NUMBER OF PERSONS EMPLOYED IN PRODUCTION

DARRANG DISTRICT (Concld.)								LAKHIMPUR DISTRICT																Sl. No.
Tezpur		Chutia		Behali		Gohpur		Bihpuria		North Lakhimpur		Dhakuakhana		Dhemaji		Sadiya		Dibrugarh		Tinsukia		Moran		
X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	
71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	1
1	4	..	..	1	3	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	1
2	6	4	15	..	..	..	..	7	19	..	..	2	7	..	..	1	1	..	..	..	..	..	..	2
..	..	3	10	..	..	..	..	2	3	..	..	..	..	..	..	..	..	..	..	..	..	..	..	3
4	13	5	8	5	20	2	4	3	19	4	11	3	8	2	3	..	..	.	..	2	7	..	..	4
4	18	5	10	..	..	..	..	1	1	2	16	2	2	..	..	..	..	..	..	..	..	..	..	5
6	8	1	10	..	..	..	..	3	6	..	..	1	1	..	..	..	..	..	..	2	3	..	..	6
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	7
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	8
2	8	4	11	4	13	5	13	1	2	1	2	1	4	1	2	1	3	4	7	3	7	1	3	9
..	..	..	..	1	2		..	1	2	..	..	..	..	..	..	..	..	..	..	..	..	..	..	10
..	..	1	2	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	11
1	3	3	16	1	3	..	..	2	5	1	5	1	2	..	..	..	..	..	..	3	15	..	..	12
..	..	..	..	..	..	..	..	1	1	..	..	..	..	..	..	..	..	..	.	..	..	..	..	13
..	..	2	4	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	14
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1	3	..	..	..	..	..	..	1	2	..	..	..	..	..	..	..	..	..	..	..	..	..	..	16
..	..	..	..	..	..	..	..	..	.	..	..	..	..	..	..	..	..	..	..	..	..	..	..	17
2	3	..	..	2	4	..	..	..	..	..	..	..	..	..	..	..	..	1	1	..	..	..	..	18
2	9	2	8	1	3	..	..	..	..	..	..	..	..	..	..	1	3	..	..	2	7	..	..	19
4	5	7	31	4	9	2	3	6	12	..	..	3	6	.	..	1	3	..	..	1	2	1	1	20
1	1	1	1	..	..	..	..	..	..	..	..	.	..	..	..	..	..	..	..	..	..	..	..	21
1	2	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	22
..	..	1	2	..	..	..	..	..	..	..	..	..	..	..	.	..	..	..	..	..	..	..	..	23
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	24
..	..	1	2	1	1	..	..	..	..	.	..	..	..	..	..	..	..	..	..	..	..	..	..	25
..	..	..	..	..	..	..	..	..	.	..	..	..	..	..	..	..	..	..	..	..	..	..	..	26
..	..	..	..	2	2	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	27
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	28
1	3	..	..	..	..	..	..	1	2	1	3	..	..	..	..	..	..	..	..	..	..	..	..	29
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	30

APPENDIX

POLICE STATION-WISE LIST OF CRAFTS AND

Sl. No.	Names of Crafts	LAKHIMPUR DISTRICT (*Concld*)								NOWGONG DISTRICT													
		Bardubi		Digboi		Jaipur		Marghe-rita		Lahari-ghat		Dhing		Rupahi-hat		Kalia-bor		Mari-gaon		Raha		Now-gong	
		X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P
1	2	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116
1	Weaving	..	..	..	..	..	..	..	..	..	..	..	..	1	7	1	2	1	3	1	4	1	18
2	Carpentry	..	..	..	..	3	12	..	..	..	..	..	..	2	7	3	10	..	..	1	4	..	..
3	Pottery	..	..	1	1	..	..	..	..	..	..	..	..	1	1	..	..	..	..	..	..	..	..
4	Blacksmithy	6	14	..	..	2	4	3	4	2	3	..	..	3	14	1	3	2	6	1	5	3	6
5	Cane and Bamboo works	..	..	..	..	..	..	..	..	..	..	..	..	1	1	..	..	..	..	1	4	..	..
6	Goldsmithy	1	1	..	..	1	2	..	..	..	..	..	..	5	21	..	..	1	1	..	..	1	3
7	Tailoring	3	13	..	..	..	..	..	..	..	..	..	..	2	3	1	3	1	1	..	..	1	2
8	Rice and Flour Mills	6	14	1	1	9	21	2	6	4	9	3	6	5	12	2	4	2	5	1	3	1	1
9	Shoe repairing & Leather works	1	3	..	..	1	1	..	..	..	..	..	..	3	8	..	..	..	..	..	..	..	..
10	Brick making	..	..	..	..	..	..	1	5	..	..	..	..	..	..	..	..	..	..	..	..	1	3
11	Bakery	..	..	..	..	1	5	2	5	..	..	..	..	1	4	..	..	1	2	..	..	..	..
12	Umbrella making	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
13	Sawing of Timber	1	3	..	..	..	..	1	7	..	..	..	..	..	..	..	..	..	..	..	..	..	..
14	Soap making	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
15	Oil Seed Pressing	1	4	..	..	..	..	..	..	1	2	..	..	..	..	..	..	..	..	..	..	..	..
16	Electrical workshop	..	..	..	..	..	..	1	4	..	..	..	..	..	..	..	..	..	..	..	..	..	..
17	Agarbati Making	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
18	Motor repairing	1	1	..	..	..	..	2	10	..	..	..	..	..	..	1	2	..	..	..	..	1	1
19	Cycle & Rickshaw repairing	7	23	..	..	2	3	1	2	1	2	..	..	3	5	2	3	2	3	1	1	3	5
20	Manufacture of Musical Instruments	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
21	Cartwheel Making and Repairing	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
22	Watch Repairing	..	..	..	..	1	2	..	..	..	..	..	..	..	..	1	1	..	..	..	..	..	..
23	Repairing of Misc. Articles	..	..	..	..	..	..	1	10	..	..	..	..	..	..	..	..	..	..	1	2	..	..
24	Bell metal and Brass Making Industry	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
25	Sweetmeat and Confectionery Works	..	..	..	..	..	..	..	..	..	..	..	..	1	4	..	..	..	..	..	..	..	..
26	Tea-Chest Making	..	..	..	..	1	3	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
27	Tube-Rod Making	..	..	..	..	..	..	..	..	1	3	..	..	..	..	..	..	..	..	..	..	..	..

II (i) (Contd.)

NUMBER OF PERSONS EMPLOYED IN PRODUCTION

SIBSAGAR DISTRICT																										Sl. No.
Jamuna-mukh		Lanka		Bokakhat		Dergaon		Golaghat		Majuli		Jorhat		Teok		Titabar		Amguri		Sibsagar		Nazira				
X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P			
117	118	119	120	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140			1
..	..	..	..	..	..	..	..	1	2	2	5	..	..	1	4	..	..	2	3	..	..	2	8			1
1	2	..	..	3	9	6	20	14	33	6	21	10	35	5	13	4	8	11	39	6	17	5	11			2
1	3	..	..	..	..	..	..	1	15	..	..	..	..	..	..	..	..	1	1	..	..	..	..			3
4	13	4	19	..	..	6	14	14	48	8	14	10	52	2	4	2	13	4	15	3	6	6	20			4
..	..	..	..	..	..	..	..	..	..	..	..	1	2	..	..	..	..	..	..	..	..	..	..			5
1	1	..	..	1	1	2	4	5	17	1	1	2	4	..	..	..	..	2	4	..	..	2	2			6
3	10	..	..	..	..	1	1	3	17	1	2	2	7	1	1	..	..	1	1	..	..	1	1			7
3	10	18	69	2	8	1	2	10	44	..	..	8	28	5	11	4	12	14	30	24	55	29	58			8
1	3	..	..	..	..	..	..	1	3	..	..	..	..	.	..	..	..	1	11	..	..	..	..			9
..	..	..	..	..	..	..	..	..	..	..	..	1	3	..	..	1	2	1	3	..	..	..	..			10
3	29	1	2	2	5	..	..	3	14	..	..	1	2	1	2	..	..	..	..	..	..	1	2			11
..	..	..	..	..	..	..	..	1	1	..	..	..	..	..	..	..	..	..	..	..	..	..	..			12
..	..	..	..	..	..	..	..	1	2	..	..	3	13	..	..	..	..	..	..	..	..	..	..			13
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	1	2			14
..	..	..	..	..	..	..	..	..	..	1	5	..	..	..	..	..	..	..	..	..	..	..	..			15
..	..	..	..	1	2	..	..	1	2	..	..	1	2	..	..	1	4	..	..	..	..	1	1			16
1	2	4	26	..	..	1	15	1	6	..	..	..	..	..	..	..	..	1	3	..	..	..	..			17
1	3	..	..	2	7	..	..	..	..	..	..	1	3	1	2	..	..	1	6	..	..	1	4			18
..	..	1	3	1	2	5	9	11	26	4	9	11	21	1	1	1	1	4	6	3	5	3	5			19
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	1	1	..	..	..	..			20
1	3	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..			21
..	..	..	..	..	..	..	..	2	2	..	..	..	..	..	..	..	..	1	1	..	..	..	..			22
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	1	2	..	..	..	..			23
..	..	..	..	..	..	1	1	1	4	..	..	1	2	..	..	..	..	1	2	..	..	..	..			24
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..			25
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..			26
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..			27

APPENDIX

POLICE STATION-WISE LIST OF CRAFTS AND

CACHAR DISTRICT

Sl. No.	Names of Crafts	Sonari		Kati-gora		Bor-khola		Udar-band		Lakhi-pur		Sil-char		Sonai		Karim-ganj		Badar-pur		Pathar-kandi		Rata-bari	
		X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P
1	2	141	142	143	144	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159	160	161	162
1	Weaving	..	..	..	..	1	6	..	..	..	..	2	19	3	19	..	..	2	5	1	8	..	..
2	Carpentry	3	8	..	..	..	..	..	..	..	..	1	3	3	6	1	13	2	12	..	..	..	..
3	Pottery	..	..	1	4	1	8	..	..	..	..	2	34	1	10	..	..	..	..	..	..	..	..
4	Blacksmithy	5	11	1	5	1	2	..	..	..	..	2	5	4	8	3	8	3	19	..	..	..	..
5	Cane & Bamboo Works	..	..	..	..	..	..	..	..	..	..	..	..	..	..	1	1	..	..	1	3	..	..
6	Goldsmithy	1	3	..	..	1	1	1	2	..	..	1	2	5	9	1	4	2	3	..	..	..	..
7	Tailoring	1	1	..	..	6	25	1	34	1	2	..	..	3	13	1	30	1	1	..	..	..	..
8	Tinsmithy	..	..	..	..	..	..	1	3	..	..	..	..	..	..	..	..	..	..	..	..	..	..
9	Rice & Flour Mills	23	64	4	9	2	4	..	..	4	24	2	9	3	8	6	20	3	6	2	2	3	8
10	Shoe repairing & Leather works	2	11	..	..	1	1	..	..	..	..	..	..	1	3	1	1	..	..	..	..	..	..
11	Soap Making	1	3	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
12	Bakery	..	..	1	3	..	..	..	..	1	3	..	..	2	7	1	7	1	6	..	..	1	5
13	Umbrella Making	..	..	..	..	1	3	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
14	Sawing of Timber	..	..	..	..	1	2	..	..	..	..	..	..	..	..	..	..	2	12	..	..	..	..
15	Oil seed pressing	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
16	Electrical Workshop	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
17	Motor Repairing	..	..	..	..	..	..	..	..	..	..	..	..	..	..	1	1	..	..	..	..	1	2
18	Cycle and Rickshaw Repairing	6	13	..	..	1	1	1	2	2	5	1	9	2	4	..	..	2	19	..	..	1	2
19	Manufacture of Musical Instruments	1	1	..	..	..	..	..	..	..	..	..	..	..	..	..	..	1	5	..	..	..	..
20	Watch Repairing	1	1	..	..	..	..	..	..	..	..	1	1	..	..	..	..	1	5	..	..	1	1
21	Repairing of Misc. Articles	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
22	Bell metal and Brass Making Industry	1	6	1	2	..	..	..	..	..	..	2	19	..	..	..	..	1	3	..	..	..	..
23	Fireworks	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..
24	Bidi Making	..	..	..	..	..	..	..	..	..	..	1	3	..	..	..	..	..	..	..	..	..	..
25	Book Binding	..	..	..	..	..	..	1	1	..	..	..	..	..	..	..	..	..	..	..	..	..	..
26	Tea-Chest Making	..	..	..	..	..	..	..	..	..	..	2	8	..	..	..	..	..	..	..	..	..	..
27	Looking-Glass Making	..	..	..	..	..	..	..	..	..	..	1	4	..	..	..	..	..	..	..	..	..	..
28	Country Liqour Making	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..

II (l) (Contd.)

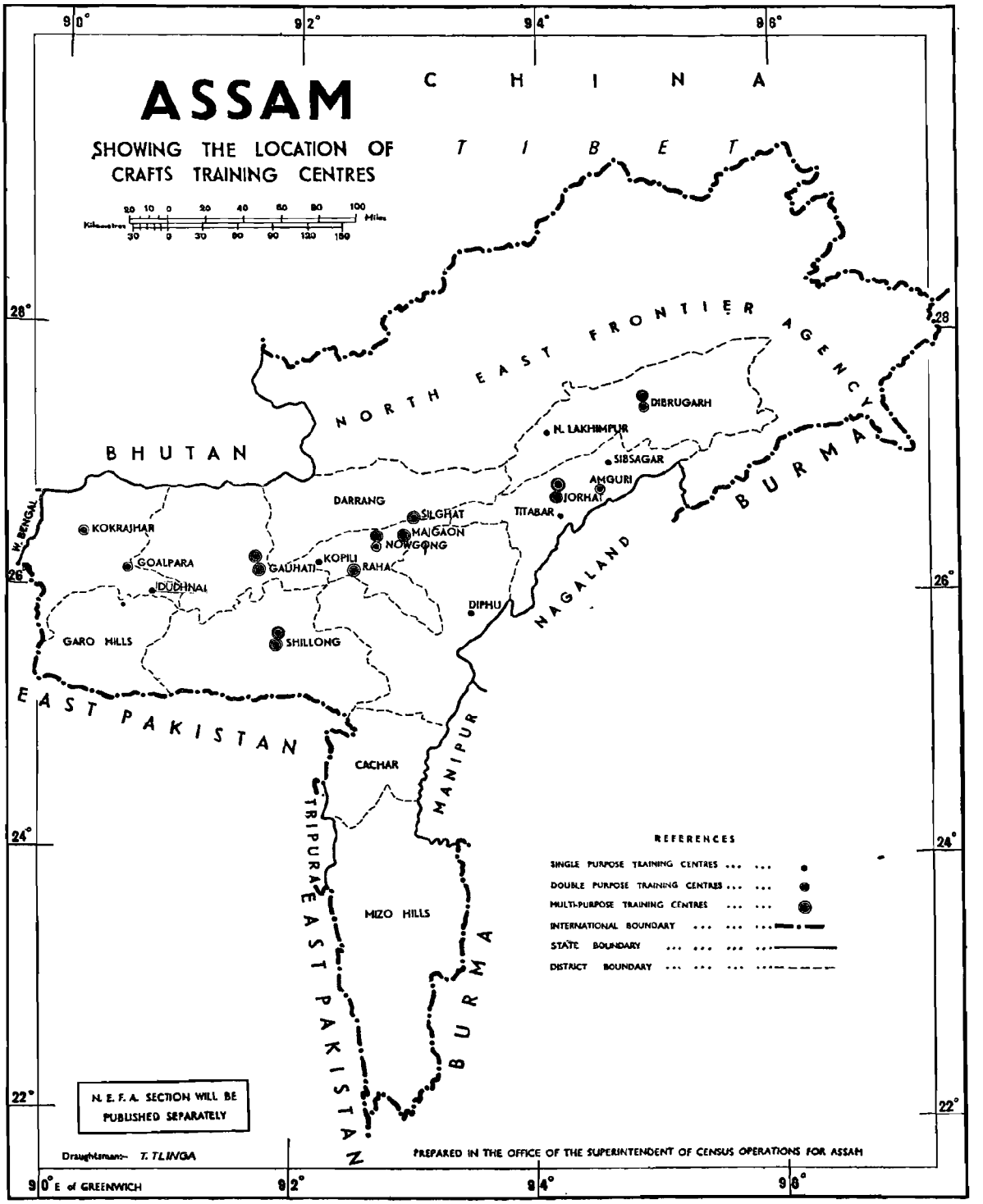
NUMBER OF PERSONS EMPLOYED IN PRODUCTION

GARO HILLS DISTRICT												UNITED KHASI & JAINTIA HILLS DISTRICT								U.M. & N.C.HILLS DISTRICT				Sl. No.
Haila-kandi		Katli-chara		Mauza II		Mauza III		Mauza VII		Mauza IX		Nong-poh		Shillong		Chera-punji		Jowai		Baitha-langso		Howra-ghat		
X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	
163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180	181	182	183	184	195	186	1
4	111	1	1	..	..	1	2	..	..	1	1	..	..	1	4	..	..	..	..	..	..	..	..	1
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2	3	..	..	..	..	..	..	..	..	2	8	..	..	..	..	1	1	..	..	..	..	..	..	3
3	12	..	..	..	..	2	6	..	..	2	6	1	2	6	23	3	9	4	7	..	..	..	..	4
..	..	..	..	1	4	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	5
3	11	2	3	..	..	..	..	..	..	1	1	..	..	..	..	1	5	..	..	..	..	1	2	6
1	1	1	3	..	..	..	..	..	..	1	6	..	..	..	..	..	..	..	..	..	..	..	..	7
..	..	1	2	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	8
4	11	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	1	3	9
1	1	..	..	..	..	..	..	..	..	1	4	..	..	..	..	..	..	..	..	..	..	..	..	10
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3	8	..	..	..	..	..	..	1	3	2	6	..	..	1	4	3	11	1	3	..	..	..	..	12
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	13
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..	..	..	..	..	..	..	..	..	..	..	..	..	..	1	2	..	..	..	..	..	..	..	..	16
..	..	..	..	..	..	..	..	..	..	..	..	..	..	1	3	1	3	..	..	..	..	..	..	17
..	..	..	..	..	..	..	..	..	..	2	5	..	..	..	..	..	..	..	..	..	..	4	12	18
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	1	1	..	..	..	..	19
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	20
..	..	..	..	..	..	..	..	..	..	1	2	..	..	..	..	2	16	..	..	..	..	..	..	21
1	1	1	1	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	22
..	..	1	1	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	23
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	24
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	25
1	9	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	26
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	27
..	..	..	..	..	..	..	..	..	..	..	..	..	..	3	8	5	19	1	2	1	8	..	..	28

APPENDIX II (i) (Concld.)

POLICE STATION-WISE LIST OF CRAFTS AND NUMBER OF PERSONS EMPLOYED IN PRODUCTION

Sl. No.	Names of Crafts	U. M. & N. C. HILLS DISTRICT (Concld.)						MIZO HILLS DISTRICT			
		Bokajan		Diphu		Haflong		Aijal		Lungleh	
		X	P	X	P	X	P	X	P	X	P
1	2	187	188	189	190	191	192	193	194	195	196
1	Weaving	..	..	..	..	..	..	1	1	1	2
2	Sericulture	..	..	..	..	2	19	..	..	..	..
3	Carpentry	2	4	..	..	..	..	6	8	3	6
4	Pottery	1	2	..	..	..	..	..	..	..	..
5	Blacksmithy	..	..	2	2	..	..	93	202	42	82
6	Tailoring	..	..	..	..	..	..	1	1	1	1
7	Tinsmithy	..	..	..	..	..	..	1	4	..	..
8	Rice & Flour Mills	..	..	1	1	..	..	..	..	..	..
9	Cycle & Rickshaw Repairing	1	1	..	..	..	..	..	..	..	..
10	Oil seed pressing	1	2	..	..	..	..	..	..	..	..


ASSAM
SHOWING THE LOCATION OF CRAFTS TRAINING CENTRES
CHINA
TIBET
NORTH EAST FRONTIER AGENCY
BHUTAN
BURMA
NAGALAND
EAST PAKISTAN
MANIPUR
TRIPURA
MIZO HILLS
GARO HILLS
W. BENGAL
DARRANG
DIBRUGARH
N. LAKHIMPUR
SIBSAGAR
AMGURI
JORHAT
TITABAR
SILGHAT
MAJGAON
NOWGONG
KOKRAJHAR
GOALPARA
GAUHATI
KOPILI
RAHA
DUDHNAI
DIPHU
SHILLONG
CACHAR
90°
92°
94°
96°
28°
26°
24°
22°
REFERENCES
SINGLE PURPOSE TRAINING CENTRES
DOUBLE PURPOSE TRAINING CENTRES
MULTI-PURPOSE TRAINING CENTRES
INTERNATIONAL BOUNDARY
STATE BOUNDARY
DISTRICT BOUNDARY
N. E. F. A. SECTION WILL BE PUBLISHED SEPARATELY
Draughtsman:- T. TLINGA
PREPARED IN THE OFFICE OF THE SUPERINTENDENT OF CENSUS OPERATIONS FOR ASSAM
90° E of GREENWICH

APPENDIX

TOWN-WISE LIST OF CRAFTS AND THE NUMBER

Sl. No.	Names of Crafts	GOALPARA DISTRICT											
		Kokrajhar		Bilasipara		Dhubri		Gauripur		Bongaigaon		Goalpara	
		X	P	X	P	X	P	X	P	X	P	X	P
1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	Weaving	..	..	..	..	1	21	..	..	..	..	1	2
2	Sericulture	..	..	..	..	..	..	..	..	..	..	..	..
3	Carpentry	2	5	..	..	..	..	..	..	1	19	..	..
4	Pottery	1	3	..	..	..	..	4	53	1	2	..	..
5	Blacksmithy	4	31	4	11	..	..	3	14	4	32	2	9
6	Goldsmithy	6	16	3	25	..	..	4	13	3	27	2	8
7	Tailoring	1	14	1	3	..	..	2	14	3	43	..	..
8	Tinsmithy	..	..	..	..	..	..	..	..	..	..	..	..
9	Rice & Flour Mills	1	3	1	1	4	11	2	5	3	15	1	2
10	Shoe Repairing & Leather Works	1	3	..	..	..	..	1	1	2	8	1	3
11	Bakery	4	9	2	3	1	3	1	3	2	9	2	3
12	Sawing of Timber	..	..	..	..	..	..	1	4	..	..	..	..
13	Printing Press	..	..	..	..	1	4	..	..	1	4	..	..
14	Soap Making	1	6	..	..	2	9	..	..	1	3	..	..
15	Oil Seed Pressing	..	..	..	..	..	..	..	..	..	..	..	..
16	Motor Repairing	2	8	2	4	2	6	..	..	..	..	..	..
17	Cycle and Rickshaw Repairing	4	10	1	3	2	4	2	6	2	5	1	2
18	Repairing of Musical Instruments	1	1	1	2	..	..	..	..	1	3	..	..
19	Gun Repairing	..	..	..	..	..	..	..	..	..	..	..	..
20	Radio Repairing	..	..	..	..	..	..	..	..	..	..	..	..
21	Cartwheel Repairing	..	..	..	..	..	..	..	..	..	..	..	..
22	Watch Repairing	2	4	..	..	..	..	..	..	..	..	..	..
23	Repairing of Misc. Articles	1	1	..	..	..	..	1	1	..	..	..	..
24	Bell metal & Brass Making Industry	2	4	..	..	..	..	5	61	3	17	2	3
25	Bidi Making	..	..	1	2	..	..	..	..	..	..	..	..
26	Book Binding	..	..	..	..	..	..	..	..	..	..	..	..
27	Sweetmeat & Confectionery Works	..	..	..	..	1	4	..	..	1	5	..	..
28	Photography and Photo Binding	..	..	..	..	..	..	..	..	1	2	..	..
29	Spices Making	..	..	..	..	..	..	..	..	..	..	..	..
30	Wire Making	..	..	..	..	..	..	..	..	..	..	..	..
31	Paper Making	..	..	..	..	..	..	..	..	..	..	..	..
32	Soda Water Making	..	..	..	..	..	..	..	..	..	..	..	..
33	Candle Making	..	..	..	..	..	..	1	3	..	..	..	..
34	Quilt Making	..	..	..	..	..	..	..	..	1	20	..	..
35	Manufacture of Sports Goods	..	..	1	2	..	..	..	..	..	..	..	..
36	Net Making	..	..	..	..	..	..	..	..	1	3	..	..
37	Manufacture of Ice-cream	..	..	..	..	1	3	..	..	..	..	..	..
38	Iron & Steel Furniture	..	..	..	..	..	..	..	..	..	..	..	..
39	Production of Musical Instrument	..	..	..	..	..	..	..	..	..	..	..	..

Note : X denotes No. of Wards or Blocks

P „ No. of Persons Employed

II (ii)

OF PERSONS EMPLOYED IN PRODUCTION

KAMRUP DISTRICT														DARRANG DISTRICT				Sl. No.
Barpeta Road		Barpeta		Nalbari		Rangia		Kamakhya		Gauhati		Pandu		Tangla		Mangaldai		
X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	
15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	1
..	..	..	..	4	11	..	..	..	..	1	2	..	..	..	..	1	2	1
..	..	..	..	..	..	..	..	..	..	1	2	..	..	..	..	..	..	2
..	..	1	2	1	10	1	4	2	8	8	20	..	..	..	..	6	15	3
..	..	..	..	2	8	..	..	..	..	2	6	..	..	..	..	1	9	4
1	8	..	..	3	17	4	17	1	2	9	32	..	..	3	6	6	29	5
1	7	..	..	2	12	1	3	..	..	10	38	..	..	..	..	5	33	6
..	..	..	..	..	..	1	5	1	33	9	30	2	4	..	..	6	12	7
..	..	2	5	1	2	..	..	..	..	..	..	..	..	..	..	..	..	8
2	4	2	7	2	8	..	..	..	..	4	15	1	4	..	..	3	8	9
..	..	..	..	..	..	..	..	..	..	2	6	..	..	..	..	2	8	10
2	5	3	62	4	11	2	8	2	8	8	29	1	4	2	7	5	19	11
..	..	..	..	..	..	..	..	..	..	2	7	..	..	..	..	..	..	12
..	..	2	8	..	..	..	..	..	..	..	..	..	..	..	..	..	..	13
1	3	..	..	..	..	..	..	1	6	4	14	..	..	..	..	1	1	14
..	..	..	..	..	..	..	..	..	..	1	3	..	..	..	..	..	..	15
..	..	..	..	3	16	2	5	2	7	6	23	..	..	1	2	1	3	16
1	5	2	16	1	7	3	13	3	13	17	56	1	1	..	..	6	17	17
..	..	..	..	..	..	..	..	2	4	2	4	..	..	..	..	4	16	18
..	..	..	..	..	..	..	..	..	..	2	5	..	..	..	..	1	1	19
..	..	..	..	..	..	..	..	..	..	4	13	..	..	..	..	..	..	20
..	..	..	..	1	8	..	..	..	..	..	..	..	..	..	..	..	..	21
..	..	1	2	..	..	..	..	..	..	3	7	1	3	..	..	2	4	22
..	..	2	5	..	..	..	..	..	..	2	6	..	..	..	..	..	..	23
..	..	1	2	..	..	..	..	..	..	3	37	..	..	..	..	..	..	24
..	..	..	..	..	..	..	..	..	..	1	4	..	..	..	..	..	..	25
..	..	..	..	1	3	1	2	..	..	4	9	..	..	..	..	..	..	26
..	..	..	..	1	3	..	..	..	..	1	1	..	..	..	..	..	..	27
..	..	..	..	2	4	..	..	..	..	3	8	..	..	..	..	..	..	28
..	..	..	..	..	..	..	..	..	..	1	3	..	..	..	..	..	..	29
..	..	..	..	..	..	..	..	..	..	1	2	..	..	..	..	..	..	30
..	..	..	..	..	..	..	..	..	..	1	4	..	..	..	..	..	..	31
..	..	..	..	..	..	..	..	..	..	1	2	1	2	..	..	..	..	32
..	..	..	..	1	2	..	..	..	..	..	..	..	..	..	..	..	..	33
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	3	9	34
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	35
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	36
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	37
..	..	..	..	..	..	..	..	..	..	..	..	..	..	1	1	..	..	38
..	..	..	..	..	..	..	..	..	..	..	..	..	..	1	1	..	..	39

APPENDIX

TOWN-WISE LIST OF CRAFTS AND THE NUMBER

Sl. No.	Names of Crafts	DARRANG DIST. (Concld.)				LAKHIMPUR DISTRICT							
		Dhekiajuli		Tezpur		North Lakhimpur		Dibrugarh		Chabua		Tinsukia	
		X	P	X	P	X	P	X	P	X	P	X	P
1	2	33	34	35	36	37	38	39	40	41	42	43	44
1	Weaving	..	..	..	..	..	..	..	..	..	..	..	..
2	Carpentry	3	13	4	14	2	14	7	41	1	2	2	9
3	Pottery	1	3	3	57	1	1	..	..	..	..	..	..
4	Blacksmithy	1	4	7	27	1	1	5	26	..	..	1	1
5	Cane & Bamboo Works	..	..	2	4	..	..	..	..	..	..	..	..
6	Goldsmithy	4	6	1	1	1	3	6	16	..	..	1	5
7	Tailoring	3	10	..	..	..	..	5	14	..	..	..	..
8	Tinsmithy	..	..	..	..	..	..	..	..	..	..	..	..
9	Rice & Flour Mills	3	9	..	..	3	12	3	20	..	..	3	8
10	Shoe repairing and Leather works	..	..	..	..	..	..	4	9	..	..	..	..
11	Bakery	1	1	2	14	..	..	4	14	1	7	..	..
12	Sawing of Timber	..	..	1	4	..	..	1	4	..	..	..	..
13	Printing Press	..	..	..	..	1	2	..	..	..	..	..	..
14	Soap making	1	4	3	11	..	..	4	11	..	..	2	4
15	Oil seed pressing	..	..	..	..	..	..	..	..	..	..	..	..
16	Electric workshop	..	..	2	5	..	..	3	14	..	..	1	6
17	Motor Repairing	..	..	5	19	3	8	11	53	..	..	3	33
18	Cycle and rickshaw repairing	3	10	5	22	..	..	14	48	..	..	2	21
19	Repairing of Musical instrument	..	..	..	..	..	..	..	..	..	..	..	..
20	Gun Repairing	..	..	..	..	..	..	..	..	..	..	..	..
21	Radio Repairing	..	..	1	3	..	..	1	8	..	..	..	..
22	Cart wheel making and Repairing	..	..	..	..	..	..	..	..	..	..	..	..
23	Watch Repairing	1	2	1	2	..	..	1	4	..	..	..	..
24	Repairing of Misc. Articles	3	8	3	7	..	..	..	..	..	..	..	..
25	Bell metal and Brass Making Industry	2	2	1	2	1	2	2	9	..	..	..	..
26	Bidi Making	..	..	..	..	..	..	..	..	..	..	..	..
27	Book Binding	..	..	2	5	..	..	..	..	..	..	..	..
28	Sweetmeat and confectionery works	1	13	1	2	..	..	3	39	..	..	..	..
29	Photography and Photo Binding	1	1	..	..	2	3	1	2	..	..	..	..
30	Soda water Making	..	..	..	..	..	..	..	..	..	..	..	..
31	Net Making	..	..	..	..	..	..	..	..	..	..	..	..
32	Manufacture of Ice-cream	..	..	..	..	..	..	..	..	..	..	1	4
33	Manufacture of pipe	..	..	..	..	..	..	..	..	..	..	..	..
34	Manufacture of perfumes	1	3	..	..	..	..	..	..	..	..	..	..
35	Production of musical instrument	1	3	..	..	..	..	..	..	..	..	..	..
36	Rickshaw Body Making	..	..	..	..	..	..	..	..	..	..	..	..

II (ii) (Contd.)

OF PERSONS EMPLOYED IN PRODUCTION

LAKHIMPUR DISTRICT (Concld.)						NOWGONG DISTRICT						SIBSAGAR DISTRICT								Sl. No.
Doom-dooma		Digboi		Naharkatiya		Dhing		Nowgong		Lumding		Dergaon		Golaghat		Jorhat		Mariani		
X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	
45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	1
..	..	..	..	..	..	1	2	1	4	..	..	1	12	1	4	1	1	..	..	1
1	3	2	14	1	4	..	..	..	..	..	..	1	17	4	16	6	30	..	..	2
1	1	..	..	1	1	..	..	..	..	..	..	..	..	..	..	..	..	..	..	3
..	..	2	2	1	1	..	..	6	17	..	..	1	15	5	13	5	17	1	6	4
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	5
1	4	1	1	2	5	..	..	3	21	..	..	1	2	2	12	3	12	1	4	6
1	4	..	..	1	6	..	..	2	8	..	..	..	..	2	11	2	9	..	..	7
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	2	6	..	..	8
..	..	1	2	1	1	2	4	6	29	..	..	1	4	1	4	6	19	..	..	9
..	..	1	1	..	..	..	..	3	6	..	..	..	..	..	..	3	7	..	..	10
1	4	1	2	..	..	..	..	1	4	..	..	..	..	2	6	..	..	1	4	11
2	6	..	..	..	..	..	..	..	..	1	1	1	2	..	..	1	2	..	..	12
..	..	..	..	..	..	..	..	1	2	..	..	..	..	..	..	1	3	..	..	13
..	..	..	..	..	..	..	..	3	22	..	..	..	..	..	..	2	6	..	..	14
..	..	..	..	..	..	..	..	1	1	..	..	..	..	2	8	..	..	..	..	15
1	4	..	..	2	7	..	..	..	..	..	..	..	..	..	..	4	21	..	..	16
1	22	..	..	2	7	..	..	5	30	1	3	..	..	7	32	5	39	..	..	17
4	22	3	11	1	10	..	..	3	11	..	..	1	12	5	13	6	44	1	3	18
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	19
..	..	..	..	..	..	..	..	..	..	..	..	1	1	1	1	..	..	..	..	20
..	..	..	..	..	..	..	..	1	6	..	..	..	..	..	..	..	..	..	..	21
..	..	..	..	..	..	..	..	..	..	..	..	1	3	..	..	..	..	..	..	22
..	..	..	..	1	2	..	..	1	4	..	..	1	4	2	4	2	7	..	..	23
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	4	14	..	..	24
..	..	..	..	..	..	..	..	1	4	..	..	1	1	2	6	3	8	..	..	25
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	1	4	26
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	27
..	..	..	..	..	..	..	..	2	15	..	..	..	..	..	..	..	..	..	..	28
..	..	..	..	..	..	..	..	..	..	..	..	..	..	1	2	1	8	..	..	29
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	1	3	..	..	30
..	..	..	..	..	..	..	..	..	..	..	..	..	..	1	4	..	..	..	..	31
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	32
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	1	1	33
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	34
..	..	..	..	..	..	..	..	1	2	..	..	..	..	..	..	..	..	..	..	35
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	2	6	..	..	36

APPENDIX

TOWN-WISE LIST OF CRAFTS AND THE NUMBER

Sl. No.	Names of Crafts	SIBSAGAR DIST. (Concld.)				CACHAR DISTRICT							
		Sibsagar		Nazira		Lakhi-pur		Silchar		Karim-ganj		Haila-kandi	
		X	P	X	P	X	P	X	P	X	P	X	P
1	2	65	66	67	68	69	70	71	72	73	74	75	76
1	Weaving	..	..	..	..	..	..	3	20	1	2	2	6
2	Sericulture	..	..	..	..	..	..	..	..	..	..	..	..
3	Carpentry	6	31	1	2	..	..	4	16	2	2	4	16
4	Pottery	..	..	..	..	.	..	1	1	..	..	..	..
5	Blacksmithy	2	2	..	..	1	39	2	3	2	6	2	4
6	Cane and Bamboo Works	..	..	..	..	..	..	1	2	..	..	..	..
7	Goldsmithy	..	..	1	2	..	..	5	57	6	15	..	..
8	Tailoring	..	..	1	4	..	..	4	28	1	6	..	..
9	Tinsmithy	1	2	..	..	..	..	2	8	..	..	..	..
10	Rice and Flour Mills	4	8	2	10	..	..	1	4	1	4	..	..
11	Shoe repairing and Leather Works	..	..	1	1	..	..	2	3	..	..	..	..
12	Bakery	..	..	..	..	..	..	3	35	1	17	1	10
13	Umbrella Making	..	..	..	..	..	..	1	3	..	..	..	..
14	Sawing of Timber	..	..	..	..	..	..	5	5	3	8	..	..
15	Printing Press	..	..	..	..	..	..	3	23	3	4	1	4
16	Soap Making	1	7	..	..	..	..	2	12	2	16	..	..
17	Oil seed Pressing	..	..	..	..	..	..	..	..	..	..	..	..
18	Dairy Products	..	..	..	..	..	..	..	..	..	..	..	..
19	Electrical Workshop	.	..	1	3	..	..	..	..	..	..	..	..
20	Motor Repairing	3	16	..	..	..	..	4	22	2	10	1	4
21	Cycle & Rickshaw Repairing	1	1	1	4	..	..	6	43	2	14	1	14
22	Repairing of Musical Instrument	..	..	..	..	..	..	1	2	1	1	1	2
23	Gun Repairing	..	..	..	..	..	..	..	..	..	.	..	..
24	Radio Repairing	..	..	..	..	..	..	1	1	..	..	..	..
25	Cart Wheel Making and Repairing	..	..	..	..	..	..	..	..	..	..	..	..
26	Watch Repairing	..	..	..	..	..	..	3	12	1	1	1	1
27	Repairing of Misc. Articles.	..	..	..	..	.	..	1	1	..	..	..	..
28	Bell Metal and Brass Making Industry	..	..	..	.	..	..	3	10	1	9	1	3
29	Book Binding	..	..	..	..	..	..	2	4	..	..	..	..
30	Sweetmeat and Confectionery Works	..	..	..	..	..	.	1	3	..	.	..	..
31	Match Making	..	..	..	..	..	..	..	..	..	..	..	..
32	Photography and Photo Binding	..	..	..	..	..	..	1	8	..	..	..	..
33	Quilt Making	..	..	..	..	..	..	1	3	..	..	..	..
34	Soda Water Making	..	.	..	..	..	..	..	..	..	..	..	..
35	Country Liquor Making	..	..	..	..	..	..	..	..	..	..	..	..
36	Dyeing	..	..	..	..	..	..	1	4	..	..	..	..
37	Fruit Canning	..	..	..	..	..	..	..	..	..	..	..	..

II (ii) (Concld.)

OF PERSONS EMPLOYED IN PRODUCTION

GARO HILLS				UNITED KHASI & JAINTIA HILLS DISTRICT										U. M. & N. C. HILLS				MIZO HILLS		Sl. No.
Lala		Tura		Shillong		Shillong Cantonnment		Nongthymmai		Mawlai		Jowai		Diphu		Haflong		Aijal		
X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	X	P	
77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	1
2	8	1	1	..		.	..	..	..	..	..	2	4	..	..	..	..	..	..	1
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	20	22	..	..	2
1	1	.	..	2	5	..	..	..	..	..	..	2	8	1	7	..	..	1	2	3
1	2	2	15	.	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	4
1	11	3	11	..	..	..	..	..	..	.	..	3	4	..	..	1	1	43	102	5
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	6
1	28	2	7	3	19	..	..	..	..	..	..	5	18	..	..	1	5	1	1	7
1	56	1	3	3	46	..	.	..	..	..	..	5	41	..	..	1	12	1	2	8
..	..	.		.	..	..	.	.	..	..	..	.	..	..	..	..	..	..	..	9
..	..		.	..	.	..	..	.	..	..	..	..	..	2	7	..	..	..	..	10
..	..	1	4	1	10	..	..	..	..	..	..	2	6	1	2	1	4	2	5	11
1	10	1	9	2	8	..	..	..	..	1	4	5	18	1	7	1	5	..	..	12
..		..	.	.	..	.	..	..	..	.	..	..	.	..	..	..	..	..	..	13
1	3	..	.	..	..	..	..	..	.	..	..	..	..	..	..	..	..	..	..	14
..	..	1	3	3	9	..	..	..	..	..	..	..	..	..	..	..	..	..	..	15
..	..	..	..	1	2	..	..	..	..	..	..	1	2	..	..	..	..	..	..	16
..	..	..	..	..	..	..	..	..	..	..	..	..	..	.	..	..	..	1	3	17
..	..	..	..	2	6	..	..	1	1	..	..	..	..	..	..	..	..	..	..	18
..	.	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	19
..	..	2	6	4	20	1	6	..	..	..	..	1	2	1	3	..	..	12	24	20
1	8	1	3	..	..	.	..	..	..	..	..	..	..	1	12	..	..	..	..	21
..	.	..	..	1	1	..	..	.	..	..	..	1	2	..	..	..	..	..	..	22
1	2	2	2	1	2	..	..	..	..	..	..	..	..	..	..	..	..	..	..	23
..	..	1	1	..	.	..	.	..	..	..	..	..	..	..	..	..	..	..	..	24
..	..	1	2	..	.	..	.	..	..	.	..	..	..	..	..	..	..	..	..	25
1	1	..	..	1	2	..	..	..	..	..	..	2	2	..	..	1	3	.	..	26
..	..	1	2	2	6	1	2	..	..	..	..	1	2	..	..	..	..	3	10	27
1	4	..	..	..	..	..	..	..	..	..	..	.	..	..	.	..	..	..	..	28
..	..	..	..	1	2	..	..	..	..	..	..	1	1	..	..	..	..	..	..	29
I	2	1	1	1	8	..	..	..	..	..	..	..	..	.	..	4	13	..	..	30
1	1	..	..	..	..	..	..	..	..	..	..	..	.	.	..	..	..	..	..	31
..	..	..	..	1	8	..	..	..	..	..	..	1	2	..	..	..	..	..	..	32
..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	..	33
..	..	..	..	1	3	..	..	..	.	..	..	..	..	..	..	..	..	..	..	34
..	..	..	..	2	30	..	..	.	.	2	30	1	21	..	..	..	..	..	..	35
..	..	..	..	..	..	..	..	..	..	..	..	.	..	..	..	..	..	.	..	36
..	..	..	..	..	..	..	..	1	2	..	..	..	..	.	..	..	..	..	..	37

LIST OF AGENTS FOR THE SALE OF GOVERNMENT OF INDIA PUBLICATIONS

(as on 17 February 1964)

AGARTALA—

Laxmi Bhandar Books & Scientific Sales (Rest.)

AGRA—

1. National Book House, Jeoni Mandi. (Reg.)
2. Wadhawa & Co., 45, Civil Lines (Reg.)
3. Banwari Lal Jain, Publishers, Moti Katra (Rest.)
4. English Book Depot, Sadar Bazar, Agra Cantt. (Rest.)

AHMADNAGAR—V.T. Jorakar, Prop. Rama General Stores, Navi Path. (Rest.)

AHMEDABAD—

1. Balgovind Kuber Dass & Co., Gandhi Road. (Reg.)
2. Chandra Kant Chiman Lal Vora, Gandhi Road. (Reg.)
3. New Order Book Co., Ellis Bridge (Reg.)
4. Mahajan Bros., Opp. Khadia Police Gate. (Rest.)
5. Sastu Kitab Ghar, Near Relief Talkies, Patthar Kuva, Relief Road. (Reg.)

AJMER—

1. Book Land, 663, Madar Gate. (Reg.)
2. Rajputana Book House, Station Road (Reg.)
3. Law Book House, 271, Hathi Bhata (Reg.)
4. Vijay Bros., Kutchery Road. (Rest.)
5. Krishna Bros., Kutchery Road. (Rest.)

ALIGARH—Friend's Book House, Muslim University Market. (Reg.)

ALLAHABAD—

1. Superintendent, Printing & Stationary, U.P.
2. Kitabistan, 17-A, Kamala Nehru Road. (Reg.)
3. Law Book Co., Sardar Patel Marg, P. Box 4. (Reg.)
4. Ram Narain Lal Beni Modho, 2-A, Katra Road. (Reg.)
5. Universal Book Co., 20, M.G. Road. (Reg.)
6. The University Book Agency (of Lahore), Elgin Road. (Reg.)
7. Wadhwa & Co., 23, M.G. Marg. (Rest.)
8. Bharat Law House, 15, Mahatma Gandhi Marg. (Rest.)
9. Ram Narain Lal Beni Prashad, 2-A Katra Road. (Rest.)

AMBALA—

1. English Book Depot, Ambala Cantt. (Reg.)
2. Seth Law House, 8719, Railway Road, Ambala Cantt. (Rest.)

AMRITSAR—

1. The Law Book Agency, G.T. Road, Putligarh. (Reg.)
2. S Gupta, Agent, Government Publications, Near P.O Majith Mandi. (Reg.)
3. Amar Nath & Sons, Near P.O. Majith Mandi. (Reg.)

ANAND—

1. Vijaya Stores, Station Road. (Rest.)
2. Charto Book Stall, Tulsi Sadan, Stn. Road. (Rest.)

ASANSOL—

D.N. Roy & R.K. Roy, Booksellers, Atwal Building. (Rest.)

BANGALORE—

1. The Bangalore Legal Practitioner Co-op. Society Ltd. Bar Association Building. (Reg.)
2. S.S. Book Emporium, 118, Mount Joy Road. (Reg.)
3. The Bangalore Press, Lake View, Mysore Road, P.O. Box 507. (Reg.)
4. The Standard Book Depot, Avenue Road. (Reg.)
5. Vichara Sahitya Private Ltd., Balepet. (Reg.)
6. Makkala Pustaka Press, Balamandira, Gandhinagar. (Reg.)
7. Maruthi Book Depot, Avenue Road. (Rest.)
8. International Book House, (P.) Ltd., Mahatma Gandhi Road. (Reg.)
9. Navakarnataka Pubns. Private Ltd., Majestic Circle. (Rest.)

BAREILLY—

Agarwal Brothers, Bara Bazar. (Reg.)

BARODA—

1. Shri Chandrakant Mohan Lal Shah, Raopura (Rest.)
2. Good Companions Booksellers, Publishers & Sub-Agent. (Rest.)
3. New Medical Book House, 540, Madan Zampa Road. (Rest.)

BEAWAR—

The Secretary, S.D. College, Co-operative Stores Ltd. (Rest.)

BELGHARIA—

Granthlok, Antiquarian Booksellers & Publishers (24-Parganas), 5/1, Amlica Mukherjee Road. (Reg.)

BHAGALPUR—

Paper Stationary Stores, D.N. Singh Road. (Reg.)

BHOPAL—

1. Superintendent, State Government Press.
2. Lyall Book Depot, Mohd. Din Bldg., Sultania Road. (Reg.)
3. Delite Books, Opp. Bhopal Talkies. (Rest.)

BHUBANESWAR-

Ekamra Vidyabhaban, Eastern Tower, Room No. 3. (Rest.)

BIJAPUR--

Shri D.V. Deshpande, Recognised Law Booksellers, Prop. Vinod Book Depot, Near Shiralshetti Chowk. (Rest.)

BIKANER—

Bhandani Bros. (Rest.)

BILASPUR—

Sharma Book Stall, Sadar Bazar. (Rest.)

BOMBAY—

1. Superintendent, Printing and Stationery, Queens Road.
2. Charles Lambert and Co., 101, Mahatma Gandhi Road. (Reg.)
3. Co-operator's Book Depot, 5/32, Ahmed Sailor Building, Dadar. (Reg.)
4. Current Book House, Maruti Lane, Raghunath Dadaji Street. (Reg.)
5. Current Technical Literature Co. P. Ltd., India House, 1st Floor. (Reg.)
6. International Book House Ltd., 9 Ash Lane, M.G. Road. (Reg.)
7. Lakkani Book Depot, Girgaum. (Reg.)
8. Elpees Agencies, 24, Bhangwadi, Kalbadevi. (Reg.)
9. P.P.H. Book Stall, 190-B, Khetwadi Main Road. (Reg.)
10. New Book Co., 188-190, Dr. Dadabhai Naoroji Road. (Reg.)
11. Popular Book Depot, Lamington Road. (Reg.)
12. Sunder Das Gian Chand, 601, Girgaum Road, Near Princess Street. (Reg.)
13. D.B. Taraporewala Sons and Co., (P) Ltd., 210, Dr. Dadabhai Naoroji, Road. (Reg.)
14. Thacker and Co., Rampart Row. (Reg.)
15. N.M. Tripathi Private Ltd., Princess Street. (Reg.)
16. The Kothari Book Depot, King Edward Road. (Reg.)
17. P.H. Rama Krishna and Sons, 147, Rajaram Bhuvan, Shivaji Park Road, No. 5. (Rest.)
18. C. Jamnadas and Co., Booksellers, 146-C, Princess Street. (Reg.)
19. Indo Nath & Co., A-6 Daula Nagar, Borivli (Reg.)
20. Minerva Book Shop, Shop No. 1/80, N. Subhas Road. (Reg.)
21. Academic Book Co., Association Building, Girgaum Road. (Rest.)
22. Dominion Publishers, 23, Bell Building, Sir P.M. Road. (Rest.)
23. Bombay National History Society, 91, Walkeshwar Road. (Rest.)
24. Dowamadeo and Co., 16, Naziria Building, Ballard Estate. (Rest.)
25. Asian Trading Co., 310, the Miraball P.B. 1505. (Rest.)

CALCUTTA—

1. Chatterjee and Co., 3/1, Bacharam Chatterjee Lane, (Reg.)
2. Das Gupta and Co. Ltd., 54/3, College Street, (Reg.)
3. Hindu Library, 69-A, Bolaram De Street. (Reg.)
4. S.K. Lahiri and Co. Private Ltd., College St. (Reg.)
5. M.C. Sarkar and Sons Private Ltd., 14, Bankim Chatterjee Street. (Reg.)
6. W. Newman and Co. Ltd., 3,Old Court House St. (Reg.)
7. Oxford Book and Stationery Co., 17, Park Street. (Reg.)
8. R. Chambray and Co., Ltd., Kent House, P. 33, Mission Row Extension (Reg.)
9. S.C. Sarkar and Sons Private Ltd., I.C. College Square (Reg.)
10. Thacker Spink and Co. (1933) P. Ltd., 3, Esplanade East. (Reg.)
11. Firma K.L. Mukhopadhya, 6/1A, Banchha Ram Akrar Lane. (Reg.)
12. K.K. Roy, P. Box, No. 10,210, Calcutta-19. (Rest.)
13. Sm. P.D. Uppadhyay, 77, Muktaram Babu Street. (Rest.)
14. Universal Book Dist., 8/2, Hastings Street. (Rest.)
15. Modern Book Depot, 9, Chowringhee Centre. (Rest.)
16. Soor and Co., 125, Canning Street. (Reg.)
17. S. Bhattacharjee, 49, Dharamtala Street. (Rest.)
18. Mukherjee Library, 10, Sarba Khan Road. (Reg.)
19. Current Literature Co., 208, Mahatma Gandhi Road. (Reg.)
20. The Book Depository, 4/1, Madan Street (1st Floor) (Rest.)
21. Scientific Book Agency, Netaji Subhas Road. (Rest.)
22. Reliance Trading Co., 17/1, Banku Bihari Ghose Lane, District Howrah. (Rest.)
23. Indian Book Dist. Co., 6512, Mahatma Gandhi Road. (Rest.)

CALICUT—

Touring Book Stall (Rest.)

CHANDIGARH—

1. Superintedent, Government Printing and Stationery, Punjab.
2. Jain Law Agency, Flat No. 8, Sector No. 22. (Reg.)
3. Rama News Agency, Bookseller, Sector No. 22 (Reg.)
4. Universal Book Store, Booth, 25, Sector 22 D (Reg.)
5. English Book Shop, 34 Sector 22D (Rest.)
6. Mehta Bros., 15-Z, Sector 22 B. (Rest.)
7. Tandan Book Depot, Shopping Centre, Sector 16 (Rest.)
8. Kailash Law Publishers, Sector 22 B (Rest.)

CHHINDWARA—

The Verma Book Depot. (Rest.)

COCHIN-

Saraswat Corporation Ltd., Palliarakav Road. (Reg.)

CUTTACK—

1. Press Officer, Orissa Sectt.
2. Cuttack Law Times (Reg.)
3. Prabhat K. Mahapatra Mangalabag, P.B. 35 (Reg.)
4. D.P. Sur & Sons, Mangalabag. (Rest.)
5. Utkal Stores, Balu Bazar. (Rest.)

DEHRA DUN--

1. Jugal Kishore & Co., Rajpur Road. (Reg.)
2. National News Agency, Paltan Bazar. (Reg.)
3. Bishan Singh and Mahendra Pal Singh, 318, Chukhuwala. (Reg.)
4. Utam Pustak Bhandar, Paltan Bazar. (Rest.)

DELHI—

1. J.M. Jain & Brothers, Mori Gate. (Reg.)
2. Atma Ram & Sons, Kashmere Gate. (Reg.)
3. Federal Law Book Depot, Kashmere Gate. (Reg.)
4. Bahri Bros., 188, Lajpat Rai Market. (Reg.)
5. Bawa Harkishan Dass Bedi (Vijaya General Agencies) P.B. 2027, Ahata Kedara, Chamalian Road. (Reg.)
6. Book Well, 4, Sant Narankari Colony, P.B. 1565 (Reg.)
7. Imperial Publishing Co., 3,Faiz Bazar, Daryaganj. (Reg.)
8. Metropolition Book Co., 1 Faiz Bazar, (Reg.)
9. Publication Centre, Subzimandi (Reg.)
10. Youngman & Co., Nai Satak (Reg.)
11. Indian Army Book Depot, 3, Daryaganj (Reg.)
12. All India Educational Supply Co., Shri Ram Buildings, Jawahar Nagar. (Rest.)
13. Dhanwant Medical & Law Book House, 1522, Lajpat Rai Market. (Rest.)
14. University Book House, 15, U.B. Bangalore Road, Jawahar Nagar. (Rest.)
15. Law Literature House, 2646, Balimaran (Rest.)
16. Summer Brothers, P.O. Birla Lines. (Rest.)
17. Universal Book & Stationery Co., 16, Netaji Subhas Marg. (Reg.)
18. B. Nath & Bros., 3808 Charkhawalan (Chowari Bazar). (Rest.)
19. Rajkamal Prakashan (P.) Ltd., 8, Faiz Bazar. (Reg.)
20. Premier Book Co., Printers, Publishers & Booksellers, Nai Sarak. (Rest.)
21. Universal Book Traders, 80, Gokhale Market. (Reg.)
22. Tech. & Commercial Book Coy., 75, Gokhale Market. (Rest.)
23. Saini Law Publishing Co., 1416, Chabiganj, Kashmere Gate. (Rest.)
24. G.M. Ahuja, Booksellers & Stationers, 309, Nehru Bazar (Rest.)
25. Sat Narain & Sons, 3141, Mohd. Ali Bazar, Mori Gate. (Reg.)
26. Kitab Mahal (Wholesale Div.) P. Ltd., 28, Faiz Bazar. (Reg.)
27. Hindu Sahitya Sansar, Nai Sarak (Rest.)
28. Munshi Ram Manohar Lal, Oriental Booksellers & Publishers, P.B. 1165, Nai Sarak (Rest.)
29. K.L. Seth, Suppliers of Law, Commercial Tech. Books, Shanti Nagar, Ganeshpura. (Rest.)
30. Adarsh Publishing Service, 5A/10, Ansari Road (Rest.)

DHANBAD—

1. Ismag Co-operative Stores Ltd., P.O. Indian School of Mines (Reg.)
2. New Sketch Press, Post Box 26 (Rest.)

DHARWAR—

1. The Agricultural College Consumers Co-op. Society. (Rest.)
2. Rameshraya Book Depot, Subhas Road. (Rest.)
3. Karnatakaya Sahitya Mandira of Publishers and Booksellers.

ERNAKULAM-

1. Pai & Co., Cloth Bazar Road. (Rest.)
2. South India Traders C/o. Constitutional Journal (Reg.)

FEROZEPUR—

English Book Depot, 78, Jhoke Road (Reg.)

GAUHATI—

Mokshada Pustakalaya (Reg.)

GAYA—

Sahitya Sadan, Gautam Budha Marg. (Reg.)

GHAZIABAD—

Jayana Book Agency (Rest.)

GORAKHPUR—

Vishwa Vidyalaya Prakashan, Nakhes Road. (Reg.)

GUDUR—

The General Manager, The N.D.C. Publishing & Printing Society Ltd. (Rest.)

GUNTUR—

Book Lovers Private Ltd., Kadriguda, Chowrasta (Reg.)

GWALIOR—

1. Superintendent, Printing & Stationery, M.B.
2. Loyal Book Depot, Patankar Bazar, Laskhar. (Reg.)
3. M.C. Deftari, Prop. M.B. Jain Bros., & Book sellers, Sarafa, Laskhar (Rest.)

HUBLI—

Pervaje's Book House, Koppikar Road (Reg.)

HYDERABAD—

1. Director, Government Press
2. The Swaraj Book Depot, Lakdikapul (Reg.)
3. Book Lovers Private Ltd. (Rest.)
4. Labour Law Publications, 873, Sultan Bazar. (Rest.)

IMPHAL—

Tikendia & Sons, Bookseller (Rest.)

INDORE—

1. Wadhawa & Co., 56, M.G. Road. (Reg.)
2. Swarup Brother's Khajuri Bazar. (Rest.)
3. Madhya Pradesh Book Centre, 41, Ahilya Pura (Rest.)
4. Modern Book House, Shiv Vilas Palace (Rest.)
5. Navyug Sahitya Sadan, Publishers & Booksellers, 10, Khajuri Bazar. (Rest.)

JABALPUR—

1. Modern Book House, 286, Jawaharganj. (Reg.)
2. National Book House, 135, Jai Prakash Narain Marg. (Reg.)

JAIPUR—

1. Government Printing and Stationery Department, Rajasthan.

2. Bharat Law House, Booksellers & Publishers, Opp. Prem Prakash Cinema. (Reg.)
3. Garg Book Co., Tripolia Bazar. (Reg.)
4. Vani Mandir, Sawai Mansingh Highway. (Reg.)
5. Kalyan Mal & Sons, Tripolia Bazar. (Rest.)
6. Popular Book Depot, Chaura Rasta. (Reg.)
7. Krishna Book Depot, Chaura Rasta. (Rest.)
8. Dominion Law Depot, Shah Building, P.B. No. 23.

JAMNAGAR—

Swadeshi Vastu Bhandar. (Reg.)

JAMSHEDPUR—

1. Amar Kitab Ghar, Diagonal Road, P.B. 78. (Reg.)
2. Gupta Stores, Dhatkidih. (Reg.)
3. Sanyal Bros., Booksellers & News Agents, Bistapur Market. (Rest.)

JAWALAPUR—

Sahyog Book Depot. (Rest.)

JHUNJHUNU—

1. Shashi Kumar Sarat Chand. (Rest.)
2. Kapram Prakashan Prasaran, 1/90, Namdha Niwas Azad Marg. (Reg.)

JODHPUR—

1. Dwarka Das Rathi, Wholesale Books and News Agents. (Reg.)
2. Kitab-Ghar, Sojati Gate. (Reg.)
3. Choppra Brothers, Tripolia Bazar. (Reg.)

JULLUNDUR—

1. Hazooria Bros., Mai Hiran Gate. (Rest.)
2. Jain General House, Bazar Bansanwala. (Reg.)
3. University Publishers, Railway Road. (Rest.)

KANPUR—

1. Advani & Co., P. Box. 100, The Mall. (Reg.)
2. Sahitya Niketan, Shradhanand Park. (Reg.)
3. The Universal Book Stall, The Mall. (Reg.)
4. Raj Corporation, Raj House P.B. 200, Chowk. (Rest.)

KARUR—

Shri V. Nagaraj Rao, 26, Srinivasapuram. (Rest.)

KODARMA—

The Bhagwati Press, P.O. Jhumri Tilaiya, Dt. Hazaribagh. (Reg.)

KOLHAPUR—

Maharashtra Granth Bhandar, Mahadwar Road. (Rest.)

KOTA—

Kota Book Depot. (Rest.)

KUMTA—

S.V. Kamat, Booksellers & Stationers N. (Kanara.) (Reg.)

LUCKNOW—

1. Soochna Sahitya Depot (State Book Depot).
2. Balkrishna Book Co. Ltd., Hazratganj. (Reg.)
3. British Book Depot, 84, Hazratganj. (Reg.)
4. Ram Advani, Hazratganj, P.B. 154. (Reg.)
5. Universal Publishers, (P.) Ltd., Hazratganj. (Reg.)
6. Eastern Book Co., Lalbagh Road. (Reg.)
7. Civil & Military Educational Stores, 106/B Sadar Bazar. (Rest.)
8. Acquarium Supply Co., 213, Faizabad Road. (Rest.)
9. Law Book Mart, Amin-Ud-Daula Park. (Rest.)

LUDHIANA—

1. Lyall Book Depot, Chaura Bazar. (Reg.)
2. Mohindra Brothers, Katcheri Road. (Rest.)
3. Nanda Stationery Bhandar, Pustak Bazar. (Rest.)
4. The Pharmacy News, Pindi Street. (Rest.)

MADRAS—

1. Superintendent, Government Press, Mount Road.
2. Account Test Institute, P.O. 760, Egmore. (Reg.)
3. C. Subbian Chetty & Co., Triplicane. (Reg.)
4. K. Krishnamurthy, Post Box. 384. (Reg.)
5. Presidency Book Supplies, 8, Pycrofts Road, Triplicane. (Reg.)
6. P. Vardhachary & Co., 8, Linghi Chetty Street. (Reg.)
7. Palani Prachuram, 3, Pycrofts Road, Triplicane. (Reg.)
8. NCBH Private Ltd., 199, Mount Road. (Rest.)
9. V. Sadanand, The Personal Bookshop, 10, Congress Building, 111, Mount Road. (Rest.)

MADURAI—

1. Oriental Book House, 258, West Masi Street. (Reg.)
2. Vivekananda Press, 48, West Masi Street. (Reg.)

MANDYA SUGAR TOWN—

K.N. Narimhe Gowda & Sons. (Rest.)

MANGALORE—

U.R. Shenoye Sons, Car Street, P. Box 128. (Reg.)

MANJESHWAR—

Mukenda Krishna Nayak. (Rest.)

MATHURA—

Rath & Co., Tilohi Building, Bengali Ghat. (Rest.)

MEERUT—

1. Prakash Educational Store, Subhas Bazar. (Reg.)
2. Hind Chitra Press, West Kutchery Road. (Reg.)
3. Loyal Book Depot, Chhipi Tank. (Reg.)
4. Bharat Educational Stores, Chhipi Tank. (Rest.)
5. Universal Book Depot, Booksellers & News Agents. (Rest.)

MONGHYR—

Anusandhan, Minerva Press Building. (Rest.)

MUSSOORIE—

1. Cambridge Book Depot, The Mal. (Rest.)
2. Hind Traders. (Rest.)

MUZAFFARNAGAR—

1. Mittal & Co., 85—C, New Mandi (Rest.)
2. B.S. Jain & Co., 71, Abupura. (Rest.)

MUZAFFARPUR—

1. Scientific & Educational Supply Syndicate. (Reg.)
2. Legal Corner, Tikmanio House, Amgola Road. (Rest.)
3. Tirhut Book Depot. (Rest.)

MYSORE—

1. H. Venkataramiah & Sons, New Statue Circle. (Reg.)
2. Peoples Book House, Opp. Jagan Mohan Palace. (Reg.)
3. Geeta Book House, Booksellers & Publishers Krishnamurthipuram. (Rest.)
4. News Paper House, Lansdowne Building. (Rest.)
5. Indian Mercantile Corporation, Toy Palace Ramvilas. (Rest.)

NADIAD—

R.S. Desay, Station Road. (Rest.)

NAGPUR—

1. Superintendent, Government Press & Book Depot. (Reg.)
2. Western Book Depot, Residency Road. (Reg.)
3. The Assistant Secretary, Mineral Industry Association, Mineral House. (Rest.)

NAINITAL—

Coural Book Depot, Bara Bazar. (Rest.)

NANDED—

1. Book Centre, College Law General Books, Station Road. (Rest.)
2. Hindustan General Stores, Paper & Stationary Merchants P.B. No. 51. (Rest.)
3. Sanjoy Book Agency, Vazirabad. (Rest.)

NEW DELHI—

1. Amrit Book Co., Connaught Circus. (Reg.)
2. Bhawani & Sons, 8. F., Connaught Place. (Reg.)
3. Central News Agency, 23/90, Connaught Circus. (Reg.)
4. Empire Book Depot, 278, Aliganj. (Reg.)
5. English Book Stores, 7—L, Connaught Circus P.O.B.328. (Reg.)
6. Faqir Chand & Sons, 15—A, Khan Market. (Reg.)
7. Jain Book Agency., C—9, Prem House, Connaught Place. (Reg.)
8. Oxford Book & Stationary Co., Scindia House. (Reg.)
9. Ram Krishna & Sons (of Lahore) 16/B, Connaught Place. (Reg.)
10. Sikh Publishing House, 7--C, Connaught Place. (Reg.)
11. Suneja Book Centre, 24/90, Connaught Circus. (Reg.)
12. United Book Agency, 31, Municipal Market, Connaught Circus. (Reg.)
13. Jayana Book Depot, Chhaparwala Kuan, Karol Bagh. (Reg.)
14. Navayug Traders, Desh Bandhu Gupta Road, Dev Nagar. (Reg.)
15. Saraswati Book Depot, 15, Lady Harding Road. (Reg.)
16. The Secretary, Indian Met. Society, Lodi Road. (Reg.)
17. New Book Depot, Latest Book, Periodicals, Sty. & Novelles, P.B. 96, Connaught Place. (Reg.)
18. Mehra Brothers, 50—G, Kalkaji, (Reg.)
19. Luxmi Book Stores, 42, Janpath. (Rest.)
20. Hindi Book House, 82, Janpath. (Rest.)
21. People Publishing House (P.) Ltd., Rani Jhansi Road. (Reg.)
22. R.K. Publishers, 23, Beadon Pura, Karol Bagh. (Rest.)
23. Sharma Brothers., 17, New Market, Moti Nagar. (Rge.)
24. Aapki Dukan, 5/5777, Dev Nagar. (Rest.)
25. Sarvodaya Service, 66A—1, Rohtak Road. P.B. 2521. (Rest.)
26. H. Chandson, P.B. No. 3034. (Rest.)
27. The Secretary, Federation of Association of Small Industry of India, 23-B/2, Rohtak Road. (Rest.)
28. Standard Booksellers & Stationers, Palam Enclave. (Rest.)
29. Laxmi Book Depot, 57, Regarpura. (Rest.)
30. Sant Ram Booksellers, 16 New Municipal Market, Lody Colony.

PANJIM—

1. Singhals Book House. P.O.B. 70, Near the Church (Rest.)
2. Sagoon Gaydev Dhoud, Booksellers, 5—7, Rua, Idc Jameria (Rest.)

PATHANKOT—

The Krishna Book Depot, Main Bazar. (Rest.)

PATIALA—

1. Superintendent, Bhupendra State Press.
2. Jain & Co., 17; Shah Nashin Bazar. (Reg.)

PATNA—

1. Superintendent, Government Printing (Bihar)
2. J.N.P. Agarwal & Co., Padri—Ki—Haveli Ragunath Bhavan (Reg.)
3. Luxmi Trading Co., Padri—Ki—Haveli. (Reg.)
4. Moti Lal Banarsi Dass, Bankipore. (Reg.)
5. Bengal Law House, Chowhatta (Rest.)

PITHORAGARH—

Maniram Punetha & Sons. (Rest.)

PONDICHERRY—

M/s. Honesty Book House, 9, Rue Duplix. (R)

POONA—

1. Deccan Book Stall, Deccan Gymkhana (Reg.)
2. Imperial Book Depot, 266, M.G. Road. (Reg.)
3. International Book Service, Deccan Gymkhana (Reg.)
4. Raka Book Agency, Opp. Natu's Chawl, Near Appa Balwant Chowk. (Reg.)
5. Utility Book Depot, 1339, Shivaji Nagar. (Rest.)

PUDUKOTTAI—

Shri P.N. Swaminathan Sivam & Co., East Main Road. (Rest.)

RAJKOT—

Mohan Lal Dossabhai Shah, Booksellers and Sub-Agents. (Reg.)

RANCHI—

1. Crown Book Depot, Upper Bazar. (Reg.)
2. Pustak Mahal, Upper Bazar. (Rest.)

REWA—

Superintendent, Government State Emporium V.P.

ROURKELA—

The Rourkela Review. (Rest.)

SAHARANPUR—

Chandra Bharata Pustak Bhandar, Court Road. (Rest.)

SECUNDERABAD—

Hindustan Diary Publishers, Market Street. (Reg.)

SILCHAR—

Shri Nishit Sen, Nazirpatti (Rest.)

SIMLA—

1. Superintendent, Himachal Pradesh Government.
2. Minerva Book Shop, The Mall. (Reg.)
3. The New Book Depot, 79, The Mall (Reg.)

SINNAR—

Shri N.N. Jakhadi, Agent, Times of India, Sinnar (Nasik). (Rest.)

SHILLONG—

1. The Officer-in-Charge, Assam Government B.D.
2. Chapala Bookstall, P.B. No. 1. (Rest.)

SONEPAT—

1. United Book Agency (Reg.)

SRINAGAR—

1. The Kashmir Bookshop, Residency Road. (Reg.)

SURAT—

1. Shri Gajanen Pustakalaya, Tower Road. (Rest.)

TIRUCHIRAPALLI—

1. Kalpana Publishers, Wosiur (Reg.)
2. S. Krishnaswami & Co., 35, Subhash Chander Bose Road. (Reg.)
3. Palamiappa Bros. (Rest.)

TRIVANDRUM·

1. International Book Depot, Main Road. (Reg.)
2. Reddear Press & Book Depot, P.B. No.4. (Rest.)

TUTICORIN—

1. Shri K. Thiagarajan, 10-C, French Chapal Road. (Rest.)

UDAIPUR—

1. Jagdish & Co., Inside Surajapole (Rest.)
2. Book Centre, Maharana, Bhopal Consumers' Co-op. Society Ltd. (Rest.)

UJJAIN—

Manak Chand Book Depot, Sati Gate. (Rest.)

VARANASI—

1. Students, Friends & Co., Lanka (Rest.)
2. Chowkhamba Sanskrit Series Office, Gopal Mandir Road., P.B. 8. (Reg.)
3. Glob Book Centre (Rest.)
4. Kohinoor Stores, University Road, Lanka. (Reg.)
5. B.H.U. Book Depot.

VELLORE—

A. Venkatasubhan, Law Booksellers. (Reg.)

VIJAYAWADA—

The Book & Review Centre, Eluru Road, Governpet. (Rest.)

VISAKHAPATNAM—

1. Gupta Brothers, Vizia Building (Reg.)
2. Book Centre, 11/97, Main Road. (Reg.)
3. The Secretary, Andhra University, General Co-op. Stores Ltd. (Rest.)

VIZIANAGARAM—

Sarda & Co. (Rest.)

WARDHA—

Swarajeya Bhandar, Bhorji Market. (Reg.)

For Local Sale

1. Government of India Kitab Mahal, Janpath, Opp. India Coffee House, New Delhi.
2. Government of India Book Depot, 8, Hastings Street, Calcutta.
3. High Commissioner for India in London, India House, London, W.C. 2.

Railway Bookstall Holders

1. S/S. A.H. Wheeler & Co., 15, Elgin Road, Allahabad.
2. Gahlot Bros., K.E.M. Road, Bikaner.
3. Higginbothams & Co. Ltd., Mount Road, Madras.
4. M. Gulab Singh & Sons Private Ltd. Mathura Road, New Delhi

Foreign

1. S/S. Education Enterprise Private Ltd., Kathumandu, (Nepal).
2. S/S. Aktie Bologat, C. E. Fritzes Kungl, Hovobokhandel, Fredsgation-2, Box 1656, Stockholm-16 (Sweden).
3. Reise-und Verkehrsverlag Stuttgart, Post 730, Guten-Bergstra 21, Stuttgart No. 11245, Stuttgart den (Germany West.)
4. Shri Iswar Subramanyam 452, Reversite Driv Apt. 6, New York, 27 NWY.
5. The Proprietor, Book Centre, Lakshmi Mansions, 49, The Mall, Lahore (Pakistan).

On S. and R. Basis

1. The Head Clerk, Govt. Book Depot, Ahmedabad.
2. The Asst Director, Extension Centre, Kapileswar Road, Belgaum.
3. The Employment Officer, Employment Exchange, Dhar.
4. The Asstt. Director, Footwear Extension Centre, Polo Ground No. 1, Jodhpur.
5. The O.I/C., Extension Centre, Club Road, Muzaffarpur.
6. The Director, Indian Bureau of Mines, Govt. of India, Minstry of Mines & Fuel, Nagpur.
7. The Asstt. Director, Industrial Extension Centre, Nadiad (Gujarat).
8. The Head Clerk, Photozincographic Press, 5, Finance Road, Poona.
9. Government Printing and Stationery, Rajkot.
10. The O. I/C. Extension Centre, Industrial Estate, Kokar, Ranchi.
11. The Director, S.I.S.I. Industrial Extension Centre, Udhna, Surat.
12. The Registrar of Companies, Narayani Building, 27, Brabourne Road, Calcutta-1.
13. The Registrar of Companies, Kerala, 50, Feet Road, Ernakulam.
14. The Registrar of Companies, H. No. 3-5-83 Hyderguda, Hyderabad.
15. The Registrar of Companies, Assam, Manipur and Tripura, Shillong.
16. Registrar of Companies, Sunlight Insurance Building, Ajmeri Gate Extension, New Delhi.
17. Registrar of Companies, Punjab and Himachal Pradesh Link Road, Jullundur City.
18. Registrar of Companies, Bihar, Jamal Road, Patna-1.
19. Registrar of Companies, Raj & Ajmer, Shri Kamta Prasad House, 1st Floor, 'C' Scheme, Ashok Marg, Jaipur.
20. The Registrar of Companies, Andhra Bank Building, 6, Linghi Chetty Street, P.B. 1530, Madras.
21. The Registrar of Companies, Mahatma Gandhi Road, West Cott. Building, P.B. 334, Kanpur.
22. The Registrar of Companies, Everest 100, Marine Drive, Bombay.
23. The Registrar of Companies, 162, Brigade Road, Bangalore.
24. The Registrar of Companies, Gwalior.
25. Asst. Director, Extension Centre, Bhuli Road, Dhanbad.
26. Registrar of Companies, Orissa, Cuttack Chandi, Cuttack.
27. The Registrar of Companies, Gujarat State, Gujarat Samachar Buiding, Ahmedabad.
28. Publication Division, Sale Depot, North Block, New Delhi.
29. The Development Commissioner, Small Scale Industries, New Delhi.
30. The O. I/C., University Employment Bureau, Lucknow.
31. O I/C., S.I.S.I. Extension Centre, Malda.
32. O. I/C., S.I.S.I. Extension Centre, Habra, Tabaluria, 24 Parganas.
33. O. I/C., S.I.S.I. Model Carpentry Workshop, Piyali Nagar, P.O. Burnipur.
34. O. I/C., S.I.S.I. Chrontanning Extension Centre, Tangra 33, North Topsia Road, Calcutta-46.
35. O. I/C., S.I.S.I. Extension Centre (Footwear), Calcutta.
36. Asst. Director, Extension Centre, Hyderabad.
37. Asstt. Director, Extension Centre, Krishna Distt. (A.P.)
38. Employment Officer, Employment Exchange, Jhabua.
39. Dy. Director Incharge, S.I.S.I. C/o. Chief Civil Admn. Goa, Panjim.
40. The Registrar of Trade Unions, Kanpur.
41. The Employment Officer, Employment Exchange, Gopal Bhavan, Mornia.
42. The O. I/C., State Information Centre, Hyderabad.
43. The Registrar of Companies, Pondicherry.
44. The Assistant Director of Publicity and Information. Vidhana Saubha (P.B. 271) Bangalore.